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M. Ram Murty

A Second Course in Analysis

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A Second Course in Analysis

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This therefore is Mathematics:
She reminds you of the invisible forms of the soul; She gives life to her own discoveries; She awakens the mind and purifies the intellect; She brings light to our intrinsic ideas.

—Proclus (412–485 CE)

Preface

This book is based on graduate courses given by author since 2007 at Queen's University. The course was meant to bring the typical graduate student (in pure mathematics) up to speed and orient his or her mind toward the research frontier. This is a difficult task. On the one hand, one cannot begin again a program of instruction covered in at least half a dozen undergraduate courses. On the other hand, if the course is pitched at too high a level, there will be a high dropout rate and the student is prone to say "I used to like mathematics until I took this course!" Therefore, a balance must be achieved in the process of instruction.

For the most part, we may assume our student was a teenager when he or she began their college education and drilled in the routine theorems of single variable and multi-variable calculus, real and complex analysis, probability and measure theory. But now at the graduate level, not only is a synthesis of these topics appropriate, but it is essential for the formation of the research mind. Such a mind has to re-examine many of the subtle notions and to have some understanding of how mathematics evolved over the centuries. Perhaps, through such a presentation, the student may find the necessary subliminal suggestions that form the intuitive basis of understanding of mathematics at a deeper level.

In this context, the student will understand that the process of mathematical research follows a basic template involving six steps. First, one **identifies** the need to quantify or make precise a particular idea. This is a process of pattern recognition and requires the student to have a wealth of examples already learned from his or her undergraduate days. Concepts have an evolutionary history, and they often arise as a response to a particular need to understand phenomenon in the physical world. For instance, calculus arose from the study of physical motion. Multivariable calculus arose from a study of motion in three dimensions and a deeper study of electrostatics and fluid dynamics. Mathematical rigor and the need to establish foundations for mathematics arose to avoid pathologies and logical contradictions. Thus, the process of identification leads to a precise mathematical **definition**. This is the source of power in mathematics. What is often referred by the multitude at large in hazy and nebulous terms is made precise and quantified by the mathematician. Having made the definition, the mathematician should then provide

examples in which the mathematical principle that has been identified and defined is highlighted and the student can **analyze** them and see a common thread that unites the seemingly diverse array of examples. Such an intuitive understanding leads to the formulation of a **theorem** or even a collection of theorems. Having the theorem in hand not only allows one to understand all the disparate examples in one unifying comprehensive glance, but it allows for further applications and **extensions**. This is the predictive power of mathematics. The six-step process of higher mathematical learning at the graduate level can then be summarized by the appropriate acronym "**ideate**."

This text assumes therefore that the reader is a graduate student who has had the standard regimen of mathematical courses at the undergraduate level. Often, in such a phase, one gains only information and not intuition that is essential for the research mathematician. It is hoped that this text can form the basis for a semester-long or year-long graduate course in analysis so as to prepare the student for a career as a research mathematician. The author has tested the material over a span of two decades and can certify that indeed the presentation here takes the student through the sacred rites needed to enter the sanctum sanctorum of the temple of mathematics.

Kingston, Ontario M. Ram Murty

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Chapter 1
Background



1.1 Axioms of Set Theory

The concept of number lies at the heart of mathematics. The subject of mathematical analysis, in the formal sense of the word, began when mathematicians initiated the exploration of infinity as a precise mathematical idea. This perhaps is the greatest event of the nineteenth century, on par with the discovery of zero as a mathematical idea. Both zero and infinity are now essential to the understanding of limits and the related processes of real and complex analysis.

Because these two ideas were poorly understood, mathematicians in the nineteenth century introduced the axiomatic method with the theory of sets as the foundation for all mathematics. In this approach, Georg Cantor (1845–1918) was definitely the leading figure. Sadly, his work received considerable opposition at first, but is now universally accepted. It is not an exaggeration to say that research into the foundations of mathematics has given us a larger understanding of the nature of mathematical logic, both its strengths and its limitations.

Set theory is the standard foundation of all mathematics. Mathematical proofs use set-theoretic ideas in one form or another. Below, we give a quick overview of the axioms of set theory so that the student is introduced to the relevant vocabulary. One need not become overly preoccupied with these formalities for otherwise a serious study of these axioms will take us into the realm of mathematical logic which is not the province of this short survey. Rather, we indicate the main themes that are relevant to the later chapters, very much similar to a quick sight-seeing tour of an ancient city. There will undoubtedly be many side streets to explore and since time is of the essence, we need to be content with a one-hour tour highlighting the significant monuments. For a gentle introduction to mathematical logic, we refer the reader to [1].

The basic objects of set theory are **classes**, some of which are **sets**. An informal definition of a set is that it is an unordered collection of objects. The basic relation between them is membership indicated by the symbol $\in$. Thus $x \in A$ means that x is a member of A. Its negation, x is not a member of A, is written $x \notin A$. Already,

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with the notion of a set and the relation of membership, we can easily run into logical contradictions if we do not formulate proper axioms for the construction of sets. The famous "Russell paradox" (named after the British philosopher and mathematician Bertrand Russell (1872–1970)) shows that the "set of all sets" is not a set. Indeed, if it were a set, it would be a proper member of itself by definition, which is a contradiction. The paradox is often posed as the following riddle: a barber shaves anyone who does not shave himself; does the barber shave himself? The contradiction arises from self-reference and so it was quickly realized that to avoid logical contradictions in mathematics, one must formulate precise axioms for the construction of sets. In a formal system of axiomatic set theory, the notions of "class" and "$\in$" are undefined terms. There are several well-developed axiomatic systems such as the Zermelo–Frankel system and the von Neumann–Gödel–Bernays system. Our treatment below follows the latter.

The underlying idea of any of these systems is that we construct new sets from old sets in a systematic manner, with the existence of the empty set as a basic axiom. This is how we avoid the Russel paradox.

The words "and" and "not" are used as propositional connectives. We usually write these words out and do not use symbols for them, though some logicians do. To say that "Proposition P implies Proposition Q" we write "$P \implies Q$". If $P \implies Q$ **and** $Q \implies P$, we write $P \iff Q$. The usual logical quantifiers are: $\forall$ which means "for all" and $\exists$ which means "there exists". The first one is sometimes called the **universal quantifier** and the second one as the **existential quantifier**. The statement, "For all x, the proposition $F(x)$ holds" is denoted $\forall x\ (F(x))$ with the parentheses used to designate the proposition. Such a statement is called a **universal proposition**. A statement of the form "There exists x such that $F(x)$ holds" is written $\exists x\ (F(x))$. Such a statement is called an **existential proposition**. Using class variables like x, y, A, B and so on, together with the basic relation of membership $\in$, quantifiers, connectives and parentheses, we can construct "well-formed" formulas. If variables are quantified, they are called "bound" and should only appear after the quantifier. Variables without quantifiers are called "free". For example,

$$\forall x \exists y((x \in y) \implies (x \in z))$$

is well-formed in which x and y are bound variables and z is free, whereas

$$\exists x((x \in z) \implies \forall z(z \in y))$$

is not well-formed since z precedes its quantifier.

As noted earlier, it is not our goal here to develop axiomatic set theory. On the other hand, the mature mathematician cannot ignore the need for foundations and must have some idea of the basic axioms and notation. So we content ourselves with a cursory overview of the basic axioms and set the stage for both notation and structure of logical proofs.

The essential idea of set theory is that sets are built up in a progressive hierarchy and there are precise rules as to how to form new sets from old sets. We begin with the **axiom of extensionality** which allows us to determine when two sets are equal.

Axiom 1: (extensionality) For any classes A and B, $A = B$ if and only if

$$\forall x(x \in A \iff x \in B).$$

This axiom can be taken as the definition of equality.

Axiom 2: (existence of the empty set) There is a set $\varnothing$ such that

$$\forall x(x \notin \varnothing).$$

The axiom of extensionality implies that the empty set $\varnothing$ is unique.

Axiom 3: (unordered pairs) For all sets x, y there is a set $\{x, y\}$ such that

$$u \in \{x, y\} \iff (u = x \quad \text{or} \quad u = y).$$

If $x = y$, we obtain the **singleton** $\{x\}$: $u \in \{x\}$ if and only if $u = x$. Again, these sets are unique by extensionality.

As defined earlier, a variable is called **free** if there are no quantifiers defining it. Otherwise, the variable is said to be **bounded**. This concept is used in the next axiom and allows us to form new sets from old using well-formed formulas. In fact, the next axiom is more appropriately called an **axiom schema** or a collection of axioms.

Axiom 4: (selection) Given any well-formed formula $P(x)$ containing one free variable x, there is a class A such that for any set x, we have $x \in A$ if and only if $P(x)$ holds.

We often write

$$A = \{x : P(x)\}$$

and read this as "A is the class of all (sets) x such that $P(x)$ holds."

The distinction between sets and classes in the last axiom is important as noted earlier. Without it, we fall into "Russell's paradox". Indeed, let $R = \{x : x \notin x\}$. If R were a set, then $R \in R$ implies $R \notin R$ and if $R \notin R$, then $R \in R$, a contradiction in both cases. Technically, R is not a set but a proper class consisting of all sets which we call "the universe" and the theory contains no logical contradiction. The previous axioms do allow us to construct new sets from old ones in a systematic manner. Thus, a set may consist of other sets that have already been well-defined.

Axiom 5 : (union) For any set x, there is a set $\cup x$ such that $y \in \cup x$ if and only if $y \in z$ for some $z \in x$.

We sometimes write $\cup_{y \in x} y$ for $\cup x$ in the last axiom. Thus,

$$x \cup y = \cup\{x, y\} = \{z : z \in x \quad \text{or} \quad z \in y\}.$$

For any well-formed formula $Q(x)$ with a free variable x,

$$(\forall x \in A)\, Q(x)$$

means

$$(\forall x)(x \in A \implies Q(x)).$$

The relation of A being a subset of B is defined by the symbol

$$(\forall x \in A)\ (x \in B)$$

and we write this as $A \subset B$ or equivalently $B \supset A$.

Axiom 6: (power set axiom) For any set x, there is a set 2^x such that $y \in 2^x$ if and only if $y \subset x$.

The reader should understand that the symbol 2^x is pure symbology for the set of all subsets of x and that no exponentiation is meant since x is a set. It will always be clear from the context which meaning is indicated.

For any two classes A and B, we define the **intersection** as

$$A \cap B = \{x : x \in A \quad \text{and} \quad x \in B\}.$$

If A is a set, then $A \cap B$ is a set for any class B by Axiom 6 and the definition of a set. If x is a non-empty set, its intersection is defined by

$$\cap x = \{y : (\forall u \in x) \implies y \in u\}.$$

This is easily seen to be a set. We can also write $\cap x$ as

$$\bigcap_{y \in x} y.$$

We can use set theory to define an **ordered pair** (x, y) as the set of the form

$$\{\{x\}, \{x, y\}\}.$$

Indeed, for any ordered pair Q, its **first member** is the unique x such that $\{x\} \in Q$ and its **second member** is the unique y such that $Q = \{\{x\}, \{x, y\}\}$ where x is the first member.

A **relation** is any set of ordered pairs. For any relation E, we define the **inverse relation** by

$$E^{-1} := \{(y, x) : (x, y) \in E\}.$$

From the definition of an ordered pair, we are led to the definition of a **function** as a set f of ordered pairs such that $(x, y) \in f$ and $(x, z) \in f$ implies $y = z$. We

write $f(x) = y$ to mean $(x, y) \in f$ for any given function f. Thus, a function is a special kind relation. In other words, all functions are relations but not all relations are functions.

If f is a function, f^{-1} is not necessarily a function. A function f is called **one-to-one** if and only if f^{-1} is a function.

This now allows us to construct the **Cartesian product** of two sets X and Y, denoted $X \times Y$, which consists of ordered pairs (x, y) with $x \in X$ and $y \in Y$.

With the axioms listed so far, we cannot prove the existence of an infinite set. The next axiom allows us to do this along with our earlier axioms.

Axiom 7: (axiom of infinity) There is a set $N \neq \varnothing$ such that for all $x \in N$, $x \cup \{x\} \in N$.

This axiom allows us to construct the natural numbers using the empty set because we can now define inductively the sequence of sets

$$\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}, \ldots$$

and define the non-negative integers using the "zero symbol" 0 to designate the empty set and define

$$1 := \{0\}, \quad 2 := \{0, 1\}, \quad 3 := \{0, 1, 2\}, \ldots$$

We could have also stated Axiom 7 in logical notation as: $\exists N$ such that $\varnothing \in N$ and $\forall x \in N$,

$$x \cup \{x\} \in N.$$

To prevent $x \cup \{x\}$ from being equal to x, and in general to keep order among sets and prevent unwanted closed cycles for the membership relation, we need a further axiom:

Axiom 8: (axiom of regularity) For any set $A \neq \varnothing$, $(\exists X \in A)\ X \cap A = \varnothing$.

This allows us to prove:

Theorem 1.1 *For all x and y, $x \notin x$ and if $x \in y$, then $y \notin x$.*

Proof By Axiom 8, $x \cap \{x\} = \varnothing$ since $x \cap \{x\} = x \cap_{y \in x} y$. If $x \in x$, then clearly $x \in x \cap \{x\}$ which implies $x \in \varnothing$, a contradiction. This proves the first half of our assertion. For the second half, if $x \in y$ then by Axiom 8, the set $\{x, y\}$ has a member disjoint from it. This member cannot be x for otherwise $x \in x$, a contradiction by the first half of the theorem. If the member is y, then $y \cap \{x, y\} = \varnothing$ is also a contradiction since $x \in y$. □

Exercises

1. Recall that an ordered pair (a, b) can be defined as the set $\{\{a\}, \{a, b\}\}$. Show that $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

2. Define the ordered triple (a, b, c) to be the ordered pair $((a, b), c)$ where the ordered pair is defined as usual. Show that

$$(a, b, c) = (a', b', c')$$

   if and only if $a = a'$, $b = b'$ and $c = c'$.

3. If $x \in y \in z \in w$, prove that $w \notin x$.

4. Which of the following are functions and why?

   (a) $\{(1, 2), (2, 3), (3, 1)\}$;
   (b) $\{(1, 2), (2, 3), (2, 1)\}$;
   (c) $\{(2, 1), (3, 1), (1, 2)\}$;
   (d) $\{(x, y) \in \mathbb{R}^2 : x = y^2\}$;
   (e) $\{(x, y) \in \mathbb{R}^2 : y = x^2\}$.

5. For any relation V (that is, any set of ordered pairs), define the **domain** of V as

$$\{x : (x, y) \in V \text{ for some } y\}$$

   and the **range** of V as

$$\{y : (x, y) \in V \text{ for some } x\}.$$

   Find the domain and range for each relation in the previous question whether or not it is a function.

1.2 Constructing Numbers from Sets

Mathematicians and philosophers of the nineteenth century pondered deeply into the nature of a number. The question of "what is a number?" is not a simple one. But since mathematicians decided to give foundations of mathematics using the axiomatic method and sets as the basic building blocks, we are led to define numbers using sets. We follow Richard Dedekind (1831–1916) and Giuseppe Peano (1858–1932) in the following construction. It was as late as 1888 and 1889 when this construction was described in two papers written independently by Dedekind and Peano.

We construct a sequence of sets to represent the natural numbers. As noted earlier, zero is represented by the empty set. We have already described the construction of

the natural numbers using the empty set. For each natural number n, the **successor** of n is denoted $n + 1$ (and sometimes as n') and defined as

$$n \cup \{n\}.$$

Thus, each natural number n is a set with n elements, namely

$$\{0, 1, 2, \ldots, n - 1\}.$$

We designate the set of natural numbers by the symbol $\mathbb{N}$. (It is a matter of personal convenience whether to include zero as a natural number or not. In this discussion, zero is a natural number. In other settings, it may not be. There is no universal convention regarding this and the student is expected to understand depending on the context. Some authors use the term "whole numbers" to indicate that zero is included in the discussion.)

The arithmetic operations on $\mathbb{N}$ are now defined recursively. Addition is defined as a function from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$:

$$+(m, n) := m + n$$

where $m + n$ is defined recursively by $0 + n = n$ and $m' + n = (m + n)'$. A similar definition is given for multiplication $\times$ by defining $0 \times n := 0$ and

$$m' \times n := (m \times n) + n.$$

We also define $m \times n$ as simply mn which is the familiar symbology.

An **equivalence relation** on a set S is a subset R of $S \times S$ satisfying

1. (reflexive axiom) $(a, a) \in R \; \forall a \in S$,
2. (symmetry axiom) $(a, b) \in R \iff (b, a) \in R$,
3. (transitive axiom) $(a, b) \in R$, and $(b, c) \in R$ implies $(a, c) \in R$.

The notion of an equivalence relation is an abstraction of our concept of equality, or at least what we implicitly expect of the notion of equality. It is more suggestive to write the equivalence relation, not as a subset of $S \times S$ as indicated above, but rather more symbolically as $\sim$ so that our axioms become

1. (reflexive) $a \sim a \; \forall a \in S$,
2. (symmetry) $a \sim b$ if and only if $b \sim a$,
3. (transitive) $a \sim b$ and $b \sim c$ implies $a \sim c$.

Equivalence relations play a fundamental role in all of mathematics. They allow us to understand aspects of sets by grouping them using certain properties.

To construct negative integers, we define an **equivalence relation** on $\mathbb{N} \times \mathbb{N}$. We write

$$(m, n) \sim (j, k) \iff m + k = j + n.$$

Intuitively, we think of (m, n) as $m - n$ so that it becomes evident that our definition is now in terms of concepts that have been defined earlier. This is very similar to how the ancients worked with negative numbers that appeared in an equation. They usually moved them to the other side so that the equation became an equation of non-negative numbers. However, with our set-theoretic definition, we have reached a more fundamental and higher level of abstraction. Thus, with our equivalence relation above on the natural numbers, we define the set of integers as the set of equivalence classes of such ordered pairs. It is now easy to see that the following lemma holds:

Lemma 1.1 *If (j, k) is an ordered pair of non-negative integers, then exactly one of the following statements holds:*

(a) *(j, k) is equivalent to $(m, 0)$ for a unique non-negative integer m;*
(b) *(j, k) is equivalent to $(0, m)$ for a unique non-negative integer m;*
(c) *(j, k) is equivalent to $(0, 0)$.*

Sometimes, we denote by $[(j, k)]$ the equivalence class of (j, k). With this lemma in place, we now denote by m the set of pairs of non-negative integers equivalent to $(m, 0)$, by $-m$ the set of pairs equivalent to $(0, m)$ and by 0 the set of pairs equivalent to $(0, 0)$. We denote these equivalence classes by $\mathbb{Z}$. This gives us a set theoretic construction of the set of integers.

We can define the operations of addition and multiplication by setting

$$[(j_1, k_1)] + [(j_2, k_2)] = [(j_1 + j_2, k_1 + k_2)],$$

$$[(j_1, k_1)] \times [(j_2, k_2)] = [(j_1 j_2 + k_1 k_2,\ j_1 k_2 + j_2 k_1)].$$

This latter definition is best understood if we recall that the symbol (j, k) represents $j - k$ so that the left hand side of the above equation is

$$(j_1 - k_1)(j_2 - k_2) = j_1 j_2 + k_1 k_2 - (j_1 k_2 + j_2 k_1).$$

One needs to check that these definitions are "well-defined" in the sense that they are independent of the representatives chosen for the equivalence class. We leave that to the student as an exercise (see exercises below).

In this way, we have now extended the notion of addition and multiplication from the set of natural numbers to the set of integers. Subtraction of integers can be defined by

$$[(j_1, k_1)] - [(j_2, k_2) = [(j_1, k_1)] + (-1)[(j_2, k_2)],$$

where -1 represents the equivalence class (0, 1). All of these definitions correspond to our usual notion of addition, subtraction and multiplication. Their virtue lies in their purely set-theoretic formulation.

We can also order the set of integers in the usual way. Thus,

$$j_1 + k_2 < k_1 + j_2 \iff [(j_1, k_1)] < [(j_2, k_2)]$$

and

$$j_1 + k_2 \leq k_1 + j_2 \iff [(j_1, k_1)] \leq [(j_2, k_2)].$$

This corresponds to our usual notion of "less than" and "less than or equal to".

Finally, we can define the **absolute value** on the set of integers by setting

$$|k| = \begin{cases} k & 0 < k, \\ 0 & k = 0 \\ -k & k < 0 \end{cases}$$

We can now construct the rational numbers $\mathbb{Q}$ from the set of integers. We do this by defining an equivalence relation on the set $\mathbb{Z} \times \mathbb{Z}_{>0}$ by stating that two pairs (j_1, k_1) and (j_2, k_2) are equivalent if and only if $j_1 k_2 = j_2 k_1$. Intuitively, we think of (j_1, k_1) as representing the "fraction" j_1/k_1 and examining what we would mean by $j_1/k_1 = j_2/k_2$ by reducing it to notions already defined. The set of rational numbers $\mathbb{Q}$ is then defined as the set of such equivalence classes.

The expected operations of addition and multiplication are now evident:

$$[(j_1, k_1)] + [(j_2, k_2)] = [(j_1 k_2 + j_2 k_1, k_1 k_2)]$$

$$[(j_1, k_1)][(j_2, k_2)] = [(j_1 j_2, k_1 k_2)].$$

Again, these definitions are easily verified to be well-defined. Finally, we can now define "division". If $[(j_1, k_1)], [(j_2, k_2)] \in \mathbb{Q}$ with $j_2 \neq 0$, we define

$$\frac{[(j_1, k_1)]}{[(j_2, k_2)]} := [(j_1 k_2,\ j_2 k_1)].$$

These operations satisfy the familiar laws of associativity, commutativity and distributivity. Subtraction of rational numbers then can be written as:

$$[(j_1, k_1)] - [(j_2, k_2)] = [(j_1, k_1)] + [(-1, 1)][(j_2, k_2)].$$

The ordering of rational numbers can also be written as

$$[(j_1, k_1)] < [(j_2, k_2)] \quad \iff \quad j_1 k_2 < j_2 k_1.$$

$$[(j_1, k_1)] \leq [(j_2, k_2)] \quad \iff \quad j_1 k_2 \leq j_2 k_1.$$

These definitions agree with our usual notions of ordering of the rational numbers.

Finally, the definition of absolute value can be extended as:

$$|[(j, k)]| = \begin{cases} [(j, k)] & \text{if } [(0, 1)] < [(j, k)], \\ [(0, 1)] & \text{if } [(j, k)] = [(0, 1)], \\ -[(j, k)] & \text{if } [(j, k)] < [(0, 1)]. \end{cases}$$

Again, our familiar properties of the absolute value of rational numbers hold. With this foundational construction in place, we can conveniently represent the equivalence class of (j, k) as simply the fraction j/k and continue to work with these numbers as we were (hopefully) taught from childhood.

In the next sections, we construct the real numbers from this axiomatic framework.

Exercises

1. Let $[(j_1, k_1)]$, $[(j_2, k_2)]$ be two elements of $\mathbb{Z}$. Show that the addition

$$[(j_1, k_1)] + [(j_2, k_2)] = [(j_1 + j_2, k_1 + k_2)]$$

is well-defined. That is, prove that for any $(j_1,', k_1') \in [(j_1, k_1)]$ and $(j_2', k_2') \in [(j_2, k_2)]$, we have that $(j_1' + j_2', k_1' + k_2')$ is equivalent to $(j_1 + j_2, k_1 + k_2)$.

2. For $j_1, j_2, k \in \mathbb{Z}$, prove the distributive law:

$$(j_1 + j_2) \cdot k = j_1 k + j_2 k.$$

3. Show that the relations $<$ and $\leq$ on $\mathbb{Z}$ have the following properties:

(a) $[(0, j)] < [(0, 0)]$ for all $j \in \mathbb{Z}_{>0}$;
(b) $[(0, j)] < [(k, 0)]$ for all $j, k \in \mathbb{Z}_{>0}$;
(c) $[(0, j)] < [(0, k)]$, $j, k \in \mathbb{Z}_{>0}$ if and only if $k < j$;
(d) $[(0, 0)] < [(j, 0)]$ for all $j \in \mathbb{Z}_{>0}$;
(e) $[(j, 0)] < [(k, 0)]$, $j, k \in \mathbb{Z}_{\geq 0}$ if and only if $j < k$;
(f) $[(0, j)] \leq [(0, 0)]$ for all $j \in \mathbb{Z}_{\geq 0}$;
(g) $[(0, j)] \leq [(k, 0)]$ for all $j, k \in \mathbb{Z}_{\geq 0}$;
(h) $[(0, j) \leq [(0, k)]$ for $j, k \in \mathbb{Z}_{\geq 0}$ if and only if $k \leq j$;
(i) $[(0, 0)] \leq [(j, 0)]$ for all $j \in \mathbb{Z}_{\geq 0}$;
(j) $[(j, 0)] \leq [(k, 0)]$ $j, k \in \mathbb{Z}_{\geq 0}$ if and only if $j \leq k$.

1.3 Set-Theoretic Construction of the Real Numbers

The rational number system is insufficient to "measure" all the lengths that arise in the "real world." This was discovered by the ancient school of Pythagoras which viewed the world through a strange mix of mathematics and mysticism. The aphorism "all is number" seems to have been the underlying mantra of the Pythagoreans. In our modern digital world, this mantra imposes its universality more than in any earlier age.

Pythagoras seems to have been a contemporary of the Buddha in India, of Confucius and Lao-Tzu in China. He seems to have traveled widely in Egypt, Babylon and India. The Pythagoreans believed in reincarnation, and this was wedded to their strict adherence to vegetarianism for by eating meat they might be eating a friend! They are credited to the discovery of both the words: philosophy and mathematics. The word "philosophy" literally means "the love of wisdom" and the word "mathematics" means "that which is learned." Philosophy then, for the Pythagoreans, was seen as a byproduct of mathematics.

The celebrated theorem of Pythagoras dealing with the length of the hypotenuse of a right-angled triangle was known to earlier cultures stretching far back in time to Babylon, China and India. It is possible that Pythagoras learned of this theorem during his extensive peregrinations. But the Pythagoreans could see the mysticism inherent in the theorem, for the rule applied to all right angled triangles and there was an element of universality in it and in mathematics in general.

But when the Pythagoreans declared "all is number" they were referring to whole numbers and it is accurate to say that the concept of number had not been clearly articulated at that time. With the underlying idea that whole numbers govern the universe, their theory of ratio and proportion was in for a rude shock. Legend has it that the Pythagorean who discovered that $\sqrt{2}$ is an irrational number (that is, a number which is not rational) was drowned!

The irrationality of $\sqrt{2}$ represents a major episode in the history of mathematics, and we owe this to the unfortunate Pythagorean who lost his life by discovering it. The proof is by contradiction and invokes the rudimentary arithmetical idea of a prime number, a concept first enunciated in Euclid's *Elements*. A natural number is called a **prime number** if it has no proper divisors. Thus, the sequence of prime numbers begins as $2, 3, 5, 7, 11, \ldots$ (The number 1 is not included in the list of prime numbers. It is called a **unit**, a concept which generalizes to rings.) Two natural numbers a and b are said to be **coprime** if they have no common prime factor. When we write a fraction a/b in "lowest terms", we are writing it such that a and b are coprime. We sometimes write $(a, b) = 1$ to indicate that a and b are coprime. This is not to be confused with the concept of an ordered pair introduced earlier.

The irrationality of $\sqrt{2}$ is established by using the **method of contradiction**. It is instructive to see all the details. If $\sqrt{2}$ is a rational number, then we can write

$$\sqrt{2} = \frac{a}{b}, \qquad (a, b) = 1.$$

Upon squaring both sides, we see

$$2b^2 = a^2$$

which implies that a is even since the left hand side of the equation is patently an even number. Writing $a = 2c$, we now get

$$b^2 = 2c^2$$

and so b is even. But this contradicts the coprimality of a and b.

This was the theorem that led to the demise of the heroic Pythagorean for it violated the aphorism "all is number" because the ancients viewed numbers as only whole numbers. They also allowed for rational numbers since these can be viewed as representing ratio and proportion of whole quantities. This innocuous theorem opens the door for the discovery of real numbers. For it signals the lack of the "least upper bound property" in the realm of the rational numbers.

The student should keep in mind that so far, we have only constructed the universe of rational numbers and thus any further definition must be given only in terms of rational numbers. A set A of rational numbers is said to be **bounded** if there is a rational number M such that $|x| \leq M$ for all $x \in A$. An **upper bound** for A is any rational number u such that $x \leq u$ for all $x \in A$. A **lower bound** for A is a rational number ℓ such that $\ell \leq x$ for all $x \in A$. A rational number v is called a **least upper bound** for A if $v \leq u$ for every upper bound u of A. A rational number m is called a **greatest lower bound** for A if $\ell \leq m$ for every lower bound ℓ of A. Thus, a set A is bounded if and only if it has both an upper bound and a lower bound.

Several questions now arise. Given a set A of rational numbers, does it always have a least upper bound? Does it have a greatest lower bound? If so, are they unique?

The theorem discovered by the Pythagoreans can be explained as follows. Nascent in this example are many of the fundamental properties we would expect of the real numbers. Essential to the understanding is the familiar fact that the sequence of rational numbers $1/n$ gets smaller and smaller and tends to zero.

If we consider the set

$$A = \{x \in \mathbb{Q} : x^2 \leq 2\}$$

then A has no least upper bound. Indeed, if u is a least upper bound, $x \leq u$ for any $x \in A$. By definition, u is a rational number and so cannot first equal $\sqrt{2}$. If $u < \sqrt{2}$, then we can choose a natural number N such that

$$\frac{1}{N} < \sqrt{2} - u,$$

so that

$$u + \frac{1}{N} < \sqrt{2}$$

contradicting that u is an upper bound for the set A. If $u > \sqrt{2}$, then we can similarly find N such that

$$u - \frac{1}{N} > \sqrt{2}$$

which again contradicts that u is the least upper bound for A. Thus, A does not have a least upper bound in the realm of rational numbers.

Two ways of constructing the real numbers were proposed independently by Richard Dedekind (1831–1916) and Augustin Cauchy (1789–1857). Each method has its virtues. The method of Dedekind, using what are called **Dedekind cuts** is

closer to the axiomatic foundations we have been discussing and is suggested by the example discussing $\sqrt{2}$ above. The method of Cauchy using what are now called **Cauchy sequences** has a wider applicability. In our example above, an essential role is played by the **ordering** of the rational numbers. The method of Dedekind cuts uses this idea in a fundamental way. It coincides with our intuitive and visual idea of the real numbers as being "points on the number line stretching from minus infinity to plus infinity."

A **Dedekind cut** in $\mathbb{Q}$ is a partition of $\mathbb{Q}$ into two sets A and B with the following properties:

1. A and B are both non-empty;
2. $A \cup B = \mathbb{Q}$;
3. A is "closed downwards", that is, if $r \in A$ and $q < r$, then $q \in A$;
4. B is "closed upwards", that is, if $q \in B$ and $q < r$, then $r \in B$;
5. A contains no greatest element, that is, there is no $q \in A$ such that $r \leq q$ for all $r \in A$.

The set $\mathbb{R}$ of **real numbers** is then defined as the set of all Dedekind cuts.

For example, the irrational number $\sqrt{2}$ is defined by the Dedekind cut

$$\mathbb{Q} = A \cup B, \qquad A = \{x \in \mathbb{Q} : x^2 < 2\} \qquad B = \mathbb{Q} \backslash A.$$

Rational numbers are also defined by Dedekind cuts. Indeed, the rational number q is defined by the cut

$$A = \{x \in \mathbb{Q} : x < q\}, \qquad B = \mathbb{Q} \backslash A.$$

For example, the number zero is represented by $A \cup B$ with A being the set of all nonzero negative rational numbers and B being the set of all nonzero positive rational numbers. As the Dedekind cut $A \cup B$ is uniquely defined by A, with B being the complement of A in the set of rationals, we may as well associate A to each real number, or even think of A **as** the real number. One would then have to define how one would work with this definition from the perspective of addition, multiplication and so on. This is not difficult to do. For instance, we have

$$A_1 < A_2 \iff A_1 \subset A_2, \qquad A_1 \leq A_2 \iff A_1 \subseteq A_2.$$

Addition is simple enough:

$$A_1 + A_2 = \{a_1 + a_2 : a_1 \in A_1, a_2 \in A_2\}.$$

Multiplication is defined (rather cumbersomely) by setting $A_1 \cdot A_2$ to be

$$\begin{array}{ll} \{a_1 \cdot a_2 : a_1 \in A_1, a_2 \in A_2, a_1, a_2 \geq 0\} \cup \{q \in \mathbb{Q} : q \leq 0\} & \text{if } A_1, A_2 \geq 0; \\ -(A_1 \cdot (-A_2)), & \text{if } A_1 \geq 0, A_2 < 0; \\ -((-A_1) \cdot A_2), & \text{if } A_1 < 0, A_2 \geq 0; \\ (-A_1) \cdot (-A_2), & \text{if } A_1, A_2 < 0. \end{array}$$

The absolute value function is then

$$|A| = \begin{cases} A \text{ if } A > 0 \\ 0 \text{ if } A = 0 \\ -A \text{ if } A < 0. \end{cases}$$

One can show that all the usual operations with numbers satisfy the expected properties of commutativity, associativity and distributivity.

By contrast, the method of Cauchy sequences begins with a definition of convergence. In some ways, it is motivated by our discussion of $\sqrt{2}$ earlier. If A is the Dedekind cut representing $\sqrt{2}$, then the least upper bound of A "is" $\sqrt{2}$. Thus, the number $\sqrt{2}$ can be thought of as a limit of a sequence of rational numbers. But there can be many such sequences and so one needs a more formal approach. The underlying observation is that a limit of a sequence of rational numbers need not be rational.

This is perhaps best illustrated by Euler's series for e. Students of calculus are familiar with

$$e = \sum_{j=0}^{\infty} \frac{1}{j!}$$

and the right hand side can be thought of as a limit of a sequence of rational numbers

$$S_n = \sum_{j=0}^{n} \frac{1}{j!}.$$

The irrationality of e is easily proved by the method of contradiction. Suppose that $e = a/b$ for two coprime natural numbers a and b. Then

$$b!e = \sum_{j=0}^{b} \frac{b!}{j!} + \sum_{j=b+1}^{\infty} \frac{b!}{j!} \tag{1.1}$$

and the second term on the right hand side is **strictly less** than the geometric series

$$\sum_{k=1}^{\infty} \frac{1}{(b+1)^k} = \frac{1}{b}$$

which is less than or equal to 1. But the left hand side of (1.1) is a positive integer, the first term on the right hand side of (1.1) is a positive integer and the second term is not an integer, which is a contradiction.

What this example shows is that the "limit" of the sequence of partial sums S_n (each of which is a rational number) is the irrational number e.

The reader is familiar with the usual notion of convergence. We say a sequence of real numbers x_n **converges** to L if given $\epsilon > 0$ there exists N such that

$$|x_n - L| < \epsilon \qquad \forall\, n \geq N.$$

One may want to view real numbers as "limits of sequences of rational numbers." But in our formal construction of numbers, several difficulties arise with this definition. The first is that as we have so far only constructed the rational numbers, we need to take the x_n's to be all rational numbers. Secondly, the limit L need not be a rational number as the example with e shows. This motivates the formal definition of a **Cauchy sequence of rational numbers**.

A sequence q_n in $\mathbb{Q}$ is called a **Cauchy sequence** if given $\epsilon > 0$, there is an N such that

$$|q_n - q_m| < \epsilon \qquad \forall\, m, n \geq N.$$

Now it is easy to see that any convergent sequence is a Cauchy sequence. For future reference, we record this below.

Theorem 1.2 *Any convergent sequence q_n is a Cauchy sequence.*

Proof Suppose that q_n converges to L. Then, choosing N such that

$$|q_n - L| < \frac{\epsilon}{2} \qquad \forall\, n \geq N,$$

we have for all $m, n \geq N$, by the triangle inequality

$$|q_m - q_n| \leq |q_m - L| + |L - q_n| < \epsilon.$$

□

Later, we will show that every Cauchy sequence converges. Thus, a sequence is Cauchy if and only if it is convergent and the two notions are equivalent. But the advantage of the notion of a Cauchy sequence is that the limit value L is not mentioned in the definition.

The construction of the real numbers is now brought about by defining an equivalence relation on the set of all Cauchy sequences of rational numbers. Since we have constructed rational numbers, we are now allowed to construct the set of all Cauchy sequences of rational numbers. On this set, we say two Cauchy sequences $\{q_n\}$ and $\{r_n\}$ are **equivalent** if for any $\epsilon > 0$, there exists a natural number N such that

$$|q_n - r_n| < \epsilon \qquad \forall\, n \geq N.$$

This is easily seen to define an equivalence relation on the set of all Cauchy sequences of rational numbers. The set of real numbers $\mathbb{R}$ is then defined as the set of all equivalence classes. We will denote the equivalence class of the sequence $\{q_n\}$ by $[\{q_n\}]$.

We leave as an exercise (see Exercise 4 below) to verify that if $\{q_n\}$ and $\{r_n\}$ are two Cauchy sequences, then so are $\{q_n + r_n\}$ and $\{q_n \cdot r_n\}$. Thus, the usual operations of addition and multiplication are easily defined:

$$[\{q_n\}] + [\{r_n\}] := [\{q_n + r_n\}], \qquad [\{q_n\}] \cdot [\{r_n\}] := [\{q_n \cdot r_n\}].$$

One needs to verify that these are **well-defined** by checking that the definition does not depend on the choice of the representative of the equivalence class.

The order relation of the real numbers can also be defined. Thus, viewing real numbers as equivalence classes or Cauchy sequences of rational numbers, we define $[\{q_n\}] < [\{r_n\}]$ if there is an N such that $q_n < r_n$ for all $n \geq N$. We also write $[\{q_n\}] \leq [\{r_n\}]$ if either $[\{q_n\}] < [\{r_n\}]$ or $[\{q_n\}] = [\{r_n\}]$. Again, these order relations have the expected properties.

Finally, the absolute value function can also be defined. We set

$$|[\{q_n\}]| := [\{|q_n|\}]$$

and the reader can easily verify that (indeed) the right hand side is a Cauchy sequence if $\{q_n\}$ is Cauchy.

The absolute value has the usual properties, the most notable being the **triangle inequality**:

$$|[\{q_n\}] + [\{r_n\}]| \leq |[\{q_n\}]| + |[\{r_n\}]|.$$

Having given this formal definition of the real numbers using Cauchy sequences, it is convenient to drop the sequence notation and just represent the real numbers as single letters. Thus, if $a, b \in \mathbb{R}$, the triangle inequality is the familiar

$$|a + b| \leq |a| + |b|.$$

It is now convenient to visualize the set of real numbers in the usual way as points of the line stretching from $-\infty$ to $+\infty$ and introduce the following notation. If a and b are real numbers such that $a \leq b$, then the **interval** $[a, b]$ indicates the set of real numbers $\{x : a \leq x \leq b\}$. The **half-open** interval $(a, b]$ represents $\{x : a < x \leq b\}$. Similarly $[a, b)$ means $\{x : a \leq x < b\}$ and the **open interval** (a, b) is $\{x : a < x < b\}$.

By its very construction, the set of real numbers has the property of **completeness** in that every Cauchy sequence converges. The reader will also observe that the construction depended only on the property of the absolute value viewed as a **metric** to measure the distance between two rational numbers. This signals a wider applicability of the method of Cauchy sequences valid more generally in metric spaces which we discuss later.

With the construction of the real numbers, we have restored the least upper bound property and can now define that a sequence of numbers (rational or real) x_n **converges** to a number $L \in \mathbb{R}$ if given any $\epsilon > 0$, there is an N such that

$$|x_n - L| < \epsilon, \qquad \forall\, n \geq N.$$

A non-empty subset A of $\mathbb{R}$ is said to be **bounded below** if there is a number $s \in \mathbb{R}$ such that $s \leq a$ for all $a \in A$. Any such number s is called a **lower bound** for A. Analogously, A is said to be **bounded above** if there is a number $t \in \mathbb{R}$ such that $a \leq t$ for all $a \in A$. Any such number t is called an **upper bound** for A. The set A is said to be **bounded** if it is both bounded below and bounded above and **unbounded** if it is not bounded.

The construction of the real numbers using Dedekind cuts shows that every non-empty subset of $\mathbb{R}$ that is bounded above has a **least upper bound** denoted sup A. The least upper bound is also called the **supremum**. Every non-empty subset which is bounded below has a **greatest lower bound** denoted inf A. The greatest lower bound is also called the **infimum**. It is clear that if A and B are non-empty subsets of $\mathbb{R}$ such that $A \subseteq B$, then sup $A \leq$ sup B (see Exercise 5). The following theorem gives a convenient characterization of the supremum.

Theorem 1.3 *Let A be a non-empty subset of $\mathbb{R}$ that is bounded above. Then $s =$ sup A if and only if*

(i) $a \leq s$ for all $a \in A$ and
(ii) for any $\epsilon > 0$, $A \cap (s - \epsilon, s] \neq \varnothing$.

Proof Let $s = \sup A$. Then s is an upper bound for A so that (i) holds. Now if (ii) were false, there would be an $\epsilon > 0$ such that $A \cap (s - \epsilon, s] = \varnothing$. But then $s - \epsilon$ would also be an upper bound for A, a contradiction. Conversely, suppose that (i) and (ii) hold. Then (i) implies s is an upper bound for A. If s were not the least upper bound for A, there would be a $t < s$ such that $a \leq t$ for all $a \in A$. Taking $\epsilon = (s - t)/2 > 0$ in (ii) we get that $A \cap (s - \epsilon, s] \neq \varnothing$. But $s - \epsilon = t + \epsilon$ so that $A \cap (t + \epsilon, s] \neq \varnothing$ which is a contradiction since $a \leq t$ for all $a \in A$. □

Theorem 1.3 can also be stated as follows. If A is a non-empty subset of $\mathbb{R}$ that is bounded above and $s = \sup A$, then either $s \in A$ or $A \cap (s - \epsilon, s) \neq \varnothing$ for all $\epsilon > 0$. A similar theorem can be written for the infimum of a set (see Exercise 6.)

Exercises

1. If p is a prime number, show that $\sqrt{p}$ is irrational.

2. Show that
$$\sum_{n=2}^{\infty} \frac{1}{n!} < 1.$$

3. Show that if $e_n = \pm 1$, then the number
$$\sum_{n=1}^{\infty} \frac{e_n}{n!}$$
is an irrational number.

4. Show that if $\{q_n\}$ and $\{r_n\}$ are two Cauchy sequences, then so are $\{q_n + r_n\}$ and $\{q_n \cdot r_n\}$.

5. If A and B are non-empty subsets of $\mathbb{R}$ such that $A \subseteq B$, then show that $\sup A \leq \sup B$.

6. Let A be a non-empty subset of $\mathbb{R}$ that is bounded below. Then $s = \inf A$ if and only if (i) $s \leq a$ for all $a \in A$ and (ii) for any $\epsilon > 0$, $A \cap [s, s + \epsilon) \neq \varnothing$.

1.4 Sequences of Real Numbers

A sequence x_n of real numbers is said to be **bounded** if there exists an $M > 0$ such that $|x_n| \leq M$ for all n. If a sequence of numbers is not convergent, it is said to be **divergent**. If the sequence is not bounded above, it is said to **diverge to** ∞. If the sequence is not bounded below, it is said to **diverge to** $-\infty$.

Thus, every convergent sequence of real numbers is bounded. A sequence x_n is said to be **increasing** if $x_m \leq x_n$ whenever $m < n$. It is said to be **strictly increasing** if $x_m < x_n$ whenever $m < n$. A similar definition applies for **decreasing** and **strictly decreasing** sequences. A sequence is said to be **monotonic** if it is either increasing or decreasing. We use the terms **monotonic increasing** and **monotonic decreasing** according as the monotonic sequence is increasing or decreasing.

Theorem 1.4 *Every bounded monotonic sequence is convergent.*

Proof Suppose first that a_n is a bounded monotonic increasing sequence. Then $L = \sup\{a_n : n \geq 1\}$ is finite. By Theorem 1.3, we see that given any $\epsilon > 0$ there is an m such that $L - \epsilon < a_m \leq L$. But as our sequence is increasing, $a_m \leq a_n \leq L$ for all $n \geq m$. Thus
$$|L - a_n| < \epsilon, \qquad \forall\, n \geq m.$$

In other words, the sequence converges to L. The case when the sequence is monotonic decreasing is similar. □

Given a sequence of numbers x_n, a **subsequence** is a sequence of the form

$$x_{n_k}, \qquad k = 1, 2, \ldots$$

with n_k being a sequence of strictly increasing natural numbers. The reader can easily check that if x_n is a convergent sequence with limit L then any subsequence is also convergent with the same limit L. (See Exercise 1 below.)

The following theorem is fundamental in the theory. It was discovered independently by Bernard Bolzano (1781–1848) and Karl Weierstrass (1815–1897). Bolzano studied mathematics, physics, philosophy and theology at the University of Prague and became a Catholic priest in 1804. He then taught at the University of Prague where he was chair of philosophy of religion and later was the dean of the philosophy department. However, his pacifist views and strong opposition to militarism led to his alienation from both academics and church leaders. He was dismissed from the university in 1819 and exiled to the countryside when he refused to retract his views. Much of his mathematical and philosophical work was recognized only after his death. Bolzano is now remembered for the introduction of rigor in mathematics, especially with regard to the $\epsilon - \delta$ definition of continuity, the greatest lower bound property of the real numbers and the intermediate value theorem.

Weierstrass trained to be a high-school teacher of mathematics obtaining his teaching certificate in 1841 at the age of twenty-six. For more than a dozen years, he taught at various high schools until 1854, when his paper on abelian functions which he published in Crelle's Journal had brought him instant recognition from mathematicians at the University of Berlin that he was offered a professorship there when he was nearly forty years old. Until his retirement in 1890, he developed the theory of infinite series and infused mathematical rigor into analysis. Weierstrass discovered the theorem below independently and it was only much later that mathematical historians recognized that Bolzano had foreseen it and proved it much earlier. Today, we honor both mathematicians by referring to it as the Bolzano–Weierstrass theorem.

Theorem 1.5 (Bolzano-Weierstrass) *Every bounded sequence has a convergent subsequence.*

Proof Since our sequence x_n is bounded, let m and M be such that $m \leq x_n \leq M$ for all n. We bisect interval $I_0 = [m, M]$ using the midpoint $c = (m + M)/2$ and consider the two intervals $[m, c]$ and $[c, M]$. For at least one of these intervals (call it I_1), there are infinitely many n such that x_n lies in that interval I_1. Choose n_1 such that $x_{n_1} \in I_1$. Writing $I_1 = [m_1, M_1]$ we iterate the procedure to successively obtain a sequence of nested intervals I_k with the length of I_k being $(M - m)/2^k$ and $x_{n_k} \in I_k$. The sequence m_k is monotonic increasing and by Theorem 1.4 converges to a limit L (say). Since $(M_k - m_k) = (M - m)/2^k$, we see that the monotonic decreasing sequence M_k also converges to L. Thus the subsequence x_{n_k} converges to L. □

This theorem allows us to define the important notions of **lim sup** and **lim inf** of any sequence of real numbers. Indeed, given a sequence of numbers x_n, if it is not bounded above, we write

$$\limsup_{n\to\infty} x_n = +\infty.$$

If it is not bounded below, we write

$$\liminf_{n\to\infty} x_n = -\infty.$$

If the sequence is bounded, let T be the set of all convergent subsequences. By Theorem 1.5, this is a non-empty set so we define:

$$\limsup_{n\to\infty} x_n = \sup T, \qquad \liminf_{n\to\infty} x_n = \inf T.$$

There are several characterizations of lim sup and lim inf that are convenient in many applications. We record these below.

Theorem 1.6 *Let L be a real number and a_n a sequence of real numbers. Then,*

$$L = \limsup_{n\to\infty} a_n$$

if and only if

(a) for each $\epsilon > 0$, there is a positive integer N such that $a_n < L + \epsilon$ for all $n \geq N$;
(b) for each $\epsilon > 0$, the inequality $L - \epsilon < a_n$ holds for infinitely many n,

Proof Suppose first that $\limsup_{n\to\infty} a_n = L$. As L is a real number, the sequence is bounded above by M (say). From the definition of lim sup, we also see that L is the limit of a convergent subsequence and that it is the largest of such limits. If (a) did not hold, then for some $\epsilon > 0$, we have $a_n > L + \epsilon$ for infinitely many n so that by the Bolzano–Weierstrass theorem there would be a subsequence converging to a limit point in $[L + \epsilon, M]$ contradicting the definition of L. Therefore (a) holds. If (b) did not hold, then for some $\epsilon > 0$, we have $L - \epsilon < a_n$ holds for only finitely many n. Thus, there is an N such that $L - \epsilon \geq a_n$ for all $n \geq N$. But then $\limsup_{n\to\infty} a_n \leq L - \epsilon$ contradicting our definition of lim sup. □

In an analogous fashion, the reader can show (see Exercise 2):

Theorem 1.7 *Let L be a real number and a_n a sequence of real numbers. Then,*

$$L = \liminf_{n\to\infty} a_n$$

if and only if

(a) for each $\epsilon > 0$, there is a positive integer N such that $a_n > L - \epsilon$ for all $n \geq N$;
(b) for each $\epsilon > 0$, the inequality $L + \epsilon > a_n$ holds for infinitely many n.

We can now return to a general discussion of Cauchy sequences of real numbers. A sequence x_n of real numbers is called a **Cauchy sequence** if for every $\epsilon > 0$, there is an N such that

$$|x_n - x_m| < \epsilon \qquad \forall\, m, n \geq N.$$

We are now in a position to prove the important theorem:

Theorem 1.8 *A sequence x_n of real numbers is convergent if and only if it is a Cauchy sequence.*

Proof The fact that a convergent sequence is a Cauchy sequence was proved earlier (see Theorem 1.2). Now suppose our sequence is Cauchy. We want to show that it converges. By taking $\epsilon = 1$ in the definition of a Cauchy sequence, we see that x_n is bounded. By the Bolzano–Weierstrass theorem, there is a convergent subsequence x_{n_k}. Putting

$$L = \lim_{k \to \infty} x_{n_k}$$

we see that given $\epsilon > 0$, there is a K such that

$$|x_{n_k} - L| < \frac{\epsilon}{2} \qquad \forall\, k \geq K.$$

As our sequence is Cauchy, there is an N such that

$$|x_n - x_m| < \frac{\epsilon}{2} \qquad \forall\, n, m \geq N.$$

As n_k is a strictly increasing sequence of natural numbers, we choose k such that $n_k \geq N$ and $k \geq K$. Now, by the triangle inequality,

$$|x_n - L| \leq |x_n - x_{n_k}| + |x_{n_k} - L| < \epsilon$$

provided $n \geq N$. In other words, x_n converges to L. This completes the proof. □

The precise understanding of the concept of a limit allows us to give a formal definition of **continuity**. A function $f : [a, b] \to \mathbb{R}$ is said to be **continuous** at $c \in [a, b]$ if given any $\epsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(c)| < \epsilon \qquad \text{whenever} \quad |x - c| < \delta.$$

Intuitively, $f(x)$ is close to $f(c)$ whenever x is sufficiently close to c. A function is said to be **continuous** on $[a, b]$ if it is continuous at every point of the interval $[a, b]$. At the point a we speak of **right continuity** and at b we speak of **left continuity** because technically, we can only take our limit either from the right or left respectively.

It is convenient to introduce the notation $B_r(c)$ to be the "open ball" or the open interval of radius r:

$$\{x \in \mathbb{R} : |x - c| < r\}.$$

With this notation, we can reformulate the notion of continuity as follows. A function $f : [a, b] \to \mathbb{R}$ is **continuous** at c if and only if for every $\epsilon > 0$, the inverse image under f of any open ball of radius ϵ contains an open ball $B_\delta(c)$ for some $\delta > 0$. This reformulation allows us to generalize the notion of continuity in a wider context, namely to the setting of metric spaces and later to topological spaces.

Exercises

1. If x_n is a convergent sequence with limit L, then any subsequence is also convergent with the same limit L.

2. Prove Theorem 1.7.

3. For any bounded sequence x_n, show that the sequences defined as

$$M_n = \sup\{x_k : k \geq n\}, \qquad m_n = \inf\{x_k : k \geq n\}$$

are monotonic sequences and hence convergent.

4. For any bounded sequence x_n show that

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup\{x_k : k \geq n\}$$

and

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \inf\{x_k : k \geq n\}.$$

5. Show that the sequence

$$a_n := 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + (-1)^{n-1}\frac{1}{n}$$

is Cauchy and hence converges. (We will see in the next section that it converges to $\log 2$.)

6. Let $a_1 = \sqrt{2}$ and define recursively $a_{n+1} = \sqrt{2 + a_n}$.
   (a) Show by induction that $a_n < 2$ for all $n \geq 1$.
   (b) Prove that a_n is a strictly increasing sequence.
   (c) Show that $\lim_{n\to\infty} a_n = 2$.

7. Prove that $\limsup_{n\to\infty} \sin n = 1$ and $\liminf_{n\to\infty} \sin n = -1$. [Hint: You may use the fact that π is irrational.]

1.5 Infinite Series

Given a sequence a_k of real numbers, the formal expression

$$\sum_{k=1}^{\infty} a_k$$

is called an **infinite series**. The nth *partial sum* is defined as

$$s_n := \sum_{k=1}^{n} a_k.$$

If the limit

$$\lim_{n\to\infty} s_n$$

exists and equals L, we say the series **converges** to L. Otherwise, we say the series **diverges**.

The geometric series

$$\sum_{k=0}^{\infty} r^k$$

plays an important role in mathematics. Its partial sums are easily seen to be (see Exercise 1)

$$s_n = 1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r},$$

provided $r \neq 1$. It is clear that if $|r| < 1$, the geometric series converges to

$$\frac{1}{1 - r},$$

and diverges otherwise.

The reader is undoubtedly familiar with some standard tests for the convergence of an infinite series. We review these here. The **comparison test** is by far the most useful:

Theorem 1.9 (Comparison test) *Let a_k and b_k be sequences of real numbers. If $|a_k| \leq b_k$ and $\sum_{k=1}^{\infty} b_k$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.*

Proof We show that the sequence of partial sums

$$s_n = \sum_{k=1}^{n} a_k,$$

is a Cauchy sequence and hence converges. Indeed, as $\sum_{k=1}^{\infty} b_k$ converges, the sequence of its partial sums is Cauchy so that given $\epsilon > 0$, there is a positive integer N such that

$$\left| \sum_{k=m}^{n} b_k \right| < \epsilon, \qquad \forall\, n \geq m \geq N.$$

Hence

$$|s_n - s_m| \leq \left| \sum_{k=m}^{n} b_k \right| < \epsilon \qquad \forall\, n \geq m \geq N.$$

□

Theorem 1.10 (Ratio test) *Suppose that a_k is a sequence of nonzero real numbers and*

$$\limsup_{k\to\infty} \left| \frac{a_{k+1}}{a_k} \right| < 1.$$

Then the series $\sum_{k=1}^{\infty} a_k$ converges.

Proof Let L be

$$\limsup_{k\to\infty} \left| \frac{a_{k+1}}{a_k} \right|.$$

Let r be any real number such that $L < r < 1$. Then, by the definition of limsup (see Theorem 1.6), there exists an N such that

$$\left| \frac{a_{k+1}}{a_k} \right| < r, \qquad \forall\, k \geq N.$$

An easy induction now shows that

$$|a_{N+j}| \leq |a_N| r^j, \qquad j = 1, 2, \ldots$$

By the comparison test, our series converges. □

In the above theorem, if $L = 1$, the result is inconclusive. If

$$\lim_{k\to\infty} \left| \frac{a_{k+1}}{a_k} \right| > 1,$$

and the sequence a_k is strictly positive for sufficiently large k, the student can verify that the series diverges.

Theorem 1.11 (Root test) *If a_k is a sequence of positive real numbers, and*

$$L := \limsup_{k\to\infty} a_k^{1/k} < 1,$$

then the series $\sum_{k=1}^{\infty} a_k$ converges. If $L > 1$, the series diverges. If $L = 1$, the test is inconclusive.

Proof If $L < 1$, then as before let r be such that $L < r < 1$. By the property of limsup (see Theorem 1.6) we see $a_k < r^k$ for all k sufficiently large. Hence, our series converges by the comparison test. If $L > 1$, then again by Theorem 1.6, we have $a_k^{1/k} > 1$ for infinitely many k. Thus, $a_k > 1$ for infinitely many k and our series diverges. This completes the proof. □

Theorem 1.12 (Integral test) *Suppose that a_k is a sequence of non-negative real numbers and that f is a continuous decreasing function on $[1, \infty)$ such that $f(k) = a_k$ for all $k \geq 1$. Then, the infinite series $\sum_{k=1}^{\infty} a_k$ converges if and only if the integral*

$$\int_1^{\infty} f(x)dx$$

converges.

Proof Since f is decreasing, it is clear by examining the graph of $f(x)$ that

$$a_{k+1} = f(k+1) \leq \int_k^{k+1} f(x)dx \leq f(k) = a_k$$

for all $k \geq 1$. Summing this inequality from $k = 1$ to infinity gives the result. □

Thus, for example, we immediately deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^s}$$

converges if and only if $s > 1$.

It is convenient to introduce the "big O" notation. Given two sequences of numbers a_n and b_n, with the b_n's non-negative, we write $a_n = O(b_n)$ if there is a constant C that $|a_n| \leq Cb_n$ for all $n \geq 1$. A similar notation is applicable for functions. Given two functions $f(x)$ and $g(x)$ defined on an interval $[a, b]$ and with $g(x)$ non-negative, we write $f(x) = O(g(x))$ if there is a constant C such that $|f(x)| \leq Cg(x)$ for all $x \in [a, b]$. The interval need not be finite. Thus, our proof of the integral test allows us to deduce that for $s > 1$, (see Exercise 4 below)

$$\sum_{n>N} \frac{1}{n^s} = O(N^{1-s}).$$

The "little o" notation is also convenient. We write $a_n = o(b_n)$ if $a_n/b_n \to 0$ as n tends to infinity. Similarly, we write $f(x) = o(g(x))$ as $x \to c$ to mean that $f(x)/g(x) = 0$ as $x \to c$. Depending on the context, the appearance of c may be suppressed if it is clear what it is. In practice, it is often as x tends to infinity or x tends to zero. If it is not evident, one can of course spell it out for clarity.

The following example illustrates that the idea is capable of further refinement.

Example 1.1 Let

$$a_k = \frac{1}{k} - \int_k^{k+1} \frac{dx}{x}.$$

Since $1/x$ is a decreasing function of x, we see that $a_k > 0$. On the other hand,

$$a_k = \int_k^{k+1} \left(\frac{1}{k} - \frac{1}{x}\right) dx = \int_k^{k+1} \frac{x-k}{kx} dx.$$

The numerator in the integrand is at most 1 in the interval $[k, k+1]$ and so the integrand is at most $1/k^2$. Consequently, $a_k \leq 1/k^2$. Therefore, by the comparison test, the series

$$\sum_{k=1}^{\infty} \left(\frac{1}{k} - \int_k^{k+1} \frac{dx}{x}\right)$$

converges to a finite limit γ, often called **Euler's constant**. In fact, our estimate on a_k shows that

$$\sum_{k=1}^{n} a_k = \gamma + O(1/n).$$

In other words,

$$\sum_{k \leq n} \frac{1}{k} = \log(n+1) + \gamma + O(1/n),$$

giving us a refinement on the familiar divergence of the harmonic series.

These tests of convergence that we have so far adumbrated are not sufficient to deal with alternating series such as

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}.$$

The following theorem discusses this situation.

Theorem 1.13 (Alternating series test) *If a_k is a decreasing sequence of positive real numbers tending to zero, then the series*

$$\sum_{k=1}^{\infty}(-1)^{k-1}a_k$$

converges.

Proof Let us consider the partial sums

$$s_n = \sum_{k=1}^{n}(-1)^{k-1}a_k.$$

Then

$$s_{2n} = (a_1 - a_2) + \cdots + (a_{2n-1} - a_{2n})$$

is easily seen to be positive and greater than s_{2n-2} so that s_{2n} is increasing. Moreover,

$$s_{2n} = a_1 - (a_2 - a_3) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n} \leq a_1,$$

so that our sequence is bounded. By Theorem 1.4, this sequence converges to a finite limit. A similar analysis applies to the partials sums of odd index:

$$s_{2n+1} = a_1 - (a_2 - a_3) + \cdots + (a_{2n} - a_{2n+1}).$$

We can see that s_{2n+1} is a decreasing sequence which is bounded below by $a_1 - a_2$ since we can write

$$s_{2n+1} = a_1 - a_2 + (a_3 - a_4) + \cdots + (a_{2n-1} - a_{2n}) + a_{2n+1}.$$

Again by Theorem 1.4, the sequence converges. But now, as

$$s_{2n+1} = s_{2n} + a_{2n+1},$$

and a_{2n+1} tends to zero, we deduce the series converges. □

Our theorem now implies that the alternating series

$$\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k}$$

converges. The partial sums of this series were shown to be Cauchy in an exercise in Sect. 1.4. However, the special nature of the series can be used to explicitly evaluate it as the following example shows.

Example 1.2 We observe that

$$\frac{1}{k} = \int_0^1 x^{k-1}dx,$$

so that

$$\sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} = \sum_{k=1}^{n} (-1)^{k-1} \int_0^1 x^{k-1} dx.$$

Interchanging the integral and sum on the right-hand side and noting the geometric series

$$\sum_{k=1}^{n} (-1)^{k-1} x^{k-1} = \frac{1 - (-x)^n}{1 + x},$$

we find

$$\sum_{k=1}^{n} \frac{(-1)^{k-1}}{k} = \int_0^1 \frac{1 - (-x)^n}{1 + x} dx = \int_0^1 \frac{dx}{1 + x} - \int_0^1 \frac{(-x)^n}{1 + x} dx.$$

The first integral is $\log 2$, and the second integral is bounded by

$$\int_0^1 x^n dx = \frac{1}{n + 1},$$

so that

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \log 2.$$

Exercises

1. If $r \neq 1$, show that

$$1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

2. If the series $\sum_{k=1}^{\infty} a_k$ converges, show that $\lim_{k \to \infty} a_k = 0$.

3. If $\sum_{k=1}^{\infty} |a_k|$ converges, show that $\sum_{k=1}^{\infty} a_k$ converges.

4. For $s > 1$, show that

$$\sum_{n > N} \frac{1}{n^s} = O(N^{1-s}).$$

5. Show that $\log(n + 1) = \log n + O(1/n)$.

6. Show that the series

$$\sum_{k=2}^{\infty} \frac{1}{k(\log k)^s}$$

converges if and only if $s > 1$.

7. Show that[1]

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots .$$

[Hint: Follow the template of Example 1.2 by noting that $1/(2k+1) = \int_0^1 x^{2k} dx$.]

1.6 Sequences of Functions

Let I be an interval in $\mathbb{R}$ and suppose that we have a sequence of functions $f_n(x)$ (sometimes written simply as f_n) defined on I. We say that the sequence **converges pointwise** if

$$\lim_{n\to\infty} f_n(x)$$

exists for each $x \in I$. The series

$$\sum_{n=1}^{\infty} f_n(x)$$

is said to converge pointwise on I if the sequence of partial sums converges pointwise for each point x in I. Thus, the sequence of functions f_n converges pointwise to f if for every $\epsilon > 0$, and every $x \in I$, there is an N such that

$$|f_n(x) - f(x)| < \epsilon \qquad \forall\, n \geq N.$$

The reader should understand that in the definition of pointwise convergence, N would depend on both ϵ and x. For example, observe that

$$\lim_{n\to\infty} x^n = \begin{cases} 1 & x = 1 \\ 0 & 0 \leq x < 1. \end{cases}$$

So if $f_n(x) = x^n$ for $x \in [0, 1]$ and if

[1] This seems to have been first discovered by Madhava in fourteenth-century India. He is also credited to have discovered the familiar Taylor series for sine and cosine functions more than 250 years before the advent of Sir Isaac Newton. In fact, the Kerala school of mathematics led by Madhava is now credited to have discovered much of what we would now describe as precalculus. See p. 224 of [2]. The formula in this exercise was rediscovered several centuries later by G. Leibniz. We therefore refer to the series as the Madhava–Leibniz series.

$$f(x) = \begin{cases} 1 & x = 1 \\ 0 & 0 \le x < 1, \end{cases}$$

then $\lim_{n\to\infty} f_n(x) = f(x)$ pointwise. Note that even though each $f_n(x)$ is a continuous function, the limit function is not continuous. In many applications, this phenomenon is not useful. Thus, pointwise convergence is too weak a notion of convergence for a sequence of functions. This motivates the definition of **uniform convergence**. As before, let I be an interval and f_n a sequence of functions defined on I. We say the sequence **converges uniformly** to f on I if for every $\epsilon > 0$, there is an N such that

$$|f_n(x) - f(x)| < \epsilon \qquad \forall\, n \ge N, \quad \text{and} \quad \forall\, x \in I.$$

The important thing to note here is that N does not depend on x and applies for all $x \in I$, hence the term **uniform** is applicable to emphasize this property.

In a similar vein, we say

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly to $f(x)$ if the sequence of partial sums converges uniformly to f. This property is sufficiently strong to prevent the pathology observed earlier as the following theorems elucidate.

Theorem 1.14 *Suppose that f_n is a sequence of continuous functions defined on an interval I. If there is a function f on I such that $\lim_{n\to\infty} f_n = f$ uniformly on I, then f is continuous.*

Proof Let $x_0 \in I$. We want to show that f is continuous on I. That is, given $\epsilon > 0$, we want to find $\delta > 0$ such that

$$|f(x) - f(x_0)| < \epsilon \qquad \text{if} \quad |x - x_0| < \delta.$$

Noting that by the triangle inequality

$$|f(x) - f(x_0)| \le |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)|$$

for all $x \in I$ and all n, we deduce by the uniform convergence property, that there is an N such that

$$|f_n(x) - f(x)| < \epsilon \qquad \forall\, n \ge N \quad \text{and} \quad \forall x \in I.$$

With this choice of N, we have

$$|f(x) - f(x_0)| \le \epsilon + |f_N(x) - f_N(x_0)| + \epsilon$$

for all x in I. Now, as $f_N(x)$ is continuous on I, there is a $\delta > 0$ such that

$$|f_N(x) - f_N(x_0)| < \epsilon \qquad \text{when} \quad |x - x_0| < \delta.$$

With this choice of δ we deduce the desired result. □

Theorem 1.15 *Suppose that f_n is a sequence of continuous functions defined on an interval $I = [a, b]$. If there is a function f on I such that $\lim_{n\to\infty} f_n = f$ uniformly on I, then*

$$\lim_{n\to\infty} \int_a^b f_n(x)dx = \int_a^b f(x)dx.$$

Proof By Theorem 1.14, the limit function f is continuous and so all the integrals in the statement of the theorem exist. Now, given $\epsilon > 0$, we want to show that

$$\left| \int_a^b f_n(x)dx - \int_a^b f(x)dx \right| < \epsilon \qquad \forall\, n \geq N.$$

Since f_n converges to f uniformly on I, there is an N such that

$$|f_n(x) - f(x)| < \epsilon \qquad \forall\, n \geq N \quad \forall\, x \in I.$$

Thus, for all $n \geq N$, we have

$$\left| \int_a^b f_n(x)dx - \int_a^b f(x)dx \right| \leq \int_a^b |f_n(x) - f(x)|dx \leq \epsilon(b - a)$$

for all $n \geq N$. This completes the proof. □

For the next theorem, we introduce a new class of functions. We write $C^1[a, b]$ to denote the space of **continuously differentiable** functions defined on $[a, b]$. It is understood that at the end points, a and b, we require only the right derivative and the left derivative respectively. For functions in this space, we sometimes say f is C^1 (the domain being understood). More generally, we say f is C^r if it is r-times continuously differentiable.

Theorem 1.16 *Suppose that f_n is a sequence of functions defined on an interval $[a, b]$ such that*

(a) f_n is C^1 for all n;
(b) there is a function g defined on $[a, b]$ such that $\lim_{n\to\infty} f_n' = g$ uniformly on $[a, b]$;
(c) there is a point $c \in [a, b]$ for which the limit $\lim_{n\to\infty} f_n(c)$ exists.

Then, there is a differentiable function f on $[a, b]$ such that $\lim_{n\to\infty} f_n = f$ uniformly on $[a, b]$ and $f' = g$.

Proof We first note that g is continuous by (a) and (b) and an application of Theorem 1.14. Hence g is integrable. The desired function f is the primitive of g and by the fundamental theorem of calculus

$$f(x) = f(a) + \int_a^x g(t)dt = \lim_{n\to\infty} f_n(a) + \int_a^b g(t)dt,$$

which suggests that we first show that

$$\lim_{n\to\infty} f_n(a)$$

exists and define f by

$$f(x) := \lim_{n\to\infty} f_n(a) + \int_a^x g(t)dt. \tag{1.2}$$

Indeed, again by the fundamental theorem of calculus, with c as in (c),

$$f_n(c) - f_n(a) = \int_a^c f_n'(t)dt,$$

so that

$$\lim_{n\to\infty} f_n(a) = \lim_{n\to\infty} \left(f_n(c) - \int_a^c f_n'(t)dt \right). \tag{1.3}$$

Condition (b) and an application of Theorem 1.15 shows that

$$\lim_{n\to\infty} \int_a^c f_n'(t)dt = \int_a^c g(t)dt.$$

Therefore, the limit of the right-hand side of (1.3) exists. Therefore f is well-defined by (1.2) and by the fundamental theorem of calculus, $f' = g$. To complete the proof, we need to show that $\lim_{n\to\infty} f_n = f$ uniformly on $[a, b]$. By (b), given $\epsilon > 0$, there is an N such that

$$|f_n'(x) - g(x)| < \epsilon \qquad \forall\, x \in [a, b] \quad \forall\, n \geq N.$$

Hence,

$$\begin{aligned} |f_m(x) - f(x)| &= \left| \left(f_m(a) + \textstyle\int_a^x f_m'(t)dt \right) - \left(\lim_{n\to\infty} f_n(a) + \int_a^x g(t)dt \right) \right| \\ &\leq \quad |f_m(a) - \lim_{n\to\infty} f_n(a)| + \left| \textstyle\int_a^x (f_m'(t) - g(t))dt \right|. \end{aligned}$$

If $m \geq N$, the absolute value of the integrand is less than ϵ so that the integral is less than $(b - a)\epsilon$. Since $\lim_{n\to\infty} f_n(a)$ exists, there is an N_1 such that for $m \geq N_1$, we have $|f_m(a) - \lim_{n\to\infty} f_n(a)| < \epsilon$. Therefore if $m \geq \max(N, N_1)$, we have

$$|f_m(x) - f(x)| < \epsilon \qquad \forall\, x \in [a, b].$$

This completes the proof. □

Condition (c) of the theorem is not superfluous. For example, if $f_n(x) = n$ for all $x \in [0, 1]$, we see that both (a) and (b) are satisfied with g identically zero. But (c) is not satisfied. And our sequence of functions does not even converge pointwise on the interval.

The following is now a simple consequence of our theorem.

Corollary 1.1 *Suppose that f_n is a sequence of C^1-functions defined on an interval $[a, b]$ and assume that there are two functions f and g also defined on $[a, b]$ such that $\lim_{n\to\infty} f_n = f$ and $\lim_{n\to\infty} f_n' = g$ uniformly on $[a, b]$. Then, $f' = g$.*

A convenient test for uniform convergence of a sequence of functions is provided by the following **Weierstrass M-test**.

Theorem 1.17 (Weierstrass M-test) *Suppose that f_n is a sequence of functions defined on an interval I and assume that there is a sequence of positive real numbers M_n such that*

(a) $|f_n(x)| \leq M_n \quad \forall x \in I$ and $\forall\, n \geq 1$ and
(b) the series $\sum_{n\geq 1} M_n$ converges.

Then, the series $\sum_{n=1}^{\infty} f_n$ converges uniformly on I.

Proof Let

$$S_n(x) := \sum_{k=1}^{n} f_k(x),$$

and

$$T_n := \sum_{k=1}^{n} M_k$$

be the respective partial sums. Then, as the series $\sum_{n\geq 1} M_n$ converges, the sequence of partial sums T_n is Cauchy. That is, given $\epsilon > 0$, there is an N such that

$$|T_m - T_n| < \epsilon \qquad \forall\, m \geq n \geq N.$$

This means that for each x, the sequence $S_n(x)$ is also Cauchy because

$$|S_m(x) - S_n(x)| = \left| \sum_{k=n+1}^{m} f_k(x) \right| \leq \sum_{k=n+1}^{m} |f_k(x)| \leq \sum_{k=n+1}^{m} M_k = |T_m - T_n| < \epsilon$$

for all $x \in I$. Therefore the sequence $S_n(x)$ converges pointwise to a limit $f(x)$ (say). We need to show this convergence is uniform. Indeed,

$$\begin{aligned} \left| f(x) - \textstyle\sum_{k=1}^{n} f_k(x) \right| &= |f(x) - S_m(x) + S_m(x) - S_n(x)| \\ &\leq |f(x) - S_m(x)| + |S_m(x) - S_n(x)| \\ &\leq \quad |f(x) - S_m(x)| + |T_m - T_n| \end{aligned}$$

for any $m \geq n$. Given $\epsilon > 0$, we have an N such that $|T_m - T_n| < \epsilon$ for $m \geq n \geq N$ so that for all $x \in I$,

$$\left| f(x) - \sum_{k=1}^{n} f_k(x) \right| \leq |f(x) - S_m(x)| + \epsilon$$

whenever $m \geq n \geq N$. Letting m tend to infinity, we see that the first term on the right-hand side of the inequality tends to zero. This completes the proof. □

The concept of uniform convergence suggests that it is convenient to introduce the following spaces. Let I be an interval and define $B(I)$ to be the space of bounded functions on I and $C(I)$ the space of all continuous functions on I. If I is closed and bounded $C(I) \subseteq B(I)$. Both of these spaces are infinite-dimensional vector spaces over $\mathbb{R}$ as the reader can verify using basic calculus. We can make both of these spaces into metric spaces by defining the **sup norm**

$$||f|| := \sup\{|f(x)| : \ x \in I\}.$$

Metric spaces are reviewed in Sect. 1.8. The student can verify that given two functions, $f, g \in B(I)$, the **distance** $||f - g||$ defines a metric. It is now clear from the above discussion that a sequence of functions f_n in $B(I)$ converges to $f \in B(I)$ if and only if $\lim_{n\to\infty} f_n = f$ uniformly on I.

Exercises

1. Show that each of the following sequence of functions converges pointwise on [0, 1] to the zero function and determine if the convergence is uniform on [0, 1].
   (a) $f_n(x) = nxe^{-nx}$;
   (b) $f_n(x) = nxe^{-n^2x}$;
   (c) $f_n(x) = nxe^{-nx^2}$.
2. Show that the series
   $$\sum_{n=1}^{\infty} \frac{\sin nx}{n^3}$$
   converges uniformly to a differentiable function $f(x)$ on $[0, 2\pi]$ and
   $$f'(x) = \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}.$$
3. Let $f_n(x) = nx/(nx + 1)$ for $x \in [0, 1]$ and $n \geq 1$.

(a) Compute the pointwise limit $\lim_{n\to\infty} f_n(x)$ for $x \in [0, 1]$. Is the convergence uniform?
(b) Compute

$$\lim_{n\to\infty} \int_0^1 f_n(x)dx \qquad \text{and} \qquad \int_0^1 \lim_{n\to\infty} f_n(x)dx.$$

Are they equal?

4. Let

$$f_n(x) = \frac{x}{1+n^2x^2}$$

for $n \geq 1$ and $x \in [-1, 1]$. Prove that

$$|f_n(x)| \leq \frac{1}{2n} \qquad \forall n \geq 1 \quad \text{and} \quad \forall x \in [-1, 1].$$

Deduce that $\lim_{n\to\infty} f_n = 0$ uniformly on $[-1, 1]$.

5. With $f_n(x)$ as in the previous exercise, show that

$$f_n'(x) = \frac{1-n^2x^2}{(1+n^2x^2)^2}.$$

Deduce that

$$\lim_{n\to\infty} f_n'(x) = \begin{cases} 1, & x = 0 \\ 0, & 0 < |x| \leq 1. \end{cases}$$

Deduce that $f_n'(x)$ does not converge uniformly on $[-1, 1]$.

1.7 Power Series

With x a real variable, and c a real number, an infinite series of the form

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + \cdots$$

is called a **power series**. Such series have an ubiquitous role in mathematics, ranging from approximation theory to the solution of differential equations. As we shall see later, they play a vital role in the development of complex analysis.

There is no loss of generality if we consider the series of the form

$$\sum_{n=0}^{\infty} a_n x^n \tag{1.4}$$

as can be seen by a simple change of variable. Thus, for the sake of elegance and simplicity, we will formulate our results with $c = 0$.

By a simple application of the root test, the series converges if

$$\limsup_{n\to\infty} |a_n|^{1/n}|x| < 1.$$

In other words, the series converges for $|x| < R$ where

$$R = \frac{1}{\limsup_{n\to\infty} |a_n|^{1/n}}, \tag{1.5}$$

where we understand $1/0$ to be infinity and $1/\infty$ to be zero. This motivates the definition of the **radius of convergence** as the largest number R such that the series (1.4) converges for $|x| < R$ and R is given by the formula (1.5).

An important application of the theory of power series is the expansion of a function as a **Taylor series**. One can also view the theory as giving polynomial approximations to C^∞ functions. The idea is simple enough and is an application of repeated integration by parts.

Without any loss of generality, suppose that f is a C^∞ function defined on $[-1, 1]$. By the fundamental theorem of calculus

$$f(x) - f(0) = \int_0^x f'(t)dt.$$

Integrating by parts, we deduce

$$f(x) - f(0) = \Big[f'(t)(t-x)\Big]_{t=0}^{t=x} - \int_0^x f''(t)(t-x)dt.$$

In other words,

$$f(x) = f(0) + f'(0)x + \int_0^x f''(t)(x-t)dt.$$

The key observation here is that the primitive of 1 is $t + c$ for any constant c and we have chosen the constant to be $-x$ in this calculation. Iterating this procedure, it is now evident that we have the following theorem.

Theorem 1.18 (Taylor's theorem) *Suppose that f is a function defined on an open interval I and that c is a point in I. Suppose that $f^{(n+1)}$ exists and is continuous on I. Then,*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k + R_n(x),$$

where

$$R_n(x) = \frac{1}{n!}\int_c^x (x-t)^n f^{(n+1)}(t)dt. \tag{1.6}$$

Moreover, there is a point $\xi \in I$ such that

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-c)^{n+1}. \tag{1.7}$$

Proof It is clear that our discussion preceding the statement of the theorem (replacing zero by c) leads to the first assertion. Indeed,

$$f(x) - f(c) = \int_c^x (x-t)^0 f'(t)dt = (x-c)f'(c) + \int_c^x (x-t)f''(t)dt,$$

by the fundamental theorem of calculus and an integration by parts. Iterating this process n times leads to the first assertion with $R_n(x)$ given by (1.21). For the estimation of the remainder, let us suppose first that $c < x$ (the case $c > x$ being similar), and put

$$m = \inf\{f^{n+1}(t) : c \le t \le x\}, \qquad M = \sup\{f^{(n+1)}(t) : c \le t \le x\}.$$

Then,

$$\frac{m}{n!}\int_c^x (x-t)^n dt \le R_n(x) \le \frac{M}{n!}\int_c^x (x-t)^n dt.$$

The integral is easily evaluated to be

$$\frac{(x-c)^{n+1}}{n+1}.$$

Consequently,

$$m \le \frac{R_n(x)(n+1)!}{(x-c)^{n+1}} \le M.$$

By the intermediate value theorem, we see that there is a $\xi \in [c, x]$ such that (1.7) holds. This completes the proof. □

It is worth remarking that in Theorem 1.18, the term

$$\sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$$

is called the nth-order **Taylor polynomial**. Equation (1.7) is called the Lagrange form for the remainder. For $n = 0$, the theorem reduces to the usual mean value theorem.

Theorem 1.18 allows us to derive polynomial approximations of some familiar functions such as e^x and the trigonometric functions like $\sin x$ and $\cos x$.

Example 1.3 We have for $x \in (0, 2\pi)$,

$$\left| \sin x - \sum_{k=0}^{n} \frac{(-1)^k x^{2k+1}}{(2k+1)!} \right| \leq \frac{|x|^{2n+2}}{(2n+2)!}.$$

Indeed, by Taylor's theorem, the remainder $R_{2n+1}(x)$ is equal to

$$\sin^{(2n+2)}(\xi) \frac{x^{2n+2}}{(2n+2)!},$$

for some $\xi \in (0, 2\pi)$ and the assertion is now evident.

Exercises

1. If P is a polynomial, show that $\lim_{n\to\infty} |P(n)|^{1/n} = 1$.

2. If
$$\sum_{n=0}^{\infty} a_n x^n$$
has radius of convergence R, and P and Q are polynomials such that $Q(n) \neq 0$ for all $n \geq 0$, show that
$$\sum_{n=0}^{\infty} \frac{P(n)}{Q(n)} a_n x^n$$
also has radius of convergence R. In particular,
$$\sum_{n=0}^{\infty} n a_n x^{n-1} \qquad \text{and} \qquad \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$$
both have radius of convergence R.

3. Show that the Taylor series of $(1+x)^r$ when r is a real number is given by
$$\sum_{k=0}^{\infty} \binom{r}{k} x^k,$$
where
$$\binom{r}{k} = \frac{r(r-1)\cdots(r-k+1)}{k!}.$$

Show further that the power series converges for $|x| < 1$.

4. Using the previous exercise, show that the Taylor series of $(1-4x)^{-1/2}$ about $x = 0$ is given by
$$\sum_{n=0}^{\infty} \binom{2n}{n} x^n.$$
Show that the series converges for $x \in (-1/4, 1/4)$.

5. Prove that
$$\sum_{n=0}^{\infty} \binom{2n}{n} \frac{(-1)^n}{4^n} = \frac{1}{\sqrt{2}}.$$

6. Let r be a fixed non-negative integer. Show that the power series
$$J_r(x) := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+r}}{2^{2n+r} n!(n+r)!},$$
has radius of convergence $R = \infty$. ($J_r(x)$ is called the **Bessel function of the first kind**.)

7. With $J_r(x)$ as in the previous exercise, show that $J_r(x)$ is a solution $y(x)$ to **Bessel's differential equation**
$$x^2 y'' + x y' + (x^2 - r^2) y = 0.$$
(Bessel functions have many applications in the study of the propagation of electromagnetic waves through cylindrical waveguides.)

8. (Generalized mean value theorem for integrals) If f and g are two continuous functions on an interval $[a, b]$ and if f is non-negative there, show that there is a $\xi \in [a, b]$ such that
$$\int_a^b f(t)g(t)dt = g(\xi) \int_a^b f(t)dt.$$

9. (Second mean value theorem for integrals) Let $f(x)$ be a bounded, monotonic decreasing, non-negative, differentiable function on $[a, b]$ and let $g(x)$ be a bounded integrable function. Show that for some $\xi \in [a, b]$, we have
$$\int_a^b f(x)g(x)dx = f(a) \int_a^{\xi} g(x)dx.$$

[Hint: define the function

$$G(t) = \int_a^t g(x)dx.$$

Then integrate by parts

$$\int_a^b f(x)g(x)dx$$

and apply the previous exercise.]

1.8 Metric Spaces and Euclidean Spaces

The construction of the real numbers from the set of rational numbers used two fundamental ideas: the concept of a metric and Cauchy sequences. It is convenient to define both of these ideas in a more general context.

Given a set X, a **pseudo-metric** for X is a function $d : X \times X \to \mathbb{R}^+$, the set of non-negative real numbers satisfying the following properties:

1. $d(x, x) = 0$ for all $x \in X$;
2. $d(x, y) = d(y, x)$ for all $x, y \in X$ (symmetry);
3. $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$ (triangle inequality).

If in addition $d(x, y) = 0$ implies $x = y$, then d is called a **metric**. We sometimes write (X, d) to indicate either the pseudo-metric space or metric space accordingly. The usual absolute value $|x - y|$ for $x, y \in \mathbb{R}$ defines a metric and more generally for m-dimensional space $\mathbb{R}^m$, we define the distance between $\mathbf{x} = (x_1, \ldots, x_m)$ and $\mathbf{y} = (y_1, \ldots, y_m)$, as

$$|\mathbf{x} - \mathbf{y}| := \sqrt{\sum_{j=1}^{m} (x_j - y_j)^2}.$$

Occasionally, we also use the notation $||\mathbf{x}||$ to denote the length of the vector $\mathbf{x}$. The fact that this distance function satisfies the triangle inequality is not totally trivial and is equivalent to the Cauchy–Schwarz inequality. (Note that there is no "t" in the spelling of Schwarz here.) Essentially, we need to show that

$$|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|, \qquad \forall\ \mathbf{x}, \mathbf{y} \in \mathbb{R}^m. \tag{1.8}$$

It is convenient to introduce the **dot product** (or sometimes called the **inner product**):

$$\mathbf{x} \cdot \mathbf{y} := \sum_{j=1}^{m} x_j y_j. \tag{1.9}$$

The reader can verify that the distributive property holds: $\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}$. Then, it is easily seen that $|\mathbf{x}| = \sqrt{\mathbf{x} \cdot \mathbf{x}}$. so that squaring the triangle inequality (1.8), it is equivalent to showing

$$(\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) \leq \mathbf{x} \cdot \mathbf{x} + \mathbf{y} \cdot \mathbf{y} + 2|\mathbf{x}||\mathbf{y}|,$$

so that we are reduced to showing

$$\mathbf{x} \cdot \mathbf{y} \leq |\mathbf{x}||\mathbf{y}|,$$

which is the Cauchy–Schwarz inequality.

This ubiquitous Cauchy–Schwarz inequality is easily proved as follows. Clearly

$$2x_j y_j \leq x_j^2 + y_j^2.$$

The homogeneity of the left-hand side shows that for any $\lambda \neq 0$, we can replace x_j by λx_j and y_j by y_j/λ to get

$$2x_j y_j \leq \lambda^2 x_j^2 + \lambda^{-2} y_j^2.$$

We sum this over $j = 1$ to m to deduce

$$2\sum_{j=1}^{m} x_j y_j \leq \lambda^2 \sum_{j=1}^{m} x_j^2 + \lambda^{-2} \sum_{j=1}^{m} y_j^2. \tag{1.10}$$

We choose λ so that we can minimize the right-hand side. Thus, setting

$$\lambda^4 = \left(\sum_{j=1}^{m} y_j^2\right)\left(\sum_{j=1}^{m} x_j^2\right)^{-1},$$

and inserting this into (1.10) gives the result.

Yet another proof of the Cauchy–Schwarz inequality can be deduced from the geometric meaning of the dot product which is a simple consequence of the cosine law that is in turn derived from the celebrated Pythagorean theorem.

The oldest source of the theorem seems to be rooted in the Sulva Sutras of Baudhayana dated as being written around 800 BCE. It has been suggested by many historians that Pythagoras, who was a contemporary of the Buddha and Lao-Tzu, learned of the theorem from his travels in India and later formed a religious and mystical cult around the theorem. In his book, "The Message of Plato", Urwick writes that Pythagoras may have actually been an Indian teacher of Vedanta philosophy, "whose very name was only the Greek form of the Indian title Pitta Guru, or

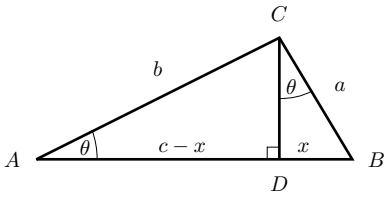


Fig. 1.1 Bhaskaracharya's proof of the Pythagorean theorem

Father-teacher."[2] Some Indian connection seems evident from all written accounts of Pythagoras and his school with its belief in reincarnation and its insistence on vegetarianism and moral practices as a means for attaining higher knowledge. In his research article highlighting the influence of Indian philosophy on Greek thought, Marlow writes that the philosophy of Pythagoras and Plato "is as unlike anything in Greek thought as it is like the Hindu mysticism of the Upanisads."[3]

The Indian penchant for abstraction, with its legendary discovery of zero and the decimal system, demonstrates a striking difference between the Greek geometric (and visual or sensory) approach to mathematics and the Indian abstract (or transcendental) approach. This is evidenced by the short and elegant proof of the Pythagorean theorem by Bhaskaracharya (1114–1185 CE). By contrast, Euclid's proof constructs squares on each side and proceeds to subdivide them, comparing them with areas of numerous triangles on each side. The proof is more geometric and visual.

Here is Bhaskara's short proof of the Pythagorean theorem. Consider the right-angled triangle ABC with the right angle at C along with the perpendicular CD to AB (see Fig. 1.1). The triangles ABC, ACD and CBD are similar.

Thus, comparing the smaller triangles to the big triangle ABC, we get

$$\cos\theta = \frac{c-x}{b} = \frac{b}{c},$$

so that $b^2 = c^2 - cx$. Similarly,

$$\sin\theta = \frac{x}{a} = \frac{a}{c},$$

so that $a^2 = cx$. Putting these two equations together gives us the familiar Pythagorean theorem:

$$c^2 = a^2 + b^2,$$

or alternatively,

$$c = \sqrt{a^2 + b^2}.$$

[2] See p. 14 of The Message of Plato by Edward J. Urwick, Methuen and Company Limited, 36 Essex Street W.C., London, 1920.

[3] See p. 39 of A.N. Marlow, Hinduism and Buddhism in Greek Philosophy, Philosophy East and West, Vol. 4, No. 1, (Apr., 1954), pp. 35–45.

The Pythagorean theorem lies at the heart of measuring distance in Euclidean space. An easy induction to n dimensions shows that the length of the vector $(x_1, x_2, \ldots, x_n)$ in $\mathbb{R}^n$ is

$$\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

But the concept of a vector was slow in coming. William Rowan Hamilton (1805–1865) is said to be the first mathematician who defined the notion of a vector in the context of developing his theory of quaternions. Throughout his life, he was obsessed in developing his quaternionic theory and in doing so, was led to give a precise definition of a vector. Both vectors and quaternions have revolutionized mathematics, but it was the idea of a vector that has had a more profound impact. Hamilton's work, however, was confined to four dimensions. It was Hermann Grassmann (1809–1877) who gave us the modern version of a vector in higher-dimensional space.

Grassmann, interestingly, never had an academic position in a university. He was a high-school teacher who also had expertise in languages. In particular, he was a Sanskrit scholar who compiled the first translation of the Rig Veda into German. It was his mastery of Sanskrit that inspired him to introduce the term "matrix" in mathematics. In doing this, he was fully aware of the Latin roots, *mater* and the Sanskrit root *mātr*, both signifying "mother" or more accurately, "the womb," because he felt that the theory of matrices is the "womb of mathematics" from which everything emerges.

Grassmann saw mathematics more as a "theory of forms" rather than as a "theory of measurement." It would be accurate to say that the modern viewpoint is that mathematics is both. But Grassmann's emphasis on forms led to fundamental abstractions leading to the concepts of vectors and matrices.

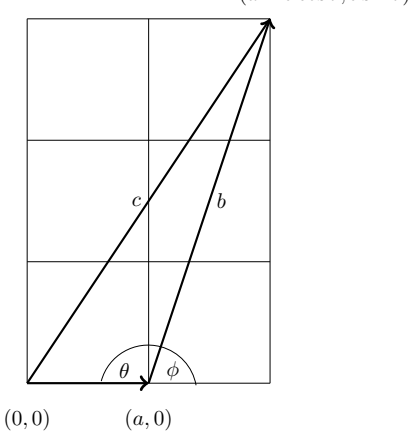
The concept of a vector is inevitable if one wants to study motion in three-dimensional space. In fact, it would be fair to say that multivariable calculus partly arose motivated by a need to study three-dimensional motion, just as one-variable calculus arose to describe linear motion, or motion in the plane. Vectors represent forces. They have a magnitude and a direction. For the novice, the formal operations with vectors and matrices seem ad hoc. A familiar case in point is the dot product of two vectors, which lies at the heart of the definition of matrix multiplication:

$$\mathbf{a} = (a_1, a_2, \ldots, a_n), \quad \mathbf{b} = (b_1, b_2, \ldots, b_n), \qquad \mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n.$$

Often, in the development of mathematics, we forget the physical origins of concepts and present these ideas very abstractly and thus lose the visual aspect of the theory as stressed by Grassmann.

The familiar cosine law which generalizes the Pythagorean theorem motivates the definition of the dot product. Indeed, given two vectors $\mathbf{a}$ and $\mathbf{b}$, let θ be the angle between them. Let us consider first the simple case where the vectors are two-dimensional. Without any loss of generality, we may coordinatize choosing the plane spanned by these two vectors as indicated in Fig. 1.2. The cosine law is easily deduced by calculating the distance from the origin to $(a - b\cos\theta, b\sin\theta)$. We find

Fig. 1.2 Cosine law



$$c^2 = a^2 + b^2 - 2ab\cos\theta.$$

From Fig. 1.2, the student can easily see that the vector sum $\mathbf{a} + \mathbf{b}$ has coordinates

$$(a + b\cos\phi, b\sin\phi).$$

Since $\theta + \phi = \pi$, we find either by looking at the graph of the cosine and sine functions, or by using the addition formulas, that

$$a + b\cos\phi = a - b\cos\theta, \quad b\sin\phi = b\sin\theta,$$

and our claim is now evident.

Thus, given two vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^2$ with angle θ between them (see Fig. 1.3), the length of the vector $\mathbf{c} = \mathbf{b} - \mathbf{a}$ is by the cosine law,

$$||\mathbf{b} - \mathbf{a}||^2 = ||\mathbf{b}||^2 + ||\mathbf{a}||^2 - 2||\mathbf{a}||\,||\mathbf{b}||\cos\theta, \tag{1.11}$$

since we may choose, without any loss of generality, our coordinate x-axis along the vector $\mathbf{a}$.

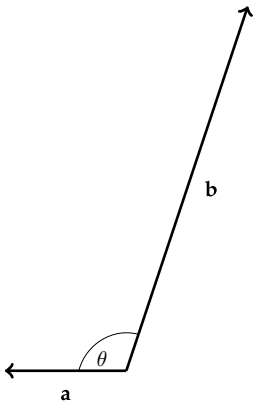
On the other hand, if $\mathbf{a} = (a_1, a_2)$ and $\mathbf{b} = (b_1, b_2)$, then $\mathbf{b} - \mathbf{a} = (b_1 - a_1, b_2 - a_2)$ so that by the Pythagorean theorem, we have

$$||\mathbf{b} - \mathbf{a}||^2 = (b_1 - a_1)^2 + (b_2 - a_2)^2 = b_1^2 + b_2^2 + a_1^2 + a_2^2 - 2(a_1b_1 + a_2b_2). \tag{1.12}$$

The right hand side of (1.12) is

$$||\mathbf{b}||^2 + ||\mathbf{a}||^2 - 2(\mathbf{a}\cdot\mathbf{b}).$$

Fig. 1.3 Dot product



Comparing this with (1.11) gives us

$$\mathbf{a} \cdot \mathbf{b} = ||\mathbf{a}||\,||\mathbf{b}|| \cos\theta$$

which gives us the visual interpretation of the dot product.

This is easily generalized to higher dimensions since two linearly independent vectors determine a plane and the lengths and angle between the two vectors are both independent of the coordinatization: If θ is the angle between the two vectors $\mathbf{a}$ and $\mathbf{b}$, then

$$\mathbf{a} \cdot \mathbf{b} = ||\mathbf{a}||\,||\mathbf{b}|| \cos\theta,$$

and the Cauchy–Schwarz inequality is now immediate since $\cos\theta$ is bounded by 1.

In physics, the dot product arises in many contexts such as giving a mathematical description of work. The physical definition of "work" is the dot product of force and displacement. In mathematical notation,

$$W = \mathbf{F} \cdot \mathbf{d}.$$

In an introductory course in physics, one often confines the study of motion to a single dimension. However, motion is three-dimensional and the notions of vectors and dot products are inevitable. For example, the kinetic energy of a particle with mass m and velocity v is often defined as $\frac{1}{2}mv^2$. However, if the particle is moving in space and velocity is seen as a vector $\mathbf{v}$, then the kinetic energy is

$$\frac{1}{2}m(\mathbf{v} \cdot \mathbf{v}).$$

Energy does not have a direction, but velocity does as it is a vector.

From the geometric meaning of the dot product, we see that two vectors **x** and **y** are orthogonal if and only if their dot product is zero. It is important to keep the geometric meaning of vectors in mind. It adds another dimension to our understanding.

It is convenient to recall here the **cross-product** of two vectors in $\mathbb{R}^3$. Formally, given two vectors

$$\mathbf{a} = (a_1, a_2, a_3), \quad \mathbf{b} = (b_1, b_2, b_3)$$

in $\mathbb{R}^3$, the cross-product is by definition

$$\mathbf{a} \times \mathbf{b} := (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1).$$

The formal definition lacks any hint of its importance and meaning. In physics, the concept arises to describe **torque**. If **a** represents the displacement of a particle from a fixed point to a movable point, and **b** is the force applied at the movable point, then the cross-product $\mathbf{a} \times \mathbf{b}$ is the torque exerted by the force about the fixed point. The above opaque definition is better remembered using determinants. One writes symbolically,

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

where **i**, **j**, **k** are the unit vectors (1, 0, 0), (0, 1, 0) and (0, 0, 1) respectively. Expanding the determinant using the first row leads to the earlier definition so that the "determinant expression" serves as a useful mnemonic for the cross-product. A straightforward and tedious computation shows that

$$|\mathbf{a} \times \mathbf{b}|^2 = (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2$$

so that

$$|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2 - |\mathbf{a}|^2|\mathbf{b}|^2 \cos^2\theta. \tag{1.13}$$

Consequently,

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta,$$

where θ is the angle between the two vectors **a** and **b**. This formula implies a geometric interpretation of the magnitude of the cross-product vector. It is the area of the parallelogram spanned by the two vectors **a** and **b**. The direction of the cross-product is given by the familiar "right-hand rule," where if you align the fingers of your right hand along the vector **a** and bend your fingers around in the direction of rotation from **a** to **b**, your thumb will point in the direction of $\mathbf{a} \times \mathbf{b}$. We also note that the cross-product of **a** and **b** is zero if and only if they are parallel.

With this interlude on the dot product and cross-product, we return to our general discussion and consider sequences in $\mathbb{R}^m$. It will be clear that much of this discussion applies to a general metric space. For the sake of clarity, we will confine our attention to a discussion of sequences in $\mathbb{R}^m$. They will be sequences of vectors $\mathbf{x}_n$ and we can

write

$$\mathbf{x}_n = (x_{n1}, \ldots, x_{nm}).$$

A sequence is said to **converge** to $\mathbf{L} \in \mathbb{R}^m$, if given $\epsilon > 0$ there exists N such that

$$|\mathbf{x}_n - \mathbf{L}| < \epsilon \qquad \forall\, n \geq N.$$

We then write $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{L}$. The reader can easily verify that convergence of a sequence of vectors is equivalent to the convergence of the component sequences (see Exercise 1).

The notion of Cauchy sequences is $\mathbb{R}^m$ is analogous (as it is in any metric space). The student can show again that a sequence in $\mathbb{R}^m$ is Cauchy if and only if it converges (see Exercise 3).

A sequence $\mathbf{x}_n$ in $\mathbb{R}^m$ is said to be **bounded** if there is a number M such that $|\mathbf{x}_n| \leq M$ for all n. Again, the reader can verify that a sequence in $\mathbb{R}^m$ is bounded if and only if each component sequence is bounded (Exercise 2).

The analogue of the Bolzano–Weierstrass theorem goes through.

Theorem 1.19 *Every bounded sequence in $\mathbb{R}^m$ has a convergent subsequence.*

Proof Let $\mathbf{x}_n$ be a bounded sequence is $\mathbb{R}^m$. We first apply Theorem 1.5 to the sequence x_{n1} in $\mathbb{R}$ and obtain a convergent subsequence $x_{n_k,1}$. We again apply Theorem 1.5 to the bounded sequence $x_{n_k,2}$ to get another convergent subsequence. Admittedly, the notation is cumbersome but it is clear that after m applications of Theorem 1.5 in this way, we derive a convergent subsequence of vectors, as desired. □

Exercises

1. Writing $\mathbf{x}_n = (x_{n1}, \ldots, x_{nm})$ and $\mathbf{L} = (L_1, \ldots, L_m)$, show that $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{L}$ if and only if
$$\lim_{n\to\infty} x_{nj} = L_j \qquad 1 \leq j \leq m.$$

2. Show that a sequence $\mathbf{x}_n$ is bounded if and only if each component sequence x_{nj} is bounded for $1 \leq j \leq m$.

3. Show that a sequence $\mathbf{x}_n$ in $\mathbb{R}^m$ is Cauchy if and only if it converges.

4. (Cauchy–Schwarz inequality for integrals) Let f and g be integrable on $[a, b]$. Prove that
$$\left| \int_a^b f(t)g(t)dt \right|^2 \leq \left(\int_a^b f^2(t)dt \right) \left(\int_a^b g(t)^2 dt \right).$$

5. A **contraction** on $\mathbb{R}^d$ is a function $f : \mathbb{R}^d \to \mathbb{R}^d$ such that there is a number $K < 1$ satisfying
$$|f(\mathbf{u}) - f(\mathbf{v})| \leq K|\mathbf{u} - \mathbf{v}|,$$
for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$. Prove that f is continuous.

6. Let f be a contraction as in the previous exercise. The following exercises establish the **contraction mapping theorem**: any contraction has a unique fixed point.
   (a) Prove that f has at most one fixed point. [Hint: suppose there were two.]
   (b) Let $\mathbf{x}_1$ be any point in $\mathbb{R}^d$. Define a sequence recursively by $\mathbf{x}_{n+1} = f(\mathbf{x}_n)$ for $n \geq 1$. Show that for $m \geq 2$,
   $$|\mathbf{x}_{m+1} - \mathbf{x}_m| < K^{m-1}|\mathbf{x}_2 - \mathbf{x}_1|.$$
   (c) Obtain an estimate for $|\mathbf{x}_n - \mathbf{x}_m|$ in terms of K and $|\mathbf{x}_2 - \mathbf{x}_1|$ for any two positive integers m, n.
   (d) Show that $\mathbf{x}_n$ is a Cauchy sequence in $\mathbb{R}^d$.
   (e) Show that the vector
   $$\mathbf{x} := \lim_{n\to\infty} \mathbf{x}_n$$
   is a fixed point of f

7. For each number a put $f_a(x) = ax$. For which numbers a is f_a a contraction on $\mathbb{R}$? When f_a is a contraction, what is the fixed point?

8. For what intervals $[0, r]$ with $r \leq 1$, is the map $f : [0, r] \to [0, r]$ given by $x \mapsto x^2$ a contraction?

1.9 The Heine–Borel Theorem

Much of our discussion regarding analysis on the real line extends to $\mathbb{R}^m$. For example in $\mathbb{R}^2$, the **closed rectangle** $[a, b] \times [c, d]$ is

$$\{(x, y) : \ x \in [a, b], \ y \in [c, d]\},$$

and more generally we can speak of closed rectangles in $\mathbb{R}^m$ (even though they are not rectangles per se, but rather generalizations of them). A similar definition can be made about **open rectangles** $(a, b) \times (c, d)$ in $\mathbb{R}^2$ and analogously in higher dimensions. A subset $U \subset \mathbb{R}^m$ is called **open** if for each point $\mathbf{x} \in U$ there is an open rectangle A such that $\mathbf{x} \in A \subset U$. This is equivalent to the definition that there is an open ball $B_r(\mathbf{x})$ contained in U. A subset C of $\mathbb{R}^m$ is **closed** if its complement is

open. A set A is said to be **compact** if every open cover of A can be reduced to a finite subcover. The reader should be familiar with the following theorem of Heine and Borel.

The theorem has a history somewhat reminiscent of the Bolzano–Weierstrass theorem in that it was first discovered in 1872 by H.E. Heine (1821–1881) and later rediscovered in 1895 by Emile Borel (1871–1956).[4]

Theorem 1.20 (Heine–Borel theorem) *The closed interval* $[a, b]$ *is compact.*

Proof If $\mathcal{O}$ is an open cover of $[a, b]$, let A be the set

$$\{x : a \leq x \leq b \text{ and } [a, x] \text{ is covered by a finite number of open sets in } \mathcal{O}\}.$$

Clearly $a \in A$ and A is bounded above by b. We want to show that $b \in A$. We will do this by showing that the least upper bound α (say) of A lies in A and that $\alpha = b$.

First, $\alpha > a$ because as a is contained in some open set U of $\mathcal{O}$, there is an open ball of radius r centered at a (which is necessarily of the form $[a, r)$ contained in U Since α is the least upper bound of A, and $\alpha \leq b$, α lies in some open set V of $\mathcal{O}$. But then there is an open ball of radius r centered at α that lies in V. If $\alpha < b$, this would give us a contradiction to the definition of α and so $\alpha = b$. □

It is not difficult to show that if $A \subset \mathbb{R}^m$ and $B \subset \mathbb{R}^n$ are compact, then $A \times B \subset \mathbb{R}^{m+n}$ is compact. In particular, a closed rectangle is compact. In fact, in $\mathbb{R}^m$, compact sets are characterized by the fact that they are closed and bounded (see Exercise 4).

Exercises

1. Show that a closed rectangle is a closed set.

2. If $B \subset \mathbb{R}^m$ is compact and $\mathbf{x} \in \mathbb{R}^n$, show that $\{\mathbf{x}\} \times B \subset \mathbb{R}^{n+m}$ is compact.

3. If B is compact and $\mathcal{O}$ is an open cover of $\{\mathbf{x}\} \times B$, show that there is an open set $U \subset \mathbb{R}^n$ containing $\mathbf{x}$ such that $U \times B$ is covered by a finite number of sets in $\mathcal{O}$.

4. Show that a closed and bounded subset of $\mathbb{R}^m$ is compact and conversely.

5. In $\mathbb{R}^m$, show that the union of any (even infinite) number of open sets is open. Prove that the intersection of finitely many open sets is open, and find a counterexample to show how this fails for infinitely many open sets.

6. If A is a closed set that contains every rational number $r \in [0, 1]$, show that $[0, 1] \subset A$.

[4] See p. 618 of Boyer [3].

1.10 Vector-Valued Functions

A function f from $\mathbb{R}^n$ to $\mathbb{R}^m$ is called a **vector-valued function** of n variables. We write $f : \mathbb{R}^n \to \mathbb{R}^m$ and the component functions are designated $f_i(\mathbf{x})$ for $1 \leq i \leq m$ and $\mathbf{x} = (x_1, \ldots, x_n)$. If $n = 1$, and f is continuous, then one calls f a **curve**, so that the component functions are functions of one-variable. (In practice, a curve is a map from an interval in $\mathbb{R}$ into $\mathbb{R}^m$.) If $f : \mathbb{R}^n \to \mathbb{R}^m$ and $m = 1$, one calls f a **scalar field**. When $m = n$, we speak of f as a **vector field**.

If I is an interval $[a, b]$ in $\mathbb{R}$, and $X : I \to \mathbb{R}^m$ is a curve, with component functions $x_1(t), \ldots, x_m(t)$ which are differentiable, it is convenient to visualize the domain as a time parameter and the curve being traced out in m-dimensional space as t moves from a to b. By the usual Pythagorean theorem and the familiar limit process, the **length** of the curve traced out in time t is

$$s(t) := \int_a^t \sqrt{x_1'(u)^2 + \cdots + x_m'(u)^2}\, du, \tag{1.14}$$

as the reader can easily verify. For example, the circle of radius r is parameterized by

$$(x(t), y(t)) = (r\cos t, r\sin t), \qquad 0 \leq t \leq 2\pi.$$

The circumference of the circle is readily seen to be $2\pi r$ by formula (1.14).

A curve, being a function of a single variable, can easily be studied using one-variable calculus. If $\mathbf{r} : I \to \mathbb{R}^m$ is a curve C whose component functions are differentiable, then $\mathbf{r}'(t)$ is the vector of component derivatives and often called the **velocity vector** or **tangent vector**. The vector

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

is called the **unit tangent vector** to the curve C. It is again a vector-valued function of a single variable. The magnitude of the rate of change of the tangent vector with respect to the length of the curve is called the **curvature** of C. More precisely, the curvature κ is given by

$$\kappa := \left|\frac{d\mathbf{T}}{ds}\right|,$$

where $s(t)$ is the curve length given by (1.14).

Using the chain rule, one can rewrite this formula in a more amenable form. Indeed,

$$\mathbf{T}'(t) = \frac{d\mathbf{T}}{dt} = \frac{d\mathbf{T}}{ds}\frac{ds}{dt},$$

so that

$$\kappa = \left|\frac{d\mathbf{T}}{ds}\right| = \frac{|\mathbf{T}'(t)|}{\left|\frac{ds}{dt}\right|}.$$

By (1.14),

$$\frac{ds}{dt} = |\mathbf{r}'(t)|.$$

Combining these formulas, we obtain

$$\kappa = \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|}.$$

The notion of a continuous function is easily extended to vector-valued functions of several variables. Let A be an open subset of $\mathbb{R}^n$. The function $f : A \to \mathbb{R}^m$ is said to be **continuous** at $\mathbf{a} \in A$ if

$$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = f(\mathbf{a}).$$

We can also define this without using limits. We say f is continuous if and only if $f^{-1}(U)$ is open for every open set U in $\mathbb{R}^m$. The following important theorem states that the continuous image of a compact set is compact.

Theorem 1.21 *If $f : A \to \mathbb{R}^m$ is continuous and $A \subset \mathbb{R}^n$ is compact, then $f(A)$ is compact.*

Proof Let $\mathcal{O}$ be an open cover of $f(A)$. For each open set U of $\mathcal{O}$, there is an open set V_U such that $f^{-1}(U) = V_U \cap A$, by the continuity of f. The collection of V_U's is an open cover of A which can be reduced to a finite subcover $V_{U_1}, \ldots, V_{U_n}$ (say) since A is compact. Thus, $U_1, \ldots, U_n$ cover $f(A)$. □

Exercises

1. The straight line in $\mathbb{R}^3$ is parameterized by the vector-valued function

$$\mathbf{r}(t) = (at + b, ct + d, et + f)$$

for some constants a, b, c, d, e, f. Show that the curvature of the line is zero.

2. Noting that the circle of radius a in $\mathbb{R}^2$ is parameterized by

$$\mathbf{r}(t) = (a\cos t, b\sin t),$$

show that the curvature of the circle of radius a is $1/a$.

3. Given a curve $\mathbf{r}(t)$, assume that the unit tangent vector to this curve is differentiable. Define the **principal unit normal vector** $\mathbf{N}(t)$ to be

$$\frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}.$$

Show that

$$\mathbf{N}(t) = \frac{1}{\kappa}\frac{d\mathbf{T}}{ds},$$

where κ denotes the curvature.

4. Under the same conditions as in the previous exercise, show that $|\mathbf{r}(t)|$ is constant if and only if $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal to each other, for all values of t in the domain of $\mathbf{r}(t)$.

5. If A is a compact subset of $\mathbb{R}^d$ and $f : A \to \mathbb{R}$ is continuous, show that f attains a maximum value and a minimum value on A. In other words, there are points $\mathbf{u}$ and $\mathbf{v}$ such that

$$f(\mathbf{u}) \leq f(\mathbf{x}) \leq f(\mathbf{v}),$$

for all $\mathbf{x}$ in A.

1.11 Derivatives of Multivariable Functions

The concept of a derivative from one-variable calculus is generalized to the multidimensional case in several ways. There is the directional derivative, partial derivative and the Jacobian matrix. The Jacobian emanates from the idea that the derivative is really a linear transformation. Let us first consider the case $f : \mathbb{R}^n \to \mathbb{R}$. Given a vector $\mathbf{u} \in \mathbb{R}^n$, we can define the **directional derivative** of f in the direction $\mathbf{u}$ at $\mathbf{x}_0$, sometimes denoted $f_{\mathbf{u}}(\mathbf{x}_0)$ by the limit,

$$\lim_{h\to 0} \frac{f(\mathbf{x}_0 + h\mathbf{u}) - f(\mathbf{x}_0)}{h},$$

provided the limit exists.

In the case that the range space is one-dimensional, it is convenient to consider **partial derivatives**. Indeed, if $f : \mathbb{R}^n \to \mathbb{R}$, and $\mathbf{a} = (a_1, \ldots, a_n) \in \mathbb{R}^n$, the limit

$$\lim_{h\to 0} \frac{f(a_1, \ldots, a_i + h, \ldots, a_n) - f(a_1, \ldots, a_n)}{h}$$

is called the ith **partial derivative** of f at $\mathbf{a}$ and denoted $D_i f(\mathbf{a})$. We can continue taking partial derivatives, provided they exist. For example, we can consider $D_j(D_i f)$, sometimes denoted $D_{j,i} f$. There are various theorems that ensure that $D_{j,i} f = D_{i,j} f$. For instance, if both of these are continuous on an open set containing $\mathbf{a}$, then we have $D_{i,j} f(\mathbf{a}) = D_{j,i} f(\mathbf{a})$ (see Exercise 7). The equality is true assuming weaker hypotheses which we do not discuss here. The function $D_{i,j} f$ is called a **second-order partial derivative** or sometimes a **mixed partial derivative**. In the case of functions of two variables $f(x, y)$, it is convenient to write f_x and f_y for $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ respectively. Thus f_{xy} would be $(f_x)_y$ and so on. A similar tradition is sometimes adopted for functions of three or more variables.

The directional derivatives and the partial derivatives are related. Namely, if $\mathbf{e}_i$ is an element of the standard basis of $\mathbb{R}^n$ with 1 in the ith component and all other components being zero, then $D_i f(\mathbf{a}) = f_{\mathbf{e}_i}(\mathbf{a})$.

For functions whose range is contained in $\mathbb{R}$, we can speak about the maximum and minimum values. The reader can verify that if $A \subset \mathbb{R}^n$ and $f : A \to \mathbb{R}$ is differentiable and the maximum value occurs at $\mathbf{a}$, then $D_i f(\mathbf{a}) = 0$ for every i.

Historically, multivariable calculus emerged in the study of electromagnetism and in particular, in the derivation of Maxwell's equations. This is analogous to the development of one-variable calculus that arose from Newton's study of motion and gravitation. Partly motivated by this context and for other practical reasons, the study of scalar fields has been given detailed attention. Thus, given a continuous function $f : \mathbb{R}^n \to \mathbb{R}$, we define the **gradient** of f, denoted ∇f as a function from $\mathbb{R}^n$ to $\mathbb{R}^n$ given by

$$\nabla f = (D_1 f, \ldots, D_n f).$$

Thus, ∇f is a vector field. Points $\mathbf{x} \in \mathbb{R}^n$ where $\nabla f(\mathbf{x}) = 0$ are called **critical points**.

If $f(x, y)$ is a function $f : \mathbb{R}^2 \to \mathbb{R}$, of two variables and we have a curve $z(t) = (x(t), y(t)) \in \mathbb{R}^2$, we can study f along this curve and it can be viewed as a function of a single variable. That is, we can consider

$$F(t) = f(x(t), y(t)).$$

We want to calculate $F'(t_0)$. Writing $(x(t_0), y(t_0)) = (x_0, y_0)$, we have from one-variable calculus,

$$F'(t_0) = \lim_{t \to t_0} \frac{F(t) - F(t_0)}{t - t_0} = \lim_{t \to t_0} \frac{f(x, y) - f(x_0, y_0)}{t - t_0}.$$

We write the numerator in the last limit as

$$[f(x, y) - f(x_0, y)] + [f(x_0, y) - f(x_0, y_0)].$$

We apply the mean value theorem to each of the bracketed terms:

$$f(x, y) - f(x_0, y) = f_x(\xi, y)(x - x_0), \quad f(x_0, y) - f(x_0, y_0) = f_y(x_0, \eta)(y - y_0),$$

for suitable $\xi \in [x(t_0), x(t)]$ and $\eta \in [y(t_0), y(t)]$. Thus,

$$F'(t_0) = \lim_{t \to t_0} f_x(\xi, y) \frac{x - x_0}{t - t_0} + f_y(x_0, \eta) \frac{y - y_0}{t - t_0}$$

which equals

$$f_x(x_0, y_0)x'(t_0) + f_y(x_0, y_0)y'(t_0).$$

In other words,

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

In general, if $f : \mathbb{R}^n \to \mathbb{R}$ is a function of n-variables, we can study the function f along a curve

$$t \mapsto \mathbf{x}(t) = (x_1(t), \ldots, x_n(t)),$$

as a function of a single variable and we have

$$\frac{df}{dt} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\frac{dx_i}{dt}.$$

This can be rewritten as a dot product:

$$\frac{df}{dt} = \nabla f(\mathbf{x}) \cdot \mathbf{x}'(t), \tag{1.15}$$

and we see the analogy to the one-variable version of the chain rule. Thus, (1.15) is often referred to as the chain rule for functions of several variables, though it is only a special case of the general chain rule for the derivative of the composition of functions $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}^r$ (see Exercise 3 at the end of this section). This observation allows us to prove Taylor's theorem in several variables.

We say a subset S of $\mathbb{R}^n$ is **star-shaped** with respect to the point $\mathbf{c}$ if for every $\mathbf{x}$ in S the line segment joining $\mathbf{x}$ to $\mathbf{c}$ lies in S.

Lemma 1.2 *Let f be a C^∞ function on an open subset U of $\mathbb{R}^n$ which is star-shaped with respect to a point $\mathbf{c} = (c_1, \ldots, c_n)$ in U. Then there are C^∞-functions $g_1, \ldots, g_n$ on U such that*

$$f(\mathbf{x}) = f(\mathbf{c}) + \sum_{i=1}^{n} (x_i - c_i) g_i(\mathbf{x}), \qquad g_i(\mathbf{c}) = \frac{\partial f}{\partial x_i}(\mathbf{c}).$$

Proof Since U is star-shaped with respect to $\mathbf{c}$, we have for any $\mathbf{x}$ in U that the line segment

$$\mathbf{c} + t(\mathbf{x} - \mathbf{c}), \quad 0 \le t \le 1$$

lies in U. Thus, $f(\mathbf{c} + t(\mathbf{x} - \mathbf{c}))$ is defined for $0 \le t \le 1$. By the chain rule,

$$\frac{d}{dt} f(\mathbf{c} + t(\mathbf{x} - \mathbf{c})) = \sum_{i=1}^{n} (x_i - c_i) \frac{\partial f}{\partial x_i}(\mathbf{c} + t(\mathbf{x} - \mathbf{c})).$$

We integrate both sides with respect to t for $0 \le t \le 1$ to get

$$f(\mathbf{c} + t(\mathbf{x} - \mathbf{c}))\Big|_{t=0}^{t=1} = \sum_{i=1}^{n} (x_i - c_i) \int_0^1 \frac{\partial f}{\partial x_i}(\mathbf{c} + t(\mathbf{x} - \mathbf{c}))dt.$$

Let

$$g_i(\mathbf{x}) = \int_0^1 \frac{\partial f}{\partial x_i}(\mathbf{c} + t(\mathbf{x} - \mathbf{c}))dt.$$

Then g_i is C^∞ and we obtain

$$f(\mathbf{x}) = f(\mathbf{c}) + \sum_{i=1}^{n} (x_i - c_i) g_i(\mathbf{x}).$$

Moreover,

$$g_i(\mathbf{c}) = \int_0^1 \frac{\partial f}{\partial x_i}(\mathbf{c})dt = \frac{\partial f}{\partial x_i}(\mathbf{c}).$$

□

The lemma shows that the linear polynomial

$$f(\mathbf{x}) = f(\mathbf{c}) + \sum_{i=1}^{n} (x_i - c_i) \frac{\partial f}{\partial x_i}(\mathbf{c}),$$

is a good approximation to $f(\mathbf{x})$ in a neighborhood of $\mathbf{c}$. To derive the general Taylor formula for multivariable functions, it is convenient to introduce the operator

$$\mathbf{h} \cdot \nabla := \sum_{i=1}^{n} h_i \frac{\partial}{\partial x_i},$$

where $\mathbf{h} = (h_1, \ldots, h_n)$. We will then write

$$(\mathbf{h} \cdot \nabla) f(\mathbf{c}) := \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}\Big|_{\mathbf{x}=\mathbf{c}}.$$

More generally, we can introduce powers of this operator $(\mathbf{h}\cdot\nabla)^r$. For instance,

$$(\mathbf{h}\cdot\nabla)^2 := \sum_{i,j=1}^{n} h_i h_j \frac{\partial^2}{\partial x_i \partial x_j}.$$

For a function $f : \mathbb{R}^n \to \mathbb{R}$ which has partial derivatives up to order r, we write

$$(\mathbf{h}\cdot\nabla)^r f(\mathbf{c}) := (\mathbf{h}\cdot\nabla)^r f\Big|_{\mathbf{x}=\mathbf{c}}.$$

With this understanding, we can now state:

Theorem 1.22 (Taylor's theorem with remainder) *Let f be a real-valued function defined on an open subset U of $\mathbb{R}^n$ which is star-shaped with respect to a point $\mathbf{c}$ having continuous partial derivatives up to order $r+1$. Then*

$$f(\mathbf{x}) = f(\mathbf{c}) + \sum_{j=1}^{r} \frac{((\mathbf{x}-\mathbf{c})\cdot\nabla)^j}{j!} f(\mathbf{c}) + \frac{((\mathbf{x}-\mathbf{c})\cdot\nabla)^{r+1}}{(r+1)!} f(\mathbf{c}+\xi(\mathbf{x}-\mathbf{c})),$$

for some $\xi \in (0,1)$

Proof We apply the one-variable Taylor's theorem (see Theorem 1.18) to the function

$$g(t) := f(\mathbf{c}+t\mathbf{h}).$$

It is now easy to see that

$$g^{(j)}(t) = (\mathbf{h}\cdot\nabla)^j f(\mathbf{c}+t\mathbf{h})$$

and the result now follows from Theorem 1.18. □

It is instructive to review here the situation in the special case $n=2$ and $m=1$, that is, for scalar valued functions of two variables. The reader should be familiar with the following.

Theorem 1.23 *Let $f(x,y)$ be continuous and have continuous partial derivatives of order at least two in a region A of $\mathbb{R}^2$ including (x_0,y_0). A sufficient condition that (x_0,y_0) is a relative maximum is that*

$$0 < \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}_{(x,y)=(x_0,y_0)} \quad \text{and} \quad \frac{\partial^2 f}{\partial x^2}(x_0,y_0) < 0.$$

Proof By Taylor's theorem for two variables, we have in a neighborhood of (x_0,y_0),

$$f(x_0+h, y_0+k) = f(x_0,y_0) + \frac{1}{2}(h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) + o((h^2+k^2)^{3/2})$$

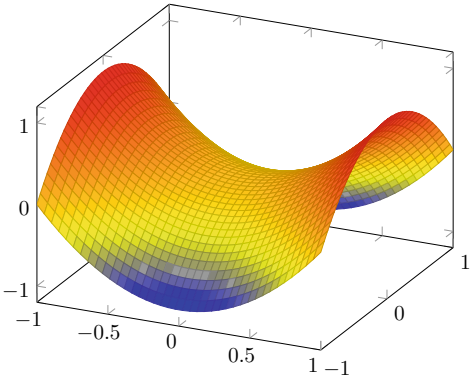


Fig. 1.4 Saddle point at (0, 0)

where the second derivatives on the right are evaluated at $(x_0 + \theta h, y_0 + \theta k)$ for some $0 < \theta < 1$. The second expression on the right-hand side is a quadratic form in h, k and by completing the squares, we can write

$$f(x_0 + h, y_0 + k) - f(x_0, y_0) = \frac{1}{2} f_{xx} \left[\left(h + \frac{f_{xy}}{f_{xx}} k \right)^2 + \left(\frac{f_{xx} f_{yy} - f_{xy}^2}{f_{xx}^2} \right) k^2 \right].$$

By hypothesis, there is a neighborhood of (x_0, y_0) in which $f_{xx} < 0$. Also by hypothesis, the coefficient of k^2 is positive at (x_0, y_0) so that the whole expression in the square brackets is positive in a neighborhood of (x_0, y_0). Whence, the entire expression on the right-hand side is negative in a neighborhood of (x_0, y_0). Therefore, $f(x_0, y_0)$ is a local maximum. □

A similar result holds for relative minima (see Exercise 8). Critical points that do not correspond to either a relative maxima or minima are called **saddle points**.

The notion of a saddle point is best illustrated by considering the function $f(x, y) = x^2 - y^2$. Clearly, $(0, 0)$ is a critical point. However, neither a local maximum nor a local minimum occurs at $(0, 0)$. This can also be seen visually in Fig. 1.4.

The appearance of the quadratic form signals a generalization in the case of more than two variables. Accordingly, we introduce the **Hessian** matrix. Given a function $f : A \subset \mathbb{R}^n \to \mathbb{R}$ which has derivatives at least to order two, we define the **Hessian** to be the $n \times n$ matrix

$$H = \left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right).$$

Sometimes, we write $H(f)$ for this matrix. It is then easy to see that in a neighborhood of $\mathbf{c}$, we have from Taylor's Theorem 1.22 that upon writing $\mathbf{h} = \mathbf{x} - \mathbf{c}$, we have

$$f(\mathbf{x}) = f(\mathbf{c}) + (\mathbf{h} \cdot \nabla f)(\mathbf{c}) + \frac{1}{2} \mathbf{h}^T [H(f)(\mathbf{c})] \mathbf{h} + o(|\mathbf{h}|^3),$$

where $\mathbf{h}^T$ denotes the transpose of $\mathbf{h}$.

The reader will recall that a matrix A is said to be **positive definite** if $\mathbf{h}^T A\mathbf{h} > 0$ for all $\mathbf{h} \neq 0$. It is **negative definite** if $\mathbf{h}^T A\mathbf{h} < 0$ for all $\mathbf{h} \neq 0$.

An analysis similar to the one for two variables leads us to the following:

Theorem 1.24 *Let $A \subset \mathbb{R}^n$ be open and $f : A \to \mathbb{R}$ be a function of class C^2. Suppose that $\mathbf{c}$ in A is a critical point of f.*

(a) If the Hessian $H(f)(\mathbf{c})$ is positive definite, then f has a local minimum at $\mathbf{c}$.
(b) If the Hessian $H(f)(\mathbf{c})$ is negative definite, then f has a local maximum at $\mathbf{c}$.
(c) If $\det H(f)(\mathbf{c}) \neq 0$, and $H(f)(\mathbf{c})$ is neither positive nor negative definite, then f has a saddle point at $\mathbf{c}$.

If the determinant of the Hessian of f at c is zero, the test is inconclusive. In such a case, the point c is said to be **degenerate** and we must use other methods to determine whether or not it is an extremum of f.

There is another way to generalize the notion of a derivative in the multidimensional case. It is motivated by Taylor's theorem. Indeed, given a differentiable function $f : \mathbb{R} \to \mathbb{R}$, we have

$$f(a+h) = f(a) + f'(a)h + o(h),$$

as $h \to 0$. Thus, the linear map $h \mapsto f'(a)h$ can be viewed as a good linear approximation to the difference $f(a+h) - f(a)$. Now, given $f : \mathbb{R}^n \to \mathbb{R}^m$, we say it is **differentiable** at $\mathbf{a} \in \mathbb{R}^n$ if there exists a linear transformation $\lambda : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + \lambda(\mathbf{h}) + o(|\mathbf{h}|)$$

as $h \to 0$. The linear transformation λ depends on $\mathbf{a}$ and so we denote it by $Df(\mathbf{a})$ and call it the **derivative** of f at $\mathbf{a}$. It is easy to prove that this linear transformation is unique (see Exercise 1).

It is convenient to consider the matrix $Df(\mathbf{a})$ with respect to the standard basis of $\mathbb{R}^n$ and $\mathbb{R}^m$. In this way, we can represent the derivative as an $m \times n$ matrix called the **Jacobian matrix** at $\mathbf{a}$ and denoted $f'(\mathbf{a})$. If we write

$$f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))$$

then

$$Df(\mathbf{a}) = \begin{pmatrix} \dfrac{\partial f_1}{\partial x_1}(\mathbf{a}) & \cdots & \dfrac{\partial f_1}{\partial x_n}(\mathbf{a}) \\ \vdots & & \vdots \\ \dfrac{\partial f_m}{\partial x_1}(\mathbf{a}) & \cdots & \dfrac{\partial f_m}{\partial x_n}(\mathbf{a}) \end{pmatrix},$$

where the ijth entry is

$$\frac{\partial f_i}{\partial x_j}(\mathbf{a}).$$

The **Jacobian determinant** is the determinant of the Jacobian matrix. We use the notation

$$\frac{\partial(f_1, \ldots, f_n)}{\partial(x_1, \ldots, x_n)}$$

to designate this determinant.

This generalized derivative enjoys some of the properties shared by the one-variable counterpart. For instance, if f is a constant function, then the derivative is zero (see Exercise 4). If f is a linear transformation, then $Df(a) = f$ (see Exercise 2).

If A is an open set of $\mathbb{R}^n$ and f is defined only on A, we say f is **differentiable** on A if it is differentiable at each point of A. The reader can verify that the usual chain rule and product rules are valid (see Exercises 3 and 6).

As with many aspects of multivariable calculus, the chain rule is best understood first for scalar valued functions of several variables, and in particular two variables. It may amplify our understanding to reduce the calculation to single variable calculus by considering parametric curves. Indeed, let $f(x, y)$ be a differentiable function of two variables and consider $g(t) = f(x(t), y(t))$ where t ranges over an interval in $\mathbb{R}$. By one-variable calculus,

$$g'(t) = \lim_{h \to 0} \frac{f(x(t+h), y(t+h)) - f(x(t), y(t))}{h}.$$

In other words, as $h \to 0$,

$$f(x(t+h), y(t+h)) = f(x(t), y(t)) + g'(t)o(h).$$

By definition of differentiability,

$$f(x(t+h), y(t+h)) = f(x(t), y(t)) + \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + o(\Delta x + \Delta y),$$

where $\Delta x = x(t+h) - x(t)$ and $\Delta y = y(t+h) - y(t)$. But $\Delta x = x'(t)o(h)$ and $\Delta y = y'(t)o(h)$. Dividing through by h and letting $h \to 0$, we deduce

$$g'(t) = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

It is convenient to view this equation as representing the infinitesimal change of f in terms of the infinitesimal changes of x and y respectively and write it symbolically as the **total differential**

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy,$$

with a corresponding generalization to higher variables.

There are certain notions in multivariable calculus that only pertain to dimensions 2 or 3, given that these concepts arose in the context of physics. For example, in $\mathbb{R}^3$, if f is the vector field (f_1, f_2, f_3), we define the vector field **curl** by

$$\text{curl } f = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}, \frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}, \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right).$$

An easy way to remember this is to make symbolic use of the cross-product notation:

$$\nabla \times f = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}, \frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}, \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right).$$

As noted earlier, the notion has its roots in physics and was introduced to measure the **rotation** of a vector field in $\mathbb{R}^3$. Thus, if $\text{curl} f = 0$, we say the vector field f is **irrotational** at that point or has no rotation.

A vector field F is called **conservative** if $F = \nabla f$ for some scalar valued function f, which is referred to in the literature as a **potential function**. If F has continuous first-order partial derivatives, then it is a consequence of Stokes theorem (see below) that F is conservative if and only if curl $F = 0$.

It is not hard to see that if F is conservative, then curl $F = 0$ if F has continuous second-order partial derivatives. Indeed, if $F = \nabla f = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z})$, we have

$$\text{curl } F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ f_x & f_y & f_z \end{vmatrix} = (f_{zy} - f_{yz}, f_{xz} - f_{zx}, f_{yx} - f_{xy}) = \mathbf{0}.$$

We record without proof, the following.

Theorem 1.25 *Suppose F is a vector field whose component functions have continuous first-order partial derivatives. Then,* curl $F = 0$ *if and only if F is conservative.*

Generally, given a function $f : \mathbb{R}^n \to \mathbb{R}^m$ we can write

$$f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x})),$$

so that we obtain **component functions** $f_i : \mathbb{R}^n \to \mathbb{R}$ and we can therefore consider the partial derivatives of each of these component functions. It is now easily seen that if f is differentiable at $\mathbf{a}$, then $f'(\mathbf{a})$ is the $m \times n$ matrix of partial derivatives $D_j f_i(\mathbf{a})$ which is the Jacobian matrix. If $m = 1$, we see that the Jacobian matrix is the row vector ∇f, the gradient of f.

If f is a scalar field (that is, $m = 1$), there is a nice relationship between the directional derivative and the gradient of f. The reader can verify that if $\mathbf{u}$ is a vector of length 1,

$$f_{\mathbf{u}}(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{u}.$$

Exercises

1. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $\mathbf{a}$ show that the derivative $\lambda = Df(\mathbf{a})$ is unique. [Hint: Note that for a real number t, $\lambda(t\mathbf{h}) = t\lambda(\mathbf{h})$ by the linearity of λ.]

2. Show that if $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $\mathbf{a}$, then it is continuous at $\mathbf{a}$.

3. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $\mathbf{a}$ and $g : \mathbb{R}^m \to \mathbb{R}^r$ is differentiable at $f(\mathbf{a})$, show that $D(g \circ f)(a) = Dg(f(\mathbf{a})) \circ D(f(\mathbf{a}))$.

4. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is a constant function, then show that $Df(\mathbf{a}) = 0$ for every $\mathbf{a} \in \mathbb{R}^n$.

5. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then $Df(a) = f$.

6. If $f, g : \mathbb{R}^n \to \mathbb{R}$ are differentiable at $\mathbf{a}$, then

$$D(f \cdot g)(\mathbf{a}) = g(\mathbf{a})Df(\mathbf{a}) + f(\mathbf{a})Dg(\mathbf{a}).$$

7. Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable. If $D_{i,j}f$ and $D_{j,i}f$ are continuous in an open set containing $\mathbf{a}$ then $D_{i,j}f(\mathbf{a}) = D_{j,i}f(\mathbf{a})$.

8. Let $f(x, y)$ be continuous and have continuous partial derivatives of order at least two in a region A of $\mathbb{R}^2$ including (x_0, y_0). A sufficient condition that (x_0, y_0) is a relative minimum is that

$$\left|\begin{matrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{matrix}\right|_{(x,y)=(x_0,y_0)} > 0 \quad \text{and} \quad \frac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0.$$

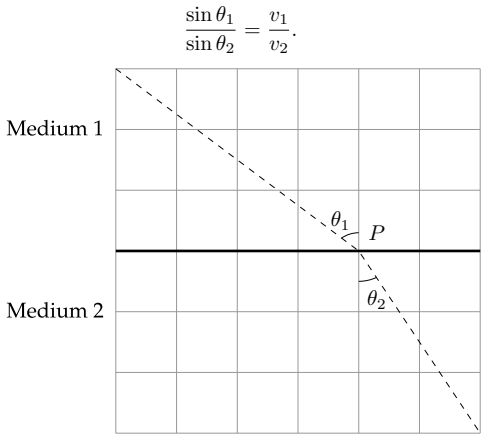
9. Find the relative maxima and minima of $f(x, y) = x^3 + y^3 - 3x - 12y + 20$.

10. Let f be a C^∞ function of two variables. Suppose that $f(0, 0) = 0$ and that $f(ta, tb) = t^2 f(a, b)$ for all real numbers t and all vectors (a, b). Show that

$$f(\mathbf{x}) = \frac{1}{2}(\mathbf{x} \cdot \nabla)^2 f(0, 0).$$

11. A ray of light travels at a constant speed in a uniform medium. In different media (such as air and water), light travels at different speeds. When light passes from one medium into another, light is refracted as shown in the figure below. If the

Fig. 1.5 Snell's law



speed of light is v_1 in medium 1 and v_2 in medium 2, then by Fermat's principle of least time, light will strike the boundary between medium 1 and medium 2 at a point P so that the total time is minimized. With the angles θ_1 and θ_2 shown in the figure, derive **Snell's law** (Fig. 1.5):

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}.$$

1.12 The Inverse Function Theorem

From single-variable calculus, the student is familiar with the following theorem: if $f : \mathbb{R} \to \mathbb{R}$ is continuously differentiable in an open set containing a and $f'(a) \neq 0$, then there is an open set V with $a \in V$ and an open set W with $f(a) \in W$ such that $f : V \to W$ has a continuous inverse $f^{-1} : W \to V$ which is differentiable. Moreover, for $y \in W$, we have

$$(f^{-1})'(y) = [f'(f^{-1}(y)]^{-1}.$$

The reason for this is clear. Indeed, if $f'(a) > 0$, there is an open interval V containing a such that $f'(x) > 0$ for $x \in V$. Thus, f is increasing on V and is therefore one-to-one with an inverse function f^{-1} defined on some open interval W containing $f(a)$. It is not difficult to show that f^{-1} is differentiable and that

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

Indeed, as $f(f^{-1}(y)) = y$, we have by the chain rule (differentiating with respect to y)

$$f'(f^{-1}(y))(f^{-1})'(y) = 1.$$

A similar statement can be made if $f'(a) < 0$. There is an analogous theorem for higher dimensions, and this is the content of the general inverse function theorem. To this end, we begin with the following lemma.

Lemma 1.3 *Let $A \subset \mathbb{R}^n$ be a rectangle and let $f : A \to \mathbb{R}^n$ be continuously differentiable. If there is a number M such that $|D_j f_i(\mathbf{x})| \leq M$ for all $x \in A^o$, then*

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq n^{3/2} M |\mathbf{x} - \mathbf{y}|,$$

for all $\mathbf{x}, \mathbf{y} \in A$.

Proof Suppose first $n = 2$. Then

$$f_i(y_1, y_2) - f_i(x_1, x_2) = [f_i(y_1, y_2) - f_i(y_1, x_2)] + [f_i(y_1, x_2) - f_i(x_1, x_2)].$$

Applying the mean value theorem to each of the bracketed terms on the right-hand side, we obtain

$$f_i(y_1, y_2) - f_i(y_1, x_2) = (y_2 - x_2) D_2 f_i(z_{i2}),$$

$$f_i(y_1, x_2) - f_i(x_1, x_2) = (y_1 - x_1) D_1 f_i(z_{i1}),$$

for some z_{ij}. Thus,

$$|f_i(y_1, y_2) - f_i(x_1, x_2)| \leq M(|y_1 - x_1| + |y_2 - x_2|) \leq \sqrt{2} M |\mathbf{y} - \mathbf{x}|,$$

by Exercise 1. Now,

$$|f(\mathbf{y}) - f(\mathbf{x})| \leq \sum_{i=1}^{2} |f_i(\mathbf{y}) - f_i(\mathbf{x})| \leq 2^{3/2} M |\mathbf{y} - \mathbf{x}|.$$

For the general case, we have

$$f_i(\mathbf{y}) - f_i(\mathbf{x}) = \sum_{j=1}^{n} [f_i(y_1, \ldots, y_j, x_{j+1}, \ldots, x_n) - f_i(y_1, \ldots, y_{j-1}, x_j, \ldots, x_n)].$$

Applying the mean value theorem for each component, we get

$$|f_i(\mathbf{y}) - f_i(\mathbf{x})| \leq \sum_{j=1}^{n} M |y_j - x_j| \leq \sqrt{n} M |\mathbf{y} - \mathbf{x}|.$$

Again by Exercise 1 below, we see

$$|f(\mathbf{y}) - f(\mathbf{x})| \leq \sum_{i=1}^{n} |f_i(\mathbf{y}) - f_i(\mathbf{x})| \leq n^{3/2} M |\mathbf{y} - \mathbf{x}|.$$

□

Theorem 1.26 (The inverse function theorem) *Suppose that*

$$f : \mathbb{R}^n \to \mathbb{R}^n$$

is continuously differentiable in an open set containing $\mathbf{a}$ *and* $\det f'(\mathbf{a}) \neq 0$. *Then there is an open set* V *containing* $\mathbf{a}$ *and an open set* W *containing* $f(\mathbf{a})$ *such that* $f : V \to W$ *has a continuous inverse* $f^{-1} : W \to V$ *which is differentiable and for all* $y \in W$ *satisfies*

$$(f^{-1})'(y) = [f'(f^{-1}(y))]^{-1}.$$

Proof Let λ be the linear transformation $Df(\mathbf{a})$. As $\det f'(\mathbf{a}) \neq 0$, λ is non-singular. Now $D(\lambda^{-1} \circ f)(\mathbf{a}) = D(\lambda^{-1})(f(\mathbf{a})) \circ Df(\mathbf{a})$. As λ^{-1} is a linear transformation, $D(\lambda^{-1}) = \lambda^{-1} = (Df(\mathbf{a}))^{-1}$. Thus, $D(\lambda^{-1} \circ f)(\mathbf{a})$ is the identity linear transformation. If the theorem is true for $\lambda^{-1} \circ f$, then it is clearly true for f, so we may assume without any loss of generality that λ is the identity. Thus, whenever $f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a})$ we have

$$\frac{|f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) - \lambda(\mathbf{h})|}{|\mathbf{h}|} = 1.$$

But as f is differentiable at $\mathbf{a}$ with derivative λ, this limit is zero as $\mathbf{h}$ tends to zero. This means we cannot have $f(\mathbf{x}) = f(\mathbf{a})$ for $\mathbf{x}$ arbitrarily close to, but unequal to, $\mathbf{a}$. Thus, there is a closed rectangle U containing $\mathbf{a}$ in its interior such that

$$f(\mathbf{x}) \neq f(\mathbf{a}) \quad \text{if} \quad \mathbf{x} \in U \quad \text{and} \quad \mathbf{x} \neq \mathbf{a}. \tag{1.16}$$

Since f is continuously differentiable in an open set containing $\mathbf{a}$, we can also assume that

$$\det f'(\mathbf{x}) \neq 0 \quad \text{for} \quad \mathbf{x} \in U \tag{1.17}$$

and

$$|D_j f_i(\mathbf{x}) - D_j f(\mathbf{a})| \leq 1/2n^{3/2} \quad \forall i, j \quad \text{and} \quad x \in U.$$

This last condition combined with Lemma 1.3 applied to $g(\mathbf{x}) = f(\mathbf{x}) - \mathbf{x}$ gives for $\mathbf{x}_1, \mathbf{x}_2 \in U$ that

$$|f(\mathbf{x}_1) - \mathbf{x}_1 - (f(\mathbf{x}_2) - \mathbf{x}_2)| \leq \frac{1}{2}|\mathbf{x}_1 - \mathbf{x}_2|.$$

Since (by the triangle inequality)

$$|\mathbf{x}_1 - \mathbf{x}_2| - |f(\mathbf{x}_1) - f(\mathbf{x}_2)| \le |f(\mathbf{x}_1) - \mathbf{x_1} - (f(\mathbf{x}_2) - \mathbf{x}_2)| \le \frac{1}{2}|\mathbf{x}_1 - \mathbf{x}_2|$$

we deduce

$$|\mathbf{x}_1 - \mathbf{x}_2| \le 2|f(\mathbf{x}_1) - f(\mathbf{x}_2)| \quad \text{for} \quad \mathbf{x}_1, \mathbf{x}_2 \in U. \tag{1.18}$$

Now $f(\partial U)$ is a compact set which by (1.16), does not contain $f(\mathbf{a})$. Therefore, there is a number $d > 0$ such that $|f(\mathbf{a}) - f(\mathbf{x})| \ge d$ for $x \in \partial U$. Let

$$W = \{\mathbf{y} : |\mathbf{y} - f(\mathbf{a})| < d/2\}.$$

If $\mathbf{y} \in W$ and $\mathbf{x} \in \partial U$, then

$$|\mathbf{y} - f(\mathbf{a})| < |\mathbf{y} - f(\mathbf{x})|. \tag{1.19}$$

We will show that for any $\mathbf{y}$ in W there is a unique $\mathbf{x}$ in the interior of U such that $f(\mathbf{x}) = \mathbf{y}$. To prove this, consider the function $g : U \to \mathbb{R}$ given by

$$g(\mathbf{x}) = |\mathbf{y} - f(\mathbf{x})|^2 = \sum_{i=1}^{n} (y_i - f_i(\mathbf{x}))^2.$$

This function is continuous and therefore has a minimum on U. If $\mathbf{x} \in \partial U$, then by (1.19), we have $g(\mathbf{a}) < g(\mathbf{x})$. Therefore, the minimum does not occur on the boundary of U. Thus, there is a point $\mathbf{x}$ in the interior of U such that $D_j g(\mathbf{x}) = 0$ for all j. In other words,

$$\sum_{i=1}^{n} 2(y_i - f_i(\mathbf{x}))D_j f_i(\mathbf{x}) = 0, \qquad \text{for all } j.$$

By (1.17), the matrix $(D_j f_i(x))$ has nonzero determinant for all $x \in U$. Therefore, $y_i = f_i(x)$ for all i. In other words, $\mathbf{y} = f(\mathbf{x})$. This proves the existence of $\mathbf{x}$. Uniqueness follows from (1.18). If $V = U^o \cap f^{-1}(W)$, we have shown that the function $f : V \to W$ has an inverse $f^{-1} : W \to V$. We can rewrite (1.18) as

$$|f^{-1}(\mathbf{y_1}) - f^{-1}(\mathbf{y_2})| \le 2|\mathbf{y_1} - \mathbf{y_2}| \quad \mathbf{y_1}, \mathbf{y_2} \in W. \tag{1.20}$$

This implies that f^{-1} is continuous. To complete the proof, we need to show that f^{-1} is differentiable. Let $u = Df(\mathbf{x})$. We will show that f^{-1} is differentiable at $\mathbf{y} = f(\mathbf{x})$ with derivative u^{-1}. By the definition of the derivative,

$$f(\mathbf{x}_1) = f(\mathbf{x}) + u(\mathbf{x}_1 - \mathbf{x}) + r(\mathbf{x}_1 - \mathbf{x})$$

where

$$\lim_{\mathbf{x}_1 \to \mathbf{x}} \frac{|r(\mathbf{x}_1 - \mathbf{x})|}{|\mathbf{x}_1 - \mathbf{x}|} = 0. \tag{1.21}$$

Therefore

$$u^{-1}(f(\mathbf{x}_1) - f(\mathbf{x})) = \mathbf{x}_1 - \mathbf{x} + u^{-1}(r(\mathbf{x}_1 - \mathbf{x})).$$

Since every $\mathbf{y}_1 \in W$ is of the form $f(\mathbf{x}_1)$ for some $\mathbf{x}_1 \in V$, we can rewrite this as

$$f^{-1}(\mathbf{y}_1) = f^{-1}(\mathbf{y}) + u^{-1}(\mathbf{y}_1 - \mathbf{y}) - u^{-1}(r(f^{-1}(\mathbf{y}_1) - f^{-1}(\mathbf{y}))),$$

so to complete the proof, it suffices to show that the last term is $o(|\mathbf{y}_1 - \mathbf{y}|)$ as $\mathbf{y}_1 \to \mathbf{y}$. As u^{-1} is a linear transformation (see Exercise 5 below) , it suffices to show that

$$(r(f^{-1}(\mathbf{y}_1) - f^{-1}(\mathbf{y}))) = o(|\mathbf{y}_1 - \mathbf{y}|).$$

But this is immediate as f^{-1} is continuous. Indeed, writing

$$\frac{|r(f^{-1}(\mathbf{y}_1) - f^{-1}(\mathbf{y}))|}{|\mathbf{y}_1 - \mathbf{y}|} = \frac{|r(f^{-1}(\mathbf{y}_1) - f^{-1}(\mathbf{y}))|}{|f^{-1}(\mathbf{y}_1) - f^{-1}(\mathbf{y})|} \cdot \frac{|f^{-1}(\mathbf{y}_1) - f^{-1}(\mathbf{y})|}{|\mathbf{y}_1 - \mathbf{y}|}$$

we see by (1.20), the second factor is bounded by 2 and the first factor approaches zero by (1.21) because $f^{-1}(\mathbf{y}_1) \to f^{-1}(\mathbf{y})$ by the continuity of f^{-1}. □

Exercises

1. Prove that if $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, then

$$\sum_{j=1}^{n} |x_j| \leq \sqrt{n}||\mathbf{x}||.$$

2. If $T : \mathbb{R}^m \to \mathbb{R}^n$ is a linear transformation, show that there is a number M such that

$$|T(\mathbf{h})| \leq M|\mathbf{h}| \qquad \forall\, \mathbf{h} \in \mathbb{R}^m.$$

3. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable such that $f'(a) \neq 0$ for all $a \in \mathbb{R}$. Show that f is one-to-one on all of $\mathbb{R}$.

4. Determine if the following functions are locally C^1-invertible at the given point.

   (a) $f(x, y) = (x^2 - y^2, 2xy)$ at $(x, y) \neq (0, 0)$;

   (b) $f(x, y) = (x^3y + 1, x^2 + y^2)$ at $(1, 2)$;

   (c) $f(x, y) = (x/(x^2 + y^2), y/(x^2 + y^2))$ at $(x, y) \neq (0, 0)$;

   (d) $f(x, y) = (x + x^2 + y, x^2 + y^2)$ at $(x, y) = (5, 8)$.

5. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be given by $f(x, y) = (e^x \cos y, e^x \sin y)$. Show that f is invertible in a neighborhood of each point of $\mathbb{R}^2$.

6. With f as in the previous exercise, show that f has no global inverse. That is, there is no function $g : \mathbb{R}^2 \to \mathbb{R}^2$ such that $g(f(\mathbf{x})) = \mathbf{x}$.

7. Let $A \subset \mathbb{R}^n$ be an open set and $f : A \to \mathbb{R}^n$ a continuously differentiable one-to-one function such that $\det f'(\mathbf{x}) \neq 0$ for all $\mathbf{x}$. Show that $f(A)$ is an open set and $f^{-1} : f(A) \to A$ is differentiable.

8. Consider the function

$$f(x) = \begin{cases} \frac{x}{2} + x^2 \sin \frac{1}{x} & x \neq 0 \\ \\ 0 & x = 0 \end{cases}$$

to show that the continuity of the derivative cannot be eliminated from the hypothesis of Theorem 1.26.

1.13 The Implicit Function Theorem

The implicit function theorem and the inverse function theorem have a long and luminous history that the student can find described in [4]. They seem to have roots in the works of Isaac Newton (1642–1727) and Gottfried Leibniz (1646–1716). Though Joseph Louis Lagrange (1736–1813) found a version of the theorem which is essentially the inverse function theorem discussed below, it was Augustin Louis Cauchy (1789–1857) who studied the theorem with sufficient mathematical rigor and so is acknowledged today as its discoverer.

We begin with a motivating example. Suppose we are given the function $f(x, y) = x^2 + y^2 - 1$. If we choose a, b such that $f(a, b) = 0$ and $a \neq \pm 1$, then there are open intervals A containing a and B containing b such that if $x \in A$, there is a unique $y \in B$ with $f(x, y) = 0$. We can therefore define a function $g : A \to \mathbb{R}$ by the condition that $g(x) \in B$ and $f(x, g(x)) = 0$. If $b > 0$, then $g(x) = \sqrt{1 - x^2}$ (See Fig. 1.6). For our function, there is another number b_1 such that $f(a, b_1) = 0$. There will also be another interval B_1 containing b_1 such that when $x \in A$, we have $f(x, h(x)) = 0$ for a unique $h(x) \in B_1$. In this case, $h(x) = -\sqrt{1 - x^2}$. Both g and h are differentiable, and these functions are said to be defined **implicitly** by the equation $f(x, y) = 0$. For $a = \pm 1$, it is impossible to find any such function g defined on an open interval containing a.

We would like a simple criterion for deciding when such a function can be found for any general continuously differentiable function f of several variables. This is supplied by the **implicit function theorem**. More generally, we ask the following. Given $f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$ and $f(a_1, \ldots, a_n, b) = 0$, when can we find for each

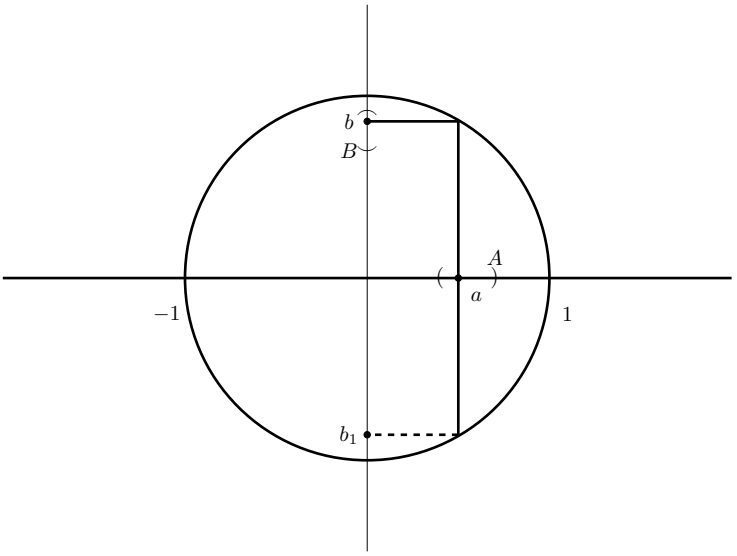


Fig. 1.6 Graph of $x^2 + y^2 - 1 = 0$

$(x_1, \ldots, x_n)$ near $(a_1, \ldots, a_n)$ a unique y such that $f(x_1, \ldots, x_n, y) = 0$? We can even be a bit more general. Given a set of functions

$$f_i : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}, \qquad 1 \leq i \leq m$$

that satisfy

$$f_i(a_1, \ldots, a_n, b_1, \ldots, b_m) = 0, \qquad 1 \leq i \leq m,$$

when can we find, for each $(x_1, \ldots, x_n)$ near $(a_1, \ldots, a_n)$ a unique $(y_1, \ldots, y_m)$ near $(b_1, \ldots, b_m)$ which satisfies

$$f_i(x_1, \ldots, x_n, y_1, \ldots, y_m) = 0, \qquad 1 \leq i \leq m.$$

This is the content of the following implicit function theorem which uses the inverse function theorem in an essential way.

Theorem 1.27 (The implicit function theorem) *Suppose* $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ *is continuously differentiable in an open set containing* $(\mathbf{a}, \mathbf{b})$ *with* $\mathbf{a} \in \mathbb{R}^n$ *and* $\mathbf{b} \in \mathbb{R}^m$ *respectively. Suppose further that* $f(\mathbf{a}, \mathbf{b}) = 0$. *Let* M *be the* $m \times m$ *matrix*

$$(D_{n+j} f_i(\mathbf{a}, \mathbf{b})), \qquad 1 \leq i, j \leq m.$$

If $\det M \neq 0$*, then there is an open set* $A \subset \mathbb{R}^n$ *containing* $\mathbf{a}$ *and an open set* $B \subset \mathbb{R}^m$ *containing* $\mathbf{b}$ *such that for each* $\mathbf{x} \in A$*, there is a unique* $g(\mathbf{x}) \in B$ *such that* $f(\mathbf{x}, g(\mathbf{x})) = 0$*. Moreover, the function* g *is differentiable.*

Proof Define $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n \times \mathbb{R}^m$ by $F(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, f(\mathbf{x}, \mathbf{y}))$. Then $\det F'(\mathbf{a}, \mathbf{b}) = \det M \neq 0$. By the inverse function theorem, there is an open set $W \subset \mathbb{R}^n \times \mathbb{R}^m$ containing $F(\mathbf{a}, \mathbf{b}) = (\mathbf{a}, \mathbf{0})$ and an open set in $\mathbb{R}^n \times \mathbb{R}^m$ containing $(\mathbf{a}, \mathbf{b})$ which we may take to be of the form $A \times B$ such that $F : A \times B \to W$ has a differentiable inverse $h : W \to A \times B$. Clearly, h is of the form $h(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, k(\mathbf{x}, \mathbf{y}))$ for some differentiable function k (since F is of this form). Let $p : \mathbb{R}^n \times \mathbb{R}^m$ be the projection map $p(\mathbf{x}, \mathbf{y}) = \mathbf{y}$. Then $p \circ F = f$. Therefore

$$f(\mathbf{x}, k(\mathbf{x}, \mathbf{y})) = f \circ h(\mathbf{x}, \mathbf{y}) = (p \circ F) \circ h(\mathbf{x}, \mathbf{y}) = p \circ (F \circ h)(\mathbf{x}, \mathbf{y}) = p(\mathbf{x}, \mathbf{y}) = \mathbf{y},$$

since $F \circ h$ is the identity map. Therefore, $f(\mathbf{x}, k(\mathbf{x}, \mathbf{0})) = \mathbf{0}$. In other words, we can define $g(\mathbf{x}) = k(\mathbf{x}, \mathbf{0})$. This completes the proof. □

Exercises

1. The equations relating Cartesian coordinates and polar coordinates are given by the familiar
$$x = r\cos\theta, \quad y = r\sin\theta.$$
Using the inverse function theorem, show that we can solve (locally) for r and θ uniquely in terms of x and y as long as we are away from the origin.

2. The cylindrical coordinates in $\mathbb{R}^3$ are given by
$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z.$$
Determine the points of $\mathbb{R}^3$ for which we can solve (locally) for r, θ and z in terms of Cartesian coordinates.

3. Spherical coordinates in $\mathbb{R}^3$ are given by (Fig. 1.7)
$$x = r\sin\phi\cos\theta, \quad y = r\sin\phi\sin\theta, \quad z = r\cos\phi.$$
Determine the points in $\mathbb{R}^3$ for which we can solve (locally) for r, ϕ, θ in terms of x, y and z.

4. Let $y = g(x)$ be an implicit function satisfying $f(x, g(x)) = 0$ with both f, g being continuously differentiable real-valued functions defined on $\mathbb{R}^2$ and $\mathbb{R}$

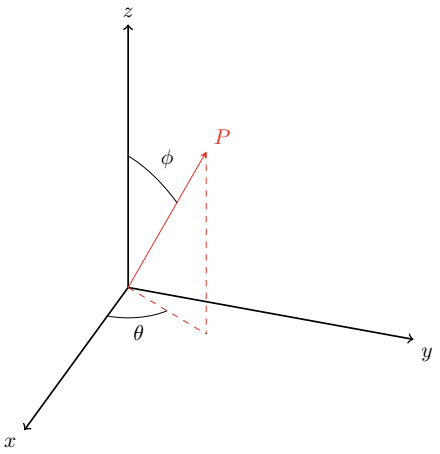


Fig. 1.7 Spherical coordinates

respectively. Show that

$$g'(x) = -\frac{D_1 f(x, g(x))}{D_2 f(x, g(x))},$$

whenever $D_2 f(x, g(x)) \neq 0$.

5. Generalize the previous exercise to functions of three variables as follows. Suppose that the equation $F(x, y, z) = 0$ implicitly defines a function $z = f(x, y)$ where f is differentiable. Show that

$$\frac{\partial z}{\partial x} = -\frac{D_1 F}{D_3 F}, \qquad \frac{\partial z}{\partial y} = -\frac{D_2 F}{D_3 F},$$

provided $D_3 F \neq 0$.

1.14 The Lagrange Multiplier Method

The **Lagrange multiplier method** gives conditions for finding the maxima or minima of a scalar field subject to a side condition. Suppose we want to find the maximum or minimum of a function $f : \mathbb{R}^n \to \mathbb{R}$ subject to the side condition $g(\mathbf{x}) = 0$ for some differentiable $g : \mathbb{R}^m \to \mathbb{R}$. Let C be any curve given by $\mathbf{r} : [0, 1] \to \mathbb{R}^n$ lying on the hypersurface defined by $g(\mathbf{x}) = 0$. Thus, if $\mathbf{r}(t) = (x_1(t), \ldots, x_n(t))$, then $g(r(t)) = 0$. Now if f has an extremum at $\mathbf{x}_0$ (say), and C passes through $\mathbf{x}_0$, then setting $h(t) := f(\mathbf{r}(t))$, we see that h also has an extremum at t_0 where t_0 is such that $r(t_0) = \mathbf{x}_0$. Thus, by the chain rule, we deduce that

$$0 = h'(t_0) = \nabla f(\mathbf{r}(t_0)) \cdot \mathbf{r}'(t_0).$$

In other words, $\nabla f(\mathbf{x}_0)$ is orthogonal to the tangent vector $\mathbf{r}'(t_0)$ for every curve C lying on $g = 0$ passing through $\mathbf{x}_0$. But if $g(\mathbf{x}) = 0$, we see that for any $\mathbf{u}$,

$$0 = D_{\mathbf{u}} g = \nabla g \cdot \mathbf{u},$$

which means that ∇g is also orthogonal to $\mathbf{r}'(t_0)$. Therefore, ∇f and ∇g must be parallel at the extreme point. In other words, there is a λ such that

$$\nabla f(\mathbf{x}_0) = \lambda \nabla g(\mathbf{x}_0).$$

This is the essential idea in the Lagrange multiplier method. It was discovered by Joseph Louis Lagrange (1736–1813) who made fundamental contributions not only to analysis but also to number theory and group theory.

We summarize our discussion formally in the following theorem and leave the proof to the reader.

Theorem 1.28 (Lagrange multiplier method) *Let U be an open set of $\mathbb{R}^n$ and suppose that $g : U \to \mathbb{R}$ is a continuously differentiable function on U. Let S be the set of points $\mathbf{x}$ in U such that $g(\mathbf{x}) = 0$ and $\nabla g(\mathbf{x}) \neq 0$. Let $f : U \to \mathbb{R}$ be continuously differentiable on U and assume that $\mathbf{x}_0$ is a point of S such that $\mathbf{x}_0$ is an extreme point for f on S. That is, $\mathbf{x}_0$ is an extremum for f subject to the constraint g. Then, there is a number λ such that*

$$\nabla f(\mathbf{x}_0) = \lambda \nabla g(\mathbf{x}_0).$$

We can consider the situation where there are more constraints. Suppose we are interested in finding the extrema of a function $f : \mathbb{R}^n \to \mathbb{R}$ subject to the constraints

$$g_i(\mathbf{x}) = 0, \qquad 1 \leq i \leq k,$$

with $k < n$. Let S be the hypersurface defined by this system of constraints. If $f|_S$ has an extremum at $\mathbf{x}_0$, with $\nabla g_1(\mathbf{x}_0), \ldots, \nabla g_k(\mathbf{x}_0)$ linearly independent, then there are scalars $\lambda_1, \ldots, \lambda_k$ such that

$$\nabla f(\mathbf{x}_0) = \lambda_1 \nabla g_1(\mathbf{x}_0) + \cdots + \lambda_k \nabla g_k(\mathbf{x}_0).$$

This is deduced by a reasoning similar to the idea indicated above.

Exercises

1. Find a formula for the surface area of an open box with length x, width y and height z. If the volume V is fixed, determine the minimum surface area.

2. The unit hypersphere in $\mathbb{R}^n$ centered at the origin is defined by the equation

$$x_1^2 + x_2^2 + \cdots + x_n^2 = 1.$$

   Find the pair of points $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ on the unit sphere that maximize and minimize the function

$$f(x_1, \ldots, x_n, y_1, \ldots, y_n) = \sum_{j=1}^{n} x_j y_j.$$

   What are the maximum and minimum values of f?

3. Let $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ be any points in $\mathbb{R}^n$. Noting that the vectors $\mathbf{u}$ and $\mathbf{v}$ with

$$u_i = \frac{x_i}{|\mathbf{x}|}, \qquad v_i = \frac{y_i}{|\mathbf{y}|},$$

   lie on the unit hypersphere, use the previous exercise to derive the Cauchy–Schwarz inequality:

$$|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}|\,|\mathbf{y}|.$$

4. Use the Lagrange multiplier method to show that the distance from the point (x_0, y_0) to the line $ax + by = d$ is

$$\frac{|ax_0 + by_0 - d|}{\sqrt{a^2 + b^2}}.$$

5. Show that the distance from the point $(x_0, y_0, z_0) \in \mathbb{R}^3$ to the plane

$$ax + by + cz = d$$

   is

$$\frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

6. Show that the maximum of

$$f(x, y, z) = x^2 y^2 z^2$$

subject to the constraint $x^2 + y^2 + z^2 = a^2$ is $a^6/27$.

7. Use the previous question to show that for all x, y, z,

$$(x^2y^2z^2)^{1/3} \leq \frac{x^2 + y^2 + z^2}{3}.$$

8. Prove the **arithmetic mean - geometric mean inequality**: for any positive numbers $x_1, x_2, \ldots, x_n$, we have

$$(x_1x_2 \cdots x_n)^{1/n} \leq \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

9. **Heron's formula** for the area of triangle whose sides have length x, y, z is given by

$$\sqrt{s(s-a)(s-b)(s-c)},$$

where s is the semiperimeter (i.e., equal to $(x + y + z)/2$). If the perimeter is fixed, show that the triangle with the largest area is the equilateral triangle.

10. Let A be the $n \times n$ symmetric matrix with real entries a_{ij} so that $a_{ij} = a_{ji}$. Show that the global maximum value of the quadratic form

$$\sum_{i,j=1}^{n} a_{ij}x_ix_j$$

subject to the constraint

$$x_1^2 + x_2^2 + \cdots + x_n^2 = 1$$

is equal to the largest eigenvalue of A. Show that the global minimum value of the quadratic form subject to the same constraint is equal to the smallest eigenvalue of A. (Recall that the eigenvalues of a real symmetric matrix are real.)

1.15 Level Sets and Tangent Spaces

Given a function $f : \mathbb{R}^n \to \mathbb{R}^m$, it is often difficult to visualize the function due to many dimensions. It is therefore convenient to consider **level sets** which are defined as follows. For each $\mathbf{a} \in \mathbb{R}^m$, the inverse image $f^{-1}(\mathbf{a})$ is called the **level set** at $\mathbf{a}$.

Suppose now f is differentiable at $\mathbf{a}$. Then

$$h(\mathbf{x}) = f(\mathbf{a}) + f'(\mathbf{a})(\mathbf{x} - \mathbf{a}),$$

(where $f'(a)$ is the Jacobian matrix $D(f)$ evaluated at a) is the closest linear approximation to $f(\mathbf{x})$ near $\mathbf{a}$. The level set $h^{-1}(f(\mathbf{a}))$ also contains $\mathbf{a}$ and we call this the **tangent space** at $\mathbf{a}$. One can visualize this tangent space as the vector space of all tangent vectors to all possible curves on the surface through the point $\mathbf{a}$. Clearly, the tangent space at $\mathbf{a}$ consists of all vectors $\mathbf{v}$ such that

$$f'(\mathbf{a}) \cdot \mathbf{v} = 0.$$

If C is a curve parameterized by the map $t \mapsto (x_1(t), \ldots, x_n(t))$ for $a \leq t \leq b$, then, we see that the **line integral** along C is

$$\int_C f(x_1, \ldots, x_n) ds := \int_a^b f(x_1(t), \ldots, x_n(t))(x_1'(t)^2 + \cdots + x_n'(t)^2)^{1/2} dt.$$

In effect, we are integrating along the curve and this can be seen as a generalization of (1.14). One can integrate with respect to the coordinate parameters as well and consider generally,

$$\int_C f(x_1, \ldots, x_n) dx_i, \qquad 1 \leq i \leq n.$$

Thus, if C is parameterized as before, then

$$\int_C f(x_1, \ldots, x_n) dx_i = \int_a^b f(x_1(t), \ldots, x_n(t)) x_i'(t) dt.$$

Exercises

1. In $\mathbb{R}^3$, **quadric surfaces** are described by equations of the form

   $$Ax^2 + Bxy + Cxz + Dy^2 + Eyz + Fz^2 + Gx + Hy + Iz + J = 0$$

   with $A, B, C, D, E, F, G, H, I, J$ being fixed real numbers. An example of a quadric surface is the **ellipsoid** given by

   $$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

   Show that the volume of this ellipsoid is $\frac{1}{4}\pi^2 abc$.

2. Find equations for the tangent plane to the surface

   $$x^2yz + 3y^2 - 2xz^2 - 8z = 0$$

at the point $(1, 2, -1)$.

3. Show that the surface
$$x^2 - 2yz + y^3 = 4$$
is perpendicular to any member of the family of surfaces
$$x^2 + (4a - 2)y^2 - az^2 + 1 = 0$$
at the point of intersection $(1, -1, 2)$ for any real value a.

1.16 Changing Variables in Integrals

The first application of the Jacobian matrix is the generalization to the multivariable setting of the change of variable method of one-variable integral calculus. The reader will recall that if $g : [a, b] \to \mathbb{R}$ is continuously differentiable, and $f : \mathbb{R} \to \mathbb{R}$ is continuous, then
$$\int_{g(a)}^{g(b)} f(x)dx = \int_a^b f(g(x))g'(x)dx.$$

If g is one-to-one, then this can be rewritten as
$$\int_{g(a,b)} f(x)dx = \int_{(a,b)} f(g(x))|g'(x)|\,dx.$$

The generalization of this formula to higher dimensions involves the Jacobian matrix. The situation is summarized by the following theorem whose proof can be found in many books such as [5].

Theorem 1.29 *Let $A \subset \mathbb{R}^n$ be an open set and $g : A \to \mathbb{R}^n$ a one-to-one, continuously differentiable function such that* $\det g'(\mathbf{x}) \neq 0$ *for all* $\mathbf{x} \in A$. *If $f : g(A) \to \mathbb{R}$ is integrable, then*
$$\int_{g(A)} f(\mathbf{x})d\mathbf{x} = \int_A f(g(\mathbf{x}))|\det g'(\mathbf{x})|\,d\mathbf{x}.$$

Alternately, this theorem can also be written as follows (which may be more familiar to the student). For ease of notation, we specialize to the case $n = 2$. Let R be a region in the xy-plane which is mapped in a one-to-one and onto fashion to the region R' in the uv-plane via the transformation
$$x = F(u, v) \quad y = G(u, v).$$

Then,

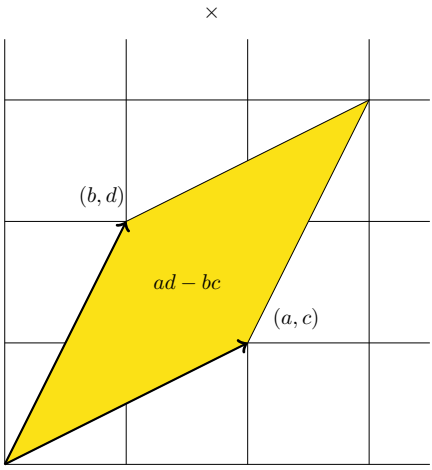


Fig. 1.8 Area of the parallelogram

$$\iint_R f(x, y)dxdy = \iint_{R'} f(F(u, v), G(u, v))\frac{\partial(F, G)}{\partial(u, v)}dudv. \tag{1.22}$$

The essential idea of the proof is understood by noting how an infinitesimal volume element changes under a linear transformation. We illustrate this in the case of $\mathbb{R}^2$. The student will recall that if we have two vectors

$$\mathbf{v} = \begin{pmatrix} a \\ c \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} b \\ d \end{pmatrix},$$

the area of the parallelogram spanned by them is given by the absolute value of the determinant $ad - bc$. Indeed, the area is (see (1.13) and Fig. 1.8 below)

$$\sqrt{|\mathbf{v}|^2|\mathbf{w}|^2 - (\mathbf{v} \cdot \mathbf{w})^2}.$$

Computing this directly using vector coordinates leads to the square root of

$$(a^2 + c^2)(b^2 + d^2) - (ab + cd)^2$$

which is easily seen to be

$$(ad - bc).$$

This is the determinant of the 2×2 matrix whose columns are $\mathbf{v}$ and $\mathbf{w}$.

With this observation, it is possible to provide the following intuitive proof. Suppose we make the change of variables

$$x = F(u, v), \quad y = G(u, v).$$

The following is suggested by the chain rule. The infinitesimal changes Δx and Δy in x and y are given by:

$$\Delta x = F_u \Delta u + F_v \Delta v, \quad \Delta y = G_u \Delta u + G_v \Delta v.$$

In the xy coordinate system, the area of the rectangle $\Delta x \Delta y$ is transformed in the uv-coordinate system to the area of the parallelogram spanned by the vectors

$$\begin{pmatrix} F_u \\ F_v \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} G_u \\ G_v \end{pmatrix}.$$

In our notation, this area is the Jacobian determinant

$$\frac{\partial(F, G)}{\partial(u, v)}.$$

Thus, if the region R is transformed into R', it is now intuitively clear that (1.22) holds. A more rigorous proof can be found in [5]. In the case $n = 2$, we will show how this is done in the section on Green's theorem.

A classical application of the above theorem is to the evaluation of the probability integral

$$I = \int_{-\infty}^{\infty} e^{-x^2} dx.$$

Indeed, we have

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2 - y^2} dx dy.$$

The integration is over $\mathbb{R}^2$ which we can also parameterize using polar coordinates:

$$x = r \cos\theta, \quad y = r \sin\theta, \qquad r > 0, \quad 0 \leq \theta \leq 2\pi.$$

The Jacobian matrix is easily calculated to be

$$\begin{pmatrix} \cos\theta & -r \sin\theta \\ \sin\theta & r \cos\theta \end{pmatrix},$$

so that our double integral becomes

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta.$$

The inner integral is evidently $1/2$ so that $I^2 = \pi$. Hence $I = \sqrt{\pi}$.

Even though the area of a circle can be calculated using elementary calculus, it is instructive to observe that the above integral I allows us to find a formula for the

area of a circle of radius r which leads us to a generalization to higher dimensions. To this end, we can calculate our integral without invoking any Jacobians but rather appealing to geometric intuition. Indeed,

$$\pi = I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}dxdy,$$

and the integral on the right can be viewed as integrating

$$e^{-x^2-y^2}$$

over expanding circles of radius r with r ranging from zero to infinity. In this expansion, the infinitesimal change in area as the circle expands is given by $2\pi r dr$. In other words, we are integrating over the infinitesimal change in the circumference as r ranges from zero to infinity. Thus,

$$I^2 = 2\pi \int_0^{\infty} e^{-r^2} r dr = \pi.$$

This point of view will be useful in the next section.

There is another way to evaluate the integral using only single variable calculus. We have

$$I^2 = 4\int_0^{\infty}\int_0^{\infty} e^{-(x^2+y^2)}dxdy.$$

We change $x = ty$ in the inner integral and interchange the order, which we can do because of absolute convergence:

$$I^2 = 4\int_0^{\infty}\int_0^{\infty} e^{-y^2(1+t^2)}\, y\, dy\, dt.$$

The inner integral is easily evaluated and we find:

$$I^2 = 4\int_0^{\infty}\left[\frac{-e^{-y^2(1+t^2)}}{2(1+t^2)}\Big|_{y=0}^{y=\infty}\right] dt = 2\int_0^{\infty}\frac{dt}{1+t^2} = 2\big[\arctan t\big]_0^{\infty} = \pi.$$

This idea will be used later when we discuss the beta function and its functional relation to the gamma function.

Exercises

1. Let $\mathbf{a} = (a_1, a_2, a_3)$, $\mathbf{b} = (b_1, b_2, b_3)$ and $\mathbf{c} = (c_1, c_2, c_3)$ be three independent vectors spanning a **parallelepiped** in $\mathbb{R}^3$. Show that the volume of this

parallelepiped is given by the absolute value of

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}).$$

2. With **a**, **b** and **c** as in the previous exercise, show that

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

3. With **a**, **b** and **c** as in the previous exercise, show that

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}.$$

4. Show that

$$\iint_R \sqrt{x^2 + y^2} dx dy = \frac{2}{3}\pi a^3,$$

where R is the circular region $x^2 + y^2 \leq a^2$.

5. With R being the region in the previous exercise, show that

$$\iint_R e^{-x^2+y^2} dx dy = \pi(1 - e^{-a^2}).$$

6. By making the transformation $x + y = u$ and $y = uv$, show that

$$\iint_T e^{y/(x+y)} dx dy = \frac{e-1}{2}$$

where T is the region defined by $x, y \geq 0$ and $x + y \leq 1$.

7. Let R be the region bounded by $x + y = 1$, $x = 0$ and $y = 0$. Under the transformation $x - y = u$ and $x + y = v$, show that R is transformed into the region $|u| \leq v \leq 1$ in the uv-plane.

8. With R as in the previous exercise, show that

$$\iint_R \cos\left(\frac{x-y}{x+y}\right) dx dy = \frac{\sin 1}{2}.$$

9. Show that

$$\iiint_R \frac{dx dy dz}{(x^2 + y^2 + z^2)^{3/2}} = 4\pi \log(a/b),$$

where R is the region bounded by the two spheres $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + z^2 = b^2$, with $a > b > 0$.

10. Let R be the region $x^2 + xy + y^2 \leq 1$ in $\mathbb{R}^2$. Show that

$$\iint_R e^{-(x^2+xy+y^2)} dxdy = \frac{2\pi(e-1)}{\sqrt{3}},$$

by making the transformation

$$x = u\cos a - v\sin a, \quad y = u\sin a + v\cos a,$$

with an appropriate choice of a so as to eliminate the xy term in the integrand.

11. Let S be the square $[0, 1] \times [0, 1]$. Show that

$$\iint_S (1 - x^2y^2)^{-1} dxdy = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.$$

By making the transformation

$$x = \frac{\sin u}{\cos v}, \quad y = \frac{\sin v}{\cos u}, \qquad u, v > 0, \quad u + v \leq \pi/2,$$

show that the integral equals $\pi^2/8$. (This exercise is due to E. Calabi.)

1.17 Volume and Surface Area of the Hypersphere

We can apply this perspective to calculate the volume and surface areas of the n-dimensional hypersphere of radius R. The reader will recall that this hypersphere is the locus of points $(x_1, \ldots, x_n)$ in $\mathbb{R}^n$ such that

$$x_1^2 + x_2^2 + \cdots + x_n^2 \leq R^2.$$

Denoting its volume by $V_n(R)$, we see that it is given by the n-dimensional integral

$$V_n(R) = \int \cdots \int_{x_1^2+\cdots+x_n^2 \leq R^2} dx_1 \cdots dx_n.$$

Replacing x_i by Rx_i, we find

$$V_n(R) = C_n R^n,$$

where C_n is the volume of the hypersphere (denoted S^{n-1}) of radius 1. If we let $S_{n-1}(R)$ denote the surface area of the n-sphere of radius R, we have

$$V_n(R) = \int_0^R S_{n-1}(r)dr,$$

so that by the fundamental theorem of calculus,

$$S_{n-1}(R) = \frac{dV_n(R)}{dR} = nC_nR^{n-1}.$$

Thus, the surface area of S^{n-1} is nC_n. We can also perform this integration by integrating the infinitesimal change in the surface area $d\omega_{n-1}$ as r ranges from zero to infinity so that

$$nC_n = \int_{S^{n-1}} d\omega_{n-1}.$$

This idea was alluded to in the previous section. To determine C_n, we use the following trick:

$$\pi^{n/2} = \left(\int_{-\infty}^{\infty} e^{-x^2}dx\right)^n = \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} e^{-(x_1^2+\cdots+x_n^2)}dx_1\cdots dx_n.$$

The n-fold integral on the right can be viewed as an integration over the surface area element $d\omega_{n-1}$ (say) of the hypersphere

$$x_1^2 + \cdots + x_n^2 = r^2$$

integrated over the radial component. Thus,

$$\pi^{n/2} = \int_0^{\infty} e^{-r^2}\left(\int_{S^{n-1}} r^{n-1}d\omega_{n-1}\right)dr.$$

In other words,

$$nC_n\int_0^{\infty} e^{-r^2}r^{n-1}dr = \pi^{n/2}.$$

The integral on the left-hand side is really a special value of the Γ-function (which will be studied in more detail in Chap. 4). Defining

$$\Gamma(s) := \int_0^{\infty} e^{-t}t^{s-1}dt, \qquad s > 0,$$

we find (after an easy change of variable) that

$$\int_0^\infty e^{-r^2} r^{n-1} dr = \frac{1}{2}\Gamma(n/2).$$

Therefore,

$$C_n = \frac{2\pi^{n/2}}{n\Gamma(n/2)},$$

so that

$$V_n(R) = \frac{2\pi^{n/2} R^n}{\Gamma(n/2)}.$$

We thus have formulas for the volume and surface area of the n-dimensional sphere of radius R.

1.18 Green's Theorem

The generalization of the fundamental theorem of calculus to $\mathbb{R}^2$ is called Green's theorem after George Green (1793–1841) who first discovered it. It seems that Green was largely self-educated and the son of a miller. He received very little formal education and spent most of his time working in his father's bakery and teaching himself mathematics. In 1828, he published an essay on electricity and magnetism in which he proved the theorem now named after him. It is also here that he introduced the term "potential function" as well as other functions which we now call Green's functions used in the study of partial differential equations. He was finally admitted to Cambridge University at the age of 40 and managed to publish several papers before his untimely death. His original essay was largely overlooked and many of its new contributions were rediscovered in 1846 by Sir William Thomson (1824–1907) also known as Lord Kelvin. It was again rediscovered in Russia by Mikhail Ostrogradski (1801–1861) where Green's theorem is called Ostrogradski's theorem to this day.

The first formal statement of the theorem appears in a postscript to a letter to Stokes written by Kelvin on July 2, 1850. This letter forms the cover page of Spivak's little book, Calculus on Manifolds [5]. We now understand that Green's theorem is a special case of Stokes' theorem. The latter along with Gauss's divergence theorem are succinctly formulated using the calculus of differential forms.

Katz [6] writes that the theorems attributed to Green, Gauss and Stokes were actually discovered implicitly much earlier. Special cases appear in the works of Lagrange, Laplace, Cauchy and Riemann. All of them were interested in higher-dimensional versions of the fundamental theorem of calculus to tackle specific problems motivated by physics. They range from the study of heat, electricity, magnetism, elasticity and fluid dynamics. But since most of these papers were quite long, the theoretical aspects that now can be identified as special cases of Stokes theorem were buried in these earlier works. We refer the reader to [6] for this enchanting history.

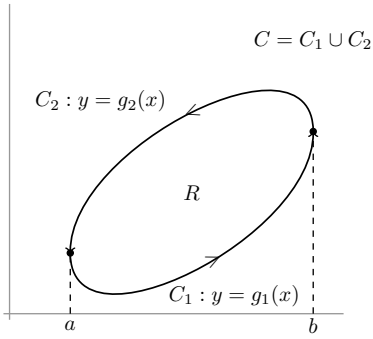


Fig. 1.9 Region R

To state and prove Green's theorem, we need first the notion of a simple closed curve. A curve $\mathbf{x} : [a, b] \to \mathbb{R}^n$ is called **closed** if $\mathbf{x}(a) = \mathbf{x}(b)$. It is called a **simple closed curve** if it does not intersect itself except at the endpoints.

Theorem 1.30 (Green's theorem) *Let C be a piecewise smooth, simple closed curve in the plane with positive orientation and let R be the region enclosed by C together with C. Suppose that $M(x, y)$ and $N(x, y)$ are continuous and have continuous first-order partial derivatives in some open region D that contains R. Then,*

$$\int_C M(x, y)dx + N(x, y)dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA.$$

Proof (Sketch) We will assume that there are two differentiable functions $g_1(x)$ and $g_2(x)$ such that the region R can be described as

$$R = \{(x, y) : a \leq x \leq b \text{ and } g_1(x) \leq y \leq g_2(x)\}$$

as illustrated in Fig. 1.9.

We can divide our contour C into two oriented parts C_1 and C_2 as indicated in the figure where C_1 is the bottom portion and C_2 is the top portion of the curve defined by $y = g_1(x)$ and $y = g_2(x)$ respectively. Thus,

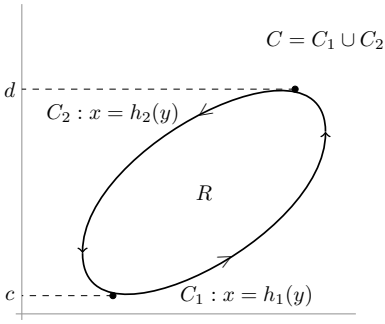
$$\int_C M(x, y)dx = \int_{C_1} M(x, y)dx + \int_{C_2} M(x, y)dx.$$

Taking into account the orientation of C_2, we find

$$\int_C M(x, y)dx = \int_a^b [M(x, g_1(x)) - M(x, g_2(x))]dx.$$

On the other hand, let us note that

Fig. 1.10 Region R



$$\iint_R \frac{\partial M}{\partial y} dA = \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial M}{\partial y} dy dx$$

and by the fundamental theorem of calculus, we get

$$\iint_R \frac{\partial M}{\partial y} dA = \int_a^b M(x, y)\Big|_{y=g_1(x)}^{y=g_2(x)} dx = \int_a^b [M(x, g_2(x)) - M(x, g_1(x))] dx.$$

Thus,

$$\int_C M(x, y) dx = -\iint_R \frac{\partial M}{\partial y} dA.$$

In a similar way, we can write our region R as

$$R = \{(x, y) : c \leq y \leq d \text{ and } h_1(y) \leq x \leq h_2(y)\}.$$

Proceeding as before, we find

$$\int_C N(x, y) dy = \iint_R \frac{\partial N}{\partial x} dA.$$

Putting everything together gives Green's theorem. To convey the essential idea of the proof, we had assumed that our contour was of the form indicated in the figures. However, it is not difficult to modify our proof to consider the more general case. We discuss the necessary modifications below. □

One important corollary of Green's theorem is the following formula for computing the area of R:

Corollary 1.2 *Suppose that C is a piecewise smooth, simple closed curve enclosing the region R with area A. Then*

$$2A = \int_C x dy - y dx.$$

Proof We take $M(x, y) = -y$ and $N(x, y) = x$ in Green's theorem to deduce the result. □

To illustrate the use of the corollary, we show that the area of an ellipse defined by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

is πab. The ellipse is a simple closed curve which can be defined parametrically by

$$C = \{(x, y) : x = a\cos t,\ y = b\sin t,\ 0 \le t \le 2\pi\}.$$

Our line integral becomes

$$\int_C x dy - y dx = \frac{1}{2}\int_0^{2\pi} [(a\cos t)(b\cos t) - (b\sin t)(-a\sin t)]dt = \pi ab.$$

As mentioned earlier, in our sketch of the proof of Green's theorem, we assumed a simple shape for our region so that we can execute the main idea of the proof quickly. However, our proof admits no loss of generality in that any piecewise smooth, simple, closed curve admits a decomposition into regions of the type we considered and one invokes the theorem for each piece and adds up the results. We illustrate the idea with the figure below.

The cycle C in the figure can then be decomposed into two oriented cycles C_1 and C_2 which share the line segment traversed indicated in Fig. 1.10, in opposite directions, and thus the contribution along this is cancelled. In other words, we can derive Green's theorem for more general regions by applying this idea. We will not go into any further details (Fig. 1.11).

As an application of Green's theorem, we derive the change of variables integration formula in the case $n = 2$. We will use Green's theorem and Corollary 1.2. Indeed by the usual subdivision of the region of integration on the right-hand side into rectangles R, it suffices to prove the theorem for rectangles R and f being a constant function. That is, we need to show

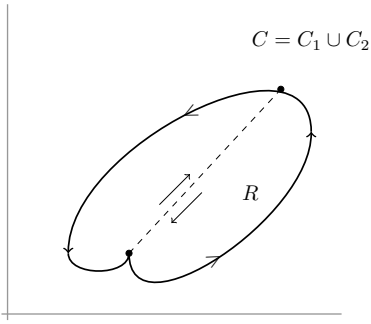


Fig. 1.11 A typical region R

$$\iint_R |\det g'(x, y)| dxdy = \iint_{g(R)} dxdy.$$

Letting S denote the region $g(R)$ and C its boundary, we have by Corollary 1.2,

$$2\iint_S dxdy = \int_C x\, dy - y\, dx.$$

We change variables to evaluate the integral on the right-hand side: $x = F(u, v)$, $y = G(u, v)$. Then,

$$dy = \frac{\partial G}{\partial u}du + \frac{\partial G}{\partial v}dv, \qquad dx = \frac{\partial F}{\partial u}du + \frac{\partial F}{\partial v}dv.$$

Denoting by C' the image of C under this transformation, we have that our integral is

$$\int_{C'} \left(F\frac{\partial G}{\partial u} - G\frac{\partial F}{\partial u}\right) du + \left(F\frac{\partial G}{\partial v} - G\frac{\partial F}{\partial v}\right) dv.$$

If we denote by R' the region bounded by C' in the uv-plane, we have by Green's theorem that this integral is

$$\iint_{R'} \Big[\frac{\partial}{\partial u}\left(F\frac{\partial G}{\partial v} - G\frac{\partial F}{\partial v}\right) - \frac{\partial}{\partial v}\left(F\frac{\partial G}{\partial u} - G\frac{\partial F}{\partial u}\right)\Big] dudv.$$

This easily simplifies to

$$\iint_{R'} \Big|\frac{\partial(x, y)}{\partial(u, v)}\Big| dudv.$$

Exercises

1. Evaluate
$$\int_C y^2 dx + x^2 dy,$$
where C is the closed path formed by the square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(1, 0)$ oriented counterclockwise.
2. Given an n-sided polygon in the plane whose vertices are
$$(a_1, b_1), (a_2, b_2), \ldots, (a_n, b_n)$$
arranged counterclockwise around the polygon, show that the area inside the polygon is given by

$$\frac{1}{2}\left(\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} + \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} + \cdots + \begin{vmatrix} a_{n-1} & b_{n-1} \\ a_n & b_n \end{vmatrix} + \begin{vmatrix} a_n & b_n \\ a_1 & b_1 \end{vmatrix}\right).$$

[Hint: parameterize each of the line segments of the polygon.]

3. If R is a region to which Green's theorem applies and C is its oriented boundary so that R is always on the left as we travel along C, show that the area of R is given by either of the two integrals:

$$\int_C x\,dy = -\int_C y\,dx.$$

1.19 Theorems of Gauss and Stokes

There are several standard higher-dimensional versions of Green's theorem that the student has undoubtedly seen in a second course in calculus. We recall these theorems (without proofs below).

The first is the **divergence theorem**. Recall that a smooth surface S is said to be **orientable** if, to each point on S, we can assign a unit normal vector in such a way that the resulting vector field of normals is continuous everywhere. Such a surface then is said to be **oriented**. Spheres and ellipsoids are common examples of orientable surfaces. The Möbius strip, obtained by taking a strip of paper twisting it around and then joining the ends, is not orientable. The following theorem is usually attributed to Gauss.

Theorem 1.31 (The divergence theorem) *Let* $\mathbf{f} = (f_1, f_2, f_3)$ *be a vector field defined and continuously differentiable in a domain* D *in* $\mathbb{R}^3$. *Let* R *be a region in* D *that is bounded by a piecewise smooth, closed orientable surface* S. *Then,*

$$\iint_S \mathbf{f} \cdot \mathbf{n}\, d\sigma = \iiint_R \nabla \cdot \mathbf{f}\, dV,$$

where S *is oriented so that at each point on* S, $\mathbf{n}$ *is the outwardly directed unit normal.*

Given a continuously differentiable vector field $\mathbf{f} = (f_1, f_2, f_3)$ in $\mathbb{R}^3$, the function

$$\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

is called the **divergence** of $\mathbf{f}$ and denoted div $\mathbf{f}$. It is also symbolically denoted by $\nabla \cdot \mathbf{f}$ as in the theorem.

The divergence theorem is not difficult to prove once appropriate notation is introduced. It is relatively simple to give a proof along the lines that we gave for Green's theorem keeping in mind that we are now working in three dimensions. Without going into too much detail, here is the idea.

We can rewrite the equation to be proved as:

$$\iint_S f_1 dydz + f_2 dzdx + f_3 dxdy = \iiint_R \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) dV.$$

For simplicity, we first assume that any line parallel to the coordinate axes cuts S in at most two points. The reader may find it convenient to visualize our surface as that of an oblate football. We can then refer to the "lower" portion and "upper" portion of the surface when we view it from either the xy-plane, or yz-plane or the xz-plane. Let us first look at the xy-plane. We can write $z = g_1(x, y)$ and $z = g_2(x, y)$ for the upper and lower portions of S denoted S_2 and S_1 respectively. If we denote the projection of S onto the xy-plane as R_1, then noting that $dV = dxdydz$,

$$\iiint_R \frac{\partial f_3}{\partial z} dV = \iint_{R_1} \left(\int_{g_1(x,y)}^{g_2(x,y)} \frac{\partial f_3}{\partial z} dz \right) dy\, dx,$$

and we immediately see that it is equal to

$$\iint_{R_1} [f_3(x, y, g_2) - f_3(x, y, g_1)] dy\, dx.$$

If $\mathbf{n}_2$ is the normal to S_2, it is easily verified that $dydx = \mathbf{k} \cdot \mathbf{n}_2\, dS_2$. Similarly, if $\mathbf{n}_1$ is the unit normal to the lower portion S_1, then $dydx = -\mathbf{k} \cdot \mathbf{n}_1 dS_1$ since the normal vector $\mathbf{n}_1$ makes an obtuse angle with $\mathbf{k}$. Combining these observations yields the equations

$$\iint_{R_1} f_3(x, y, g_2) = \iint_{S_2} f_3\, \mathbf{k} \cdot \mathbf{n}_2\, dS_2,$$

$$\iint_{R_1} f_3(x, y, g_1) = -\iint_{S_2} f_3\, \mathbf{k} \cdot \mathbf{n}_1\, dS_2.$$

Therefore

$$\iint_{R_1} [f_3(x, y, g_2) - f_3(x, y, g_1)] dy\, dx = \iint_S f_3\, \mathbf{k} \cdot \mathbf{n}\, dS.$$

In other words, putting everything together, we have

$$\iiint_R \frac{\partial f_3}{\partial z} dV = \iint_S f_3\, \mathbf{k} \cdot \mathbf{n}\, dS.$$

A similar analysis projecting the surface onto the other two coordinate planes gives

$$\iiint_R \frac{\partial f_1}{\partial x} dV = \iint_S f_1\, \mathbf{i} \cdot \mathbf{n}\, dS.$$

$$\iiint_R \frac{\partial f_2}{\partial z} dV = \iint_S f_2 \mathbf{j} \cdot \mathbf{n}\, dS.$$

Adding up both sides of our equations now gives the divergence theorem.

Suitable modifications can be made now to consider the more difficult case when lines parallel to the coordinate axes meet the surface in more than two points. As in the case of our proof of Green's theorem, we subdivide the region into subregions that satisfy our conditions above and then add up the final result, keeping in mind that we have oriented surfaces and that cancellations take place in the appropriate way.

Perhaps an important theme in both Green's theorem and Gauss's divergence theorem is how the notion of oriented surface and boundary impose themselves in the course of the proof. It is this essential ingredient that is formalized in the theory of differential forms.

One interesting application of the divergence theorem is a simple formula for the volume a region R bounded a piecewise smooth closed oriented surface S. If $\mathbf{n}$ is the unit normal function at each point of S, then

$$\text{Volume of } R = \frac{1}{3} \iint_S \mathbf{r} \cdot \mathbf{n}\, d\sigma$$

where $\mathbf{r}$ is the position vector (x, y, z). This formula is analogous to the one we obtained earlier using Green's theorem for the area enclosed by a simple closed curve. Since $\mathbf{r} = (x, y, z)$, we see that $\nabla \cdot \mathbf{r} = 3$ and the result is immediate from the divergence theorem.

As highlighted before, much of the development of multivariable calculus was motivated by questions arising in physics. For instance, in fluid mechanics, the divergence of a vector field F at a point (x, y, z) corresponds to the net flow of fluid out of a small box centered at (x, y, z). The gradient of F has the following meaning. If $\text{div } F(x, y, z) = \nabla \cdot F(x, y, z) > 0$, there is more fluid going out of the box than into the box in which we call (x, y, z) a **source**. If $\text{div } F(x, y, z) = \nabla \cdot F(x, y, z) < 0$, there is more fluid entering into the box than that which goes out of the box, in which case we call (x, y, z) a **sink**. If $\nabla \cdot F(x, y, z) = 0$ in a region D, we say that the vector field is **incompressible** or **source-free** in D.

The divergence of a vector field is then a measure of the "flow" of the vector field. For example, for the vector field $F(x, y) = (x, y)$, the divergence is 2 indicating that the flow is outward. For the vector field $F(x, y) = (y, -x)$, the divergence is zero so that the net flow is zero.

The third theorem in the "holy trinity" of theorems from vector calculus is Stokes theorem.

Theorem 1.32 (Stokes theorem) *Let S be an oriented surface parameterized by a one-to-one parameterization $r : D \subseteq \mathbb{R}^2 \to \mathbb{R}^3$, where D is a region to which Green's theorem applies. Let ∂S be the positively oriented piecewise smooth boundary of S. Then*

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S},$$

for any C^1-vector field $\mathbf{F}$ *on* S.

We will not give the proof here but only indicate that it follows similar lines of argument as in Green's theorem and our sketch of the divergence theorem. An important consequence of the Stokes theorem is:

Theorem 1.33 *Let F be a vector field in $\mathbb{R}^3$ whose component functions have continuous first-order partial derivatives in all of $\mathbb{R}^3$. Then, the following are equivalent:*

(1) F is conservative.

(2) $\int_C F \cdot dr$ is independent of the path for any piecewise smooth curve C in $\mathbb{R}^3$.

(3) $\int_C F \cdot dr = 0$ for any closed piecewise smooth curve C.

(4) $\nabla \times F = 0$.

(5) F is a gradient field; that is, $F = \nabla f$ for some potential function f.

As remarked earlier, vector calculus developed as a result of the study of electromagnetism. In this context, it may help the students in their understanding if we explain briefly, the celebrated Maxwell equations.

If $\mathbf{E}$ denotes the electric field and $\mathbf{B}$ is the magnetic field for electromagnetic waves propagating through a vacuum, Maxwell's laws are (with c denoting the speed of light):

$$\operatorname{div} \mathbf{E} = 0 \quad \operatorname{curl} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$$\operatorname{div} \mathbf{B} = 0 \quad \operatorname{curl} \mathbf{B} = \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}$$

Exercises

1. Let $C(t) = (x(t), y(t))$ for $a \leq t \leq b$ be a curve with $x(t)$, $y(t)$ being continuously differentiable. The **right normal vector** at t is defined as

$$N(t) = \left(\frac{dy}{dt}, -\frac{dx}{dt} \right).$$

Show that $N(t)$ is perpendicular to the curve.

2. Let A be a region in $\mathbb{R}^2$ which is the interior of a simple closed curve oriented counterclockwise. Let $F = (p, q)$ be a vector field on A. Let G be the vector field $G = (-q, p)$. Let C be the curve in the previous exercise.

   (a) By applying Green's theorem to G, show that

   $$\int_C (-qdx + pdy) = \iint_A \left(\frac{\partial p}{\partial x} + \frac{\partial q}{\partial y}\right) dydx.$$

   (b) Show that the left side of the equation in (a) is

   $$\int_a^b F \cdot N(t)dt,$$

   and thus deduce the divergence theorem.

3. Let $F(x, y) = (y, -x)$. Let C be the circle of radius 1 oriented counter clockwise. Show that

   $$\int_C F \cdot \mathbf{n}\, ds = 0.$$

4. Recall that a region is called **simply connected** if every closed path can be continuously shrunk to a point. Let $P(x, y)$ and $Q(x, y)$ be continuous and continuous first-order partial derivatives at each point of a simply connected region R. Show that a necessary and sufficient condition that

   $$\int_C Pdx + Qdy = 0$$

   around every closed path C in R is that

   $$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

   at every point of R.

5. With notation as in the previous exercise, we say $Pdx + Qdy$ is an **exact differential** if there is a differentiable function $\phi(x, y)$ such that

   $$d\phi = Pdx + Qdy.$$

   Prove that a necessary and sufficient condition that $Pdx + Qdy$ is an exact differential is that

   $$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

1.20 Differential Forms

The fundamental theorem of calculus generalizes to higher dimensions and this is again often called **Stokes theorem**. It is neatly stated using the language of **differential forms**. We give a cursory overview of the formalism to suggest to the student that it is profitable to study differential topology in greater detail as a unifying theory of advanced calculus to the setting of manifolds. An excellent reference is the concise book by Spivak [5]. The student may also find [7–9] helpful.

To motivate our discussion toward a "calculus of differentials", let us look at the following "example". Consider the double integral

$$\int\int f(x, y)dxdy,$$

and let us effect the change of variables

$$x = x(u, v), \qquad y = y(u, v)$$

so that

$$dxdy = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} dudv.$$

If we formally set $x = y$, we see that the determinant is zero and so this suggests $dxdx = 0$. Interchanging x and y changes the sign of the determinant so that $dxdy = -dydx$. These observations suggest a search for a suitable "algebra" of differential forms that satisfy these two properties. Such an algebra leads to an elegant synthesis of Green's theorem and Stokes theorem and in a single formula leads to grand generalization to higher dimensions, which can be seen as the ultimate fundamental theorem of calculus. Below, we give a lightning overview of the theory of differential forms. The student can find a detailed treatment in [5].

Let V be a vector space over $\mathbb{R}$ and denote by V^k the k-fold product

$$V \times \cdots \times V.$$

A function $T : V^k \to \mathbb{R}$ is called a **multilinear** map (or sometimes a k-linear map) if it is linear in each of the arguments. Such multilinear maps are also called **k-tensors**. We will denote by $\mathfrak{T}^k(V)$ the set of all k-tensors on V. This set is easily seen to be a vector space over $\mathbb{R}$. Given $S \in \mathfrak{T}^r(V)$ and $T \in \mathfrak{T}^s(V)$, we define the **tensor product** $S \otimes T \in \mathfrak{T}^{r+s}(V)$ by

$$S \otimes T(v_1, \ldots, v_r, v_{r+1}, \ldots, v_{r+s}) := S(v_1, \ldots, v_r)T(v_{r+1}, \ldots, v_{r+s}).$$

Note that the order of the factors S and T matters here so the operation $\otimes$ is not commutative. The reader can easily verify the following properties:

$$(S_1 + S_2) \otimes T = S_1 \otimes T + S_2 \otimes T,$$

$$S \otimes (T_1 + T_2) = S \otimes T_1 + S \otimes T_2,$$

$$(aS) \otimes T = S \otimes (aT) = a(S \otimes T), \quad a \in \mathbb{R}$$

$$(S \otimes T) \otimes U = S \otimes (T \otimes U).$$

The last property allows us to write unambiguously $S \otimes T \otimes U$ and a similar property applies to higher-order products. The space $\mathfrak{T}^1(V)$ is simply the **dual space** of functionals. The tensor product allows us to describe the space of k-tensors similarly. Indeed, if V is n-dimensional over $\mathbb{R}$, with basis $v_1, \ldots, v_n$ and corresponding dual basis $\phi_1, \ldots, \phi_n$ so that $\phi_i(v_j) = \delta_{ij}$ (where δ_{ij} is the usual **Kronecker delta function**, equal to 1 if $i = j$ and zero otherwise), then the set of k-fold tensors

$$\phi_{i_1} \otimes \cdots \otimes \phi_{i_k}, \qquad 1 \leq i_1, \ldots, i_k \leq n,$$

is a basis for $\mathfrak{T}^k(V)$ and consequently, $\mathfrak{T}^k(V)$ has dimension n^k.

The familiar dot product of two vectors is an example of a 2-tensor. In fact, one defines generally an **inner product** on a vector space V over $\mathbb{R}$ to be a 2-tensor which is **symmetric**, that is, $T(v, w) = T(w, v)$, and **positive definite**, that is $T(v, v) > 0$ for all vectors $v \neq 0$.

Another example of a tensor perhaps familiar to readers is the determinant function. This is really an n-tensor on the vector space $\mathbb{R}^n$. To explain this connection, we define a k-tensor to be **alternating** if

$$T(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_k) = -T(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_k) \qquad \forall\, v_1, \ldots, v_k \in V$$

where we have interchanged v_i and v_i on the right-hand side and left all other v's fixed in their position. In other words, the sign of the tensor changes if we apply a transposition to the order of the arguments, a property shared by the determinant function. The set of alternating tensors is denoted $\Lambda^k(V)$. The reader will recall that for the permutation group S_k on k letters, the sgn function assigns $+1$ if the permutation is even and -1 if it is odd. This function allows us to construct an alternating tensor from any tensor as follows. Given $T \in \mathfrak{T}^k(V)$, we define

$$Alt(T)(v_1, \ldots, v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} (\text{sgn } \sigma) T(v_{\sigma(1)}, \ldots, v_{\sigma(k)}).$$

We leave the verification that this is indeed an alternating tensor to the student who we assume is familiar with basic linear algebra and has already demonstrated this property for the determinant function. The proof here is similar.

If T is alternating, then $Alt(T) = T$. Just as we defined the tensor product of two tensors, we can define the **wedge product**, denoted $\omega \wedge \eta$, of two alternating tensors $\omega \in \Lambda^r(V)$ and $\eta \in \Lambda^s(V)$ as

$$\omega \wedge \eta = \frac{(r+s)!}{r!s!} Alt(\omega \otimes \eta).$$

The reason for the coefficient will be apparent later. The wedge product has the usual properties of distributivity and associativity. It has the important property that $\omega \wedge \eta = (-1)^{rs} \eta \wedge \omega$. In particular, if r is odd, $\omega \wedge \omega = 0$. If r or s is even, then $\omega \wedge \eta = \eta \wedge \omega$. If r and s are both odd, then $\omega \wedge \eta = -\eta \wedge \omega$.

We can construct a basis for $\Lambda^k(V)$ using the wedge product. In fact, the set of all

$$\phi_{i_1} \wedge \cdots \wedge \phi_{i_k}, \qquad 1 \leq i_1 < i_2 < \cdots < i_k \leq n$$

is a basis for $\Lambda^k(V)$ with V being an n-dimensional vector space over $\mathbb{R}$. It therefore has dimension

$$\binom{n}{k}.$$

Thus, if V has dimension n, the space $\Lambda^n(V)$ is one-dimensional and spanned by the determinant function. Therefore, it is not difficult to deduce that if $v_1, \ldots, v_n$ is a basis for V and $w_1, \ldots, w_n$ are n vectors such that

$$w_i = \sum_{j=1}^{n} a_{i,j} v_j,$$

then for $\eta \in \Lambda^n(V)$, we have

$$\eta(w_1, \ldots, w_n) = \det(a_{ij}) \eta(v_1, \ldots, v_n). \tag{1.23}$$

If $v_1, \ldots, v_n$ and $w_1, \ldots, w_n$ are two bases of V, the **change of basis matrix** $A = (a_{ij})$ is non-singular and consequently, $\det A$ is either positive or negative. This allows us to define an equivalence relation on the set of bases of V. Two bases are in the same class if and only if their change of basis matrix has positive determinant. It is evident that this partitions the bases into two groups. Indeed, if $v_1, \ldots, v_n$ is a basis, *all* other bases are obtained by applying a non-singular matrix to this basis and the sign of the determinant determines the class. Either of these two groups is called an **orientation** for V. Thus, an orientation is an equivalence class of bases and we denote the orientation to which the basis $v_1, \ldots, v_n$ belongs to by $[v_1, \ldots, v_n]$. If $e_1, \ldots, e_n$ is the standard basis for $\mathbb{R}^n$, we define the **usual orientation** as $[e_1, \ldots, e_n]$.

Equation 1.23 also tells us that we can partition n-tensors of an n-dimensional vector space V into two groups using the following idea. Suppose we have an inner product T on V and that $v_1, \ldots, v_n$ and $w_1, \ldots, w_n$ are two bases of V which are orthonormal with respect to a 2-tensor T and $w_i = \sum_{j=1}^{n} a_{ij} v_j$. Then,

$$\delta_{ij} = T(w_i, w_j) = \sum_{k,\ell=1}^{n} a_{ik} a_{j\ell} T(v_k, v_\ell)$$

$$= \sum_{k=1}^{n} a_{ik} a_{jk}.$$

In other words, if A^t is the transpose of A, the above equation says $AA^t = I$ so that $\det A$ is ± 1.

From (1.23), we deduce that if $\omega \in \Lambda^n(V)$ satisfies $\omega(v_1, \ldots, v_n) = \pm 1$ then $\omega(w_1, \ldots, w_n) = \pm 1$. Thus, if an orientation for V has been given, there is a unique $\omega \in \Lambda^n(V)$ such that $\omega(v_1, \ldots, v_n) = 1$ whenever $v_1, \ldots, v_n$ is an orthonormal basis lying in the given orientation. This unique ω is called the **volume element** of V determined by the orientation and the inner product T. Thus, det is the volume element determined by the usual dot product on $\mathbb{R}^n$ and the usual orientation given by the standard basis.

The reader may already be familiar with vector-valued functions. As discussed earlier, these are sometimes called **vector fields**. It is now convenient to have a slightly refined version of this definition.

For a fixed vector $p \in \mathbb{R}^n$, we consider the set of all pairs $(p, \mathbf{v})$ with $\mathbf{v} \in \mathbb{R}^n$ and turn this set into a vector space over $\mathbb{R}$ in the obvious way:

$$(p, \mathbf{v}) + (p, \mathbf{w}) = (p, \mathbf{v} + \mathbf{w}), \qquad a \cdot (p, \mathbf{v}) = (p, a\mathbf{v})$$

for $a \in \mathbb{R}$. Essentially, the vector p can be thought of as a "book-keeping" device. We call this the **tangent space** of $\mathbb{R}^n$ at p and denote it by $T_p(\mathbb{R}^n)$. Clearly, an analogous construction can be made for any vector space V over $\mathbb{R}$. We can endow $T_p(\mathbb{R}^n)$ with an inner product in the obvious way as well as the usual orientation.

A **vector field** is a function F of $\mathbb{R}^n$ such that $F(p) \in T_p(\mathbb{R}^n)$. If

$$(e_1)_p, \ldots, (e_n)_p$$

is the standard basis of $T_p(\mathbb{R}^n)$, our function can then be written as

$$F(p) = F_1(p)(e_1)_p + \cdots + F_n(p)(e_n)_p.$$

We thus obtain **component functions** $F_i : \mathbb{R}^n \to \mathbb{R}$. The vector field is called continuous, differentiable, and so on according as the component functions are continuous, differentiable and so on. Of course, these definitions can be made for any function F defined on an open set of $\mathbb{R}^n$.

We define the **divergence** of F, denoted div F as

$$\sum_{i=1}^{n} D_i F_i,$$

where D_i denotes differentiation with respect to the ith variable.

The set of all vector fields on $\mathbb{R}^n$ can be made into a vector space in the obvious way: Given two vector fields F, G, we define $F + G$ to be the vector field that assigns to p the vector

$$F(p) + G(p)$$

If we introduce the symbolism

$$\nabla = \sum_{i=1}^{n} D_i \cdot e_i,$$

then we can write symbolically, div $F = \langle \nabla, F \rangle$.

These definitions are motivated by what can be described as the algebraic view of vector calculus. We have already seen that the derivative is a linear transformation between two vector spaces. In order to speak about "higher derivatives", we would need to topologize these vector spaces of linear transformations. On the other hand, the concept of partial derivatives and component functions allows us to descend back into familiar Euclidean spaces.

It is convenient to consider alternating k-tensors on $T_p(\mathbb{R}^n)$. These are called **differential k-forms** or simply **k-forms**. In a way, differential forms generalize the notion of functions. Indeed, 0-forms are simply functions $f : \mathbb{R}^n \to \mathbb{R}$. The wedge product allows us to speak about the algebra of differential forms in the following sense.

If $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable, then the total differential (that we met earlier in Sect. 1.11):

$$df := D_1 f \cdot dx_1 + \cdots + D_n f \cdot dx_n$$

is a 1-form. Thus, the map $f \mapsto df$ changes 0-forms into 1-forms. This observation allows us to construct an explicit basis for the vector space of k-forms, namely,

$$dx_{i_1} \wedge \cdots \wedge dx_{i_k}, \qquad i_1 < i_2 < \cdots < i_k$$

is a basis for $\Lambda^k(\mathbb{R}^n)$. Every k-form ω can be written as a linear combination

$$\sum_{i_1 < \cdots < i_k} \omega_{i_1,\ldots,i_k} dx_{i_1} \wedge \cdots \wedge dx_{i_k}$$

and we say ω is C^r if the functions $\omega_{i_1,\ldots,i_k}$ are all C^r. If ω is a differentiable k-form, then we define $d\omega$ as

$$d\omega := \sum_{i_1 < \cdots < i_k} d\omega_{i_1,\ldots,i_k} \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}$$

which is a $(k+1)$-form. This is the algebra of differential forms that we have been looking for. The d-operator is called the **exterior derivative** operator.

On the geometric side, we define a **singular n-cube** in $A \subset \mathbb{R}^n$ to be a continuous function $c : [0,1]^n \to A$. The **standard n-cube** is $I_n : [0,1]^n \to \mathbb{R}^n$ given by $I_n(\mathbf{x}) = \mathbf{x}$ for $\mathbf{x} \in [0,1]^n$. We can consider the $\mathbb{Z}$-module generated by the singular n-cubes. These are simply finite formal sums of the form

$$\sum_j n_j c_j,$$

with $n_j \in \mathbb{Z}$ and c_j's are singular n-cubes. Such sums are called **n-chains**. In particular, any singular n-cube is also an n-chain.

For each singular n-chain c in A, we will define an $(n-1)$-chain in A called the **boundary** of c and denoted ∂c. For example, the boundary of I_2 may be defined as the sum of four singular 1-cubes arranged counterclockwise around the boundary of $[0,1]^2$ as indicated in the figure below.

However, it will be convenient to define it as the sum of four singular 1-cubes with the indicated coefficients:

With this as motivation, we can define generally the boundary of I_n as follows. For each i with $1 \le i \le n$, we define two singular $(n-1)$-cubes $I_n^{(i,0)}$ and $I_n^{(i,1)}$ as follows. If $\mathbf{x} \in [0,1]^{n-1}$, then

$$I_n^{(i,0)}(\mathbf{x}) = (x_1, \ldots, x_{i-1}, 0, x_i, \ldots, x_{n-1})$$

and

$$I_n^{(i,1)}(\mathbf{x}) = (x_1, \ldots, x_{i-1}, 1, x_i, \ldots, x_{n-1}).$$

We call $I_n^{(i,0)}$ the $(i,0)$-**face** of I_n and $I_n^{(i,1)}$ the $(i,1)$-face. Thus, in the case $n = 1$, $I_1^{(1,0)} = (0,0)$ and $I_1^{(1,1)} = (0,1)$ are simply two points. In the case $n = 2$ illustrated in Figs. 1.12 and 1.13, they are given as indicated in the figure below (Fig. 1.14).

Thus, the boundary of I_2 is seen to be the formal sum

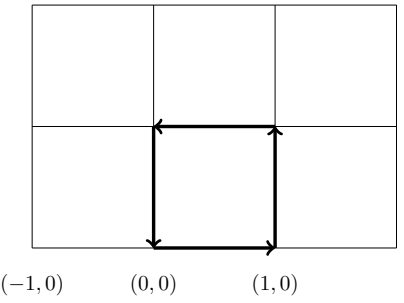


Fig. 1.12 Boundary of I_2

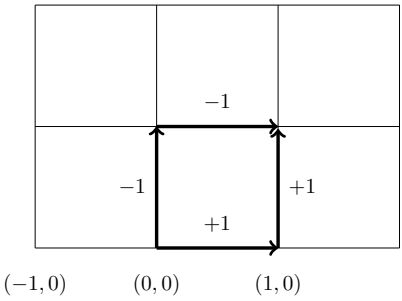


Fig. 1.13 Signed boundary of I_2

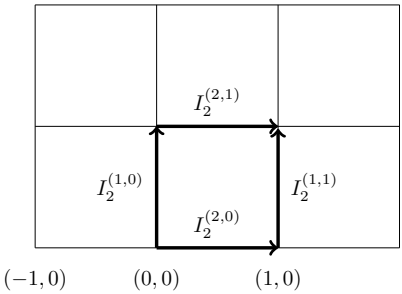


Fig. 1.14 Faces of I_2

$$I_2^{(2,0)} - I_2^{(2,1)} + I_2^{(1,1)} - I_2^{(1,0)}.$$

This partially motivates our definition of the boundary of I_n:

$$\partial I_n := \sum_{i=1}^{n} \sum_{a=0,1} (-1)^{i+a} I_n^{(i,a)}.$$

For a general singular n-cube $c : [0, 1]^n \to A \subset \mathbb{R}^n$, we first define

$$c_{(i,a)} = c \circ (I_n^{(i,a)})$$

and then define the **boundary** of c as the alternating sum

$$\partial c = \sum_{i=1}^{n} \sum_{a=0,1} (-1)^{i+a} c_{(i,a)}.$$

We then define the boundary of an n-chain $\sum_i n_i c_i$ as

$$\partial \left(\sum_i n_i c_i \right) = \sum_i n_i \partial c_i.$$

It is somewhat tedious (but not difficult) to verify that $\partial^2 = 0$.

This language and formalism allows for an elegant reformulation of Stokes theorem:

Theorem 1.34 (Stokes theorem) *If ω is a $(k-1)$-form on an open set $A \subset \mathbb{R}^n$ and c is a k-chain in A, then*

$$\int_c d\omega = \int_{\partial c} \omega.$$

This is the higher-dimensional generalization of the fundamental theorem of calculus. It has a synthetic generalization to smooth manifolds with boundary as explained in the concise book by Spivak [5].

Exercises

1. Let $S \in \mathfrak{T}^r(V)$ and $T \in \mathfrak{T}^s(V)$. Suppose $Alt(S) = 0$. Show that

$$Alt(S \otimes T) = Alt(T \otimes S) = 0.$$

2. Prove that if $\omega \in \mathfrak{T}^r(V)$, $\eta \in \mathfrak{T}^s(V)$ and $\theta \in \mathfrak{T}^t(V)$, then

$$(\omega \wedge \eta) \wedge \theta = (\omega \wedge (\eta \wedge \theta) = \frac{(r+s+t)!}{r!s!t!} Alt(\omega \otimes \eta \otimes \theta).$$

3. If $v_1, v_2, \ldots, v_{n-1} \in \mathbb{R}^n$, define $\phi : \mathbb{R}^n \to \mathbb{R}$ by

$$\phi(w) = \det \begin{pmatrix} v_1 \\ \vdots \\ v_{n-1} \\ w \end{pmatrix}.$$

   Show that there is a unique $z \in \mathbb{R}^n$ such that

$$\phi(w) = (w, z).$$

   We call z the **cross-product** of $v_1, \ldots, v_{n-1}$. This generalizes the cross-product to higher dimensions.

4. If ω and η are k-forms, show that $d(\omega + \eta) = d\omega + d\eta$.

5. If ω is a k-form and η is an ℓ-form, prove that

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta.$$

6. If ω is a k-form, show that $d(d\omega) = 0$.

7. Let $M(x, y)$ and $N(x, y)$ be C^1. Show that

$$d(Mdx + Ndy) = \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dxdy.$$

Deduce Green's theorem from Theorem 1.34.

8. If $F = (F_1, F_2, F_3)$ is a vector field in $\mathbb{R}^3$, let

$$\omega = F_1\, dy \wedge dz + F_2\, dz \wedge dx + F_3\, dx \wedge dy.$$

Show that $d\omega = (\text{div } F)dx\, dy\, dz$. Deduce the divergence theorem from Theorem 1.34.

9. If $F = (F_1, F_2, F_3)$ is a vector field in $\mathbb{R}^3$, let

$$\eta = F_1 dx + F_2 dy + F_3 dz.$$

Show that

$$d\eta = (D_2F_3 - D_3F_2)dy \wedge dz + (D_3F_1 - D_1F_3)dz \wedge dx + (D_1F_2 - D_2F_1)dx \wedge dy.$$

This observation can be used to give a proof of Theorem 1.32 using Theorem 1.34.

10. A k-form ω is called **exact** if either $\omega = 0$ or $\omega = d\eta$ for some form η. A k-form ω is called **closed** if $d\omega = 0$. Show that an exact form is closed.

11. Consider the 1-form in $\mathbb{R}^2 \backslash (0, 0)$:

$$\omega = \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy.$$

Show that ω is closed but not exact. [Thus, the converse of the previous exercise does not hold. This is because the domain is not simply connected. In a simply connected domain, every closed 1-form is exact.]

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Chapter 2
Measure Theory



2.1 Topological Spaces and Measure Spaces

By the end of the nineteenth century, it was clear that Riemann integration (about which one learns in a first course in calculus) had to be replaced by a more versatile method of integration. For instance, the characteristic function of the irrational numbers is not Riemann integrable, and yet intuition suggests that over any finite interval it should integrate to the length of that interval. It was Henri Lebesgue (1875–1941) who came up with the appropriate theory in his doctoral thesis written in 1902. There was an urgency in developing such a theory because there were other disciplines in mathematics that were struggling to find an appropriate framework, most notably, probability theory.

Metric spaces are examples of a larger universe of objects called **topological spaces**. A **topological space** is a pair $(X, \mathcal{O})$ consisting of a set X and a collection $\mathcal{O}$ of subsets of X (called **open sets**) such that

(a) $\emptyset$ and $X \in \mathcal{O}$;
(b) $U, V \in \mathcal{O}$ implies $U \cap V \in \mathcal{O}$;
(c) for any collection of open sets U_a, we have $\bigcup_a U_a \in \mathcal{O}$.

The complements of open sets are called **closed sets**. Given a topological space $(X, \mathcal{O})$, a **base** $\mathcal{U}$ for $\mathcal{O}$ is any collection of open sets such that every open set is a union of a subcollection of $\mathcal{U}$. A **neighborhood** of a point $x \in X$ is any set N (not necessarily open) such that $x \in U \subset N$ for some open U. We say that a collection $\mathcal{N}$ of neighborhoods of x is a **neighborhood base** at x iff for every neighborhood V of x, we have $x \in N \subset V$ for some $N \in \mathcal{N}$.

Every metric space gives rise to a topology. Indeed, if (X, d) is a metric space, we define the **open ball** of radius r and centered at x as

$$B_r(x) = \{y \in X : d(x, y) < r\}.$$

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The **closed ball** is denoted as $\overline{B_r(x)}$ and is defined as

$$\{y \in X : d(x, y) \leq r\}.$$

The **punctured ball** is sometimes denoted $\widehat{B}_r(x)$ and is defined as

$$\{y \in x : 0 < d(x, y) < r\}.$$

A subset A of X is then said to be **open** if for every $x \in A$, there is an $r > 0$ such that $B_r(x) \subseteq A$. The open sets of (X, d) now make X into a topological space.

It is convenient to consider the **extended real line** $[-\infty, \infty]$ which is the set of real numbers along with two new elements $+\infty$ and $-\infty$. We make this into a topological space by declaring that for any two real numbers $a, b \in \mathbb{R}$, with $a < b$, the sets (a, b), $[-\infty, a)$, $(a, \infty]$ and any union of segments of this type, as open.

As noted earlier, we can define the notion of a continuous function in the context of general topological spaces. Given two topological spaces X and Y, a function $f : X \to Y$ is said to be **continuous** if the inverse image $f^{-1}(U)$ is open in X for every open set U of Y.

One advantage in the study of metric spaces is that we can "measure" the distance between two points and this can be used as a starting point for an integration theory. However, on close examination, we can identify certain properties of the metric and abstract these properties so as to arrive at a more general notion of a measure space.

Let X be an arbitrary set. A subset $\mathscr{A}$ of the power set 2^X is called a **σ-algebra** if $X \in \mathscr{A}$, and it is closed under complements and countable unions. A map $\mu : \mathscr{A} \to [0, \infty]$ is called a **positive measure** (or simply a **measure**) if for any countable collection of disjoint sets $\{A_i\}$ of $\mathscr{A}$, we have

$$\mu\left(\bigcup_i A_i\right) = \sum_i \mu(A_i).$$

That is, μ is **countably additive**. We then speak of the triple $(X, \mathscr{A}, \mu)$ (or simply (X, μ)) as a **measure space**. One can also consider more generally **real measures** and **complex measures** where μ takes values in $\mathbb{R}$ or $\mathbb{C}$, respectively. But we will only be dealing with positive measures here.

If (X, μ) is a measure space, members of $\mathscr{A}$ are called **measurable sets**. If Y is a topological space and $f : X \to Y$, then f is called **measurable** if $f^{-1}(U)$ is measurable for every open U of Y. In Lebesgue's original theory, Y was the set of non-negative real numbers $[0, \infty]$. If $f : X \to \mathbb{R}$ is measurable, we say f is a **real measurable function**. If $f : X \to \mathbb{C}$ is measurable, we say f is a **complex measurable function**. Such a function can clearly be decomposed as $f = u + iv$ with u and v being real measurable functions. One can also show that $|f|$ is a real measurable function in such an instance.

Theorem 2.1 *Let X be a measure space. If $f_n : X \to [-\infty, \infty]$ is a sequence of measurable functions, and*

$$g = \sup_{n\geq 1} f_n, \qquad h = \limsup_{n\to\infty} f_n,$$

then g and h are measurable.

Proof For any $\alpha \in \mathbb{R}$,

$$g^{-1}((\alpha, \infty]) = \bigcup_{n=1}^{\infty} f_n^{-1}((\alpha, \infty]).$$

Thus g is measurable. It is clear that the same result holds with sup replaced by inf. Thus, as

$$h = \inf_{k\geq 1}\left(\sup_{i\geq k} f_i\right)$$

we see that h is also measurable. □

Exercises

1. Let X be a measure space and E a measurable set of X. Show that the **characteristic function** χ_E of E defined by $\chi_E(x) = 1$ if $x \in E$ and zero otherwise is a measurable function.
2. If f and g are measurable functions with range in $[-\infty, \infty]$, show that the functions
$$\max\{f, g\}, \quad \text{and} \quad \min\{f, g\}$$
are measurable functions.
3. Let X be a set. Define $\mu(E)$ for any subset E of X as follows. If E is infinite, $\mu(E) = \infty$. If E is finite, $\mu(E)$ is the cardinality of E. Show that μ defines a measure on X. (This is often called the **counting measure**.)
4. Let (X, μ) be a measure space. If
$$A_1 \subset A_2 \subset A_3 \subset \cdots$$
are measurable sets and
$$A = \bigcup_{n=1}^{\infty} A_n,$$
show that $\mu(A_n) \to \mu(A)$ as $n \to \infty$.
5. Let (X, μ) be a measure space. If
$$A_1 \supset A_2 \supset \cdots$$

are measurable sets with $\mu(A_1)$ finite, show that if

$$A = \bigcap_{n=1}^{\infty} A_n$$

then $\mu(A_n) \to \mu(A)$ as $n \to \infty$.

6. Let μ be the counting measure on the set of natural numbers. Let

$$A_n = \{n, n+1, n+2, ...\}.$$

If

$$A = \bigcap_{n=1}^{\infty} A_n,$$

then A is empty and thus has measure zero, whereas $\mu(A_n) = \infty$. Does this contradict the previous exercise?

7. If f is a real-valued function on a measurable space X such that

$$\{x : f(x) \geq r\}$$

is measurable for every rational r, then show that f is measurable.

2.2 The Lebesgue Integral

A function on a measure space X is called a **simple function** if its range is finite. Clearly, such a function s can be written as a finite linear combination of characteristic functions

$$s = \sum_{i=1}^{n} \alpha_i \chi_{A_i},$$

where $\alpha_1, ..., \alpha_n$ are the distinct values of s and $A_i = \{x : s(x) = \alpha_i\}$. It is clear that s is measurable if and only if each A_i is measurable.

For a simple function s and a measurable set E of $\mathscr{A}$, we define

$$\int_E s \, d\mu := \sum_{i=1}^{n} \alpha_i \mu(A_i \cap E).$$

It is not hard to show that for any measurable function $f : X \to [0, \infty]$, there are simple measurable functions $s_1, s_2, ...$ such that

$$0 \leq s_1 \leq s_2 \leq \cdots \leq f$$

and

$$s_n(x) \to f(x) \quad \text{as } n \to \infty$$

for every $x \in X$.

This allows us to define the Lebesgue integral. If $f : X \to [0, \infty]$ is measurable, and $E \in \mathscr{A}$, we define the **Lebesgue integral** of f over E as

$$\int_E f \, d\mu := \sup \int_E s \, d\mu,$$

where the supremum is taken over all simple measurable functions s such that $0 \leq s \leq f$. If f is simple, the two definitions of the Lebesgue integral agree.

We now come to the interesting part of the theory. The following theorems encapsulate the versatility of Lebesgue's theory to handle limit operations. Throughout, (X, μ) is a measure space.

Theorem 2.2 (Lebesgue's monotone convergence theorem) *Let $\{f_n\}$ be a sequence of measurable functions and suppose that*

(a) $0 \leq f_1(x) \leq f_2(x) \leq \cdots \leq \infty$ *for every* $x \in X$;
(b) $f_n(x) \to f(x)$ *as* $n \to \infty$ *for every* $x \in X$.

Then f is measurable and

$$\int_X f_n \, d\mu \to \int_X f \, d\mu$$

as $n \to \infty$.

Proof Since the sequence of numbers

$$\int_X f_n d\mu$$

is monotonic increasing, there is an $\alpha \in [0, \infty]$ such that

$$\int_X f_n d\mu \to \alpha$$

as $n \to \infty$. By Theorem 2.1, f is measurable and as $f_n \leq f$, we have

$$\int_X f_n \, d\mu \leq \int_X f \, d\mu$$

for every n. Consequently,

$$\alpha \leq \int_X f d\mu.$$

Let s be a simple measurable function such that $0 \leq s \leq f$ and let c be a constant with $0 < c < 1$. Put

$$E_n = \{x : f_n(x) \geq cs(x)\}, \qquad n \in \mathbb{N}$$

Then, every E_n is measurable, and we have

$$E_1 \supset E_2 \supset E_3 \cdots, \qquad \text{and} \qquad X = \bigcup_{n\geq 1} E_n.$$

To see this last equality, let $x \in X$. If $f(x) = 0$, then $x \in E_1$; if $f(x) > 0$, then $cs(x) < f(x)$ because $c < 1$, and so $x \in E_n$ for some n. Clearly,

$$\int_X f_n d\mu \geq \int_{E_n} f_n d\mu \geq \int_{E_n} cs d\mu, \qquad n \in \mathbb{N}$$

Letting $n \to \infty$ and applying Exercise 1, we deduce

$$\alpha \geq c \int_X s d\mu.$$

This holds for every $c < 1$ and so,

$$\alpha \geq \int_X s d\mu,$$

for every simple measurable s satisfying $0 \leq s \leq f$. Thus, by the definition of the Lebesgue integral, we now have

$$\alpha \geq \int_X f d\mu.$$

□

The following theorem highlights the simplicity of Lebesgue's theory.

Theorem 2.3 *If $f_n : X \to [0, \infty]$, is measurable for $n = 1, 2, \ldots$ and*

$$f(x) = \sum_{n=1}^{\infty} f_n(x) \qquad (x \in X),$$

then

$$\int_X f \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu.$$

Proof Put

$$g_n = f_1 + f_2 + \cdots + f_n.$$

Then, g_n converges monotonically to f, and by Theorem 2.2, the result is immediate. □

Theorem 2.4 (Fatou's lemma) *If $f_n : X \to [0, \infty]$ is measurable for each positive integer n, then*

$$\int_X \left(\liminf_{n\to\infty} f_n\right) d\mu \leq \liminf_{n\to\infty} \int_X f_n \, d\mu.$$

Proof Put

$$g_n = \inf_{i\geq n} f_i(x), \qquad n = 1, 2, \ldots; \quad x \in X.$$

Clearly $g_n \leq f_n$, so that

$$\int_X g_n d\mu \leq \int_X f_n d\mu, \qquad n = 1, 2, 3, \ldots$$

Moreover, $0 \leq g_1 \leq g_2 \leq \cdots$ and each g_n is measurable and by the definition of lim inf,

$$g_n(x) \to \liminf_{k\to\infty} f_k(x).$$

The monotone convergence theorem now implies the result. □

For (X, μ) a positive measure space, we define $L^1(\mu)$ to be the set of all complex measurable functions f on X for which

$$\int_X |f| \, d\mu < \infty.$$

The reader will recall that f measurable implies that $|f|$ is measurable and so the above integral is well-defined. Elements of $L^1(\mu)$ are then called **Lebesgue integrable** functions. We now come to what may be termed as the most important theorem in Lebesgue's theory.

Theorem 2.5 (Lebesgue's dominated convergence theorem) *Suppose $\{f_n\}$ is a sequence of measurable functions on X such that*

$$f(x) = \lim_{n\to\infty} f_n(x)$$

exists for every $x \in X$. If there is a function $g \in L^1(\mu)$ such that

$$|f_n(x)| \leq g(x) \qquad \forall x \in X, \quad \forall n = 1, 2, \ldots$$

then $f \in L^1(\mu)$ and

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Proof Since $|f| \leq g$ and f is measurable, we see $f \in L^1(\mu)$. Now

$$|f_n - f| \leq 2g$$

and so Fatou's lemma applied to $2g - |f_n - f|$ gives

$$\int_X 2g d\mu \leq \liminf_{n\to\infty} \int_X (2g - |f_n - f|) d\mu$$

and we deduce

$$\int_X 2g d\mu \leq \int_X 2g d\mu + \liminf_{n\to\infty} \left(- \int_X |f_n - f| d\mu \right) = \int_X 2g d\mu - \limsup_{n\to\infty} \int_X |f_n - f| d\mu.$$

Since

$$\int_X 2g d\mu$$

is finite, we can subtract it from both sides of the inequality and deduce

$$\limsup_{n\to\infty} |f_n - f| d\mu \leq 0.$$

This implies the theorem. □

Sets of measure zero play a special role in Lebesgue's theory. If a certain property holds apart from a set of measure zero, we say the property holds **almost everywhere** and abbreviate it as **a.e.** More precisely, a property P holds almost everywhere on a measurable set E of the σ-algebra $\mathscr{A}$ if there exists $N \subset \mathscr{A}$ such that $\mu(N) = 0$ and P holds on $E \backslash N$. In fact, we can define an equivalence relation on measurable functions and say two functions f and g are equivalent if $f - g = 0$ a.e. In such a case, we have for every measurable set E of X that

$$\int_E f d\mu = \int_E g d\mu.$$

For $0 < p < \infty$, and a measure space X, we define $L^p(\mu)$ to be the set of measurable functions f on X for which

$$||f||_p := \left(\int_X |f|^p d\mu \right)^{1/p} < \infty.$$

We call $||f||_p$ the L^p-norm of f. These spaces are examples of Banach spaces, and the student should review the basic properties of these spaces as adumbrated in Sect. 2.6. If $X = \mathbb{R}^k$, we denote this space as $L^p(\mathbb{R}^k)$. These are examples of complete metric spaces. Technically, $L^p(\mu)$ is not a space of functions, but rather a space of

equivalence classes of functions where two functions f and g are equivalent if they are equal a.e. However, in common parlance, we drop this subtle distinction and still continue to refer to elements of $L^p(\mu)$ as functions with this tacit understanding in the background.

The space of continuous functions with compact support on X, denoted $C_c(X)$, is dense in $L^p(\mu)$. This fact is often useful in proving many fundamental theorems concerning $L^p(\mu)$.

We also introduce the space $L^\infty(X)$ consisting of (real or complex-valued) functions f for which

$$||f||_\infty := \sup_{x \in X} |f(x)| < \infty.$$

Elements of this space are called **essentially bounded** functions.

Exercises

1. Let s be a simple function on a measure space X and for each E a measurable set, define
$$\phi(E) := \int_E s d\mu.$$
Show that ϕ defines a measure on X.
2. If $a_{ij} \geq 0$ for $i, j \geq 1$ is a double sequence non-negative real numbers, show that
$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$
3. Suppose $f : X \to [0, \infty]$ is measurable and for each E a measurable set define
$$\phi(E) := \int_E f d\mu. \tag{2.1}$$
Show that ϕ is a measure on X and that
$$\int_X g d\phi = \int_X g f d\mu$$
for every measurable g on X with range in $[0, \infty]$. (The converse of this result is called the **Radon–Nikodym theorem**: any positive measure ϕ on X is of the form (2.1) for some $f \in L^1(\mu)$. One refers to f as the **Radon–Nikodym derivative** of ϕ.)
4. Let $f : X \to [0, \infty]$ be measurable and suppose that for some measurable set E, we have

$$\int_E f d\mu = 0.$$

Show that $f = 0$ a.e. on E. [Hint: for each n, show that the set

$$\{x \in E : f(x) > 1/n\}$$

has measure zero.]

5. If $f \in L^1(\mu)$ and $\int_E f d\mu = 0$ for every measurable set E of X, show that $f = 0$ a.e. on X.

2.3 Inner Product Spaces

A vector space H over $\mathbb{C}$ is called an **inner product space** if for every pair of elements $x, y \in H$, we associate $(x, y) \in \mathbb{C}$ (called the **inner product** of x and y) satisfying the following:

(a) $\overline{(x, y)} = (y, x)$; (the bar denotes complex conjugation)
(b) $(x + y, z) = (x, z) + (y, z)$ for all $x, y, z \in H$;
(c) $(\alpha x, y) = \alpha(x, y)$ for all $x, y \in H$ and $\alpha \in \mathbb{C}$;
(d) $(x, x) \geq 0$ for all $x \in H$ with equality if and only if $x = 0$.

Property (c) implies that $(0, y) = 0$ for all $y \in H$. Properties (b) and (c) may be combined to say that the map $x \mapsto (x, y)$ is a linear functional on H. Properties (a) and (c) show that $(x, \alpha y) = \overline{\alpha}(x, y)$. Finally (a) and (b) imply that $(z, x + y) = (z, x) + (z, y)$ and (d) allows us to define the **norm** of x, denoted $||x||$ to be the non-negative square root of (x, x).

Theorem 2.6 (The parallelogram law) *In any inner product space H, we have*

$$||x + y||^2 + ||x - y||^2 = 2(||x||^2 + ||y||^2).$$

Proof We have

$$(x + y, x + y) = (x, x) + (x, y) + (y, x) + (y, y)$$

and

$$(x - y, x - y) = (x, x) - (x, y) - (y, x) + (y, y).$$

The result is obtained by adding these two equations. □

The above theorem has the following geometric interpretation in Euclidean geometry. In any parallelogram, the sum of the squares of the diagonals is equal to the sum of the squares of its sides.

Theorem 2.7 (Cauchy–Schwarz inequality) *In any inner product space, we have*

$$|(x, y)| \leq ||x||||y||.$$

Proof Without any loss of generality, we may suppose that (x, y) is real since we can multiply x by $e^{i\theta}$ for some real θ without altering $||x||$ or $||(x, y)||$. Thus, for r real,

$$0 \leq (x - ry, x - ry) = ||x||^2 - 2r(x, y) + r^2||y||^2.$$

In particular, the corresponding quadratic equation must have negative discriminant. Thus,

$$|(x, y)|^2 - (x, x)(y, y) \leq 0,$$

as desired. □

The Cauchy–Schwarz inequality is one of the most ubiquitous inequalities in mathematics. If we consider $\mathbb{R}^n$ with the usual inner product, the inequality

$$\left|\sum_{j=1}^{n} a_j b_j\right|^2 \leq \left(\sum_{j=1}^{n} |a_j|^2\right)\left(\sum_{j=1}^{n} |b_j|^2\right)$$

was discovered by Cauchy in 1821. The corresponding inequality for integrals, namely

$$\left|\int_a^b f(x)g(x)dx\right|^2 \leq \left(\int_a^b |f(x)|^2 dx\right)\left(\int_a^b |g(x)|^2 dx\right),$$

seems to have been first stated by Buniakowsky in 1859 and later independently by Schwarz in 1885. The interested student may find an exposition of the influence of this inequality in mathematics, statistics and physics in [1].

Theorem 2.8 (The triangle inequality) *For any $x, y \in H$, we have*

$$||x + y|| \leq ||x|| + ||y||.$$

Proof By the Cauchy–Schwarz inequality, we have

$$||x + y||^2 = (x + y, x + y) = (x, x + y) + (y, x + y) = (x, x) + (x, y) + (y, x) + (y, y)$$

$$\leq ||x||^2 + 2||x||||y|| + ||y||^2,$$

from which the result follows immediately.

Thus,

$$||x - z|| \leq ||x - y|| + ||y - z||,$$

for any $x, y, z \in H$. This result allows us to define a metric on H by defining the distance between x and y to be $||x - y||$. Then, all the axioms for a metric space are satisfied in view of the triangle inequality. If this metric space is **complete**, that is, every Cauchy sequence converges, then we call H a **Hilbert space**.

We now consider some examples of Hilbert spaces.

(a) For any fixed n, let $H = \mathbb{C}^n$ and define for $x = (x_1, ..., x_n)$ and $y = (y_1, ..., y_n)$,

$$(x, y) = \sum_{i=1}^{n} x_i \overline{y_i}.$$

Since $\mathbb{C}$ is complete, we see that H is a Hilbert space.

(b) If μ is any positive measure, then the space of square integrable functions, $L^2(\mu)$ becomes an inner product space if we set

$$(f, g) = \int_X f \overline{g} d\mu.$$

How do we know the integral is well-defined? This follows from first applying the Cauchy–Schwarz inequality to simple functions and then taking limits. Since $L^p(\mu)$ is a complete metric space for every $1 \leq p \leq \infty$, we see that $L^2(\mu)$ is a Hilbert space.

(c) The vector space of all continuous complex functions on $[0, 1]$ is an inner product space if

$$(f, g) = \int_0^1 f(t)\overline{g(t)} dt,$$

but is not a Hilbert space. To show this, we need to produce a sequence of continuous functions whose limit is not continuous. But this is clear upon considering the sequence continuous functions $f_n(x) = x^n$ whose limit function is not continuous (see Exercise 4).

Theorem 2.9 *For any fixed $y \in H$, the maps,*

$$x \mapsto (x, y), \quad x \mapsto (y, x), \quad x \mapsto ||x||,$$

are continuous functions on H.

Proof The Cauchy–Schwarz inequality implies

$$||(x_1, y) - (x_2, y)|| \leq ||x_1 - x_2||\, ||y||,$$

which shows that the map $x \mapsto (x, y)$ is uniformly continuous. The same argument works for $x \mapsto (y, x)$. Finally, the triangle inequality shows that

$$||x_1|| - ||x_2|| \leq ||x_1 - x_2||.$$

Interchanging x_1, x_2 shows that

$$| \, ||x_1|| - ||x_2|| \, | \leq ||x_1 - x_2||,$$

again showing that $x \mapsto ||x||$ is uniformly continuous. □

A subset M of a vector space V is called a **subspace** if M is itself a vector space with respect to the addition and scalar multiplication defined on V. Clearly, M is a subspace if and only if for all $x, y \in M$ and $\alpha \in \mathbb{C}$, we have $x + y \in M$ and $\alpha x \in M$. A **closed subspace** of H is one that is closed relative to the topology of H. Let us observe that if M is a subspace, then so is its closure.

A set E in a vector space V is said to be **convex** if for every $x, y \in E$, and $0 < t < 1$, we have

$$(1 - t)x + ty \in E.$$

It is evident that every subspace of V is convex. Moreover, if E is convex, then so are its translates.

If $(x, y) = 0$, we say x and y are **orthogonal** and sometimes write $x \perp y$. Let $x^\perp$ denote the set of all vectors orthogonal to x. If M is a subspace, we denote by $M^\perp$ to be the set of all vectors orthogonal to every vector in M. It should be clear that $x^\perp$ is a subspace of H. Moreover, it is a closed subspace since the map $x \mapsto (x, y)$ is continuous. For any subspace M, it is clear that

$$M^\perp = \bigcap_{x \in M} x^\perp.$$

Since $M^\perp$ is the intersection of closed subspaces, it is also closed.

Notice that if x and y are orthogonal, then the proof of the parallelogram shows that

$$||x + y||^2 = ||x||^2 + ||y||^2,$$

which can be viewed as a generalization of the theorem of Pythagoras.

Theorem 2.10 *Every non-empty, closed, convex set E in a Hilbert space H contains a unique element of smallest norm. That is, there is a unique $x_0 \in E$ such that*

$$||x|| \geq ||x_0||,$$

for all $x \in E$.

Proof Let δ be the infimum of $||x||$ for $x \in E$. For any $x, y \in E$, we apply the parallelogram law to $x/2$ and $y/2$ to get

$$\frac{1}{4}||x - y||^2 = \frac{1}{2}||x||^2 + \frac{1}{2}||y||^2 - ||(x + y)/2||^2.$$

Since E is convex, $(x + y)/2 \in E$. Hence,

$$||x - y||^2 \leq 2||x||^2 + 2||y||^2 - 4\delta^2. \tag{2.2}$$

From this we see that if x_0 exists, then it must be unique for $||x|| = ||y|| = \delta$ implies $||x - y||^2 = 0$. The definition of δ implies that there is a sequence of elements $y_n \in E$ such that $||y_n|| \to \delta$. Replacing x, y in (2.2) by y_n, y_m, we see that the sequence y_n is Cauchy. Since H is complete, there is an $x_0 \in H$ such that $y_n \to x_0$. As E is closed, we must have $x_0 \in E$. □

Theorem 2.11 *Let M be a closed subspace of a Hilbert space H. Then, every $x \in H$ has a unique decomposition as $y + z$ with $y \in M$ and $z \in M^\perp$. The points y and z are the nearest points to x in M and $M^\perp$, respectively. The maps $x \mapsto y$ and $x \mapsto z$ are linear, and we have*

$$||x||^2 = ||y||^2 + ||z||^2.$$

Proof Uniqueness is clear for if $y + z = y' + z'$, we have $y - y' = z' - z$, and since $M \cap M^\perp = 0$, we see that $y = y'$ and $z = z'$. To prove existence, consider the set

$$x + M = \{x - y : y \in M\}.$$

This set is closed and convex. Define z to be the element of smallest norm in $x + M$, which exists by the previous theorem. We need to show that $z \in M^\perp$. Let $w \in M$ and consider (w, z). Without loss of generality, we may suppose that $||w|| = 1$. The minimizing property of z means that

$$||z||^2 \leq ||z - \alpha w||^2 = (z - \alpha w, z - \alpha w)$$

for any $\alpha \in \mathbb{C}$. Thus,

$$0 \leq -\alpha(w, z) - \overline{\alpha}(z, w) + |\alpha|^2$$

for every $\alpha \in \mathbb{C}$. Choosing $\alpha = (z, w)$ gives $0 \leq -|(z, w)|^2$ from which we deduce that $z \in M^\perp$. Thus, $x = y + z$ with $y \in M$ and $z \in M^\perp$. To see that y and z are the nearest points to x in M and $M^\perp$, respectively, we have by the theorem of Pythagoras, for $u \in M$,

$$||x - u||^2 = ||y + z - u||^2 = ||z||^2 + ||y - u||^2,$$

which is clearly minimized for $y = u$. A similar argument shows that z is the nearest point to x in $M^\perp$. The linearity of the maps is clear, and the last assertion follows from Pythagoras. □

Corollary 2.1 *If $M \neq H$, there exists a $z \in H$, $z \neq 0$ which is orthogonal to M.*

Proof Take $x \in H$, $x \notin M$. Take z as in the theorem. □

We have already seen that the map $x \mapsto (x, y)$ is a linear functional for all $y \in H$. It is a very important result of Riesz that **all** continuous linear functionals on H are of this type.

Theorem 2.12 (Riesz representation theorem) *If L is a continuous linear functional on H, then there is a unique $y \in H$ such that $L(x) = (x, y)$, for all $x \in H$.*

Proof If $L(x) = 0$ for all $x \in H$, we can take $y = 0$. Otherwise, define $M = \{x : L(x) = 0\}$. Linearity of L implies this is a subspace. Continuity of L implies that it is closed. Thus, by the previous theorem, M has an orthogonal complement which is not the zero subspace. Thus, there is a $z \in M^{\perp}$ with $||z|| = 1$. Put

$$u = L(x)z - L(z)x.$$

Note that $L(u) = 0$ so that $u \in M$. Therefore, $(u, z) = 0$ and

$$L(x) = L(x)(z, z) = (L(x)z, z) = (u + L(z)x, z) = L(z)(x, z) = (x, \overline{L(z)}z).$$

In other words, $L(x) = (x, y)$ with $y = \overline{L(z)}z$. The uniqueness is obvious since $(x, y) = (x, y')$ for all x means that $y - y'$ is orthogonal to every $x \in H$. In particular, it is orthogonal to itself, which means that $||y - y'|| = 0$ from which we deduce that $y = y'$. □

Exercises

1. For real a, b and $x > 0$ and $y > 0$ show that

$$\frac{(a+b)^2}{x+y} \leq \frac{a^2}{x} + \frac{b^2}{y}.$$

   Replacing b by $b + c$ and y by $y + z$ deduce that

$$\frac{(a+b+c)^2}{x+y+z} \leq \frac{a^2}{x} + \frac{b^2}{y} + \frac{c^2}{z}.$$

   Iterating, deduce that for $x_i > 0$,

$$\frac{(a_1 + a_2 + \cdots + a_n)^2}{x_1 + x_2 + \cdots + x_n} \leq \frac{a_1^2}{x_1} + \frac{a_2^2}{x_2} + \cdots + \frac{a_n^2}{x_n}.$$

   Replacing a_i with $y_i z_i$ and x_i with z_i^2, deduce the Cauchy–Schwarz inequality over the reals.
2. Prove that

$$\left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right) - \left(\sum_{k=1}^{n} a_k b_k\right)^2 = \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}(a_i b_j - a_j b_i)^2.$$

3. For any set of strictly positive real numbers $a_1, ..., a_n$, show that

$$(a_1 + a_2 + \cdots + a_n)\left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}\right) \geq n^2.$$

4. Show that the space $C[0, 1]$ of real-valued continuous functions on $[0, 1]$ with the usual inner product

$$(f, g) = \int_0^1 f(t)g(t)dt,$$

is not a Hilbert space.

5. If M is a closed subspace of a Hilbert space H, prove that $(M^\perp)^\perp = M$.

2.4 Orthonormal Sets

It may be instructive to begin with an example. Consider the space $C[0, 1]$ and the function $f(x) = x - 1/2$. The functions $\phi_n(x) = e^{2\pi inx}$ are orthonormal with respect to the usual inner product. We can try to expand f in terms of these functions:

$$c_n = \int_0^1 f(x)e^{-2\pi inx}dx = -\frac{1}{2\pi in},$$

by an easy calculation. If we apply Parseval's formula, we get

$$\sum_{n\neq 0} \frac{1}{4\pi^2 n^2} = \int_0^1 (x - 1/2)^2 dx = \frac{1}{12}.$$

From this, we deduce

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

To what extent can this calculation be made rigorous?

Let us consider the general setting. A set of vectors $\{u_\alpha\}$ is **orthonormal** if $(u_\alpha, u_\beta) = 0$ for $\alpha \neq \beta$ and 1 otherwise. We define the **Fourier transform**

$$\hat{x}(\alpha) := (x, u_\alpha)$$

and we refer to these as the Fourier coefficients of x relative to the set u_α. We expect in some sense,

$$x = \sum_{\alpha\in A} \hat{x}(\alpha)u_\alpha.$$

Given x in a Hilbert space H, and F any finite subset of A, we can consider the sum

$$s_F(x) = \sum_{\alpha \in F} (x, u_\alpha) u_\alpha.$$

If s in the span of F, then we claim that

$$||x - s_F|| \leq ||x - s||$$

which is to say that s_F is the vector nearest to x in the space spanned by F. Indeed, it is easy to see that s_F is in the space spanned by F and that $x - s_F$ is orthogonal to every u_α with $\alpha \in F$ by direct computation. Thus, $x - s_F$ is orthogonal to $s - s_F$. Now, $x - s = x - s_F + s_f - s$ and by Pythagoras, we get

$$||x - s||^2 = ||x - s_F||^2 + ||s_F - s||^2,$$

from which we see the stated claim immediately. Setting $s = 0$ gives us

$$||s_F||^2 \leq ||x||^2,$$

for all finite F. We record this discussion in the following.

Theorem 2.13 *Suppose that $\{u_\alpha : \alpha \in A\}$ is an orthonormal set in a Hilbert space H and that F is a finite subset of A. Let M_F be the span of $\{u_\alpha : \alpha \in F\}$. If ϕ is a complex-valued function on A which is zero outside F, then there is a vector $y \in M_F$, namely*

$$y = \sum_{\alpha \in F} \phi(\alpha) u_\alpha,$$

which has $\hat{y}(\alpha) = \phi(\alpha)$ for every $\alpha \in A$ and

$$||y||^2 = \sum_{\alpha \in F} |\phi(\alpha)|^2.$$

If $x \in H$ and

$$s_F(x) = \sum_{\alpha \in F} \hat{x}(\alpha) u_\alpha,$$

then

$$||x - s_F(x)|| < ||x - s||,$$

for every $x \in M_F$, except for $s = s_F(x)$, and

$$\sum_{\alpha \in F} |\hat{x}(\alpha)|^2 \leq ||x||^2.$$

Proof Only the first part needs proving. But this is clear from the orthogonality relations. The last inequality is called **Bessel's inequality**. □

We want to extend this discussion to infinite F. We want to give meaning to

$$\sum_{\alpha\in A}\phi(\alpha),$$

where $\phi(\alpha)$ are non-negative numbers. To this end, we simply define the sum to be the supremum of the set of all sums over finite subsets of A. We consider the space $\ell^2(A)$, the set of functions ϕ with domain A and satisfying

$$\sum_{\alpha\in A}|\phi(\alpha)|^2<\infty.$$

This is a Hilbert space with inner product

$$(\phi,\psi)=\sum_{\alpha\in A}\phi(\alpha)\overline{\psi(\alpha)}.$$

If $\phi\in\ell^2(A)$, then it is easy to see that $\phi(\alpha)\neq 0$ for at most countably many elements of A. Indeed, the set A_n of α for which $|\phi(\alpha)|>1/n$ satisfies

$$\sum_{\alpha\in A_n}|n\phi(\alpha)|^2\le n^2\sum_{\alpha\in A}|\phi(\alpha)|^2<\infty,$$

so that A_n is finite for every n.

Recall that an **isometry** between two metric spaces X and Y is a one-to-one and onto map $f:X\to Y$ that preserves distances. That is, if d_1 and d_2 are the distance functions on X and Y, respectively, then

$$d_2(f(x),f(x'))=d_1(x,x'),\qquad\forall\,x,x'\in X.$$

We then say that the two spaces X and Y are **isometric**.

Lemma 2.1 *Suppose that*

(a) X and Y are metric spaces, X is complete,
(b) $f:X\to Y$ is continuous,
(c) X has a dense subspace X_0 on which f is an isometry, and
(d) $f(X_0)$ is dense in Y.

Then f is an isometry of X onto Y.

The important part of the conclusion is that f maps X *onto all of* Y.

Proof The fact that f is an isometry on X follows from the continuity of f and since X_0 is dense in X. To show surjectivity, let $y\in Y$ and let x_n be a sequence in

X_0 satisfying $f(x_n) \to y$, which exists since $f(X_0)$ is dense in Y. Thus, $f(x_n)$ is a Cauchy sequence. Since f is an isometry on X_0, it follows that x_n is a Cauchy sequence in X. Since X is complete, this sequence converges to some $x \in X$ and we have $f(x) = y$ by continuity. □

Theorem 2.14 *Let $\{u_\alpha : \alpha \in A\}$ be an orthonormal set in a Hilbert space H. Let P be the space of all finite linear combinations of the vectors u_α. Then,*

$$\sum_{\alpha \in A} |\hat{x}(\alpha)|^2 \leq ||x||^2, \tag{2.3}$$

holds for every $x \in H$. The map $x \mapsto \hat{x}$ is a continuous linear map from H onto $\ell^2(A)$ whose restriction to the closure $\overline{P}$ of P is an isometry of $\overline{P}$ onto $\ell^2(A)$.

Proof The inequality is immediate from Theorem 2.13. Define f on H by $f(x) = \hat{x}$. Then, the inequality shows that f maps H into $\ell^2(A)$. The linearity of this map is clear. Applying (2.3) to $x - y$ gives

$$||f(y) - f(x)|| = ||\hat{y} - \hat{x}|| \leq ||y - x||,$$

from which we see the continuity of f. Theorem 2.13 shows that f is an isometry of P onto the dense subspace of $\ell^2(A)$ consisting of those functions whose support is a finite set F of A. The theorem now follows from the previous lemma applied to $x = \overline{P}$, $X_0 = P$, $Y = \ell^2(A)$. Note that $\overline{P}$ being a closed subspace of a complete metric space is itself complete. □

The fact that the map $x \to \hat{x}$ carries H *onto* $\ell^2(A)$ is known as the **Riesz–Fischer theorem**.

We now prove the important theorem concerning orthonormal bases.

Theorem 2.15 *Let $\{u_\alpha : \alpha \in A\}$ be an orthonormal set in H. The following are equivalent.*

(a) $\{u_\alpha\}$ is a maximal orthonormal set;
(b) The set P of all finite linear combinations of elements of $\{u_\alpha\}$ is dense in H;
(c) The equality

$$\sum_{\alpha \in A} |\hat{x}(\alpha)|^2 = ||x||^2,$$

holds for every $x \in H$;
(d) Parseval's identity holds:

$$\sum_{\alpha \in A} \hat{x}(\alpha)\overline{\hat{y}(\alpha)} = (x, y),$$

holds for all $x, y \in H$.

Proof We will show that $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (a)$. If P is not dense, then its closure is not all of H and we can write

$$H = \overline{P} \oplus \overline{P}^{\perp},$$

and a non-zero vector in the complement can be added to our orthonormal set, contrary to its maximal property. (b) implies (c) follows from Theorem 2.13. To see that (c) implies (d), we use the "polarization identity":

$$4(x, y) = ||x + y||^2 - ||x - y||^2 + i||x + iy||^2 - i||x - iy||^2,$$

which is easily verified (see exercises below) to be valid in any Hilbert space. We apply this with x, y replaced by $\hat{x}$, $\hat{y}$ since $\ell^2(A)$ is also a Hilbert space. The sum in (c) is $||\hat{x}||^2$ and the sum in (d) is $(\hat{x}, \hat{y})$. Finally, to see that (d) implies (a), suppose that (a) is false and our set is not a maximal orthonormal set. Then, we can find a vector $u \neq 0$ which is orthogonal to every u_α and we can add u to this set. If $x = y = u$, we have $(x, y) = ||u||^2 > 0$. On the other hand, all the Fourier coefficients of u are zero so that (d) does not hold for this vector, a contradiction. This completes the proof. □

Exercises

1. Let $\{x_n : n = 1, 2, ...\}$ be a linearly independent set of vectors in a Hilbert space H. Show that the following construction yields an orthonormal set $\{u_n\}$ such that $\{x_1, ..., x_N\}$ and $\{u_1, ..., u_N\}$ have the same span for all N. Inductively define: $u_1 = x_1/||x_1||$, and $u_n = v_n/||v_n||$, where

$$v_n = x_n - \sum_{j=1}^{n-1} (x_n, u_j)u_j,$$

   for $n \geq 2$. This is the extension of the **Gram–Schmidt process** that the student has undoubtedly seen in a first course in linear algebra.
2. If $M = \{x : L(x) = 0\}$ where L is a continuous linear functional on H, which is not identically zero, prove that $M^{\perp}$ is a vector space of dimension 1.
3. Let H be a Hilbert space and $\{u_1, ..., u_N\}$ an orthonormal system of vectors. Let M be the span of these orthonormal vectors. Given $x \in H$, show that the vector

$$y = \sum_{i=1}^{N} (x, u_i)u_i \in M$$

   is the unique vector in M minimizing $||x - y||$ with $y \in M$.

4. In the space $L^2[-1, 1]$, let M be the subspace spanned by the functions $1, x, x^2$. Find an orthonormal basis for M.
5. Compute
$$\min_{a,b,c} \int_{-1}^{1} |x^3 - a - bx - cx^2|^2 dx.$$
6. Fix a positive integer N, put $\omega = e^{2\pi i/N}$. Prove the orthogonality relations
$$\frac{1}{N} \sum_{n=1}^{N} \omega^{nk} = 1$$
if $k \equiv 0 \pmod{N}$ and 0 otherwise. Use this to derive the identities
$$(x, y) = \frac{1}{N} \sum_{n=1}^{N} ||x + \omega^n y||^2 \omega^n,$$
that hold in every inner product space for $N \geq 3$. Show also that
$$(x, y) = \frac{1}{2\pi} \int_{-\pi}^{\pi} ||x + e^{i\theta} y||^2 e^{i\theta} d\theta.$$

2.5 Trigonometric Series

Let $\mathbb{T}$ be the unit circle. We would like to consider the Hilbert space $L^2(\mathbb{T})$ and show that functions in this space can be approximated by trigonometric polynomials. A **trigonometric polynomial** is a finite sum of the form
$$a_0 + \sum_{n=1}^{N} a_n \cos nt + b_n \sin nt.$$
Because of the formulas
$$\cos t = \frac{e^{it} + e^{-it}}{2}, \quad \text{and} \quad \sin t = \frac{e^{it} - e^{-it}}{2i},$$
we can rewrite these sums as
$$\sum_{n=-N}^{N} c_n e^{int}.$$
This is the form we will consider below. Observe that the set of functions

$$u_n(t) = e^{int},$$

forms an orthonormal family with respect to the usual inner product on $L^2(\mathbb{T})$ given by

$$(f, g) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t)\overline{g(t)}dt.$$

To show that this is a maximal orthonormal family, it suffices to show that the trigonometric polynomials are dense in $L^2(\mathbb{T})$ by the theorem of the previous section. Since $C(\mathbb{T})$ is dense in $L^2(\mathbb{T})$, it suffices to show that every continuous function can be approximated by a trigonometric polynomial. In fact, we will prove the following stronger theorem.

Theorem 2.16 *Fix $\epsilon > 0$ and $f \in C(\mathbb{T})$. There exists a trigonometric polynomial P such that*

$$|f(t) - P(t)| \leq \epsilon,$$

uniformly for all $t \in \mathbb{T}$.

We need the following lemma.

Lemma 2.2 *There exist trigonometric polynomials $Q_1, Q_2, Q_3, \ldots$ with the following properties:*

(a) $Q_k(t) \geq 0$, for all $t \in \mathbb{R}$;
(b)

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} Q_k(t)dt = 1.$$

(c) let $\eta_k(\delta) = \sup\{Q_k(t) : \delta \leq |t| \leq \pi\}$,

then

$$\lim_{k\to\infty} \eta_k(\delta) = 0,$$

for every $\delta > 0$.

Proof Put

$$Q_k(t) = c_k \left(\frac{1 + \cos t}{2}\right)^k,$$

where c_k is chosen so that (b) holds. This is the family of trigonometric polynomials we will use. Clearly (a) also holds. We need to show (c). To this end, we first note that since $Q_k(t)$ is even,

$$1 = \frac{c_k}{\pi} \int_0^{\pi} \left(\frac{1 + \cos t}{2}\right)^k dt > \frac{c_k}{\pi} \int_0^{\pi} \left(\frac{1 + \cos t}{2}\right)^k \sin t dt = \frac{2c_k}{\pi(k + 1)}.$$

Since Q_k is decreasing in $[0, \pi]$, we have

$$Q_k(t) \leq Q_k(\delta) \leq \frac{\pi(k+1)}{2}\left(\frac{1+\cos\delta}{2}\right)^k,$$

for $0 < \delta \leq |t| \leq \pi$. Since $1 + \cos\delta < 2$, the result is now immediate. □

We can now prove Theorem 2.16. To each $f \in C(\mathbb{T})$, let

$$P_k(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t-s)Q_k(s)ds.$$

Replacing s by $t - s$, and using the periodicity of f, we see that

$$P_k(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(s)Q_k(t-s)ds.$$

Since each Q_k is a trigonometric polynomial, it follows that each P_k is also a trigonometric polynomial. Now let $\epsilon > 0$. Since f is continuous, it is uniformly continuous on the compact set $\mathbb{T}$. That is, there is a $\delta > 0$, such that

$$|f(t) - f(s)| < \epsilon, \quad \text{whenever} \quad |t - s| < \delta.$$

Thus,

$$P_k(t) - f(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\{f(t-s) - f(t)\}Q_k(s)ds.$$

Thus, by the positivity of Q_k, we have

$$|P_k(t) - f(t)| \leq \frac{1}{2\pi}\int_{-\pi}^{\pi}|f(t-s) - f(t)|Q_k(s)ds + A_1 + A_2 \quad \text{(say)}$$

where A_1 is the integral over the interval $[-\delta, \delta]$, and A_2 is over the complementary set. In A_2, we use the estimate on $\eta_k(\delta)$ to see that

$$A_2 \leq 2||f||_\infty \eta_k(\delta) < \epsilon,$$

for sufficiently large k. For A_1, we have by the uniform continuity of f that $A_1 \leq \epsilon$ using (2) of Lemma 2.2. This completes the proof. □

In Sect. 3.10, we will use the Féjer kernel to give a more constructive proof of the previous theorem.

As a consequence of the theorem proved in the previous section, we have the Parseval theorem in the special case of $L^2(\mathbb{T})$. For $f \in L^2(\mathbb{T})$, we define the **Fourier coefficients** by

$$\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt.$$

For $f, g \in L^2(\mathbb{T})$, we have

$$(f, g) = \sum_{n=-\infty}^{\infty} \hat{f}(n)\overline{\hat{g}(n)}$$

and specializing to $f = g$ gives

$$||f||^2 = \sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2.$$

We can ask if the converse is true. That is, suppose that

$$\sum_{n=-\infty}^{\infty} |c_n|^2 < \infty.$$

Does there exist an $f \in L^2(\mathbb{T})$ whose sequence of Fourier coefficients coincides with the c_n. The answer is provided by the Riesz–Fischer theorem. Recall that the map $x \mapsto \hat{x}$ is an isometry from H to $L^2(A)$ in the context of Theorem 2.15. Since this map is onto, there is an $f \in L^2(\mathbb{T})$ such that $\hat{f}(n) = c_n$ for all n.

Let us look at the following example. Consider the function $f(t) = t$ for $-\pi \leq t < \pi$ and extended by periodicity on the real line. Its Fourier coefficients are easily computed:

$$\int_{-\pi}^{\pi} xe^{-inx}dx = \frac{2(-1)^n\pi}{n},$$

for $n \neq 0$ and for $n = 0$, the Fourier coefficient is zero. Thus, by Parseval's formula, we immediately deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

This gives another proof of this celebrated result of Euler.

Exercises

1. Suppose f is a continuous function of $\mathbb{R}^1$ with period 1. Prove that

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f(n\alpha) = \int_0^1 f(t)dt$$

for every irrational real number α. [Hint: do it first for $f(t) = e^{2\pi ikt}$, with $k = 0, \pm 1, \pm 2, ...$]

2. Is the series
$$\sum_{n=1}^{\infty} \frac{\cos nx}{\sqrt{n}}, \qquad x \in \mathbb{T}$$
the Fourier series of some function in $L^2(\mathbb{T})$?
3. Is the series
$$\sum_{n=1}^{\infty} \frac{\sin nx}{n^{3/2}}, \qquad x \in \mathbb{T}$$
the Fourier series of some function in $L^2(\mathbb{T})$?
4. If $f \in L^2(\mathbb{T})$ show that
$$\hat{f}(n) \to 0$$
as $|n|$ tends to infinity.
5. Find the Fourier series of
$$f(x) = x^2$$
for $x \in \mathbb{T}$ and use it to show that
$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

2.6 Banach Spaces

A complex vector space X is said to be a **normed linear space** if there is a map $|| \cdot || : X \to \mathbb{R}$ such that

(a) $||x|| \geq 0$ with equality if and only if $x = 0$;
(b) $||\lambda x|| = |\lambda| ||x||$, for every scalar λ;
(c) $||x + y|| \leq ||x|| + ||y||$, for all $x, y \in X$ (triangle inequality).

It is easily seen that we may define a metric on X by defining the distance between x and y as $||x - y||$. A **Banach space** is a normed linear space which is complete with respect to this metric. Every Hilbert space is a Banach space. However, the class of Banach spaces is a wider class than the class of Hilbert spaces, and we proceed to develop this theory. In particular, let us note that $C(\mathbb{T})$ is a Banach space relative to the sup norm.

Here are some examples. Let X be a measure space with measure μ. Consider the space $L^p(\mu)$ of all complex valued functions with norm defined by

$$||f|| := \left(\int_X |f(x)|^p d\mu \right)^{1/p},$$

for $p \geq 1$. We have to check that this satisfies the axioms for our norm. The first two are clear. The third follows from **Minkowski's inequality.** For $p = 1$, this is clear. So let us assume $1 < p < \infty$ and proceed to show that Minkowski's inequality follows from the Cauchy–Schwarz inequality.

To prove this, we begin with an elementary remark. Suppose that $\phi(t)$ is a twice differentiable function on [0, 1] such that $\phi''(t) \geq 0$. Thus, $\phi'(t)$ is increasing, and $\phi(t)$ is concave up in this interval. Thus,

$$\frac{\phi(0) + \phi(1)}{2} \geq \phi(1/2).$$

We will make use of this observation below. For any set $a_1, ..., a_n, b_1, ..., b_n$ of non-negative real numbers, we have

$$\left(\sum_{j=1}^n (a_j + b_j)^p \right)^{1/p} \leq \left(\sum_{j=1}^n a_j^p \right)^{1/p} + \left(\sum_{j=1}^n b_j^{1/p} \right)^{1/p},$$

from which the inequality for L^p-spaces follows by looking at finite sums and taking limits. To prove the inequality, let $A_j = ta_j + (1-t)b_j$ and set

$$\phi(t) = \left(\sum_{j=1}^n A_j^p \right)^{1/p}.$$

The first derivative is

$$\left(\sum_{j=1}^n A_j^p \right)^{1/p-1} \sum_{j=1}^n A_j^{p-1}(a_j - b_j).$$

The second derivative is easily seen to be

$$(p-1) \left(\sum_{j=1}^n A_j^p \right)^{1/p-2} \left\{ \left(\sum_{j=1}^n A_j^p \right) \sum_{j=1}^n A_j^{p-2}(a_j - b_j)^2 - \left(\sum_{j=1}^n A_j^{p-1}(a_j - b_j) \right)^2 \right\}.$$

The quantity in the braces is non-negative by an application of the Cauchy–Schwarz inequality. Since $p \geq 1$, we see that $\phi''(t) \leq 0$. By our initial observation, we deduce that

$$\frac{\phi(0) + \phi(1)}{2} \geq \phi(1/2),$$

which is Minkowski's inequality:

$$\left(\sum_j (a_j + b_j)^p\right)^{1/p} \leq \left(\sum_j a_j^p\right)^{1/p} + \left(\sum_j b_j^p\right)^{1/p},$$

for non-negative a_j, b_j.

By the usual limiting process for integrals, we deduce

$$\left(\int_X |f+g|^p d\mu\right)^{1/p} \leq \left(\int_X (|f|+|g|)^p d\mu\right)^{1/p} \leq ||f||_p + ||g||_p,$$

showing that $L^p(\mu)$ is a normed linear space. Technically speaking, as stated earlier, $L^p(\mu)$ does not consist of functions but rather equivalence classes of functions equal almost everywhere. One can show that this space is complete (Riesz–Fischer theorem). Thus, $L^p(\mu)$ is a Banach space.

In this context, it is useful to derive another fundamental inequality called **Hölder's inequality**. We begin with an elementary observation. Let p, q be positive real numbers such that $\frac{1}{p} + \frac{1}{q} = 1$. Then, for $a, b \geq 0$

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}. \tag{2.4}$$

To see this, let us first note that the inequality is trivially true if a or b equals zero. So, let us assume $a, b > 0$. Replacing a and b by $x^{1/p}$ and $y^{1/q}$, respectively, we need to prove that

$$x^{1/p} y^{1/q} \leq \frac{x}{p} + \frac{y}{q}.$$

Taking logarithms, this is equivalent to

$$\frac{1}{p}\log x + \frac{1}{q}\log y \leq \log\left(\frac{x}{p} + \frac{y}{q}\right).$$

But this is now immediate from the concavity of the logarithmic function. We can now prove:

Theorem 2.17 (Hölder's inequality) *Let $a_1, a_2, ..., a_n, b_1, b_2, ..., b_n$ be complex numbers and let $1 < p < \infty$, $1 < q < \infty$ satisfy $1/p + 1/q = 1$. Then,*

$$\left|\sum_{k=1}^n a_k b_k\right| \leq \left(\sum_{k=1}^n |a_k|^p\right)^{1/p} \left(\sum_{k=1}^n |b_k|^q\right)^{1/q}.$$

The case $p = q = 2$ is the Cauchy–Schwarz inequality.

Proof In (2.4), putting

$$a = \frac{|a_k|}{\left(\sum_{k=1}^n |a_k|^p\right)^{1/p}}, \qquad b = \frac{|b_k|}{\left(\sum_{k=1}^n |b_k|^q\right)^{1/q}}$$

and summing over k gives the desired result. □

By the usual limit process, Hölder's inequality translates to an integral version:

$$\left|\int_X f(x)g(x)d\mu\right| \leq \left(\int_X |f(x)|^p d\mu\right)^{1/p} \left(\int_X |g(x)|^q d\mu\right)^{1/q}$$

provided all of the integrals exist. An important consequence of this is the following.

Theorem 2.18 *Let (X, μ) be a measure space and let $1 \leq p \leq \infty$ and q be such that*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

If $f \in L^p(\mu)$ and $g \in L^q(\mu)$. Then $fg \in L^1(\mu)$ and

$$||fg||_1 \leq ||f||_p \, ||g||_q.$$

Proof For $p < \infty$, this is an immediate consequence of Hölder's inequality. For $p = \infty$, we have

$$|f(x)g(x)| \leq ||f||_\infty \, |g(x)|$$

almost everywhere. Integrating this over X gives the desired inequality in this case also. □

An important class of examples is obtained by considering linear transformations from a normed linear space X to a normed linear space Y. Let $T : X \to Y$. We say T is a **bounded linear transformation** if there is an $M < \infty$ such that

$$||T(x)|| \leq M||x||,$$

for all $x \in X$. (Note that the norm on the left hand side of the inequality is in Y and the norm on the right hand side is in X. We will continue this convention since it will be clear from the context, which norm is being used.) It is easy to see that the set of all bounded linear transformations is a vector space. We define a norm on this space by setting

$$||T|| = \sup\{||T(x)|| : x \in X, ||x|| \leq 1\}.$$

It turns out that the space of all bounded linear transformations from a normed vector space X to a Banach space Y is itself a Banach space.

In the special case that Y is the field of complex numbers, we call such linear transformations, **functionals**. If X is a Banach space, the space X^* consisting of all bounded linear functionals is again a Banach space, called the **dual space** of X. Since X^* is again a Banach space, one can take its dual X^{**}, and one can show that X can be identified with an isometric subspace of X^{**}. It is not in general true that X is isomorphic to X^{**}. However, for some important spaces, like L^p-spaces, it is true that $X \simeq X^{**}$. We refer the reader to [2] for further details.

The space $L^1(\mu)$ is of some interest in the theory of Fourier series. Recall that we proved that any $f \in L^2(\mathbb{T})$ has a Fourier series which converges to f in the L^2-norm. A natural question to ask is if for any $f \in C(\mathbb{T})$, the Fourier series converges to f. This question has stimulated extensive research in the subject. In 1876, du Bois Reymond showed that there exists a continuous function whose Fourier series diverges at a point. He asked if the Fourier series converges almost everywhere. This was finally settled by Carleson in 1966, who showed that for any L^2-function, its Fourier series converges almost everywhere. We will use the theory of Banach spaces to show that there exist continuous functions whose Fourier series diverge on a dense subset of $\mathbb{T}$.

In this context, let us consider the problem of convergence of Fourier series. The question is if the partial sums,

$$s_n(f, x) = \sum_{j=-n}^{n} \hat{f}(j) e^{ijx}$$

converge to $f(t)$. To this end, we observe that

$$s_n(f, x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) D_n(x - t) dt,$$

where $D_n(t)$ is the **Dirichlet kernel**:

$$D_n(t) = \sum_{j=-n}^{n} e^{ijt}.$$

Noting that

$$e^{it/2} D_n(t) - e^{-it/2} D_n(t) = 2i \sin(n + 1/2)t,$$

we find

$$D_n(t) = \frac{\sin(n + 1/2)t}{\sin t/2}.$$

Let us show that the L^1-norm of D_n tends to infinity as n tends to infinity. Indeed, using the inequality $|\sin t| \leq |t|$,

$$||D_n||_1 > \frac{2}{\pi}\int_0^{\pi} |\sin(n+1/2)t|\frac{dt}{t} = \frac{2}{\pi}\int_0^{(n+1/2)\pi} |\sin t|\frac{dt}{t}.$$

The last integral is easily seen to be greater than

$$\frac{2}{\pi}\sum_{k=1}^{n}\frac{1}{k\pi}\int_{(k-1)\pi}^{k\pi} |\sin t|dt = \frac{4}{\pi^2}\sum_{k=1}^{n}\frac{1}{k} \to \infty.$$

We also consider the space $L^\infty(\mu)$ consisting of essentially bounded functions. Recall that this space has norm defined by

$$||f||_\infty = \text{essential supremum of } |f|,$$

where the **essential supremum** is

$$\inf\{M : \mu(t : |f(t)| > M) = 0\}.$$

The fact that $L^\infty(\mu)$ is a normed linear space follows from the usual triangle inequality.

We will prove in the next section:

Theorem 2.19 (Banach–Steinhaus theorem) *Suppose X is a Banach space and Y is a normed linear space. Suppose that T_α with $\alpha \in A$ is a family of bounded linear transformations from X to Y. Then, either all the T_α are uniformly bounded or*

$$\sup_{\alpha\in A} ||T_\alpha(x)|| = +\infty,$$

for all x belonging to a dense subset of X.

Remark 2.1 One can show that this dense subset is of type G_δ, that is, a countable intersection of open sets.

Before we prove this theorem, we want to consider its application to the theory of Fourier series alluded to above. Consider the linear functional $T_n : C(\mathbb{T}) \to s_n(f, 0)$. Clearly, each T_n is bounded since

$$||T_n|| \le ||D_n||_1.$$

Now fix n. Put $g(t) = 1$ if $D_n(t) \ge 0$ and -1 if $D_n(t) < 0$. There exist continuous functions $f_j \in C(\mathbb{T})$ such that $-1 \le f_j \le 1$ and $f_j(t) \to g(t)$ for every t, as $j \to \infty$. By the dominated convergence theorem,

$$\lim_{j\to\infty} T_n(f_j) = \lim_{j\to\infty}\frac{1}{2\pi}\int_{-\pi}^{\pi} f_j(-t)D_n(t)dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} g(-t)D_n(t)dt = ||D_n||_1.$$

Thus, these operators are unbounded, and so by the Banach–Steinhaus theorem, there exist functions in $C(\mathbb{T})$ whose Fourier series diverges on a dense subset of $\mathbb{T}$.

Exercises

1. Let $X = \mathbb{C}^n$. For $x = (x_1, ..., x_n)$, define

$$||x||_\infty = \max_{1 \le i \le n} |x_i|.$$

   Show that this is a norm on X.
2. With X as in the previous exercise, show that

$$||x||_\infty = \lim_{p \to \infty} ||x||_p.$$

3. Let ℓ_∞ be the space of all bounded sequences. Show that for $x = \{x_n\}$,

$$||x|| := \sup_n |x_n|$$

   defines a norm on ℓ_∞.
4. Let c be the subspace of ℓ_∞ consisting of convergent sequences and c_0 the subspace of null sequences. Show that c and c_0 are closed Banach subspaces of ℓ_∞.

2.7 Baire's Theorem

The proof of the Banach–Steinhaus theorem will rely on the following theorem due to Baire.

Theorem 2.20 (Baire) *If X is a complete metric space, then the intersection of a countable collection of dense open subsets is dense in X.*

Proof Let $V_1, V_2, ...$ be dense and open in X. Let W be any open set of X. We have to show that $\cap V_n$ has a point in W for $W \neq \emptyset$. Let $B(x, r)$ denote the open ball of radius r centered at $x \in X$ and let $\overline{B}(x, r)$ be its closure. Since V_1 is dense, $W \cap V_1$ is non-empty and so we can find x_1, r_1 such that

$$\overline{B}(x_1, r_1) \subset W \cap V_1 \quad \text{and} \quad 0 < r_1 < 1.$$

We now proceed inductively. If $n \geq 2$, we can choose x_{n-1} and r_{n-1} such that $V_n \cap B(x_{n-1}, r_{n-1})$ is non-empty and so we can find x_n and r_n so that

$$\overline{B}(x_n, r_n) \subset V_n \cap B(x_{n-1}, r_{n-1}), \quad \text{and} \quad 0 < r_n < 1/n.$$

This process gives us a sequence of elements x_n which is easily seen to be Cauchy in X. As X is complete, the limit x of the sequence lies in X. Since x_i lies in the closed set $\overline{B}(x_n, r_n)$ for $i > n$, it follows that x lies in each V_n. By construction, we have $x \in W$. This completes the proof. □

We can now prove the Banach–Steinhaus theorem.

Proof of Theorem 2.19 Put $\phi(x) = \sup_{\alpha \in A} ||T_\alpha(x)||$ and let V_n be the set of $x \in X$ for which $\phi(x) > n$. Since each T_α is continuous and the norm is a continuous map, each function $x \mapsto ||T_\alpha(x)||$ is continuous on X. It is easy to see that each V_n is open. If all of these V_n's are dense, then by Baire's theorem, $\cap V_n$ is dense in X, and the second half of the theorem is established. Therefore, let us suppose that V_N fails to be dense. Then, there is an x_0 and $r > 0$ such that $B(x_0, r) \cap V_N = \emptyset$. In other words,

$$\phi(x) = \sup_{\alpha \in A} ||T_\alpha(x)|| \leq N$$

for all $x \in X$ satisfying $||x - x_0|| \leq r$ (the fact that we can take the closed disk follows by continuity). Putting $y = x - x_0$, we can rephrase this as $\phi(x_0 + y) \leq N$ for all $||y|| < r$. In particular, with $y = 0$, we have $\phi(x_0) \leq N$. Now,

$$T_\alpha(y) = T_\alpha(y + x_0) - T_\alpha(x_0),$$

so by the triangle inequality and taking sup over $\alpha \in A$, we get

$$\phi(y) \leq \phi(y + x_0) + \phi(x_0) \leq 2N.$$

Thus, if $||y|| < r$, we have $||T_\alpha(y)|| \leq 2N$. Changing y to ry shows that for $||y|| \leq 1$, we have $||T_\alpha(y)|| \leq 2N/r$ for all α. This completes the proof. □

There are several important corollaries of the Banach–Steinhaus theorem. The most notable is the following.

Corollary 2.7.1 *Let X be a Banach space and Y a normed linear space. Suppose that we have a sequence of operators $T_n : X \to Y$ such that for each x, the sequence $T_n(x)$ is bounded. Then, it follows that all these sequences are uniformly bounded.*

Although we do not develop the theory here, it is clear that many of the notions of calculus generalize to the setting of normed linear spaces. For instance, the notion of **derivative** can be defined for any map $f : X \to Y$ of normed linear spaces. Fix $a \in X$. Then, we say f is **differentiable** at a if there is a linear transformation $D_{f,a} : X \to Y$ such that

$$\lim_{h \to 0} \frac{||f(a + h) - f(a) - D_{f,a}(h)||}{||h||} = 0.$$

One can show that this linear transformation, if it exists, is unique, and thus, we can speak of $D_{f,a}$ as the **Fréchet derivative** of f at a. It is then possible to derive a general theory of calculus in a Banach space. We refer the interested reader to [2].

Exercises

1. Show that the norm of f in a Banach space can be evaluated on the boundary of the unit ball.
2. Show that the derivative in a Banach space, when it exists, is unique.
3. Let X be a complete metric space. A subset E of X is called **nowhere dense** if its closure contains no non-empty open subset of X. Any countable union of nowhere dense sets is called a set of the **first category**. All other subsets are said to be of **second category**. Show that no complete metric space is of the first category.

2.8 Hahn–Banach Theorem

Theorem 2.21 *If M is a subspace of a normed linear space X and if f is a bounded linear functional on M, then f can be extended to a bounded linear functional F on X so that $||F|| = ||f||$.*

We make some comments. To say that F is an **extension** of f means that the domain of F includes that of f and that $F(x) = f(x)$ on the domain of f. Second, the norms are computed relative to the domains of F and f, respectively. Thus,

$$||f|| = \sup\{|f(x)| : x \in M, ||x|| \leq 1\}$$

and

$$||F|| = \sup\{|F(x)| : x \in X, ||x|| \leq 1\}.$$

The third comment concerns the field of scalars. So far, we have been concerned about spaces over $\mathbb{C}$ but all of the theory is valid over $\mathbb{R}$. In fact, the Hahn–Banach theorem was originally proved for real normed spaces.

A complex function ϕ on a complex vector space V is a **complex linear functional** if

$$\phi(x + y) = \phi(x) + \phi(y), \quad \phi(\alpha x) = \alpha\phi(x), \tag{2.5}$$

for all $x, y \in V$ and all **complex** α. A real-valued function ϕ on a complex (real) vector space is a real-linear functional if (2.5) holds for all real α. If u is the real part of a complex linear functional, then u is a real-linear functional.

Proposition 2.8.1 *Let V be a complex vector space.*

(a) If u is the real part of a complex linear functional f on V, then

$$f(x) = u(x) - iu(ix), \quad (x \in V). \tag{2.6}$$

(b) If u is a real-linear functional on V and if f is defined as in (2.6), then f is a complex linear functional on V.

(c) If V is a normed linear space and f and u are related as in (2.6), then $||f|| = ||u||$.

Proof If $z = \alpha + i\beta$ with $\alpha, \beta \in \mathbb{R}$, the real part of iz is $-\beta$. Thus,

$$z = \mathrm{Re}(z) - i\,\mathrm{Re}(iz),$$

for all complex numbers z. Since f is linear,

$$\mathrm{Re}(if(x)) = \mathrm{Re}(f(ix)) = u(ix),$$

from which (2.6) follows. To prove the second part, we have to check that if f is defined as in (2.6), then it is complex linear. But this follows easily by noting that

$$f(ix) = u(ix) - iu(-x) = u(ix) + iu(x) = if(x).$$

Finally, since $|u(x)| \le |f(x)|$ we have $||u|| \le ||f||$. On the other hand, to every $x \in V$, there is a complex number α, $|\alpha| = 1$ so that $\alpha f(x) = |f(x)|$. Then,

$$|f(x)| = f(\alpha x) = u(\alpha x) \le ||u||||\alpha x|| = ||u|| \cdot ||x||$$

so that $||f|| \le ||u||$. This completes the proof. □

In the proof of the Hahn–Banach theorem, a key role is played by Zorn's lemma (which is equivalent to the axiom of choice). We recall this for the sake of completeness.

A set P is said to be a **partially ordered set** if there is a relation $\le$ on the set satisfying

(a) $a \le b$ and $b \le c$ implies $a \le c$,
(b) $a \le a$ for all $a \in P$ and
(c) $a \le b$ and $b \le a$ implies $a = b$.

A subset Q of a partially ordered set P is said to be **linearly ordered** if for every pair a, b, either $a \le b$ or $b \le a$. A subset Q is a **maximal** totally ordered subset if it is totally ordered and if adjoining any new element of P not in Q is adjoined to Q, the resulting set is no longer totally ordered. The Hausdorff maximality theorem says that every non-empty partially ordered set contains a maximal totally ordered subset. This is equivalent to Zorn's lemma and the axiom of choice.

We are now ready to prove the Hahn–Banach theorem. We assume first that X is a real normed linear space. Consequently, f is a real-linear bounded functional on

M. If $||f|| = 0$, then $F = 0$ is the desired extension. There is no loss of generality in assuming that $||f|| = 1$. Choose $x_0 \in X, x_0 \notin M$. Let M_1 be the vector space spanned by M and x_0. We first extend f to M_1. M_1 consists of vectors of the form $x + \lambda x_0$ with $x \in M$ and λ a real scalar, so that if we define $f_1(x + \lambda x_0) = f(x) + \lambda\alpha$ with α any fixed real number, then it is easy to verify that we have an extension of f to M_1. The problem is to choose α so that the norm is preserved. That is, we must ensure that

$$|f(x) + \lambda\alpha| \leq ||x + \lambda x_0||, \quad (x \in M, \lambda \in \mathbb{R}).$$

Replacing x by λx and dividing both sides of the above by $|\lambda|$, the requirement is

$$|f(x) - \alpha| \leq ||x - x_0||, \quad (x \in M).$$

That is, we must have

$$A_x := f(x) - ||x - x_0|| \leq \alpha \leq f(x) + ||x - x_0|| =: B_x.$$

There exists such an α if all the intervals $[A_x, B_x]$ have a common point. That is, if and only if

$$A_x \leq B_y$$

for all $x, y \in M$. But

$$f(x) - f(y) = f(x - y) \leq ||x - y|| \leq ||x - x_0|| + ||y - x_0||$$

from which we see that $A_x \leq B_y$ for all $x, y \in M$. Thus, there is a norm-preserving extension f_1 of f to M_1. Now let P be the collection of ordered pairs (M', f') where M' is a subspace of X containing M and f' is a real-linear extension of f to M' with $||f'|| = 1$. Partially order P by declaring that $(M', f') \leq (M'', f'')$ to mean that $M' \subset M''$ and $f''(x) = f'(x)$ for all $x \in M'$. The axioms for partial order are clearly satisfied. Moreover, by the previous discussion, P is not empty. Thus, by Hausdorff's maximality principle, there is a maximal totally ordered subcollection Ω of P. Let Φ be the collection of all M' such that $(M', f') \in \Omega$. Then Φ is totally ordered by set inclusion, so that the union $\tilde{M}$, of all members of Φ, is a subspace of X (Exercise). We extend f to $\tilde{M}$ as follows. If $x \in \tilde{M}$, then $x \in M'$ for some $M' \in \Phi$. Define $F(x) = f'(x)$ where f' is the function that occurs in the pair $(M', f') \in \Omega$. Since Ω is totally ordered, it is immaterial which $M' \in \Phi$ we choose to define $F(x)$ as long as M' contains x. It is now easy to check that F is a linear functional on $\tilde{M}$ and $||F|| = 1$. If $\tilde{M}$ were a proper subspace of X, then the construction in the first part of the proof gives us a further extension which would contradict the maximality of Ω. Thus, $\tilde{M} = X$. This completes the proof in the case of real scalars.

If now, f is a complex linear functional on the subspace M of the complex normed linear space X, let u be the real part of f and use the real Hahn–Banach theorem to extend u to a real-linear functional U on X with $||U|| = ||u||$. Now define,

$$F(x) = U(x) - iU(ix), \quad x \in X.$$

By Proposition 2.8.1, F is a complex linear extension of f and $||F|| = ||U|| = ||u|| = ||f||$, which completes the proof. □

An important consequence is:

Theorem 2.22 *If X is a normed linear space and $x_0 \in X$ with $x_0 \neq 0$, then there is a bounded linear functional f on X of norm 1 so that $f(x) = ||x_0||$.*

Proof Let $M = \{\lambda x_0\}$ and define $f(\lambda x_0) = \lambda ||x_0||$. Then, f is a linear functional of norm 1 on M. By the Hahn–Banach theorem, f can be extended to X with the norm preserved. □

The previous theorem allows us to define the notion of a **dual space**. If X is a normed linear space, let X^* be the set of all bounded linear functionals on X. With addition and scalar multiplication defined in the obvious way, we see that X^* is again a normed linear space. In fact, as $\mathbb{C}$ and $\mathbb{R}$ are complete metric spaces, X^* is a Banach space as will be shown below. It is customary to denote elements of X^* by x^* and the value of $x^* \in X^*$ at the point $x \in X$ is denoted $X^*(x)$ or by the more suggestive notation, $\langle x, x^* \rangle$.

By the previous theorem, X^* is not trivial if X is not trivial. In fact, X^* **separates points** on X. That is, given $x_1, x_2 \in X$ with $x_1 \neq x_2$, there is a functional $f \in X^*$ such that $f(x_1) \neq f(x_2)$. Indeed, take $x_0 = x_1 - x_2$ in the previous theore, and the result is immediate. We call X^* the **dual space** of X.

Another consequence is that for $x \in X$, we have

$$||x|| = \sup\{|f(x)| : f \in X^*, ||f|| = 1\}.$$

Indeed, the right hand side is certainly a lower bound for $||x||$. However, by the Hahn–Banach theorem, there is a continuous linear functional f of norm 1 with $f(x) = ||x||$ for a given x. Thus, the right hand side is also an upper bound for $||x||$. Hence, for fixed $x \in X$, the map $f \mapsto f(x)$ is a bounded linear functional on X^*. This gives us an injection

$$X \to X^{**}.$$

The study of the interplay between X and X^* forms a large part of what is called **functional analysis**.

Theorem 2.23 *If X is a Banach space, then X^* is a Banach space.*

Proof Since X^* is already a normed linear space, we have to show that it is complete. Let x_n^* be a Cauchy sequence in X^*. Then, $||x_n^* - x_m^*||$ tends to zero as n, m both tend to infinity. Thus, for any fixed $x \in X$, the sequence $x_n^*(x)$ is a Cauchy sequence of scalars since

$$|x_n^*(x) - x_m^*(x)| \leq ||x_n^* - x_m^*|| \cdot ||x||.$$

Thus, for each $x \in X$, there is a scalar $x^*(x)$ such that $x_n^*(x) \to x^*(x)$. The functional x^* defined on all of X in this way is clearly linear since

$$x^*(ax + by) = \lim x_n^*(ax + by) = \lim ax_n^*(x) + bx_n^*(y)$$

$$= a \lim x_n^*(x) + b \lim x_n^*(y) = ax^*(x) + bx^*(y).$$

Now since x_n^* is a Cauchy sequence, given $\epsilon > 0$, there is an M such that

$$|x_n^*(x) - x_m^*(x)| < \epsilon ||x||,$$

for all $n, m > M$ and all x. Letting n tend to infinity, we get

$$|x^*(x) - x_m^*(x)| < \epsilon ||x||,$$

for $m > M$. Thus,

$$|x^*(x)| = |x^*(x) - x_m^*(x) + x_m^*(x)| \leq |x^*(x) - x_m^*(x)| + |x_m^*(x)| \leq (\epsilon + ||x_m^*||)||x||,$$

so we conclude that x^* is a bounded linear functional. Also from

$$|x^*(x) - x_m^*(x)| < \epsilon ||x||, \quad m > M$$

follows the inequality $||x^* - x_m^*|| < \epsilon$ so that $x_m^* \to x^*$ in X^*.

Exercises

1. Recall that ℓ_1 consists of sequences $x = \{x_n\}$ such that

$$\sum_{n=1}^{\infty} |x_n| < \infty.$$

   Let c_0 be the space of null sequences. Fix $y \in \ell_1$. Show that $T : c_0 \to \ell_1$ defined by

$$T(x) = \sum_{n=1}^{\infty} x_n y_n$$

   defines a bounded linear functional on c_0 and that $||T|| = ||y||_1$.
2. Prove that $(c_0)^* = \ell_1$.
3. Using a construction similar to the first exercise, prove that $\ell_1^* = \ell_\infty$.
4. Show that ℓ_∞^* contains non-trivial functionals that vanish on all of c_0.

2.9 Examples of Dual Spaces

Let us first consider the dual space of $\mathbb{R}^n$. Recall that the norm of the vector $x = (x_1, ..., x_n)$ is given by

$$||x|| = \left(\sum_i x_i^2 \right)^{1/2}.$$

Let $e_1, ..., e_n$ be the standard basis vectors. Then, any linear map f is determined by the values $f(e_i)$. Thus, letting $y_i = f(e_i)$, we have

$$f(x) = \sum_i x_i y_i,$$

since $x = \sum_i x_i e_i$. By Cauchy–Schwarz, we have

$$|f(x)| = |\sum_i x_i y_i| \leq \left(\sum_i y_i^2 \right)^{1/2} ||x||.$$

We see that f is a bounded linear functional with norm bounded by

$$\left(\sum_i y_i^2 \right)^{1/2}.$$

However, choosing $x = (y_1, ..., y_n)$, we see that this bound is achieved. Hence, every bounded linear functional corresponds to a vector $(y_1, ..., y_n)$ and

$$f(x) = \sum_i x_i y_i$$

is the usual inner product. Thus, the dual space of $\mathbb{R}^n$ is again $\mathbb{R}^n$ as Banach spaces.

This example should not surprise us in view of our earlier characterization of linear functionals on a Hilbert space H. They are all of the form $x \mapsto (x, y)$. What is the norm of this functional? By Cauchy–Schwarz, we have

$$|(x, y)| \leq ||y||||x||,$$

so that the norm is bounded by $||y||$. However, choosing $x = y$, this bound is attained, and hence the norm of this functional is $||y||$.

We will now determine the dual of the space ℓ_p, consisting of all sequences (x_n) such that

$$\sum_n |x_n|^p < \infty.$$

We need to use Hölder's inequality:

$$\left\| \sum_i x_i y_i \right\| \leq \left(\sum_i |x_i|^p \right)^{1/p} \left(\sum_i |y_i|^q \right)^{1/q},$$

for any $p, q > 1$ satisfying $1/p + 1/q = 1$. The special case $p = q = 2$ is the familiar Cauchy–Schwarz inequality.

Every bounded linear functional on ℓ_p is of the form

$$f(x) = \sum_n x_n y_n,$$

where (y_n) is a sequence in ℓ_q. Using Hölder's inequality, it is easy to verify that f is a bounded linear functional on ℓ_p. Conversely, if e_i denotes the sequence which is zero everywhere except at the i-th coordinate, we see that $y_i = f(e_i)$ determines by continuity $f(x)$ since

$$f(x) = \sum_i x_i y_i.$$

We want to show that the sequence (y_i) lives in ℓ_q. To this end, define for each natural number N, the sequence x_N having for its i-th component $|y_i|^{q/p} \operatorname{sgn} y_i$ for $i \leq N$ and zero otherwise. Then,

$$||x_N|| = \left(\sum_{i=1}^N |y_i|^q \right)^{1/p},$$

and

$$f(x_N) = \sum_{i=1}^N |y_i|^{q/p+1} = \sum_{i=1}^N |y_i|^q.$$

But $|f(x_N)| \leq ||f||||x_N||$ so that

$$\left(\sum_{i=1}^N |y_i|^q \right)^{1/q} \leq ||f||,$$

for all N. Thus, the sequence (y_i) is an element of ℓ_q.

For $p = 1$, the dual of ℓ_1 is ℓ_∞, the space of bounded sequences. To see this, let as before $y_i = f(e_i)$. Define the sequence u_N by $x_i = \operatorname{sgn} y_N$ for $i = N$ and zero otherwise. Then, $||u_N|| \leq 1$ and

$$|y_N| = f(u_N) \leq ||f||||u_N|| \leq ||f||.$$

Thus, the sequence of y's is bounded. Conversely, given such an element y, we can clearly define and element of ℓ_1^* by setting

$$f(x) = \sum_i x_i y_i.$$

We caution the reader that the dual space of ℓ_∞ is NOT ℓ_1 (see Exercise 4 in the previous section and the exercises below).

The above discussion can be extended to determine the dual space of $L^p(X)$ for any measure space X. It turns out that it is $L^q(X)$ for $1 \leq p < \infty$. The proof is similar to what we have said above and makes use of the Radon-Nikodym theorem.

The dual space of a Hilbert space is itself as we have already seen.

The notion of orthogonality can be introduced in normed spaces through the dual space. The vectors $x \in X$ and $x^* \in X^*$ are said to be **orthogonal** if $\langle x, x^* \rangle = 0$. If S is a subset of a normed linear space X, the orthogonal complement of S, denoted $S^\perp$, consists of all elements $x^* \in X^*$ orthogonal to every vector of S. Recall that if in a Hilbert space H, M is a closed subspace, we have the decomposition

$$H = M \oplus M^\perp.$$

Given any vector $x \in H$, we have a unique decomposition

$$x = y + z,$$

with $y \in M$ and $z \in M^\perp$. The vector y can be characterized as the unique vector in M that comes closest to x. How close it gets to y is measured by $||z||$ which is also equal to $(x, z)/||z||$ since $z \in M^\perp$. The analogous result is the following.

Theorem 2.24 *Let x be an element in a real normed vector space and let d be its distance from the subspace M. Then,*

$$d = \inf_{y \in M} ||x - y|| = \max_{z \in M^\perp} \frac{\langle x, z \rangle}{||z||},$$

where the maximum on the right is achieved for some $z \in M^\perp$.

Proof For $\epsilon > 0$, let $y_\epsilon \in M$ be such that $||x - y_\epsilon|| \leq d + \epsilon$. Then, for any $z \in M^\perp$, we have

$$\langle x, z \rangle = \langle x - y_\epsilon, z \rangle = z(x - y_\epsilon) \leq ||z||||x - y_\epsilon|| \leq ||z||(d + \epsilon)$$

so that the right hand side of the equation in the theorem is bounded by $d + \epsilon$. Since ϵ was arbitrary, the right hand side is less than or equal to d. We now have to find a z so that $\langle x, z \rangle = d||z||$. Let N be the subspace spanned by x and M. Elements of N can be written uniquely as $u = ax + m$ with $a \in \mathbb{R}$. Define a linear functional on N by setting $f(u) = ad$. The norm of this functional is given by

$$\sup\{|f(u)|/||u|| : u \in N\} = \sup \frac{|a|d}{|a||x + m/a||} = \frac{d}{\inf ||x + m/a||} = 1.$$

By the Hahn–Banach theorem, f has an extension z to X with $||z|| = 1$. By construction, we have $z \in M^{\perp}$ and $\langle x, z\rangle = d$, which completes the proof. □

Exercises

1. Give an explicit isometric isomorphism between c^* and ℓ_1.
2. Give an explicit isometric isomorphism between c_0^* and ℓ_1.
3. For $x \in c$, define x_∞ to be $\lim_{n\to\infty} x_n$. Define a map $T : c \to c_0$ by setting $T(x)$ to be the sequence $x_\infty, x_1 - x_\infty, x_2 - x_\infty, \ldots$. Show that T is one-to-one and that $T(c) = c_0$.

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Chapter 3
Fourier Transforms



3.1 Fubini's Theorem and Convolutions

Jean Baptiste Joseph Fourier (1768–1830), the son of a tailor, was orphaned at the age of eight and brought up and educated by the clergy. In 1790, at the age of 22, he was appointed as a professor at the Ecole Polytechnique and in 1798, Napoleon took him on his campaign to Egypt. His research on the heat equation began in 1800, and by 1811, he had developed the theory now called the theory of Fourier series as the essential tool for solving it. Fourier is best known today for his 1822 book, "Théorie Analytique de la Chaleur" which was described by Kelvin as a "great mathematical poem." The theory of Fourier series and Fourier transforms that are in use today in mathematics, physics and engineering can be traced back to this remarkable work.

In the subsequent years, Fourier's theory has been extended to the context of the Lebesgue integral. The essential theorems are easily proved using Fubini's theorem discovered by Guido Fubini (1879–1943). We begin with a discussion of this theorem in the context of measure theory.

Let (X, μ) and (Y, ν) be two measure spaces. We can define a measure on $X \times Y$ as follows. Let $A \subseteq X$ be measurable and $B \subseteq Y$ be measurable. Then we call $A \times B$ a measurable rectangle and define a measure λ on $X \times Y$ by setting

$$\lambda(A \times B) = \mu(A)\nu(B).$$

One can check that λ extends uniquely to a complete measure on the σ-algebra generated by these rectangles. Recall that (X, μ) is said to be a complete measure space if it contains all the subsets of sets of measure zero. We denote λ as $\mu \times \nu$.

Now if f is an integrable function on $X \times Y$, then the functions $f(x, \cdot)$ on Y and $f(\cdot, y)$ on X are both integrable for almost all x and y. In this situation, we have the following theorem of Fubini.

Theorem 3.1 *Let f be a function on $X \times Y$, measurable with respect to the product σ-algebra defined above. If either*

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(i) $0 \leqslant f \leqslant \infty$ *or*

(ii) $\int_X \int_Y |f| d\nu d\mu < \infty$

then

$$\int_{X\times Y} f d\lambda = \int_X \left(\int_Y f d\nu \right) d\mu = \int_Y \left(\int_X f d\mu \right) d\nu.$$

In the case (ii) holds, we have that $f \in L^1(\lambda)$ and that $f(x, \cdot) \in L^1(\nu)$ for almost every $x \in X$ and $f(\cdot, y) \in L^1(\mu)$ for almost every $y \in Y$.

Throughout (unless stated otherwise)

$$d\mu = \frac{dx}{\sqrt{2\pi}}$$

where dx is the Lebesgue measure on $\mathbb{R}$.

Definition 3.1 We define the **convolution** of two functions $f, g \in L^1(\mathbb{R})$ by

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x - y) g(y) d\mu(y).$$

It is easy to see that the translation invariance of the Lebesgue measure as well as Fubini's theorem now implies that the convolution is well-defined. More precisely, we have

Theorem 3.2 *Suppose* $f, g \in L^1(\mathbb{R})$. *Then*

$$\int_{-\infty}^{\infty} |f(x - y) g(y)| d\mu(y) < \infty$$

for almost every x. *Moreover,* $f * g \in L^1(\mathbb{R})$ *and*

$$\|f * g\|_1 \leqslant \|f\|_1 \|g\|_1.$$

Proof Observe that $F(x, y) = f(x - y)g(y)$ is a measurable function on the product space $\mathbb{R}^2$. Moreover, by Fubini's theorem

$$\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} |F(x, y)| dx \right) dy = \int_{-\infty}^{\infty} |g(y)| \int_{-\infty}^{\infty} |f(x - y)| dx dy$$

and as the Lebesgue measure is translation invariant, the inner integral is $\|f\|_1$ which is a constant and as

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |F(x, y)| dx dy = \|f\|_1 \|g\|_1,$$

we conclude $\|h\|_1 \leqslant \|f\|_1 \|g\|_1$. □

We say that h is a convolution of f and g and write $h = f * g$.

Exercises

1. Let
$$f(x, y) = \begin{cases} \frac{(x^2-y^2)}{(x^2+y^2)^2} & (x, y) \in (0, 1) \times (0, 1) \\ 0 & \text{otherwise} \end{cases}$$
Show that
$$\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(x, y) dx \right) dy = -\frac{\pi}{4},$$
and
$$\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(x, y) dy \right) dx = \frac{\pi}{4}.$$
Why does Fubini's theorem not apply?

2. Prove that
$$\int_0^{\infty} \left(\frac{2y}{1+y^2} - \frac{2y}{x^2+y^2} \right) dy = 2 \log x.$$
Deduce that
$$-\frac{\log x}{1-x^2} = \int_0^{\infty} \frac{y \, dy}{(1+y^2)(x^2+y^2)}.$$

3. Show that
$$I := \int_0^1 \frac{\log x}{1-x^2} dx = \int_1^{\infty} \frac{\log x}{1-x^2} dx.$$
Deduce that
$$-2I = \int_0^{\infty} \int_0^{\infty} \frac{y \, dx \, dy}{(1+y^2)(x^2+y^2)}.$$

4. Applying Fubini's theorem in the last exercise, use the previous exercises to deduce that
$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}.$$

5. For any natural number n, show that

$$\sum_{\substack{m=1\\ m\neq n}}^{\infty} \frac{1}{m^2-n^2} = \frac{3}{4n^2}.$$

6. Set for $k > 1$,

$$\zeta(k) = \sum_{n=1}^{\infty} \frac{1}{n^k}.$$

Show that

$$\left(k+\frac{1}{2}\right)\zeta(2k) = \sum_{j=1}^{k-1} \zeta(2j)\zeta(2k-2j).$$

This shows that $\zeta(2k)$ can be recursively determined from $\zeta(2)$.[1] [Hint: write the summand as a double sum, interchange summations, and use the previous exercise.]

3.2 The Fourier Transform

For $f \in L^1(\mathbb{R})$, we define the **Fourier transform** $\hat{f}$ by

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x)e^{-ixt}d\mu(x)$$

where $i = \sqrt{-1}$. Here are some basic properties of the Fourier transform.

Theorem 3.3 *Suppose $f \in L^1(\mathbb{R})$ and $\alpha, \lambda \in \mathbb{R}$. Then*

(1) If $g(x) = f(x)e^{i\alpha x}$, then $\hat{g}(t) = \hat{f}(t-\alpha)$.
(2) If $g(x) = f(x-\alpha)$, then $\hat{g}(t) = \hat{f}(t)e^{-i\alpha t}$.
(3) If $g \in L^1(\mathbb{R})$ and $h = f * g$, then $\hat{h} = \hat{f}\hat{g}$.
(4) If $g(x) = \overline{f(-x)}$, then $\hat{g}(t) = \overline{\hat{f}(t)}$.
(5) If $g(x) = f(x/\lambda)$ where $\lambda > 0$, then $\hat{g}(t) = \lambda\hat{f}(\lambda t)$.
(6) If $g(x) = -ixf(x)$ and $g \in L^1(\mathbb{R})$, then $\hat{f}$ is differentiable with derivative $\hat{f}'(t) = \hat{g}(t)$.

Proof Property (1) is easy to see:

$$\begin{aligned}\hat{g}(t) &= \int_{-\infty}^{\infty} \left(f(x)e^{i\alpha x}\right)e^{-ixt}d\mu(x)\\ &= \int_{-\infty}^{\infty} f(x)e^{-i(t-\alpha)x}d\mu(x) = \hat{f}(t-\alpha).\end{aligned}$$

[1] These exercises are taken from the author's short note [1] in Math. Student.

To prove (2), we have

$$\hat{g}(t) = \int_{-\infty}^{\infty} f(x-\alpha)e^{-ixt}d\mu(x).$$

Putting $u = x - \alpha$, we have

$$\hat{g}(t) = \int_{-\infty}^{\infty} f(u)e^{-i(\alpha+u)t}d\mu(u),$$

since the Lebesgue measure is translation invariant. Thus

$$\hat{g}(t) = e^{-i\alpha t}\int_{-\infty}^{\infty} f(u)e^{-iut}d\mu(u) = e^{-i\alpha t}\hat{f}(t).$$

To prove (3), we apply Fubini's theorem:

$$\begin{aligned}\hat{h}(t) &= \int_{-\infty}^{\infty} h(x)e^{-ixt}d\mu(x)\\ &= \int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} f(x-y)g(y)d\mu(y)\right)e^{-ixt}d\mu(x).\end{aligned}$$

We write x as $x - y + y$ in the exponential to get

$$\begin{aligned}\hat{h}(t) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x-y)e^{-i(x-y)t}d\mu(x)g(y)e^{-iyt}d\mu(y)\\ &= \hat{f}(t)\int_{-\infty}^{\infty} g(y)e^{-iyt}d\mu(y)\\ &= \hat{f}(t)\hat{g}(t)\end{aligned}$$

because the Lebesgue measure is translation invariant.

To prove (4), put $g(x) = \overline{f(-x)}$. Then

$$\begin{aligned}\hat{g}(t) &= \int_{-\infty}^{\infty}\overline{f(-x)}e^{-ixt}d\mu(x)\\ &= \int_{-\infty}^{\infty}\overline{f(-x)e^{-i(-x)t}}d\mu(x) = \overline{\hat{f}(t)}.\end{aligned}$$

To prove (5), we have

$$\begin{aligned}\hat{g}(t) &= \int_{-\infty}^{\infty} f(x/\lambda)e^{-ixt}d\mu(x)\\ &= \int_{-\infty}^{\infty} f(x/\lambda)e^{-i(x/\lambda)\lambda t}d\mu(x).\end{aligned}$$

We change variables and put $v = x/\lambda$ so that

$$\begin{aligned}\hat{g}(t) &= \int_{-\infty}^{\infty} f(v)e^{-iv(\lambda t)}\lambda d\mu(v)\\ &= \lambda\hat{f}(\lambda t).\end{aligned}$$

Finally, to prove (6), we note that for $s \neq t$,

$$\frac{\hat{f}(s) - \hat{f}(t)}{s-t} = \int_{-\infty}^{\infty} f(x)e^{-ixt}\phi(x, s-t)d\mu(x)$$

where

$$\phi(x,u) = \frac{e^{-ixu} - 1}{u}.$$

Now

$$\begin{aligned}\phi(x,u)e^{ixu/2} &= \frac{e^{ixu/2}(e^{-ixu} - 1)}{u}\\ &= \frac{e^{-ixu/2} - e^{ixu/2}}{u} = -\frac{2i\sin(xu/2)}{u}\end{aligned}$$

so that for $u \neq 0$, we have

$$|\phi(x,u)| = \frac{|2\sin(xu/2)|}{|u|} = \left|\frac{\sin(xu/2)}{xu/2}\cdot x\right| \leqslant |x|$$

because $|(\sin\theta)/\theta| \leqslant 1$. (See Exercise 1 below)

Thus, we can apply the dominated convergence theorem to get

$$\begin{aligned}\hat{f}'(t) &= \lim_{s\to t}\frac{\hat{f}(s) - \hat{f}(t)}{s-t}\\ &= \int_{-\infty}^{\infty} f(x)e^{-ixt}\lim_{s\to t}\phi(x, s-t)d\mu(x)\\ &= \int_{-\infty}^{\infty}(-ix)e^{-ixt}f(x)d\mu(x)\\ &= \int_{-\infty}^{\infty} g(x)e^{-ixt}d\mu(x) = \hat{g}(t).\end{aligned}$$

□

Example 3.1 Compute the Fourier transform of $f(x) = e^{-x^2/2}$.

We first compute the integral (which we have seen earlier)

$$I = \int_{-\infty}^{\infty} e^{-x^2/2} \frac{dx}{\sqrt{2\pi}}.$$

Observe that

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} \frac{dxdy}{2\pi}.$$

This suggests that we switch to polar coordinates. We put $x = r\cos\theta$, $y = r\sin\theta$ with $0 \leqslant r < \infty$ and $0 \leqslant \theta \leqslant 2\pi$. The Jacobian determinant of this transformation is r and so

$$I^2 = \int_0^{\infty}\int_0^{2\pi} e^{-r^2/2} \frac{rdrd\theta}{2\pi} = \int_0^{\infty} e^{-r^2/2} rdr = \int_0^{\infty} e^{-r} dr = \left[-e^{-r}\right]_0^{\infty} = 1.$$

Therefore $I = \pm 1$. But I is the integral of a non-negative function, so we must have $I = 1$. □

Now let us compute $\hat{f}(t)$. To this end, let

$$I(u) = \int_{-\infty}^{\infty} e^{-(x+iu)^2/2} d\mu(x).$$

We computed above that $I(0) = 1$.
Let us observe that

$$\begin{aligned} I'(u) &= \frac{d}{du}\int_{-\infty}^{\infty} e^{-(x+iu)^2/2} d\mu(x) \\ &= \int_{-\infty}^{\infty} \frac{d}{du}\left(e^{-(x+iu)^2/2}\right) d\mu(x) \\ &= -\int_{-\infty}^{\infty} i(x+iu) e^{-(x+iu)^2/2} d\mu(x) \\ &= \frac{i}{\sqrt{2\pi}}\left[e^{-(x+iu)^2/2}\right]_{x=-\infty}^{x=\infty} \\ &= 0. \end{aligned}$$

The interchange of integration and differentiation can be justified (see next section) and so we deduce $I(u)$ is constant. But since $I(0) = 1$, we have $I(u) = 1$. That is

$$\int_{-\infty}^{\infty} e^{-x^2/2} e^{-iux} d\mu(x) = 1$$

which implies

$$\hat{f}(u) = \int_{-\infty}^{\infty} e^{-x^2/2} e^{-iux} d\mu(x) = e^{-u^2/2} = f(u).$$

That is, f is its own Fourier transform. In other words, when we view the Fourier transform as a linear transformation, it is an eigenvector with eigenvalue 1.

Exercises

1. Prove that
$$\left|\frac{\sin\theta}{\theta}\right| \leq 1,$$
which was used in the proof of Theorem 3.3.
2. Compute the Fourier transform of the function $f \in L^1(\mathbb{R})$ given by $f(x) = 1$ if $x \in [-1, 1]$ and zero otherwise. Show that $\widehat{f} \in L^2(\mathbb{R})$ but not in $L^1(\mathbb{R})$.
3. Calculate the Fourier transform of $f(x) = e^{-2|x|}$.
4. In a Hilbert space, show that the map $x \mapsto (x, y)$ for fixed y and the map $x \mapsto ||x||$ are continuous maps.
5. For functions $f \in L^1(\mathbb{R}^n)$, we define the Fourier transform
$$\hat{f}(t) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-i(x,t)} dx,$$
where (x, t) denotes the usual inner product, and dx is the Lebesgue measure on $\mathbb{R}^n$. If $f(x) = e^{-|x|^2/2}$, show that $\hat{f} = f$. (Here $x = (x_1, ..., x_n)$ and $|x|^2 = x_1^2 + \cdots + x_n^2$.)
6. Prove that
$$\frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt \sim \frac{1}{\sqrt{2\pi}} \frac{e^{-x^2/2}}{x},$$
as x tends to infinity.

3.3 Differentiation Under the Integral Sign

In the previous section, we computed the derivative of $I(u)$ by differentiatiing under the integral sign. We prove below a general theorem that allows us to justify that step. For this purpose, let us recall Lebesgue's dominated convergence theorem: if $\{f_n\}_{n=1}^\infty$ is a sequence of measurable functions on x such that $|f_n(x)| \leqslant g(x)$ for almost all $x \in X$ with g a measurable function and $f(x) = \lim_{n\to\infty} f_n(x)$, then
$$\int_X f(x) d\mu = \lim_{n\to\infty} \int_X f_n(x) d\mu.$$

Theorem 3.4 *Let $[a, b]$ be an interval in $\mathbb{R}$ (which may be infinite) and suppose that f is a continuous function on the rectangle $[a, b] \times [c, d]$. Set*

$$g(y) = \int_a^b f(x, y)\, dx.$$

Suppose that $\dfrac{\partial f}{\partial y}$ exists and is continuous. Further suppose if $[c, d]$ is infinite then $\dfrac{\partial f}{\partial y}$ is bounded. Then g is differentiable and

$$g'(y) = \int_a^b \frac{\partial f}{\partial y}(x, y)\, dx.$$

Proof We compute

$$\lim_{n\to\infty} \frac{g(y + 1/n) - g(y)}{1/n}$$

and show that it exists and is equal to

$$\int_a^b \frac{\partial f}{\partial y}(x, y)\, dx.$$

Indeed,

$$n\,[g(y + 1/n) - g(y)] = \int_a^b n\,[f(x, y + 1/n) - f(x, y)]\, dx.$$

Consider the family of functions

$$f_n(x, y) = n\,[f(x, y + 1/n) - f(x, y)]\,.$$

By the assumption that $\dfrac{\partial f}{\partial y}$ exists and the mean value theorem, we have that

$$f(x, y + 1/n) - f(x, y) = \frac{1}{n}\frac{\partial f}{\partial y}(x, \xi)$$

for some intermediate point $\xi \in [y, y + 1/n]$.

As $\dfrac{\partial f}{\partial y}$ is bounded, $f_n(x, y)$ is bounded. By the Lebesgue dominated convergence theorem

$$\lim_{n\to\infty} \int_a^b f_n(x, y)\, dx = \int_a^b \lim_{n\to\infty} f_n(x, y)\, dx$$

and the latter integral is precisely $\int_a^b \frac{\partial f}{\partial y}(x, y)\, dx$. □

We illustrate the theorem through the following interesting example. We want to evaluate the (conditionally convergent) integral

$$\int_0^\infty \frac{\sin x}{x} dx.$$

To this end, we consider

$$g(t) = \int_0^\infty e^{-tx} \frac{\sin x}{x} dx, \tag{3.1}$$

which is now absolutely convergent for $t > 0$. It is now easy to check that the function

$$f(x, t) = e^{-tx} \frac{\sin x}{x}$$

satisfies the hypotheses of the theorem for $[a, b] = [0, \infty)$, $[c, d] = [\epsilon, \infty)$ with $\epsilon > 0$. Differentiating under the integral sign with respect to t, we obtain

$$g'(t) = -\int_0^\infty e^{-tx} \sin x dx.$$

We integrate by parts:

$$\int_0^\infty e^{-tx} \sin x dx = -\frac{e^{-tx}}{t} \sin x \Big|_{x=0}^{x=\infty} + \int_0^\infty \frac{e^{-tx}}{t} \cos x dx.$$

The first term on the right is zero so that

$$\int_0^\infty e^{-tx} \sin x dx = \int_0^\infty \frac{e^{-tx}}{t} \cos x dx.$$

One more integration by parts shows

$$\int_0^\infty \frac{e^{-tx}}{t} \cos x dx = -\frac{e^{-tx}}{t^2} \cos x \Big|_{x=0}^{x=\infty} - \int_0^\infty \frac{e^{-tx}}{t^2} \sin x dx.$$

The first term on the right hand side is $1/t^2$ so that we deduce

$$-g'(t) = 1/t^2 + g'(t)/t^2.$$

In other words,

$$g'(t) = -\frac{1}{1 + t^2}.$$

Integrating, we obtain

$$g(t) = -\arctan t + C$$

for some constant C. By the dominated convergence theorem applied to (3.1), $g(\infty) = 0$ so that $C = \pi/2$. We conclude that for $t > 0$,

$$\int_0^\infty e^{-tx} \frac{\sin x}{x} dx = \frac{\pi}{2} - \arctan t,$$

a result which is of independent interest. Applying again the dominated convergence theorem with $t \to 0$, we find

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

In his semi-autobiography, "Surely you are joking Mr. Feynman," Richard Feynman narrates that he learned this method while in high school and it was the one trick he would use again and again and this is how he earned a reputation for evaluating integrals.

Exercises

1. Suppose $f \in L^1(\mathbb{R})$ and $f(x) > 0$ for all real x. Show that $|\hat{f}(y)| < \hat{f}(0)$ for every $y \neq 0$.
2. Find the Fourier transform of

$$f(x) = \frac{x}{(1+x^2)^2}.$$

3. Prove that

$$\int_0^1 \frac{x-1}{\log x} dx = \log 2.$$

[Hint: Consider the function

$$f(a) := \int_0^1 \frac{x^a - 1}{\log x} dx.]$$

4. Let

$$\phi(\alpha) = \int_{u_1}^{u_2} f(x, \alpha) dx, \qquad a \leq \alpha \leq b,$$

where u_1 and u_2 may depend on the parameter α. Prove that

$$\phi'(\alpha) = \int_{u_1}^{u_2} \frac{\partial f}{\partial \alpha} dx + f(u_2, \alpha)\frac{du_2}{d\alpha} - f(u_1, \alpha)\frac{du_1}{d\alpha}.$$

This is sometimes called **Leibniz's rule** for differentiating under the integral sign.

5. Let

$$\phi(a) = \int_a^{a^2} \frac{\sin ax}{x} dx, \qquad a \neq 0.$$

Show that

$$\phi'(a) = \frac{3\sin a^3 - 2\sin a^2}{a}.$$

6. Prove that

$$\int_{-\infty}^{\infty} e^{-tx^2/2} dx = \frac{\sqrt{2\pi}}{\sqrt{t}}.$$

Differentiating with respect to t, and setting $t = 1$, deduce that for even n,

$$\int_{-\infty}^{\infty} x^n e^{-x^2/2} dx = 1 \cdot 3 \cdot 5 \cdots (n-1)\sqrt{2\pi}.$$

3.4 Further Examples of Fourier Transforms

The Fourier transform can be extended to $L^1(\mathbb{R}^n)$ by setting

$$\hat{f}(t) = \int_{\mathbb{R}^n} f(x) e^{-i(x,t)}\, d\mu(x)$$

where now $d\mu(x) = \left(\frac{dx}{\sqrt{2\pi}}\right)^n$ is the product measure on $\mathbb{R}^n$ and (x, t) is the dot product of the vectors $x, t \in \mathbb{R}^n$.

It can be shown that if

$$f(x) = e^{-|x|^2/2}$$

with $x = (x_1, \cdots, x_n)$ and $|x|^2 = x_1^2 + \cdots + x_n^2$ then

$$\hat{f}(t) = e^{-|t|^2/2}.$$

This was Exercise 5 in the penultimate section.

As another example of an explicit example of the Fourier transform, we consider $H(x) = e^{-|x|}$.

Let $H_\lambda(x) = H(\lambda x)$. We want to compute the Fourier transform of $H_\lambda(x)$. Set

$$h_\lambda(t) = \int_{-\infty}^{\infty} H_\lambda(x) e^{-ixt} d\mu(x).$$

This is

$$\int_{-\infty}^{\infty} e^{-|\lambda||x|} e^{-ixt} d\mu(x) = \int_{-\infty}^{0} e^{-|\lambda||x|} e^{-ixt} d\mu(x) + \int_{0}^{\infty} e^{-|\lambda||x|} e^{-ixt} d\mu(x).$$

Let us note that

$$\begin{aligned}\int_0^\infty e^{-|\lambda|x - ixt} d\mu(x) &= \int_0^\infty e^{-x(|\lambda| + it)} d\mu(x) \\ &= \left[-\frac{e^{-x(|\lambda|+it)}}{|\lambda| + it} \right]_{x=0}^{x=\infty} \cdot \frac{1}{\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi}(|\lambda| + it)}.\end{aligned}$$

Also,

$$\int_{-\infty}^{0} e^{-|\lambda||x|} e^{-ixt} d\mu(x) = -\int_{\infty}^{0} e^{-|\lambda||x| + ixt} d\mu(x)$$

upon changing x to $-x$. This is equal to

$$\int_0^\infty e^{-|\lambda|x + ixt} d\mu(x) = \int_0^\infty e^{-x(|\lambda| - it)} d\mu(x) = \frac{1}{\sqrt{2\pi}(|\lambda| - it)}.$$

Putting the computations together, we get

$$\int_{-\infty}^{\infty} H(\lambda x) e^{-ixt} d\mu(x) = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{|\lambda| + it} + \frac{1}{|\lambda| - it} \right) = \frac{2|\lambda|}{\sqrt{2\pi}(|\lambda|^2 + t^2)}.$$

In other words,

$$h_\lambda(t) = \sqrt{\frac{2}{\pi}} \frac{|\lambda|}{\lambda^2 + t^2}.$$

Let us observe that if $\lambda > 0$,

$$\int_{-\infty}^{\infty} h_\lambda(x) d\mu(x) = \int_{-\infty}^{\infty} \sqrt{\frac{2}{\pi}} \frac{\lambda}{\lambda^2 + x^2} d\mu(x) = 2\sqrt{\frac{2}{\pi}} \int_0^\infty \frac{\lambda d\mu(x)}{\lambda^2 + x^2}.$$

Changing x to λx gives

$$\begin{aligned}\int_{-\infty}^{\infty} h_\lambda(x) d\mu(x) &= 2\sqrt{\frac{2}{\pi}} \int_0^\infty \frac{dx}{(1 + x^2)\sqrt{2\pi}} \\ &= \frac{2}{\pi} \int_0^\infty \frac{dx}{1 + x^2} = \frac{2}{\pi} \Big[\arctan x \Big]_0^\infty = 1.\end{aligned}$$

We also note for future reference that if $H(x) = e^{-|x|}$, then

$$0 < H(x) \leqslant 1$$

and for $\lambda > 0$, $\lambda \to 0$, $H(\lambda x) \to 1$.

Exercises

1. Suppose $f \in L^1(\mathbb{R})$ has a Fourier transform. For any $t_1, ..., t_n \in \mathbb{R}$ and $z_1, z_2, ..., z_n \in \mathbb{C}$, prove that
$$\sum_{j=1}^{n}\sum_{k=1}^{n} \widehat{f}(t_j - t_k) z_j \overline{z_k} \geq 0.$$

2. Calculate the Fourier transform of $x^n e^{-x^2}$.
3. Prove that
$$\int_0^\infty e^{-x^2/2} \cos(tx)\, dx = \sqrt{\frac{\pi}{2}} e^{-t^2/2}.$$

3.5 A Convolution Theorem

Proposition 3.5.1 *Let $f \in L^1(\mathbb{R})$. Then*

$$(f * h_\lambda)(x) = \int_{-\infty}^{\infty} H(\lambda t) \hat{f}(t) e^{ixt} d\mu(t).$$

Proof We have

$$\begin{aligned}(f * h_\lambda)(x) &= \int_{-\infty}^{\infty} f(x-y) h_\lambda(y) d\mu(y) \\ &= \int_{-\infty}^{\infty} f(x-y) \left(\int_{-\infty}^{\infty} H_\lambda(t) e^{-ity} d\mu(t) \right) d\mu(y) \\ &= \int_{-\infty}^{\infty} H(\lambda t) \left(\int_{-\infty}^{\infty} f(x-y) e^{-ity} d\mu(y) \right) d\mu(t)\end{aligned}$$

by an application of Fubini's theorem. We change variables in the inner integral by putting $u = x - y$ so that our integral becomes

$$\int_{-\infty}^{\infty} H(\lambda t) e^{-itx} \left(\int_{-\infty}^{\infty} f(u) e^{itu} d\mu(u) \right) d\mu(t) = \int_{-\infty}^{\infty} H(\lambda t) e^{-itx} \hat{f}(-t) d\mu(t).$$

Changing t to $-t$ gives us the final result. □

Exercises

1. Solve for f:
$$\int_{-\infty}^{\infty} f(x-t)e^{-|t|}dt = \frac{4}{3}e^{-|x|} - \frac{2}{3}e^{-2|x|}.$$

2. For $t > 1$, define
$$F(t) = \int_2^{\infty} \frac{dx}{x^t \log x}.$$
Show that
$$F(t) + \log(t-1) = O(1),$$
as $t \to 1+$.

3. For each natural number n, let g_n be the characteristic function of $[-n, n]$ and let h be the characteristic function of $[-1, 1]$. Compute the convolution $g_n * h$ explicitly. (The graph is piecewise linear.)

4. With notation as in the previous exercise, show that $g_n * h$ is the Fourier transform of a function $f_n \in L^1(\mathbb{R})$, where (apart from a multiplicative constant),
$$f_n(x) = \frac{(\sin x)(\sin nx)}{x^2}.$$

5. With f_n as in the previous exercise, show that $||f_n||_1 \to \infty$ as $n \to \infty$. Conclude that the map $f \mapsto \hat{f}$ maps $L^1(\mathbb{R})$ into a *proper* subset of C_0, the space of continuous functions that vanish at infinity.

3.6 The Inversion Theorem

The integrand on the right hand side of the equation in Proposition 3.5.1 is bounded by $|\hat{f}(t)|$ since $|H(x)| \leqslant 1$. Moreover, as $\lambda \to 0^+$, $H(\lambda t) \to 1$ and so by the Lebesgue dominated convergence theorem,

$$\lim_{\lambda \to 0^+} \int_{-\infty}^{\infty} H(\lambda t)\hat{f}(t)e^{ixt}d\mu(t) = \int_{-\infty}^{\infty} \hat{f}(t)e^{ixt}d\mu(t)$$

provided $\hat{f} \in L^1(\mathbb{R})$. What we would like to prove now is that the right hand side of the above limit is in fact equal to $f(x)$ almost everywhere. To this end, we need a few technical results and we collate them below.

We give two proofs of the inversion theorem. The first proof follows closely Rudin [2]. The second proof is shorter and makes essential use of the Lebesgue dominated convergence theorem, Fubini's theorem and the fact that $e^{-x^2/2}$ is its own Fourier transform. Though the first proof is longer, it has the merit of invoking related results of independent interest. The second proof of course is crisp and succinct.

Proposition 3.6.1 *For any f on $\mathbb{R}$ and any $y \in \mathbb{R}$, let*

$$f_y(x) := f(x - y).$$

If $1 \leqslant p < \infty$ and $f \in L^p(\mathbb{R})$, then the map $y \mapsto f_y$ is a uniformly continuous map from $\mathbb{R}$ to $L^p(\mathbb{R})$.

Proof Fix $\varepsilon > 0$. Since continuous functions with compact support are dense in $L^p(\mathbb{R})$, we can find a function g with support in $[-A, A]$ such that

$$\|f - g\|_p < \varepsilon.$$

Since continuous functions on closed intervals are uniformly continuous, we have that there exists $\delta > 0$ with $0 < \delta < A$ satisfying

$$|g(s) - g(t)| < \frac{\varepsilon}{(3A)^{1/p}} \quad \text{when } |s - t| < \delta.$$

The translate g_s of g has support $[-A + s, A + s]$ and similarly for g_t. Hence

$$\|g_s - g_t\|_p^p = \int_{-\infty}^{\infty} |g(x - s) - g(x - t)|^p \, d\mu(x)$$

has support in $I = [\min(-A + t, -A + s), \max(A + t, A + s)]$. Thus

$$\|g_s - g_t\|_p^p \leqslant (3A)^{-1}\varepsilon^p \mu(I) \leqslant \frac{\varepsilon^p(2A + |s - t|)}{3A} \leqslant \frac{\varepsilon^p(2A + \delta)}{3A} \leqslant \varepsilon^p.$$

Therefore, when $|s - t| < \delta$, $\|g_s - g_t\|_p \leqslant \varepsilon$.

Finally,

$$\begin{aligned} \|f_s - f_t\|_p &\leqslant \|f_s - g_s\|_p + \|g_s - g_t\|_p + \|g_t - f_t\|_p \\ &\leqslant \|f - g\|_p + \|g_s - g_t\|_p + \|f - g\|_p \\ &\leqslant 2\varepsilon + \varepsilon = 3\varepsilon \end{aligned}$$

This completes the proof. □

Proposition 3.6.2 *If $g \in L^\infty(\mathbb{R})$ is continuous at x, then*

$$\lim_{\lambda \to 0^+} (g * h_\lambda)(x) = g(x).$$

Proof Observe that $h_\lambda(y) = \lambda^{-1}h_1(y/\lambda)$. Noting

$$\int_{-\infty}^{\infty} h_\lambda(y)d\mu(y) = 1,$$

we have

$$\begin{aligned}(g * h_\lambda)(x) - g(x) &= \int_{-\infty}^{\infty} [g(x-y) - g(x)]\lambda^{-1}h_1(y/\lambda)d\mu(y) \\ &= \int_{-\infty}^{\infty} [g(x-\lambda s) - g(x)]h_1(s)d\mu(s)\end{aligned}$$

where we have changed variables $y = \lambda s$ in the penultimate step. Since g is in $L^\infty(\mathbb{R})$, the integrand is essentially bounded, and we can apply the Lebesgue dominated convergence theorem with $\lambda \to 0^+$. The integrand tends to zero as g is continuous at x. □

Proposition 3.6.3 *For $1 \leqslant p < \infty$, let $f \in L^p(\mathbb{R})$. Then*

$$\lim_{\lambda \to 0^+} \|f * h_\lambda - f\|_p = 0.$$

Proof It is clear that $h_\lambda \in L^q(\mathbb{R})$ for all $q \geqslant 1$ because

$$h_\lambda(t) = \sqrt{\frac{2}{\pi}}\frac{|\lambda|}{\lambda^2 + t^2}.$$

Choose q such that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Now,

$$\begin{aligned}(f * h_\lambda)(x) - f(x) &= \int_{-\infty}^{\infty} f(x-y)h_\lambda(y)d\mu(y) - \int_{-\infty}^{\infty} f(x)h_\lambda(y)d\mu(y) \\ &= \int_{-\infty}^{\infty} (f(x-y) - f(x))h_\lambda(y)d\mu(y) \\ &= \int_{-\infty}^{\infty} (f(x-y) - f(x))h_\lambda^{1/p}(y)h_\lambda^{1/q}(y)d\mu(y).\end{aligned}$$

Applying Hölder's inequality,

$$
\begin{aligned}
&|(f * h_\lambda)(x) - f(x)| \\
&\leqslant \left(\int_{-\infty}^{\infty} |f(x-y) - f(x)|^p h_\lambda(y) d\mu(y) \right)^{1/p} \left(\int_{-\infty}^{\infty} h_\lambda(y) d\mu(y) \right)^{1/q} \\
&\leqslant \left(\int_{-\infty}^{\infty} |f(x-y) - f(x)|^p h_\lambda(y) d\mu(y) \right)^{1/p}
\end{aligned}
$$

because

$$\int_{-\infty}^{\infty} h_\lambda(y) d\mu(y) = 1.$$

Taking p-th powers and integrating over x, we get

$$\|(f * h_\lambda) - f\|_p^p \leqslant \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x-y) - f(x)|^p h_\lambda(y) d\mu(y) d\mu(x).$$

Letting $g(y) = \|f_y - f\|_p^p$, we have by Fubini's theorem that the double integral is equal to

$$\int_{-\infty}^{\infty} g(y) h_\lambda(y) d\mu(y).$$

By Proposition 3.6.1, the map $y \mapsto f_y$ is a uniformly continuous map from $\mathbb{R}$ to $L^p(\mathbb{R})$. Thus g is a continuous map. Moreover,

$$|g(y)|^{1/p} \leqslant \|f_y\|_p + \|f\|_p = 2\|f\|_p$$

so that g is bounded. The previous proposition can be applied to g. That is,

$$\lim_{\lambda \to 0^+} (g * h_\lambda)(x) = g(x).$$

Applying this to $x = 0$, we get

$$\lim_{\lambda \to 0^+} \int_{-\infty}^{\infty} g(y) h_\lambda(y) d\mu(y) = \lim_{\lambda \to 0^+} \int_{-\infty}^{\infty} g(-y) h_\lambda(y) d\mu(y) = g(0) = 0.$$

This completes the proof. □

Proposition 3.6.4 *Let $1 \leqslant p \leqslant \infty$. If $f_n \in L^p(\mathbb{R})$ is a Cauchy sequence converging to $f \in L^p(\mathbb{R})$, then there is a subsequence λ_n such that f_{λ_n} converges to f almost everywhere.*

Proof Exercise.

We are now ready to prove the inversion theorem.

Theorem 3.5 *Let $f \in L^1(\mathbb{R})$ and suppose $\hat{f} \in L^1(\mathbb{R})$. Then*

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} d\mu(t) \quad a.e.$$

Proof By Proposition 3.5.1, we have

$$(f * h_\lambda)(x) = \int_{-\infty}^{\infty} H(\lambda t) \hat{f}(t) e^{ixt} d\mu(t)$$

and by the dominated convergence theorem,

$$\lim_{\lambda \to 0^+} (f * h_\lambda)(x) = \int_{-\infty}^{\infty} \hat{f}(t) e^{ixt} d\mu(t) = g(x) \quad \text{(say)}$$

because $\hat{f} \in L^1(\mathbb{R})$.

But Proposition 3.6.3 says

$$\lim_{\lambda \to 0^+} \| f * h_\lambda - f \|_1 = 0.$$

Proposition 3.6.4 says there is a subsequence $\lambda_n \to 0$ such that

$$\lim_{n \to \infty} f * h_{\lambda_n}(x) = f(x) \quad \text{a.e.}$$

But the left hand side is $g(x)$. Thus $g(x) = f(x)$ a.e. This completes the proof. □

Here is the promised short proof of the inversion theorem. Let $g_\epsilon(x) = e^{-\epsilon^2 x^2/2}$. Consider the absolutely convergent double integral:

$$I_\epsilon(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) e^{iw(x-u)} e^{-\epsilon^2 w^2/2} d\mu(u) d\mu(w). \tag{3.2}$$

By Fubini's theorem, we can evaluate the inner integral and deduce that

$$I_\epsilon(x) = \int_{-\infty}^{\infty} \hat{f}(w) e^{iwx} e^{-\epsilon^2 x^2/2} d\mu(w).$$

Since $\hat{f} \in L^1(\mathbb{R})$, we can apply the dominated convergence theorem to deduce that

$$\lim_{\epsilon \to 0} I_\epsilon(x) = \int_{-\infty}^{\infty} \hat{f}(w) e^{iwx} d\mu(w).$$

On the other hand, interchanging the order of integration in (3.2), we have

$$I_\epsilon(x) = \int_{-\infty}^{\infty} f(u) \widehat{g_\epsilon}(x-u) d\mu(u). \tag{3.3}$$

A straightforward computation using (5) of Theorem 3.3 shows that

$$\widehat{g_\epsilon}(x) = \frac{1}{\epsilon} e^{-x^2/2\epsilon^2}.$$

Changing variables by setting $u = x + v\epsilon$ in (3.3), we have

$$I_\epsilon(x) = \int_{-\infty}^{\infty} f(x + \epsilon v) e^{-v^2/2} d\mu(v).$$

Letting $\epsilon \to 0$ and applying again the dominated convergence theorem gives the limit is $f(x)$ since the probability integral equals 1. This completes the proof. □

Exercises

1. Prove Proposition 3.6.4.
2. For $t > 0$, show that

$$\int_0^t \frac{\log(1+tx)}{1+x^2} dx = \frac{1}{2}[\arctan t] \log(1+t^2).$$

Deduce that

$$\int_0^1 \frac{\log(1+x)}{1+x^2} dx = \frac{\pi \log 2}{8}.$$

[Hint: differentiate under the integral sign.]

3.7 Further Properties of the Fourier Transform

Recall that C_0 is the space of continuous functions decaying to zero at infinity. We have

Theorem 3.6 (Riemann–Lebesgue Lemma) *Let $f \in L^1(\mathbb{R})$ then $\widehat{f} \in C_0$. That is,*

$$\lim_{|t| \to \infty} \widehat{f}(t) = 0.$$

Proof Let $t_n \to t$. Then

$$\hat{f}(t_n) - \hat{f}(t) = \int_{-\infty}^{\infty} f(x)\left(e^{-it_n x} - e^{-itx}\right) d\mu(x)$$

so that

$$|\hat{f}(t_n) - \hat{f}(t)| \leqslant \int_{-\infty}^{\infty} |f(x)||e^{-it_n x} - e^{-itx}|d\mu(x)$$

which is $\leqslant 2\|f\|_1$. By the dominated convergence theorem, we have

$$\hat{f}(t_n) \to \hat{f}(t),$$

so that $\hat{f}$ is a continuous function.

Moreover, using $e^{i\pi} = -1$, we get

$$\begin{aligned}\hat{f}(t) &= \int_{-\infty}^{\infty} f(x)e^{-itx}d\mu(x) \\ &= -\int_{-\infty}^{\infty} f(x)e^{-it(x+\pi/t)}d\mu(x) \\ &= -\int_{-\infty}^{\infty} f(u-\pi/t)e^{-itu}d\mu(u)\end{aligned}$$

by a simple change of variable. Thus,

$$2\hat{f}(t) = \int_{-\infty}^{\infty} [f(u) - f(u-\pi/t)]e^{-itu}d\mu(u)$$

so that

$$|2\hat{f}(t)| \leqslant \|f - f_{\pi/t}\|_1 \to 0$$

as $t \to \infty$ by an application of Proposition 3.6.1. □

Exercises

1. Let $f \in C[a,b]$. Show that

$$\int_a^b f(x)(\sin tx)dx \to 0, \qquad \text{and} \qquad \int_a^b f(x)(\cos tx)dx \to 0,$$

as $t \to \pm\infty$.

2. For any smooth function f, show that

$$\lim_{|t|\to\infty} \int_{-\infty}^{\infty} f(x)\frac{\sin tx}{x}dx = \pi f(0).$$

3.8 The Plancherel Theorem

The main weakness of the inversion theorem is that $f \in L^1(\mathbb{R})$ does not guarantee $\hat{f} \in L^1(\mathbb{R})$. It turns out that by considering functions $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, we can guarantee $\hat{f} \in L^2(\mathbb{R})$. We will show that $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ is dense in $L^2(\mathbb{R})$. Using this fact, we can extend the concept of the Fourier transform to all functions $f \in L^2(\mathbb{R})$. Moreover, we will see that for $f \in L^2(\mathbb{R})$, we have $\|f\|_2 = \|\hat{f}\|_2$ so that the Fourier transform is an isometry on $L^2(\mathbb{R})$.

Theorem 3.7 (The Plancherel theorem) *To each $f \in L^2(\mathbb{R})$ we can associate $\hat{f} \in L^2(\mathbb{R})$ (which is sometimes called the Plancherel transform of f) such that*

(1) if $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, then

$$\hat{f}(x) = \int_{-\infty}^{\infty} f(t)e^{-ixt}d\mu(t)$$

(that is, the Plancherel transform extends the Fourier transform);
(2) $\|f\|_2 = \|\hat{f}\|_2$ (that is, the Plancherel transform is an isometry);
(3) the map $f \mapsto \hat{f}$ is a Hilbert space isomorphism;
(4) if $\phi_A(t) = \int_{-A}^{A} f(x)e^{-ixt}d\mu(x)$ and $\psi_A(x) = \int_{-A}^{A} \hat{f}(t)e^{ixt}d\mu(t)$, then

$$\|\phi_A - \hat{f}\|_2 \to 0 \text{ and } \|\psi_A - f\|_2 \to 0 \text{ as } A \to \infty.$$

Proof Fix $f \in L^2(\mathbb{R})$. Suppose first $f \in L^1(\mathbb{R})$. Define

$$\widetilde{f}(x) := \overline{f(-x)}$$

and

$$g = f * \widetilde{f} \in L^1(\mathbb{R}).$$

Thus,

$$g(x) = \int_{-\infty}^{\infty} f(x-y)\overline{f(-y)}d\mu(y) = (f_{-x}, f).$$

From the fact that the map $x \mapsto f_x$ is a uniformly continuous map from $\mathbb{R}$ to $L^p(\mathbb{R})$ (for any $1 \leqslant p < \infty$), and the fact that the inner product is a continuous map, we see that g is continuous. Moreover, by the Cauchy–Schwarz inequality,

$$|g(x)| \leqslant \|f\|_2.$$

In other words, g is bounded. We can therefore apply Proposition 3.6.2 to deduce

$$\lim_{\lambda \to 0}(g * h_\lambda)(0) = g(0) = (f, f) = \|f\|_2^2.$$

On the other hand, the definition of g implies that

$$\hat{g} = \hat{f} \cdot \hat{\bar{f}} = |\hat{f}|^2 \geqslant 0$$

so that $H(\lambda t)\hat{g}(t)$ is non-negative and monotonically decreasing as $\lambda \to 0^+$.

By the monotone convergence theorem,

$$\begin{aligned} \|f\|_2^2 &= \lim_{\lambda \to 0} (g * h_\lambda)(0) \\ &= \int_{-\infty}^{\infty} \lim_{\lambda \to 0} H(\lambda t)\hat{g}(t) d\mu(t) \\ &= \int_{-\infty}^{\infty} |\hat{f}(t)|^2 d\mu(t) = \|\hat{f}\|_2^2. \end{aligned}$$

Thus, $\hat{f} \in L^2(\mathbb{R})$.

Next, we need to show that

$$Y = \{\hat{f} : f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})\}$$

is dense in $L^2(\mathbb{R})$. We just proved that $Y \subseteq L^2(\mathbb{R})$. Since Y is a Hilbert subspace of the Hilbert space $L^2(\mathbb{R})$, we can decompose it as

$$L^2(\mathbb{R}) = Y \oplus Y^\perp.$$

It suffices to show that $Y^\perp = \{0\}$. Observe that the map

$$x \mapsto e^{i\alpha x} H(\lambda x)$$

defines a map into $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, for all $\alpha \in \mathbb{R}$ and $\lambda > 0$. Thus the Fourier transform of these functions, namely

$$t \mapsto h_\lambda(\alpha - t) = \int_{-\infty}^{\infty} e^{i(\alpha - t)x} H(\lambda x) d\mu(x),$$

defines a function into Y. If $w \in Y^\perp$, then $(v, w) = 0$ for all $v \in Y$. In particular,

$$0 = \int_{-\infty}^{\infty} h_\lambda(\alpha - t)\overline{w(t)} d\mu(t) = (h_\lambda * \overline{w})(\alpha).$$

Recall that for $f \in L^p(\mathbb{R})$, $\lim_{\lambda \to 0} \|f * h_\lambda - f\| = 0$. Thus $w = 0$ a.e. Therefore, Y is dense in $L^2(\mathbb{R})$.

Let us temporarily denote $\hat{f}$ by $\Phi(f)$. By our discussion above, Φ is an L^2-isometry from one dense subspace of L^2, namely $L^1 \cap L^2$ onto another, namely Y. Elementary Cauchy sequence arguments now imply that Φ extends to an isometry

Φ of L^2 onto L^2. This suffices to prove (1) and (2). The Hilbert space isomorphism follows from Parseval's formula:

$$(f, g) = (\hat{f}, \hat{g}).$$

To prove this, we use the following identity:

$$4(x, y) = \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2$$

which expresses the inner product in terms of norms.

Finally, to prove (4), let k_A be the characteristic function of the interval $[-A, A]$. Then $k_A f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ and $\phi_A = \widehat{k_A f}$. But $\|f - k_A f\|_2 \to 0$ as $A \to \infty$, and so taking Fourier transform and the result of (3), we get $\|\hat{f} - \phi_A\|_2 \to 0$. The other limit is proved similarly. □

Exercises

1. Prove that
$$\int_{-\infty}^{\infty} \frac{e^{itx} dt}{1 + t^2} = \pi e^{-|x|}.$$
2. Suppose $f, g \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ are such that $\hat{f}(x) = \hat{g}(x)$ a.e. Show that $f(x) = g(x)$ a.e.
3. Extend Plancherel's theorem for functions $f \in L^2(\mathbb{R}^k)$.

3.9 The Uncertainty Principle

Recall the following fundamental inequality which is ubiquitous in mathematics. We have already seen this in different settings but now we state it in the context of an inner product space. We also need to highlight the case of equality in the inequality.

Theorem 3.8 (The Cauchy–Schwarz inequality) *If V is an inner product space (over $\mathbb{R}$ or $\mathbb{C}$) then*

$$|(v, w)| \leqslant \|v\|\|w\|$$

for all $v, w \in V$ with equality if and only if v is a scalar multiple of w.

Proof Decomposing v into its components parallel and perpendicular to w, we can write

$$v = \lambda w + (v - \lambda w)$$

for some scalar λ. For $v - \lambda w$ to be perpendicular to w, we need

$$0 = (v - \lambda w, w) = (v, w) - \lambda \|w\|^2.$$

That is,

$$\lambda = \frac{(v, w)}{\|w\|^2}.$$

Now, $(v - \lambda w, v - \lambda w) \geqslant 0$ for any λ, so that

$$\|v\|^2 - \overline{\lambda}(v, w) - \lambda(w, v) + |\lambda|^2 \|w\|^2 \geqslant 0.$$

With our choice of λ in particular, we deduce

$$\|v\|^2 - \frac{|(v, w)|^2}{\|w\|^2} - \frac{|(v, w)|^2}{\|w\|^2} + \frac{|(v, w)|^2}{\|w\|^4} \|w\|^2 \geqslant 0.$$

In other words,

$$|(v, w)| \leqslant \|v\| \|w\|.$$

Clearly, equality can occur if and only of $v = \lambda w$, for any λ. □

The celebrated Heisenberg uncertainty principle, which is a cornerstone of quantum mechanics, is an immediate consequence of the Cauchy–Schwarz inequality combined with basic Fourier theory, once we understand the dictionary that translates the concepts of physics into mathematical language. When it was discovered by Werner Heisenberg (1901–1976) it created a sensation in that it demonstrated an inherent obstruction to observations in physics. Heisenberg called his new theory "matrix mechanics." A parallel theory was discovered by Erwin Schrödinger (1887–1961) at about the same time. Schrödinger called his theory "wave mechanics", and we now know that both theories are mathematically equivalent. Indeed, from a mathematical standpoint, Heisenberg's uncertainty principle states that a function F and its Fourier transform $\widehat{F}$ cannot *both* have compact support. To prove this, it is convenient to introduce the following class of functions.

Definition 3.2 We define the **Schwartz space**[2] $\mathcal{S}$ to be the space of infinitely differentiable functions $F : \mathbb{R} \to \mathbb{C}$ such that

$$|x^k F^{(\ell)}(x)| \to 0$$

as $|x| \to \infty$, for all non-negative integers k and ℓ.

[2] Notice the "t" in the spelling of Schwartz's name. The space is named after Laurent Schwartz (1915–2002), who is different from H.A. Schwarz (1848–1921) of the Cauchy–Schwarz inequality.

In the following discussion, it is convenient to use the following definition of the Fourier transform:

$$\widehat{\psi}(x) = \int_{-\infty}^{\infty} \psi(t) e^{-2\pi i t x}\, dt.$$

Theorem 3.9 (Heisenberg's uncertainty principle) *Let ψ be a Schwartz function satisfying $\|\psi\|_2 = 1$. Then*

$$\left(\int_{-\infty}^{\infty} x^2 |\psi(x)|^2\, dx\right)\left(\int_{-\infty}^{\infty} x^2 |\widehat{\psi}(x)|^2\, dx\right) \geqslant \frac{1}{16\pi^2}$$

with equality if and only if $\psi(x) = Ae^{-Bx^2}$ for some $B > 0$ and

$$|A|^2 = \sqrt{\frac{2B}{\pi}}.$$

Proof Writing

$$1 = \int_{-\infty}^{\infty} |\psi(x)|^2\, dx = \int_{-\infty}^{\infty} \psi(x)\overline{\psi(x)}\, dx.$$

We integrate by parts, to get

$$x\psi(x)\overline{\psi}(x)\Big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} x \frac{d}{dx}\left(\psi(x)\overline{\psi}(x)\right) dx.$$

Because $\psi \in \mathcal{S}$, the first term equals zero. Thus,

$$1 = -\int_{-\infty}^{\infty} x\left\{\psi(x)\overline{\psi'(x)} + \psi'(x)\overline{\psi(x)}\right\} dx$$

so that

$$1 \leqslant 2\int_{-\infty}^{\infty} |x||\psi(x)||\psi'(x)|\, dx.$$

By the Cauchy–Schwarz inequality,

$$1 \leqslant 2\left(\int_{-\infty}^{\infty} x^2 |\psi(x)|^2\, dx\right)^{1/2}\left(\int_{-\infty}^{\infty} |\psi'(x)|^2\, dx\right)^{1/2}.$$

By the inversion theorem,

$$\psi(x) = \int_{-\infty}^{\infty} \widehat{\psi}(t) e^{2\pi i t x}\, dt$$

so that

$$\psi'(x) = \int_{-\infty}^{\infty} (2\pi i t)\widehat{\psi}(t)e^{2\pi i t x}\, dt$$

In other words, ψ' is the (inverse) Fourier transform of $(2\pi i t)\widehat{\psi}(t)$.

By Parseval's formula, the L^2-norm of ψ' is equal to

$$\int_{-\infty}^{\infty} 4\pi^2 t^2 |\widehat{\psi}(t)|^2\, dt.$$

Thus

$$1 \leqslant 2(2\pi)\left(\int_{-\infty}^{\infty} x^2|\psi(x)|^2\, dx\right)^{1/2}\left(\int_{-\infty}^{\infty} x^2|\widehat{\psi}(x)|^2\, dx\right)^{1/2}$$

from which the main inequality emerges.

In our application of the Cauchy–Schwarz inequality, equality can occur if and only if

$$\psi'(x) = \lambda x\psi(x)$$

for some scalar λ. This is an ordinary differential equation which is easily solved:

$$\psi(x) = Ae^{-Bx^2}.$$

Substituting this back gives the final result. □

In quantum mechanics, $|\psi(x)|^2\, dx$ represents the probability density that a particle is at position $x \in \mathbb{R}$. Thus, the probability that such a particle lies in the interval $[a, b]$ is given by

$$\int_a^b |\psi(x)|^2\, dx.$$

Our best guess for the particle's position is then given by the expectation

$$\mu = \int_{-\infty}^{\infty} x|\psi(x)|^2\, dx$$

and the error (or uncertainty) involved in this guess is given by the variance

$$\int_{-\infty}^{\infty} (x-\mu)^2|\psi(x)|^2\, dx.$$

A similar analysis holds for the momentum of the particle. Indeed, the probability that the momentum of a particle lies in (a, b) is given by

$$\int_a^b |\widehat{\psi}(x)|^2\, dx.$$

The expectation and variance are defined as before. Without loss of generality, we can normalize our functions so that the expectations and variances of both the position and the momentum are zero. With this normalization, we see that Heisenberg uncertainty principle establishes a lower bound for the product of these variances.

In other words, any attempt to lower the error in our observation of position increases the error in estimating momentum and vice versa. Though the theorem originally arose in the context of quantum mechanics, we see that it is really a theorem about Fourier transforms.

In the terminology of physics, the uncertainty principle is written as

$$(\text{uncertainty of position}) \times (\text{uncertainty of momentum}) \geq \frac{\hbar}{16\pi^2},$$

where $\hbar$ denotes Planck's constant.

Exercises

1. The uncertainty principle can also be formulated in terms of the **Hermite operator** H defined as
$$H(f) := -\frac{d^2 f}{dx^2} + x^2 f.$$
Show that $(H(f), f) \geq (f, f)$ for all f in the Schwartz space with the usual inner product.
2. Suppose f is a continuous function in $L^1(\mathbb{R})$. Show that if f and $\hat{f}$ both have compact support, then f is identically zero.

3.10 Trigonometric Polynomials

A "polynomial" of the form

$$\sum_{n=-N}^{N} a_n e^{2\pi i n x}$$

represents a continuous function on the space $\mathbb{R}/\mathbb{Z}$ and is called a **trigonometric polynomial** in that it is a "polynomial" in $e^{2\pi i x}$. We have already seen these "polynomials" in Sect. 2.5 in our brief study of Fourier series. Here, we will introduce the impotant Féjer kernel and give a constructive proof of Theorem 2.16 using this kernel

as well as the properties of convolutions that were nascent in our earlier treatment. The following approximation theorem is due to Weierstrass.

Theorem 3.10 *Let $f \in C(\mathbb{R}/\mathbb{Z})$, that is, f is a continuous function on $\mathbb{R}/\mathbb{Z}$. Let $\varepsilon > 0$ be given. Then there is a trigonometric polynomial p such that*

$$\|f - p\|_\infty \leqslant \varepsilon.$$

To prove this, we introduce the following useful concept. Let $\varepsilon > 0$ and $0 < \delta < 1/2$. A function $f \in C(\mathbb{R}/\mathbb{Z})$ is said to be a **periodic** (ε, δ) **approximation to the identity** if

(a) $f(x) \geqslant 0 \, \forall x \in \mathbb{R}$ and $\int_0^1 f(x)\,dx = 1$;
(b) $f(x) < \varepsilon \, \forall \delta \leqslant |x| < 1 - \delta$.

In other words, the "bulk" of the continuation of $f(x)$ occurs in $[0, \delta)$ and $(1 - \delta, 1]$.

Lemma 3.1 *For every $\varepsilon > 0$ and $0 < \delta < 1/2$, there exists a trigonometric polynomial P which is an (ε, δ) approximation to the identity.*

Proof We use the **Féjer kernel**:

$$F_N(x) = \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N}\right) e^{2\pi i n x}.$$

Clearly, $F_N(x)$ is a trigonometric polynomial. It is easy to see that (see Exercise 1 below)

$$F_N(x) = \frac{1}{N} \left| \sum_{n=0}^{N-1} e^{2\pi i n x} \right|^2,$$

an identity easily proved by induction. But the sum

$$\sum_{n=0}^{N-1} e^{2\pi i n x}$$

is a geometric series which can be summed as (for $x \notin \mathbb{Z}$)

$$e^{\pi i (N-1) x} \cdot \frac{\sin N\pi x}{\sin \pi x}$$

so that

$$F_N(x) = \left(\frac{\sin N\pi x}{\sin \pi x}\right)^2 \frac{1}{N},$$

for $x \notin \mathbb{Z}$. If $x \in \mathbb{Z}$, then $F_N(x) = N$. In any case, $F_N(x) \geqslant 0$ for any x. Now

$$\int_0^1 F_N(x)\,dx = \sum_{n=-N}^{N}\left(1-\frac{|n|}{N}\right)\int_0^1 e^{2\pi inx}\,dx = 1.$$

But $|\sin \pi Nx| \leqslant 1$ so that

$$|F_N(x)| \leqslant \frac{1}{N|\sin \pi x|^2} \leqslant \frac{1}{N(\sin \pi\delta)^2}$$

for $\delta < |x| < 1-\delta$ because the sine function is increasing on $[0, \pi/2]$ and decreasing on $[\pi/2, \pi]$.

Thus, by choosing N sufficiently large, we can make

$$|F_N(x)| \leqslant \varepsilon$$

for all $\delta < |x| < 1-\delta$. This completes the proof. □

Proof of Theorem 3.10 Let $f \in C(\mathbb{R}/\mathbb{Z})$. Since f is continuous on $[0,1]$, f is bounded so that $|f(x)| \leqslant M\ \forall x \in C(\mathbb{R}/\mathbb{Z})$.

Let $\varepsilon > 0$. As f is uniformly continuous, there is a $\delta > 0$ such that

$$|f(x) - f(y)| \leqslant \varepsilon \text{ whenever } |x-y| \leqslant \delta.$$

By Lemma 3.9, we can find a trigonometric polynomial P which is an (ε, δ) approximation to the identity. Then $f * P$ is also a trigonometric polynomial because

$$\begin{aligned} f * P(x) &= \int_0^1 f(x-y)P(y)\,dy \\ &= \int_0^1 f(x-y)\sum_{n=-N}^{N} a_n e^{2\pi iny}\,dy \\ &= \sum_{n=-N}^{N} a_n \int_0^1 f(x-y)e^{2\pi iny}\,dy \\ &= \sum_{n=-N}^{N} a_n \int_0^1 f(t)e^{2\pi in(x-t)}\,dt \\ &= \sum_{n=-N}^{N} a_n \left(\int_0^1 f(t)e^{-2\pi int}\,dt\right)e^{2\pi inx} \end{aligned}$$

which we may write as

$$\sum_{n=-N}^{N} a_n \hat{f}(n)e^{2\pi inx}$$

where $\hat{f}(n) = \int_0^1 f(t)e^{-2\pi int}\,dt$ in the sense of classical Fourier series.

Thus,

$$|f(x) - (f * P)(x)| = \left| \int_0^1 f(x)P(y)\,dy - \int_0^1 f(x-y)P(y)\,dy \right|$$

because

$$\int_0^1 P(y)\,dy = 1.$$

Thus, as P is non-negative,

$$|f(x) - (f * P)(x)| \leqslant \int_0^1 |f(x) - f(x-y)|P(y)\,dy.$$

The right hand side can be split into three integrals:

$$\int_0^\delta |f(x) - f(x-y)|P(y)\,dy \quad +$$

$$\int_\delta^{1-\delta} |f(x) - f(x-y)|P(y)\,dy + \int_{1-\delta}^1 |f(x) - f(x-y)|P(y)\,dy.$$

By the uniform continuity of f, the first and the last integrals are

$$\leqslant \varepsilon \int_0^\delta P(y)\,dy + \varepsilon \int_{1-\delta}^1 P(y)\,dy \leqslant 2\varepsilon$$

because $\int_0^1 P(y)\,dy = 1$.

The middle integral is bounded by

$$M \int_\delta^{1-\delta} P(y)\,dy \leqslant M\varepsilon$$

by the property of the trigonometric polynomial P. Hence

$$\|f - f * P\|_\infty \leqslant (2M + 2)\varepsilon.$$

As M is fixed and ε is arbitrary, we can make $f * P$ arbitrarily close to f under the sup norm. This completes the proof. □

The virtue of this proof is that everything can be made explicit. Indeed, given a continuous function f, the trigonometric polynomial approximations are provided by $f * F_N$, where F_N is the Féjer kernel.

Exercises

1. Let
$$F_N(x) = \sum_{|n|\le N} \left(1 - \frac{|n|}{N}\right) e^{2\pi i n x}.$$
Show that
$$F_N(x) = \frac{1}{N}\left|\sum_{n=0}^{N-1} e^{2\pi i n x}\right|^2 = \frac{1}{N}\left(\frac{\sin \pi N x}{\sin \pi x}\right)^2.$$

2. Let f be a Schwartz function. Show that
$$\lim_{R\to\infty} \int_{-R}^{R} \left(1 - \frac{|x|}{R}\right) \hat{f}(x) e^{2\pi i x t} dx = f(t).$$

3.11 The Isoperimetric Inequality

An important application of the theory of Fourier series is to the **isoperimetric inequality**. Is there a relation between the area enclosed by a simple closed curve and its length? Here is the answer.

Theorem 3.11 (The isoperimetric inequality) *Suppose that C is a simple closed curve in $\mathbb{R}^2$ of length L. If A is the area of the region enclosed by C, then*
$$A \le \frac{L^2}{4\pi},$$
with equality if and only if C is a circle.

Proof Our proof is due to Hurwitz. The reader will recall that Green's theorem gives us the formula for the area A:
$$A = \frac{1}{2}\left|\int_C (x\,dy - y\,dx)\right|,$$
where we have parametrized the curve C by $\gamma : [0, 2\pi] \to \mathbb{R}^2$ with $\gamma(t) = (x(t), y(t))$. We first observe that without any loss of generality, we may suppose that $L = 2\pi$ because any dilation $(x, y) \mapsto (\lambda x, \lambda y)$ increases the area A by λ^2 and the length L by λ. Thus, without any loss of generality, taking $\lambda = 2\pi/L$, we see that it suffices to show that when $L = 2\pi$, then $A \le \pi$ with equality if and only C is a circle. Utilizing our formula for the length of the curve, we have

$$\frac{1}{2\pi}\int_0^{2\pi}(x'(t))^2+(y'(t))^2dt=1.$$

As our curve is closed, the functions $x(t)$, $y(t)$ are periodic with period 2π. We may write down their Fourier series:

$$x(t)=\sum_n a_n e^{int},\qquad y(t)=\sum_n b_n e^{int}.$$

Then,

$$x'(t)=\sum_n a_n ine^{int},\qquad y'(t)=\sum_n b_n ine^{int}.$$

Parseval's identity now gives

$$\sum_{n=-\infty}^{\infty}|n|^2(|a_n|^2+|b_n|^2)=1. \tag{3.4}$$

Now $x(t)$ and $y(t)$ are real valued, so that $a_{-n}=\overline{a_n}$ and $b_{-n}=\overline{b_n}$. Our formula for the area via Green's theorem combined with another application of Parseval's formula now gives

$$A=\frac{1}{2}\left|\int_0^{2\pi}\left(x(t)y'(t)-y(t)x'(t)\right)dt\right|=\pi\left|\sum_{n=-\infty}^{\infty}n\left(a_n\overline{b_n}-b_n\overline{a_n}\right)\right|.$$

Observing that

$$|a_n\overline{b_n}-b_n\overline{a_n}|\le 2|a_n||b_n|\le|a_n|^2+|b_n|^2, \tag{3.5}$$

and since $|n|\le n^2$, we obtain

$$A\le\pi\sum_{n=-\infty}^{\infty}|n|^2(|a_n|^2+|b_n|^2)\le\pi,$$

by (3.4). This is the isoperimetric inequality.

Let us now consider the case of equality. We used the inequality $|n|\le n^2$ in the above proof. But this is a strict inequality for $|n|\ge 2$. In other words, if $A=\pi$, we must have $a_n, b_n=0$ for $|n|\ge 2$, and our two Fourier series become

$$x(t)=a_{-1}e^{-it}+a_0+a_1e^{it},\quad\text{and}\quad y(t)=b_{-1}e^{-it}+b_0+b_1e^{it}.$$

Now $a_{-1}=\overline{a_1}$, $b_{-1}=\overline{b_1}$ so that the Eq. (3.4) gives

$$2(|a_1|^2+|b_1|^2)=1.$$

Since we have equality at every stage in (3.5), we deduce $|a_1| = |b_1| = 1/2$. Thus writing

$$a_1 = \frac{1}{2}e^{i\alpha}, \quad \text{and} \quad b_1 = \frac{1}{2}e^{i\beta},$$

the equality from (3.5) gives

$$1 = 2|a_1\overline{b_1} - \overline{a_1}b_1|$$

implies $|\sin(\alpha - \beta)| = 1$. Thus, $\alpha - \beta = k\pi/2$ where k is an odd integer. We deduce from this that

$$x(t) = a_0 + \cos(\alpha + t), \quad \text{and} \quad y(t) = b_0 \pm \sin(\alpha + t),$$

where the sign in $y(t)$ depends on the parity of $(k-1)/2$. These functions parametrize the circle of radius 1 centered at (a_0, b_0). This completes the proof. □

Exercises

1. If f is C^1 with period 2π, and

$$\int_0^{2\pi} f(t)dt = 0,$$

show that

$$\int_0^{2\pi} |f(t)|^2 dt \leq \int_0^{2\pi} |f'(t)|^2 dt.$$

This is called **Wirtinger's inequality**.

2. Show that equality can hold in the above inequality if and only if $f(t) = A\sin t + B\cos t$, for some constants A and B.

3.12 Weyl's Criterion and Uniform Distribution

A sequence of numbers $x_1, x_2, \cdots$ in $[0, 1]$ is said to be **uniformly distributed** (or **equidistributed**) if for every interval $[a, b] \subseteq [0, 1]$ we have

$$\lim_{N\to\infty} \frac{\#\{1 \leqslant n \leqslant N : x_n \in [a, b]\}}{N} = b - a.$$

Another way of stating the equidistribution criterion is as follows. Let $I = (a, b)$ and $\chi_I(x)$ the characteristic function of I. Then

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \chi_I(x_n) = \int_0^1 \chi_I(x)\, d\mu(x).$$

Since step functions are not continuous functions, we would like to replace this criterion with continuous functions.

To this end, let $\varepsilon > 0$. Then we can find continuous functions

$$f_\varepsilon^-(x) \leqslant \chi_I(x) \leqslant f_\varepsilon^+(x)$$

such that

$$\int_0^1 f_\varepsilon^\pm(x)\, dx = b - a \pm \varepsilon.$$

We call such functions pseudo-characteristic functions. Thus, it is easy to see that our equidistribution criterion can be replaced by

$$\frac{1}{N} \sum_{n=1}^{N} f(x_n) \to \int_0^1 f(x)\, d\mu(x)$$

for all continuous functions f.

We leave this equivalence as a simple exercise for the reader.

We are now ready to prove Weyl's criterion.

Theorem 3.12 (Weyl, 1916) *A sequence of numbers $\{x_n\}_{n=1}^\infty$ with $x_n \in [0, 1]$ is uniformly distributed if and only if*

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} e^{2\pi i m x_n} = 0 \quad \forall m \neq 0, m \in \mathbb{Z}.$$

Proof Recall that from our previous remark, we know that $\{x_n\}_{n=1}^\infty$ is equidistributed if and only if

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f(x_n) = \int_0^1 f(x)\, dx$$

for all continuous functions f. In particular, we can apply this to the continuous function

$$f_m(x) = e^{2\pi i m x}$$

and so if $\{x_n\}_{n=1}^\infty$ is equidistributed, then

$$\lim_{N\to\infty}\frac{1}{N}\sum_{n=1}^{N} f_m(x_n) = \int_0^1 e^{2\pi imx}\,dx = 0.$$

To prove the converse, we recall from the previous section that trigonometric polynomials are dense in $C[0,1]$. Given a continuous function f, let $\varepsilon > 0$ be fixed. We can find a trigonometric polynomial $P(x)$ such that

$$P(x) = \sum_{|m|\leqslant R} a_m e^{2\pi imx}$$

and

$$\sup_{x\in[0,1]} |f(x) - P(x)| \leqslant \varepsilon.$$

Thus,

$$\left|\frac{1}{N}\sum_{n=1}^{N} f(x_n) - \frac{1}{N}\sum_{n=1}^{N} P(x_n)\right| \leqslant \varepsilon$$

and

$$\lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{N}\sum_{|m|\leqslant R} a_m e^{2\pi imx_n} = \sum_{|m|\leqslant R} a_m \lim_{N\to\infty}\frac{1}{N}\sum_{n=1}^{N} e^{2\pi imx_n}.$$

The inner limits are all zero except for the case $m = 0$. On the other hand,

$$\left|\int_0^1 f(x)\,dx - \int_0^1 P(x)\,dx\right| \leqslant \varepsilon$$

and $\int_0^1 P(x)\,dx = a_0$.

From our construction in Theorem 3.10, we see that

$$a_0 = \int_0^1 f(x)\,dx.$$

Thus,

$$\lim_{N\to\infty}\frac{1}{N}\sum_{n=1}^{N} f(x_n) = \int_0^1 f(x)\,dx$$

as desired. □

Example 3.2 If $\theta \in \mathbb{Q}$, the sequence of fractional parts $\{n\theta\}_{n=1}^{\infty}$ is *not* equidistributed. If $\theta \notin \mathbb{Q}$, the sequence of fractional parts $\{n\theta\}_{n=1}^{\infty}$ is equidistributed.

We apply Weyl's criterion: indeed, if $\theta = p/q$ (say), then

$$\sum_{n=1}^{N} e^{2\pi i q(np/q)} = N$$

so that the Weyl's limit is 1 for $m = q$. This proves the first part of the assertion. For the second part, we see that

$$\sum_{n=1}^{N} e^{2\pi i m n \theta}$$

is a geometric sum easily summed to be

$$e^{2\pi i m\theta} \cdot \frac{(e^{2\pi i m\theta})^N - 1}{e^{2\pi i m\theta} - 1}$$

which for $m \neq 0$ is bounded. Thus the Weyl limit is zero. □

Exercises

1. As e is irrational, the sequence of fractional parts $\{ne\}$ is equidistributed mod 1. Show that
$$\lim_{n\to\infty} \{n!e\} = 0,$$
and deduce that the subsequence $\{n!e\}$ is not equidistributed mod 1.
2. Prove the **Euler–Maclaurin summation formula**: if $f(t)$ is a complex-valued function with continuous derivative on $1 \leq t \leq N$, with N a natural number, then
$$\sum_{n=1}^{N} f(n) = \int_1^N f(t)dt + \frac{1}{2}(f(1) + f(N)) + \int_1^N (\{t\} - \frac{1}{2}) f'(t)dt.$$
3. Show that the sequence $\{\log n\}$ is not equidistributed mod 1. [Hint: apply the Euler–Maclaurin summation formula with $f(t) = e^{2\pi i \log t}$.]
4. Let a and b be integers with $a < b$ and suppose that f is twice differentiable on $[a, b]$. For all $x \in [a, b]$ suppose that either $f''(x) \geq \delta > 0$ or $f''(x) \leq -\delta < 0$ holds. Then,
$$\left| \sum_{n=a}^{b} e^{2\pi i f(n)} \right| \leq \left(|f'(b) - f'(a)| + 2 \right) \left(\frac{4}{\sqrt{\delta}} + 3 \right).$$
[Hint: apply the Euler–Maclaurin summation formula.]
5. Let α satisfy $0 < \alpha < 1$. Prove that the sequence $\{n^\alpha\}$ is uniformly distributed mod 1.
6. Prove that the sequence $\{n \log n\}$ is uniformly distributed mod 1.

3.13 Fourier Series

Suppose $f : \mathbb{R}/\mathbb{Z} \to \mathbb{C}$ is a continuous function. We would like to investigate if f can be written as a Fourier series

$$f(x) \stackrel{?}{=} \sum_{n\in\mathbb{Z}} a_n e^{2\pi inx}.$$

If such a series exists, we see that proceeding formally

$$\int_0^1 \left(\sum_{n\in\mathbb{Z}} a_n e^{2\pi inx}\right) e^{-2\pi imx}\, dx = a_m.$$

In other words,

$$a_m = \int_0^1 f(x)e^{-2\pi imx}\, dx.$$

Motivated by this observation, we make the following definition. Given a function $f \in C(\mathbb{R}/\mathbb{Z})$, we define its n-th Fourier coefficient as

$$\hat{f}(n) := \int_0^1 f(x)e^{-2\pi inx}\, dx$$

and inquire when does the Fourier series

$$\sum_{n\in\mathbb{Z}} \hat{f}(n)e^{2\pi inx}$$

converge pointwise to $f(x)$? What condition do we need to impose on f?

To this end, we begin with the Fourier series analogue of the Riemann–Lebesgue lemma which we saw earlier in the context of Fourier transforms:

Lemma 3.2 (Riemann–Lebesgue lemma) *For $f \in C(\mathbb{R}/\mathbb{Z})$,*

$$\lim_{|n|\to\infty} \widehat{f}(n) = 0.$$

Proof We take a trigonometric polynomial P such that

$$||f - P||_\infty < \epsilon.$$

This we can do by Theorem 3.10. Then,

$$\widehat{f}(n) = \frac{1}{2\pi}\int_0^{2\pi} f(x)e^{-2\pi inx}dx = \frac{1}{2\pi}\int_0^{2\pi} (f(x) - P(x))e^{-2\pi inx}dx$$

for n sufficiently large (depending on P) because P being a trigonometric polynomial, all its Fourier coefficients are zero after some point. But then, as the integrand is less than ϵ, we see that $|\widehat{f}(n)| < \epsilon$. Since ϵ is arbitrary, this completes the proof. □

To study convergence of Fourier series, we introduce the **Dirichlet kernel**:

$$D_N(x) = \sum_{|n| \leqslant N} e^{2\pi i n x}.$$

Clearly, $D_N(x) \in C(\mathbb{R}/\mathbb{Z})$. Also

$$\int_0^1 D_N(x)\, dx = 1.$$

Finally, by summing the geometric series we find

$$D_N(x) = \frac{\sin(2N+1)\pi x}{\sin \pi x} = \cos 2N\pi x + \cot \pi x \cdot \sin 2N\pi x$$

for $x \neq 0$. For $x = 0$, $D_N(0) = 2N + 1$.

Theorem 3.13 *Suppose $f \in C(\mathbb{R}/\mathbb{Z})$ is differentiable and real valued. Then*

$$f(x) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n x}.$$

Proof Let

$$\begin{aligned} S_N(x, f) &= \sum_{|n| \leqslant N} \hat{f}(n) e^{2\pi i n x} \\ &= \sum_{|n| \leqslant N} e^{2\pi i n x} \int_0^1 f(t) e^{-2\pi i n t}\, dt \\ &= \int_0^1 f(t) \sum_{|n| \leqslant N} e^{2\pi i n (x-t)}\, dt \\ &= f * D_N(x). \end{aligned}$$

Now,

$$\begin{aligned} S_N(x, f) - f(x) &= \int_0^1 f(t) D_N(x-t)\, dt - f(x) \\ &= \int_0^1 f(x-u) D_N(u)\, du - \int_0^1 f(x) D_N(u)\, du \end{aligned}$$

so that

$$S_N(x, f) - f(x) = \int_0^1 (f(x-u) - f(x))\{\cos 2N\pi u + \cot \pi u \cdot \sin 2N\pi u\}\, du.$$

Let

$$g_1(u) = f(x-u) - f(x)$$

and

$$g_2(u) = g_1(u) \cot \pi u.$$

Both are continuous functions since f is differentiable. To say that $g_2(u)$ is a continuous function is tantamount to saying f is differentiable. To see this, note

$$\{f(x-u) - f(x)\}\frac{\cos \pi u}{\sin \pi u} = \frac{f(x-u) - f(x)}{\pi u}\frac{\pi u}{\sin \pi u}\cos \pi u \to \frac{f'(x)}{\pi} \text{ as } u \to 0.$$

Lemma 3.2 applies to give

$$|S_N(x, f) - f(x)| = |\mathrm{Re}\, \hat{g}_1(N) + \mathrm{Im}\, \hat{g}_2(N)|.$$

so that $|S_N(x, f) - f(x)| \to 0$ as $N \to \infty$, by the Riemann–Lebesgue lemma. □

Exercises

1. With D_N denoting the Dirichlet kernel,

$$D_N(x) = \frac{\sin(N + 1/2)x}{\sin(x/2)},$$

show that the Féjer kernel $F_N(x)$ can be written as

$$\frac{1}{N}\left(D_0(x) + D_1(x) + \cdots + D_{N-1}(x)\right).$$

2. Suppose that f is periodic and of class C^k. Show that

$$\hat{f}(n) = o(1/|n|^k),$$

as $|n|$ tends to infinity.

3.14 The Poisson Summation Formula

We will now study how to make the following heuristic argument rigorous. As earlier, for $f \in L^1(\mathbb{R})$ we define

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x)e^{-2\pi ixt}\,dx.$$

This is a minor variation of the Fourier transform where ixt has been replaced by $2\pi ixt$ in the exponential. This is motivated by a desire to have a more elegant formulation of the Poisson formula.

Given $f \in L^1(\mathbb{R})$, we define

$$F(x) = \sum_{n\in\mathbb{Z}} f(n+x).$$

Then $F(x+1) = F(x)\ \forall x \in \mathbb{R}$ so that F is periodic and so can be viewed as a function defined on $\mathbb{R}/\mathbb{Z}$. Assuming some "suitable" conditions, we can expand F as a Fourier series:

$$F(x) = \sum_{n\in\mathbb{Z}} \hat{F}(n)e^{2\pi inx}.$$

But,

$$\begin{aligned}\hat{F}(n) &= \int_0^1 F(x)e^{-2\pi inx}\,dx\\ &= \int_0^1 \sum_{m\in\mathbb{Z}} f(m+x)e^{-2\pi inx}\,dx\\ &= \sum_{m\in\mathbb{Z}} \int_0^1 f(m+x)e^{-2\pi inx}\,dx.\end{aligned}$$

Changing variables in the integral by setting $u = m + x$, we obtain

$$\begin{aligned}&\sum_{m\in\mathbb{Z}} \int_m^{m+1} f(u)e^{-2\pi in(u-m)}\,du\\ &= \sum_{m\in\mathbb{Z}} \int_m^{m+1} f(u)e^{-2\pi inu}\,du\\ &= \int_{-\infty}^{\infty} f(u)e^{-2\pi inu}\,du = \hat{f}(n).\end{aligned}$$

Thus,

$$\sum_{m\in\mathbb{Z}} f(m+x) = \sum_{n\in\mathbb{Z}} \hat{f}(n)e^{2\pi inx}.$$

Setting $x = 0$, we obtain what is usually termed as the **Poisson summation formula**:

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \hat{f}(n),$$

which is evidently a beautiful identity.

To what extent can this be made rigorous? Firstly, we need

$$\sum_{n\in\mathbb{Z}} f(n+x)$$

to be absolutely convergent to ensure convergence of the series. Secondly, we want to expand this as a Fourier series. To do so, it suffices to have $F(x)$ to be differentiable. This may necessitate that $f(x)$ is differentiable. We would need

$$\sum_{n\in\mathbb{Z}} \hat{f}(n)$$

to be absolutely convergent.

Now, to ensure $F(x)$ is differentiable, we need to study

$$F(x+h) - F(x) = \sum_{n\in\mathbb{Z}} \{f(n+x+h) - f(n+x)\}.$$

Since we are assuming $f(x)$ is differentiable, we have by the mean value theorem

$$f(n+x+h) - f(n+x) = hf'(\xi_{n,x})$$

for some $\xi_{n,x} \in [n+x, n+x+1]$.

In order to take the limit

$$\lim_{h\to 0} \frac{F(x+h) - F(x)}{h}$$

we would need some form of absolute convergence of

$$\sum_{n\in\mathbb{Z}} |f'(\xi_{n,x})|.$$

All of these conditions can be ensured if we impose the following conditions.

Suppose $f \in C^2(\mathbb{R})$, that is, f is twice differentiable and that for some $\delta > 0$,

$$|f(x)|, |f'(x)|, |f''(x)| \leqslant c(1+|x|)^{-1-\delta}$$

for some fixed constant $c > 0$.

Theorem 3.14 (Poisson summation formula) *Under the above conditions on f, we have*

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \hat{f}(n).$$

Proof Let us observe first that under our assumptions,

$$\sum_{n\in\mathbb{Z}} f(n) \text{ and } \sum_{n\in\mathbb{Z}} f(n+x)$$

are absolutely convergent for $0 \leqslant x \leqslant 1$. Also,

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x)e^{-2\pi ixt}\,dx.$$

Integrating by parts the integral, we see that it is for $t \neq 0$,

$$f(x)\frac{e^{-2\pi ixt}}{-2\pi it}\Big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} f'(x)\frac{e^{-2\pi ixt}\,dx}{2\pi it}.$$

Another integration by parts shows

$$|\hat{f}(t)| \leqslant \frac{1}{2\pi|t|^2}\left|\int_{|x|\geqslant 1} f''(x)e^{-2\pi ixt}\,dx\right|$$

for $|t| \geqslant 1$ (say). Our estimates on $f''(x)$ ensure absolute convergence of the integral. Therefore,

$$\sum_{n\in\mathbb{Z}} \hat{f}(n)$$

is absolutely convergent.

The differentiability of f along with our estimates on f, f' now ensure $F(x)$ is differentiable so that it admits Fourier series expansion.

With those remarks in place, the heuristic "proof" now can be made rigorous. We leave the details to the reader. □

The Poisson summation formula can be generalized to arbitrary lattices in $\mathbb{R}^n$. Such a result finds numerous applications in number theory and in particular, in the theory of modular forms. On the other hand, there are several variants of the Poisson summation formula. The following version is useful.

Theorem 3.15 (Poisson summation formula) *Let f be a continuous function on $\mathbb{R}$ such that for some $\delta > 0$, and some fixed constant c,*

$$|f(x)| \leqslant \frac{c}{(1+|x|)^{1+\delta}}.$$

Suppose further that

$$\sum_{m\in\mathbb{Z}} |\hat{f}(m)| < \infty.$$

Then

$$\sum_{m\in\mathbb{Z}} \hat{f}(m)e^{2\pi imx} = \sum_{n\in\mathbb{Z}} f(n+x).$$

We leave the proof as an exercise to the reader and turn our attention to various applications.

Recall that if $F(x) = e^{-c|x|}$ with Re (c) > 0, then

$$\hat{F}(u) = \frac{2c}{c^2 + 4\pi^2 u^2}$$

which follows easily from our earlier calculation of the Fourier transform of the continuous function $e^{-|x|}$. The conditions of the theorem are satisfied, and we obtain

$$\sum_{n=-\infty}^{\infty} e^{-c|n|} = \sum_{n=-\infty}^{\infty} \frac{2c}{c^2 + 4\pi^2 n^2}.$$

The left hand side is a geometric progression and equal to

$$1 + 2\sum_{n=1}^{\infty} e^{-cn} = 1 + \frac{2e^{-c}}{1 - e^{-c}} = \frac{1 + e^{-c}}{1 - e^{-c}} = \frac{e^c + 1}{e^c - 1}.$$

Therefore,

$$\frac{e^c + 1}{e^c - 1} = \sum_{n=-\infty}^{\infty} \frac{2c}{c^2 + 4\pi^2 n^2}. \qquad (*)$$

Putting $c = 2\pi$ gives

$$\pi\left(\frac{e^{2\pi} + 1}{e^{2\pi} - 1}\right) = \sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 1} = 1 + 2\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}.$$

Thus

$$2\sum_{n=1}^{\infty} \frac{1}{n^2 + 1} = \frac{\pi e^{2\pi} + \pi - (e^{2\pi} - 1)}{e^{2\pi} - 1} = \frac{\pi e^{2\pi} - e^{2\pi} + \pi + 1}{e^{2\pi} - 1}.$$

A famous theorem of Gelfond and Schneider states that e^{π} is transcendental. In the 1970s Chudnovski proved that π and e^{π} are algebraically independent. Thus, the sum is a transcendental number.

The celebrated Basel problem asked for the evaluation of

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

which was finally solved by Euler who showed that it equals $\pi^2/6$. This can be deduced from our result.

Indeed, from $(*)$, we have

$$\frac{1}{2c}\left(1+\frac{2}{e^c-1}\right) = \sum_{n=-\infty}^{\infty} \frac{1}{c^2+4\pi^2n^2} = \frac{1}{c^2} + 2\sum_{n=1}^{\infty} \frac{1}{c^2+4\pi^2n^2}.$$

We would like to set $c=0$ in the last sum. Therefore

$$\frac{1}{2\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2} = \lim_{c\to 0}\left\{\frac{1}{2c}\left(1+\frac{2}{e^c-1}\right)-\frac{1}{c^2}\right\}.$$

Now

$$e^c - 1 = c + \frac{c^2}{2!} + \frac{c^3}{3!} + \cdots$$

so that

$$\begin{aligned}
&\frac{1}{2c}\left(1+\frac{2}{c\left(1+\frac{c}{2}+\frac{c^2}{6}+\cdots\right)}\right)-\frac{1}{c^2}\\
&=\frac{1}{2c}\left(1+\frac{2}{c}\left(1-\frac{c}{2}-\frac{c^2}{6}+\cdots+\left(\frac{c}{2}+\frac{c^2}{6}+\cdots\right)^2+\cdots\right)\right)-\frac{1}{c^2}
\end{aligned}$$

which simplifies to

$$\begin{aligned}
&\frac{1}{2c}+\frac{1}{c^2}\left(1-\frac{c}{2}-\frac{c^2}{6}-\cdots+c^2\left(\frac{1}{2}+\frac{c}{6}+\cdots\right)^2\cdots\right)-\frac{1}{c^2}\\
&=-\frac{1}{6}+\frac{1}{4}=\frac{1}{12}.
\end{aligned}$$

Therefore,

$$\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}.$$

The method can be adapted to deduce that

$$\sum_{n=1}^{\infty} \frac{1}{n^{2k}} \in \pi^{2k}\mathbb{Q}$$

which is a celebrated theorem of Euler.

Another application of the Poisson summation formula is due to Riemann. Applying it to the function $F(t) = e^{-\pi t^2}$, we easily see that if

$$f(t) = e^{-\pi(a+t/\sqrt{x})^2}$$

then

$$\hat{f}(t) = \sqrt{x} e^{2\pi i a t\sqrt{x}} \cdot e^{-\pi t^2 x}.$$

Therefore,

$$\sum_{n=-\infty}^{\infty} e^{-\pi(a+n/\sqrt{x})^2} = \sqrt{x} \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x + 2\pi i a n\sqrt{x}}. \tag{3.6}$$

Riemann used this identity to derive the analytic continuation and functional equation for his famous zeta function:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Riemann's insight was to study $\zeta(s)$ as a function of a complex variable s and relate this study to the distribution of prime numbers. We will discuss this in more detail in the next chapter.

In this connection, it is useful to mention the **heat equation**: let $x \in \mathbb{R}/\mathbb{Z}$ and let $u(x,t)$ denote the heat at x at time t. It is known that $u(x,t)$ satisfies

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2} \tag{$\star$}$$

where κ is a constant (called the conductivity of the ring's material).

Fourier solved this equation by the method of separation of variables, by assuming that

$$u(x,t) = A(x)B(t).$$

Then ($\star$) reduces to

$$A(x)B'(t) = \kappa A''(x)B(t)$$

so that

$$\kappa \frac{A''(x)}{A(x)} = \frac{B'(t)}{B(t)} = \lambda = \text{constant}$$

since the left hand side is a function of x and the right hand side is a function of t.

We therefore get two ordinary differential equations, and they can be solved by classical methods. The general solution is of the form for $A(x)$ is

$$A(x) = e^{2\pi inx}$$

since A has period 1. For $B(t)$, we therefore get

$$B(t) = e^{-4\pi^2 n^2 t}.$$

The general solution is therefore a linear combination of such functions:

$$u(x,t) = \sum_{n=-\infty}^{\infty} a_n e^{-4\pi^2 n^2 t} e^{2\pi inx}.$$

In this way, we see that the general **theta function** appears as the solution to the heat equation.

If in addition, we have an initial condition:

$$u(x,0) = f(x)$$

for some continuous function $f(x)$, then

$$u(x,0) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi inx}$$

must be the Fourier series of f so that

$$a_n = \int_0^1 f(x) e^{-2\pi inx}\, dx = \hat{f}(x).$$

This suggests that if we put

$$\theta_t(x) = \sum_{n\in\mathbb{Z}} e^{-4\pi^2 n^2 t} e^{2\pi inx}$$

then we see that $e^{-4\pi^2 n^2 t}$ is the n-th Fourier coefficient of $\theta_t(x)$ and $a_n e^{-4\pi^2 n^2 t}$ is the n-th Fourier coefficient of $u(x,t)$. As $a_n = \hat{f}(n)$, and $e^{-4\pi^2 n^2 t} = \hat{\theta}_t(n)$, we see that

$$u(x,t) = (f * \theta_t)(x).$$

In other words, the θ-function enters in a fundamental way to resolve the general heat equation.

Exercises

1. Let
$$g(x) = \begin{cases} 1 - |x| \text{ if } |x| \leq 1 \\ 0 \quad \text{otherwise.} \end{cases}$$
Show that
$$\widehat{g}(x) = \left(\frac{\sin \pi x}{\pi x} \right)^2.$$

2. Apply Poisson's summation formula for the function g in the previous exercise to deduce that
$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+x)^2} = \frac{\pi^2}{(\sin \pi x)^2},$$
whenever x is real and unequal to an integer.

3. Show that
$$\sum_{n=-\infty}^{\infty} \frac{1}{n+x} = \frac{\pi}{\tan \pi x}$$
when x is real and unequal to an integer.

3.15 A Fourier Analytic Proof of the Central Limit Theorem

The theory of characteristic functions in probability theory is really the theory of Fourier transforms reformulated in the vernacular of that theory. The central limit theorem plays a pivotal role in probability and statistics. With a long and sinuous journey spanning two centuries, the theorem reached its culmination in the middle of the twentieth century. The purpose of this section is to highlight that indeed, the central limit theorem is a theorem about Fourier transforms. We assume the reader has a minimal background in probability theory so as to hasten our discussion. But the prerequisites are not formidable and we collect the basic ideas in the exercises at the end of this section. The reader may consult these in order to follow the narrative. However, for the sake of clarity, we summarize the dictionary of terms in language of measure theory that is perhaps familiar to the reader of this text.

In measure theory, we have a measure space Ω and measurable sets A. In probability theory, we also have a measure space Ω, often called the **sample space** with total measure 1. Measurable sets A of Ω are called **events**. The measure of A, denoted $P(A)$, is called the **probability** of the event A. A measurable function of Ω is called a **random variable**. Given a random variable X, the **probability distribution** is given by

$$F_X(x) := P(X \leq x).$$

If we can write this as

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt,$$

we call f_X the **density function** of X. Clearly,

$$\int_{-\infty}^{\infty} f_X(x)dx = 1.$$

We say X and Y are **independent** if

$$P(X \leq a; Y \leq b) = P(X \leq a)P(Y \leq b).$$

If X and Y are independent and have density functions f_X and f_Y, then $X + Y$ has density function $f_X * f_Y$ (see exercises).

We say X and Y are **identically distributed** if $f_X = f_Y$. Given a random variable X, its **expectation** denoted $E(X)$ is defined as

$$E(X) := \int_X x f_X(x)dx.$$

Sometimes, one denotes $E(X)$ as μ and calls it the **mean** of X. The **variance** of X, denoted $\text{var}(X)$, is $E((X - \mu)^2)$.

With this terminology and understanding, we are now ready to state and prove the central limit theorem.

Theorem 3.16 (The central limit theorem) *Let $X_1 X_2$, ... be independent and identically distributed random variables with $E(X_i) = 0$ and $\text{var}(X_i) = 1$ Let $S_n = X_1 + X_2 + \cdots + X_n$. Then,*

$$\lim_{n\to\infty} P(\alpha\sqrt{n} \leq S_n \leq \beta\sqrt{n}) = \frac{1}{\sqrt{2\pi}} \int_{\alpha}^{\beta} e^{-t^2/2}dt.$$

Proof We want to determine the density function of $S_n/\sqrt{n}$. Recall that if X has density f_X and Y has density f_Y, then $X + Y$ has density $f_X * f_Y$, provided X and Y are independent. Since all the X_i's are identically distributed with density function f (say) and the X_i's are independent, S_n has density function given by

$$f^{*n} := f * \cdots * f.$$

Now if X has density $f(t)$, λX has density $\lambda^{-1} f(t/\lambda)$ (Exercise). Therefore $S_n/\sqrt{n}$ has density $g^{*n}(t)$ where $g(t) = \sqrt{n} f(\sqrt{n}t)$. We want to show

$$\lim_{n\to\infty} g^{*n}(t) \to \frac{1}{\sqrt{2\pi}} e^{-t^2/2}.$$

Taking Fourier transforms of both sides, this is equivalent to

$$\widehat{g}(t)^n \to e^{-t^2/2}.$$

But $\widehat{g}(t) = \widehat{f}(t/\sqrt{n})$. Taking the Taylor expansion, we have

$$f(t/\sqrt{n}) = \widehat{f}(0) + \widehat{f}'(0)t/\sqrt{n} + \widehat{f}''(0)t^2/2n + O(1/n^{3/2}).$$

Now

$$\widehat{f}(0) = \int_{-\infty}^{\infty} f(t)dt = 1,$$

since f is a probability density function. Also,

$$\widehat{f}'(0) = -i \int_{-\infty}^{\infty} tf(t)e^{-ixt}dt\Big|_{x=0} = \int_{-\infty}^{\infty} tf(t)dt = E(X) = 0$$

with $X = X_i$ by our hypothesis. Moreover, for $X = X_i$,

$$\widehat{f}''(0) = -\int_{-\infty}^{\infty} t^2 f(t)e^{-t^2/2}dt\Big|_{x=0} = -\int_{-\infty}^{\infty} t^2 f(t)dt = -\text{var}(X) = -1.$$

Thus, our Taylor expansion simplifies to

$$1 - t^2/2n + O(1/n^{3/2}).$$

By basic calculus, we immediately see that

$$\lim_{n\to\infty} \left(1 - \frac{t^2}{2n} + O\left(\frac{1}{n^{3/2}}\right)\right)^n = e^{-t^2/2}.$$

This completes the proof. □

Exercises

1. If X and Y are independent random variables with density functions f_X and f_Y, respectively, show that $X + Y$ has density function $f_X * f_Y$.
2. Show that if X has density $f(t)$, λX has density $\lambda^{-1} f(t/\lambda)$.
3. Prove that

$$\lim_{n\to\infty} \sum_{|k-np|\leq a\sqrt{np(1-p)}} \binom{n}{k} p^k (1-p)^{n-k} = \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} e^{-t^2/2} dt.$$

[Hint: choose the random variables to be Bernoulli trials with $\Omega = \{H, T\}$ and independent random variables X_i such that $P(X_i = H) = p$ and apply the central limit theorem.]

4. Prove that

$$\lim_{n\to\infty} e^{-n} \sum_{k=0}^{n} \frac{n^k}{k!} = \frac{1}{2}.$$

[Hint: consider $X_1, ..., X_n$ independent random variables with the Poisson distribution and parameter 1.]

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Chapter 4
Complex Analysis



4.1 Basic Definitions

Let Ω be an open set of the complex plane $\mathbb{C}$. Suppose that $f : \Omega \to \mathbb{C}$ is differentiable. That is, the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists for every $z_0 \in \Omega$. Writing $z = x + iy$, we can decompose $f(z)$ as

$$u(x, y) + iv(x, y)$$

where $u(x, y)$ and $v(x, y)$ are real-valued functions of x and y.

Restricting the limiting process to the two orthogonal directions, namely the x and y directions, we get

$$\lim_{x \to x_0} \frac{f(x + iy_0) - f(x_0 + iy_0)}{x - x_0} = \lim_{y \to y_0} \frac{f(x_0 + iy) - f(x_0 + iy_0)}{i(y - y_0)}.$$

The first limit is

$$\lim_{x \to x_0} \frac{u(x, y_0) + iv(x, y_0) - u(x_0, y_0) - iv(x_0, y_0)}{x - x_0} = \left[\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right]_{x=x_0, y=y_0}.$$

The second limit is

$$-i \lim_{y \to y_0} \frac{u(x_0, y) + iv(x_0, y) - u(x_0, y_0) - iv(x_0, y_0)}{y - y_0} = -i\left[\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right]_{x=x_0, y=y_0}.$$

As these two limits are equal, we have

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$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y},$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}.$$

These are called the **Cauchy–Riemann equations**. We have shown that if f is differentiable in an open set $\Omega \subseteq \mathbb{C}$, then f must satisfy the Cauchy–Riemann equations.

The converse is also essentially true. If $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial y}$ exist, are continuous on Ω and satisfy the Cauchy–Riemann equations, then f is differentiable on Ω. This is easily seen and left as an exercise.

The reader should understand that although the definition of differentiability of f is similar to the usual derivative of a function of a real variable, the complex variable case is much richer. This is underscored by the Cauchy–Riemann equations. For instance, we see that for an analytic function $f(z) = u(x, y) + iv(x, y)$, the functions $u(x, y)$ and $v(x, y)$ of real variables satisfy **Laplace's equation**:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{and} \quad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0.$$

Functions of a real variable that satisfy Laplace's equation are called **harmonic**. Thus, the real and imaginary parts of a differentiable function are harmonic via the Cauchy–Riemann equations. Later, we will see that if f is analytic, u and v are, in addition, infinitely differentiable.

There are many synonyms for the property of differentiability of a function f defined on an open set in the complex plane. One says f is **holomorphic**, or **analytic**, or **regular**, and these words are often used interchangeably.

Given an open set Ω of $\mathbb{C}$, we let $H(\Omega)$ be the set of all holomorphic functions on Ω. The set $H(\Omega)$ forms a ring under pointwise addition and multiplication. Moreover, if $f \in H(\Omega)$ and Ω_1 is an open set containing $f(\Omega)$, then letting $g \in H(\Omega_1)$, we may consider

$$h = g \circ f.$$

We can easily see that $h \in H(\Omega)$ and that the chain rule holds:

$$h' = (g' \circ f) f'.$$

We begin our study of holomorphic functions by first considering power series. A **power series** is a series of the form

$$\sum_{n=0}^{\infty} c_n (z - z_0)^n,$$

where the c_n's are complex numbers.

For such a series, there is a unique number $R \in [0, \infty]$ such that the series converges absolutely and uniformly in the closed disk $\overline{B(z_0, R)}$ where

$$B(z_0, r) = \{z : |z - z_0| < r\},$$

and diverges if $z \notin \overline{B(z_0, R)}$. R is called **the radius of convergence**. By the classical theory of power series, we have various formulas for R:

$$\frac{1}{R} = \limsup_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right|$$

or

$$\frac{1}{R} = \limsup_{n\to\infty} |c_n|^{1/n},$$

the former using the ratio test and the latter from the root test for convergence of series.

A function $f : \Omega \to \mathbb{C}$ is **representable** by a power series in Ω if for every open ball $B(z_0, r) \subseteq \Omega$,

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

for all $z \in B(z_0, r)$.

Theorem 4.1 *If f is representable as a power series in Ω, then $f \in H(\Omega)$ and f' is also representable as a power series in Ω and hence is holomorphic. More precisely, if*

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

for $z \in B(z_0, r)$, then

$$f'(z) = \sum_{n=0}^{\infty} n c_n (z - z_0)^{n-1}$$

for all $z \in B(z_0, r)$.

Proof Without any loss of generality, we may suppose $z_0 = 0$.

Fix $w \in B(0, r) \subseteq \Omega$. Let p be such that

$$|w| < \rho < r.$$

Let

$$g(z) = \sum_{n=1}^{\infty} n c_n z^{n-1}.$$

By the root test for convergence, we see that g also converges for all $z \in \Omega$. Now consider

$$\frac{f(z) - f(w)}{z - w} - g(w) = \sum_{n=1}^{\infty} c_n \left(\frac{z^n - w^n}{z - w} - n w^{n-1} \right)$$

for $z \neq w$. Note that

$$\frac{z^n - w^n}{z - w} - n w^{n-1} = (z - w) \sum_{k=1}^{n-1} k z^{k-1} w^{n-k-1}$$

for all $n \geqslant 2$. To see this, we observe that

$$\lim_{z \to w} \left\{ \frac{z^n - w^n}{z - w} - n w^{n-1} \right\} = 0$$

so that $z - w$ is a factor of the left-hand side. Thus, we may write

$$\frac{z^n - w^n}{z - w} - n w^{n-1} = (z - w) G(z, w),$$

say. Changing z to λz, w to λw shows that

$$G(\lambda z, \lambda w) = \lambda^{n-2} G(z, w).$$

In other words, $G(z, w)$ is a homogeneous polynomial of degree $n - 2$, so we can write

$$G(z, w) = \sum_{i+j=n-2} c_{ij} z^i w^i = \sum_{k=0}^{n-2} \alpha_k z^k w^{n-2-k}.$$

But

$$\frac{z^n - w^n}{z - w} - n w^{n-1} = \sum_{k=0}^{n-1} z^k w^{n-1-k} - n w^{n-1}.$$

We now only need to compare coefficients. That is,

$$(z - w) G(z, w) = \sum_{k=0}^{n-2} \alpha_k z^{k+1} w^{n-2-k} - \sum_{k=0}^{n-2} \alpha_k z^k w^{n-1-k}.$$

We get $\alpha_0 = 1$, $\alpha_i - \alpha_{i-1} = 1$ for $1 \leqslant i \leqslant n - 3$ and finally $-\alpha_{n-2} = 1 - n$. In any case, for $|z| < \rho$,

$$\left|\sum_{k=1}^{n-1} kz^{k-1}w^{n-k-1}\right| < \sum_{k=1}^{n-1} k\rho^{n-2} = \frac{n(n-1)}{2}\rho^{n-2}$$

so that

$$\left|\frac{f(z)-f(w)}{z-w} - g(w)\right| \leqslant |z-w| \sum_{n=2}^{\infty} n^2|c_n|\rho^{n-2}.$$

By the ratio test, the series on the right-hand side converges. Therefore, we may take limits as $z \to w$ to get

$$f'(w) = g(w)$$

as desired. □

Applying the above theorem to f', we deduce $f'' \in H(\Omega)$ and so on. Thus, if f is representable as a power series, then the differentiability of f implies the existence of *all* higher derivatives. We exploit this to prove:

Theorem 4.2 *Suppose μ is a complex finite measure on a measure space X, ϕ is a complex measurable function on X and $\Omega \subseteq \mathbb{C}$ is an open set not intersecting $\phi(X)$. Then,*

$$f(z) := \int_X \frac{d\mu(\zeta)}{\phi(\zeta) - z}$$

is representable as a power series in Ω.

Proof Suppose $B(a, r) \subseteq \Omega$. Since $\Omega \cap \phi(X) = 0$, we have $|\phi(\zeta) - a| \geqslant r$ for any $\zeta \in X$. Thus, for $z \in B(a, r)$, we have

$$|z-a| < r \leqslant |\phi(\zeta) - a|$$

so that

$$\left|\frac{z-a}{\phi(\zeta)-a}\right| < 1.$$

Therefore, the geometric series

$$\frac{1}{\phi(\zeta)-a} \cdot \sum_{n=0}^{\infty} \left(\frac{z-a}{\phi(\zeta)-a}\right)^n$$

converges to

$$\frac{1}{\phi(\zeta)-a} \cdot \frac{1}{1-\frac{z-a}{\phi(\zeta)-a}} = \frac{1}{\phi(\zeta)-z}.$$

So by Fubini's theorem,

$$f(z) := \int_X \frac{1}{\phi(\zeta) - a} \cdot \sum_{n=0}^{\infty} \left(\frac{z-a}{\phi(\zeta) - a} \right)^n d\mu(\zeta) = \sum_{n=0}^{\infty} c_n (z-a)^n$$

where

$$c_n = \int_X \frac{d\mu(\zeta)}{(\phi(\zeta) - a)^{n+1}}.$$

Note that as $|z-a| < r \leqslant |\phi(\zeta) - a|$, we have

$$|c_n| \leqslant \frac{|\mu(X)|}{r^{n+1}}$$

so that our series converges absolutely. □

Exercises

1. Let $f(z) = u(x, y) + iv(x, y)$ with $z = x + iy$. If $\dfrac{\partial u}{\partial x}, \dfrac{\partial v}{\partial x}, \dfrac{\partial u}{\partial y}, \dfrac{\partial v}{\partial y}$ exist, are continuous on Ω and satisfy the Cauchy–Riemann equations, then show that f is differentiable on Ω.
2. Show that the function $f(z) = \overline{z}$ is not analytic.
3. Verify the Cauchy–Riemann equations for $f(z) = z^3$.
4. For $z = x + iy$, define the differential operator:

$$\frac{\partial}{\partial \overline{z}} := \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

Show that $f(z)$ is analytic if and only if

$$\frac{\partial f}{\partial \overline{z}} = 0.$$

5. Show that $f(z) = e^z$ is analytic in $\mathbb{C}$.
6. Let $w \in \mathbb{C}$ be a fixed complex number with $|w| < 1$. Let

$$f(z) = \frac{z - w}{1 - \overline{w} z}.$$

Show that f is regular in $|z| \leq 1$ and calculate $f(w)$ and $f'(w)$.
7. Find the radius of convergence of

$$\sum_{n=1}^{\infty} n! \frac{z^n}{n^n}.$$

8. Prove the following discrete version of integration by parts (often referred to as **partial summation**): for any two sequences of numbers a_n, b_n,

$$S_N := \sum_{n=1}^{N} a_n b_n = a_N B_N - \sum_{n=1}^{N-1} B_n (a_{n+1} - a_n),$$

where

$$B_n = \sum_{j=1}^{n} b_j.$$

Using this result, show that the power series

$$\sum_{n=1}^{\infty} \frac{z^n}{n}$$

converges for every complex number z with $|z| = 1$ and $z \neq 1$.

9. Show that the series

$$\sum_{n=1}^{\infty} \frac{z^n}{n(n+1)}$$

converges for all $|z| \leq 1$. What is its radius of convergence? Does the series

$$\sum_{n=1}^{\infty} z^n$$

converge for any z with $|z| = 1$?

10. Let n be a non-negative integer. Define for $\alpha \in \mathbb{C}$, the binomial coefficient

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}.$$

Show that the series

$$\sum_{n=0}^{\infty} \binom{\alpha}{n} z^n$$

converges for $|z| < 1$.

11. Let A be an open set in $\mathbb{C}$, and suppose that $f : A \to \mathbb{C}$ is analytic. If $z_0 \in A$ such that $f'(z_0) \neq 0$, show that there are open sets U of z_0 and V of $f(z_0)$ such that $f : U \to V$ is a bijection. Moreover, $f^{-1}(z)$ is analytic with

$$\frac{d}{dw} f^{-1}(w) = \frac{1}{f'(z)}, \qquad \text{where } w = f(z).$$

(This is the complex analytic version of the inverse function theorem of real variables.) [Hint: Apply the inverse function theorem from real variables along with the Cauchy–Riemann equations.]

4.2 Integration over Paths

We have shown that any function representable by a power series in some open subset of $\mathbb{C}$ is holomorphic there. Our next goal is to show that any holomorphic function has a power series representation.

Definition 4.1 A **curve** in a topological space X is a continuous map $\gamma : [\alpha, \beta] \to X$. We let γ^* be the image of γ. If $X = \mathbb{C}$ and the curve γ is piecewise C^1, then we call γ a **path**. A **closed** path is a path $\gamma : [\alpha, \beta] \to X$ such that $\gamma(\alpha) = \gamma(\beta)$.

Definition 4.2 If $\gamma : [\alpha, \beta] \to \mathbb{C}$ is a path and $f : \mathbb{C} \to \mathbb{C}$ is continuous on γ^*, we define the **path integral** of f as

$$\int_\gamma f = \int_\gamma f(z)dz = \int_\alpha^\beta f(\gamma(t))\gamma'(t)\,dt.$$

Note that

$$\left|\int_\gamma f\right| \leqslant \left(\max_{z\in\gamma^*} |f(z)|\right) \int_\alpha^\beta |\gamma'(t)|\,dt,$$

using the formula for the length of the path.

For any $a, b \in \mathbb{C}$, we denote by $[a, b]$ the curve parametrized by $\gamma : [0, 1] \to \mathbb{C}$ given by $\gamma(t) = a + (b - a)t$. Thus for a triangle $\triangle$ with vertices (a, b, c) in $\mathbb{C}^3$ and orientation of the boundary $\partial\triangle$ given by the order of the vertices, we have

$$\int_{\partial\triangle} f = \int_{[a,b]} f + \int_{[b,c]} f + \int_{[c,a]} f.$$

Note that this integral is invariant under cyclic permutations of the vertices (a, b, c).

Throughout, Ω is an open set in $\mathbb{C}$.

Definition 4.3 We define the **index** of z with respect to a closed path γ by

$$\mathrm{Ind}_\gamma(z) := \frac{1}{2\pi i} \int_\gamma \frac{dw}{w - z}$$

for $z \in \mathbb{C} \setminus \gamma^* = \Omega$.

We will show that this is always an integer. Intuitively, the index $\mathrm{Ind}_\gamma(z)$ measures how many times γ "winds around z." Thus, sometimes, it is also called the

winding number. Note that γ^* is compact and hence lies in a bounded disk D whose complement D^c is connected and hence lies in precisely one unbounded connected component.

Theorem 4.3 *Let γ be a closed path, and let Ω be the complement of γ^*. Then, $\mathrm{Ind}_\gamma(z) \in \mathbb{Z}\ \forall z \in \Omega$ and is constant on any connected component of Ω. Moreover, the index is zero on the unbounded component of Ω.*

Proof By our previous theorem, $\mathrm{Ind}_\gamma(z)$ is holomorphic (as it is representable as a power series) and hence continuous. Thus, constancy on any connected component will follow once we show it is integer-valued.

Let $\gamma : [\alpha, \beta] \to \gamma^*$ and fix $z \in \Omega$. Then,

$$\mathrm{Ind}_\gamma(z) = \frac{1}{2\pi i} \int_\alpha^\beta \frac{\gamma'(t)\,dt}{\gamma(t) - z}.$$

Observe that $\mathrm{Ind}_\gamma(z) \to 0$ as $|z| \to \infty$ so that once we show that $\mathrm{Ind}_\gamma(z)$ is integer-valued, it will follow that it is zero on the unbounded component. To show that $\mathrm{Ind}_\gamma(z)$ is integer-valued, recall that

$$e^w = 1 \Leftrightarrow w = 2\pi i n \quad \text{for some integer } n.$$

To see this, write $w = u + iv$, $u, v \in \mathbb{R}$. Then as

$$e^w = e^u e^{iv} = e^u(\cos v + i \sin v),$$

we see that $e^w = 1$ implies $u = 0$ and $v \in 2\pi\mathbb{Z}$.

Let

$$\phi(s) = \exp\left(\int_\alpha^s \frac{\gamma'(t)\,dt}{\gamma(t) - z}\right)$$

for $\alpha \leqslant s \leqslant \beta$. Differentiating, we get

$$\phi'(s) = \frac{\gamma'(s)}{\gamma(s) - z} \exp\left(\int_\alpha^s \frac{\gamma'(t)\,dt}{\gamma(t) - z}\right).$$

Thus,

$$\frac{\phi'(s)}{\phi(s)} = \frac{\gamma'(s)}{\gamma(s) - z}$$

which is valid except possibly on a finite set S where γ is not differentiable. Everywhere else, $\phi'(s)$ is continuous. Thus,

$$\frac{\phi'(s)}{\gamma'(s)} = \frac{\phi(s)}{\gamma(s) - z} \qquad (*)$$

is continuous on $[\alpha, \beta] \setminus S$.

Now,

$$\frac{d}{ds}\left(\frac{\phi(s)}{\gamma(s)-z}\right) = \frac{(\gamma(s)-z)\phi'(s) - \phi(s)\gamma'(s)}{(\gamma(s)-z)^2} = 0$$

by virtue of $(*)$.

Thus, $\dfrac{\phi(s)}{\gamma(s)-z}$ is constant on $[\alpha, \beta] \setminus S$.

But by continuity, this function is constant on all of $[\alpha, \beta]$ because $z \notin \gamma^*$.

Thus,

$$\frac{\phi(s)}{\gamma(s)-z} = \frac{\phi(\alpha)}{\gamma(\alpha)-z} = \frac{1}{\gamma(\alpha)-z}$$

as $\phi(\alpha) = 1$. Therefore,

$$\phi(s) = \frac{\gamma(s)-z}{\gamma(\alpha)-z}.$$

But γ is closed, so $\gamma(\alpha) = \gamma(\beta)$. Thus, $\phi(\alpha) = \phi(\beta) = 1$. Therefore, $\text{Ind}_\gamma(z)$ is integer-valued. □

Corollary 4.2.1 *If γ is a positively oriented circle with center a and radius r, then*

$$\text{Ind}_\gamma(z) = \begin{cases} 1, \text{ if } |z-a| < r; \\ 0, \text{ if } |z-a| > r. \end{cases}$$

Proof We need to only show $\text{Ind}_\gamma(z) = 1$ for $|z-a| < r$. Parametrize $\gamma : [0, 2\pi] \to \mathbb{C}$ by $\gamma(\theta) = a + re^{i\theta}$. Then as $\text{Ind}_\gamma(z)$ is constant for $|z-a| < r$,

$$\text{Ind}_\gamma(z) = \text{Ind}_\gamma(a) = \frac{1}{2\pi i}\int_0^{2\pi} \frac{ire^{i\theta}\,d\theta}{a + re^{i\theta} - a} = \frac{1}{2\pi i}\int_0^{2\pi} i\,d\theta = 1.$$

The corollary makes complete sense from our intuitive understanding of the index as a winding number.

Exercises

1. Let γ be a path. Show that the function of α defined by

$$\alpha \mapsto \int_\gamma \frac{dz}{z-\alpha}$$

for α not on the path is a continuous function of α.

2. If γ is a closed path and $z \notin \gamma^*$, show that

$$\frac{1}{2\pi i}\int_\gamma \frac{w\,dw}{w-z} = z\mathrm{Ind}_\gamma(z).$$

3. Consider the curve γ given by $\gamma(t) = (\cos t, 3\sin t)$, $0 \le t \le \pi$. Show that $I_\gamma(0) = 2$.
4. If γ is the unit circle oriented counterclockwise, show that

$$\int_\gamma \frac{\cos z}{z}dz = \int_\gamma \frac{\sin z}{z^2}dz = 2\pi i.$$

5. Prove that for a closed curve γ, we have $I_\gamma(z) = -I_{-\gamma}(z)$.
6. If γ_1, γ_2 are closed curves, show that $I_{\gamma_1+\gamma_2}(z) = I_{\gamma_1}(z) + I_{\gamma_2}(z)$.
7. Prove that

$$\int_0^\pi e^{\cos\theta}\cos(\sin\theta)\,d\theta = \pi,$$

by considering

$$\int_\gamma \frac{e^z}{z}dz$$

and γ is the unit circle oriented counterclockwise.
8. Prove that if the image of a closed curve γ lies in a simply connected region A and if $z \notin A$, then $I_\gamma(z) = 0$.

4.3 The Local Cauchy Theorem

By a **region**, we mean a non-empty connected open subset of the complex plane. The next theorem is a simple extension of the fundamental theorem of calculus to holomorphic functions.

Theorem 4.4 *If $F \in H(\Omega)$ and F' is continuous in Ω, then*

$$\int_\gamma F'(z)\,dz = F(\gamma(\beta)) - F(\gamma(\alpha))$$

for every path $\gamma : [\alpha, \beta] \to \Omega$.

Proof By the fundamental theorem of calculus,

$$\int_\gamma F'(z)\,dz = \int_\alpha^\beta F'(\gamma(t))\gamma'(t)\,dt = F(\gamma(\beta)) - F(\gamma(\alpha)).$$

□

Corollary 4.3.1 *If F is as in the theorem,*

$$\int_\gamma F'(z)\,dz = 0$$

for every closed path γ in Ω.

Corollary 4.3.2 *For any closed path γ,*

$$\int_\gamma z^n\,dz = 0$$

for all $n \geqslant 0$; this also holds for $n \leqslant -2$ if $0 \notin \gamma^*$.

We are now ready to prove what is sometimes called the Cauchy–Goursat theorem:

Theorem 4.5 (Cauchy's theorem for a triangle). *Let $\triangle$ be a triangle in the open set $\Omega \subseteq \mathbb{C}$, and let $p \in \Omega$. Suppose f is* continuous *on Ω and $f \in H(\Omega \setminus \{p\})$. Then,*

$$\int_{\partial\triangle} f(z)\,dz = 0.$$

Proof First assume $p \notin \triangle$. Let (a, b, c) be the oriented vertices of $\triangle$ (see Fig. 4.1), and let a', b', c' be the corresponding opposite midpoints (see Fig. 4.2).

Now consider the four oriented triangles $\triangle_1, \triangle_2, \triangle_3 and \triangle_4$ formed by joining these midpoints (a, c', b'), (b, a', c'), (c, b', a') and (a', b', c'), respectively. Then,

$$J := \int_{\partial\triangle} f(z)\,dz = \sum_{j=1}^{4} \int_{\partial\triangle_j} f(z)\,dz.$$

The orientation is important since the "internal" line integrals are canceled.

For one of the triangles, call it $\triangle^1$, we have

$$\left| \int_{\partial\triangle^1} f(z)\,dz \right| \geqslant \frac{|J|}{4}.$$

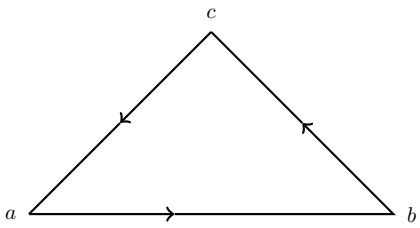


Fig. 4.1 Oriented triangle abc

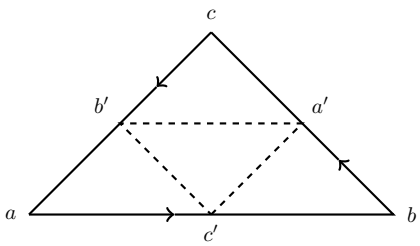


Fig. 4.2 Oriented triangles abc and $a'b'c'$

Repeating this process inductively, we get a sequence $\triangle^n$ of nested triangles such that the length of $\partial\triangle^n$ is $L/2^n$, where L is the length of $\partial\triangle$. Moreover,

$$|J| \leqslant 4^n \left| \int_{\partial\triangle^n} f(z)\, dz \right|.$$

Let $z_0 \in \bigcap_{n=1}^{\infty} \triangle^n$. We see that $z_0 \in \triangle$, $z_0 \neq p$. Thus, for any $\varepsilon > 0$, there is an $r > 0$ such that

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leqslant \varepsilon |z - z_0|$$

for $|z - z_0| < r$, because f is holomorphic in $\Omega \setminus \{p\}$.

But for n large, $|z - z_0| < r$ for $z \in \triangle^n$. Moreover, for such n, we have $|z - z_0| < L/2^n$.

By an earlier corollary,

$$\int_{\partial\triangle^n} f(z)\, dz = \int_{\partial\triangle^n} \left\{ f(z) - f(z_0) - f'(z_0)(z - z_0) \right\} dz$$

so that

$$\left| \int_{\partial\triangle^n} f(z)\, dz \right| \leqslant \varepsilon \int_{\partial\triangle^n} |z - z_0|\, dz \leqslant \frac{L\varepsilon}{2^n} \cdot \frac{L}{2^n}$$

for $z \in \triangle^n$. Thus,

$$|J| \leqslant 4^n \cdot \frac{L^2 \varepsilon}{4^n} = \varepsilon L^2.$$

Since ε was arbitrary, $J = 0$.

In the case that p is a vertex of $\triangle$, choose two points of $\partial\triangle$ very close to p and join with each other and with p, to form a triangle (see Fig. 4.3).

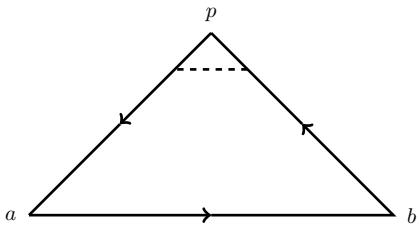


Fig. 4.3 Oriented triangle *abp*

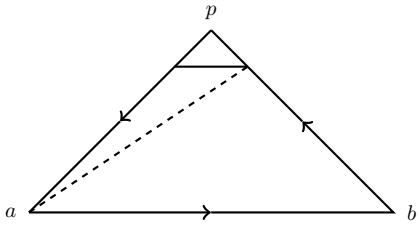


Fig. 4.4 Oriented triangle *abp*

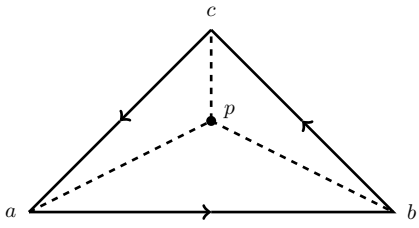


Fig. 4.5 Case when p lies inside the triangle

This splits $\triangle$ into a very small triangle containing the vertex p and a larger non-triangular region. This larger non-triangular region can be cut into two triangular regions, neither of which contain p (see Fig. 4.4).

As before, the integral over $\triangle$ is the oriented sum of these integrals over these three triangles. By our initial case, the integral over the other two integrals is zero. So, the integral over $\triangle$ is simply the integral over the remaining small triangle that contains the vertex p. Since the triangle can be made arbitrarily small (as f is continuous on Ω and hence bounded), this integral is zero.

Finally, in the general case that p lies in the interior of the triangle, join lines from the vertices of $\triangle$ to p, so as to split $\triangle$ into three triangles, each having p as a vertex (see Fig. 4.5).

By the previous calculation, the integral is again zero. □

Theorem 4.6 (Cauchy's theorem for a convex set). *Let Ω be a convex open set. Let $p \in \Omega$ and suppose f is continuous on Ω and $f \in H(\Omega \setminus \{p\})$. Then, $f = F'$ for some $F \in H(\Omega)$. Therefore,*

$$\int_\gamma f(z)\,dz = 0$$

for any closed path γ in Ω.

Proof Let $[a, z]$ denote the oriented line segment joining a to z. Fixing $a \in \Omega$, the convexity of Ω allows us to define

$$F(z) = \int_{[a,z]} f(\xi)\, d\xi$$

for all $z \in \Omega$. Now for $z, z_0 \in \Omega$, let $\triangle$ be the triangle in Ω with vertices (a, z_0, z). Then,

$$0 = \int_{\partial\triangle} f(\xi)\, d\xi = \int_{[a,z_0]} f(\xi)\, d\xi + \int_{[z_0,z]} f(\xi)\, d\xi + \int_{[z,a]} f(\xi)\, d\xi.$$

The convexity of Ω implies that the triangle lies in Ω because $a, z_0, z \in \Omega$. By our definition of $F(z)$, we see this is

$$0 = F(z_0) - F(z) + \int_{[z_0,z]} f(w)\, dw$$

so that for $z \neq z_0$

$$\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0} \int_{[z_0,z]} [f(w) - f(z_0)]\, dw.$$

Now f is continuous in Ω so that given $\varepsilon > 0$, $\exists \delta > 0$ such that

$$|f(w) - f(z_0)| < \varepsilon \text{ when } |w - z_0| < \delta.$$

Hence,

$$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| \leqslant \varepsilon \text{ for } |z - z_0| < \delta.$$

This proves that $f = F'$. That is, $F \in H(\Omega)$. By Theorem 4.4, the result follows. □

Theorem 4.7 (Cauchy's formula for a convex set) *Suppose γ is a closed path in a convex open set Ω and $f \in H(\Omega)$. If $z \in \Omega$ and $z \notin \gamma^*$, then*

$$f(z)\mathrm{Ind}_\gamma(z) = \frac{1}{2\pi i} \int_\gamma \frac{f(w)dw}{w - z}.$$

Proof Defining

$$g(w) = \begin{cases} \dfrac{f(w) - f(z)}{w - z}, & \text{for } w \in \Omega,\, w \neq z \\ \\ f'(z), & \text{if } w = z, \end{cases}$$

we see that $g \in H(\Omega \setminus \{z\})$ because f is holomorphic. Also, g is continuous in Ω including z. We can therefore apply the previous theorem to deduce

$$\int_\gamma g(w)\, dw = 0, \quad \text{for any close path } \gamma.$$

Rearranging what this means gives us the result. □

Theorem 4.8 *If $\Omega \subseteq \mathbb{C}$ is open, then any $f \in H(\Omega)$ is representable as a power series in Ω.*

Proof Let γ be a positively oriented circle in an open ball $B \subseteq \Omega$, centered at a and radius $r < R =$ radius of B. The convexity of the ball allows us to apply the previous theorem. As the $\text{Ind}_\gamma(z) = 1$, we see

$$f(z) = \frac{1}{2\pi i} \int_\gamma \frac{f(w) dw}{w - z}$$

for $z \in B(a, r)$. By our earlier theorem, this can be written as a power series.

Corollary 4.3.3 *If $f \in H(\Omega)$, then $f' \in H(\Omega)$.*

The Cauchy theorem has a useful converse.

Theorem 4.9 (Morera's theorem). *If $f : \Omega \to \mathbb{C}$ is continuous and*

$$\int_{\partial \triangle} f(z)\, dz = 0$$

for every closed triangle $\triangle \subseteq \Omega$, then $f \in H(\Omega)$.

Proof Let V be a convex open set in Ω. As in the proof of Cauchy's theorem for a convex set, we can construct

$$F(z) = \int_{[z_0, z]} f(w)\, dw$$

and again by continuity of f, we see $F(z)$ is holomorphic with $F'(z) = f(z)$. By Corollary 4.3.3, $f \in H(\Omega)$. □

Exercises

1. Evaluate
$$\int_\gamma \frac{z^2}{z-1} dz$$
where γ is a circle of radius 2 centered at 0.
2. Evaluate
$$\int_\gamma \frac{e^z}{z^2} dz$$
where γ is the unit circle.
3. Evaluate
$$\int_\gamma \frac{z^2-1}{z^2+1} dz$$
where γ is a circle of radius 2 centered at 0.

4.4 Zeros and Singularities

In earlier sections, we studied how a holomorphic function f can be expanded as a power series. In this section, we want to study the nature of the set of zeros of f. To what extent can the study of power series representations be extended to functions that are not holomorphic?

Theorem 4.10 *Suppose Ω is a region, $f \in H(\Omega)$ and $Z(f) = \{a \in \Omega : f(a) = 0\}$. Then, either $Z(f) = \Omega$ or $Z(f)$ has no limit point in Ω, in which case for each $a \in Z(f)$, there is a unique positive integer $m = m(a)$ such that $f(z) = (z-a)^m g(z)$ for $z \in \Omega$, and $g \in H(\Omega)$ with $g(a) \neq 0$. Moreover, in this case, $Z(f)$ is at most countable.*

Remark 4.1 The number $m = m(a)$ is called the **order** of f at a.

Proof Let A be the set of limit points of $Z(f)$ in Ω. Then, by the continuity of f, we have $f|_A = 0$. Thus, $A \subseteq Z(f)$. Fix $a \in Z(f)$ and take $r > 0$ with $B(a, r) \subseteq \Omega$ for which we can write

$$f(z) = \sum_{n=0}^{\infty} c_n (z-a)^n \quad \forall z \in B(a, r).$$

Suppose that not *all* c_n are zero. Then, there is a smallest m such that $c_m \neq 0$. As $f(a) = 0$, we have $m > 0$. Define

$$g(z) = \begin{cases} (z-a)^{-m} f(z), & \forall z \in \Omega \setminus \{a\} \\ c_m, & \text{if } z = a \end{cases}$$

so that $f(z) = (z-a)^m g(z)$. Also, $g \in H(\Omega \setminus \{a\})$. But the power series representation of f gives

$$g(z) = \sum_{k=0}^{\infty} (z-a)^k c_{m+k}$$

for $z \in B(a,r)$. Hence, $g \in H(B(a,r))$. Therefore, $g \in H(\Omega)$.

Finally, $g(a) \neq 0$. Therefore, by continuity, a is an isolated point in the case that not all the c_n's are zero. What happens if all the c_n's are zero? In this case, $f \equiv 0$ in the neighborhood of a. Thus, A is open. But A is the set of limit points and hence closed so $\Omega \setminus A$ is open. Thus, Ω is the disjoint union of open sets. But Ω is a connected set, so $A = \Omega$ or $A = \varnothing$. If $A = \Omega$, then $Z(f) = \Omega$ and $f = 0$. If $A = \varnothing$, then $Z(f)$ has at most finitely many points in any compact subset. Since Ω can be covered by a countable union of compact sets, $Z(f)$ is at most countable. □

Corollary 4.4.1 *Suppose Ω is a region and $f, g \in H(\Omega)$. If $f(z) = g(z)$ in some set with a limit point in Ω, then $f(z) = g(z)\ \forall z \in \Omega$.*

Proof We apply the theorem to $f - g$. □

An important consequence of this theorem is that if $f \in H(\Omega), a \in \Omega$ and $B(a,r)$ is the largest open disk centered at a and contained in Ω, then the power series representation of f converges in $B(a,r)$.

Definition 4.4 If $a \in \Omega$ and $f \in H(\Omega \setminus \{a\})$, then f is said to have an **isolated singularity** at a. If f has an isolated singularity at a and there is a function $g \in H(\Omega)$ such that $f(z) = g(z)\ \forall z \in \Omega \setminus \{a\}$, then f is said to have a **removable singularity**.

Whenever we encounter a function with a removable singularity, we often replace it with its analytic extension.

Let us define $B^o(a,r) := B(a,r) \setminus \{a\}$ which is the **punctured disk** of radius r centered at a. The following theorem is often called Riemann's removable singularity theorem.

Theorem 4.11 *Suppose $f \in H(\Omega \setminus \{a\})$ and f is bounded in $B^o(a,r)$ for some $r > 0$. Then, f has a removable singularity.*

Proof Define

$$h(z) = \begin{cases} (z-a)^2 f(z), & \forall z \in \Omega \setminus \{a\} \\ \\ 0, & \text{if } z = a. \end{cases}$$

It is clear that $h \in H(\Omega \setminus \{a\})$. To show $h \in H(\Omega)$, we compute

$$\lim_{z \to a} \frac{h(z) - h(a)}{z - a} = \lim_{z \to a} \frac{(z-a)^2 f(z)}{z - a} = \lim_{z \to a} (z-a) f(z) = 0$$

since f is bounded in $B^o(a,r)$. Thus, $h \in H(\Omega)$. So, we can write

$$h(z) = \sum_{n=2}^{\infty} c_n (z-a)^n$$

because $h(a) = h'(a) = 0$. This suggests we define $f(a) = c_2$ to get

$$f(z) = \sum_{n=0}^{\infty} c_{n+2} (z-a)^n.$$

This gives the desired holomorphic extension of f to Ω. □

The following theorem classifies the possible types of isolated singularities.

Theorem 4.12 *If $a \in \Omega$, $f \in H(\Omega \setminus \{a\})$, then one of the following occurs:*

(1) f has a removable singularity.
(2) There are complex numbers $c_1, \ldots, c_m$ with $m > 0$ and $c_m \neq 0$ such that

$$f(z) - \sum_{k=1}^{m} \frac{c_k}{(z-a)^k}$$

has a removable singularity at a.
(3) If $r > 0$ and $B(a, r) \subseteq \Omega$, then $f(B^o(a, r))$ is dense in the plane.

Remark 4.2 In case of (3), f is said to have an **essential singularity** at a. If (2) holds, f is said to have a **pole of order** m at a. In this case,

$$Q_a(z) := \sum_{k=1}^{m} \frac{c_k}{(z-a)^k}$$

is called the **principal part** of f at a.

Since the function

$$g(z) = f(z) - Q_a(z)$$

is holomorphic for $z \in \Omega \setminus \{a\}$, it can be expanded as a power series

$$\sum_{k=0}^{\infty} c_k (z-a)^k$$

about a, and we get the **Laurent series expansion**

$$f(z) = \sum_{k=-m}^{\infty} c_k (z-a)^k$$

for f about a.

Proof Suppose (3) does not hold. Then, there is an $r > 0$, $\delta > 0$ and $w \in \mathbb{C}$ such that $|f(z) - w| > \delta$ for *all* $z \in B^o(a, r)$.

Then defining

$$g(z) = \frac{1}{f(z) - w} \quad \text{for } z \in B^o(a, r),$$

we see $g \in H(B^o(a, r))$. Moreover, g is bounded in this punctured disk as

$$|g(z)| = \frac{1}{|f(z) - w|} < \frac{1}{\delta}.$$

Hence, by our previous theorem g extends to a holomorphic function in $B(a, r)$. There are now two cases to consider.

Case 1. $g(a) \neq 0$. Then, there exists $c > 0$ such that $|g(z)| \geqslant c$ for all $z \in \overline{B(a, r_1)}$ with $0 < r_1 < r$. Thus,

$$|f(z) - w| \leqslant \frac{1}{c} \quad \forall z \in \overline{B(a, r_1)}.$$

Hence, f is bounded, and again by our previous theorem, f has a removable singularity at a. So (1) holds.

Case 2. $g(a) = 0$. Then, we can write

$$g(z) = (z - a)^m g_1(z)$$

for some $g_1 \in H(B(a, r))$ with $g_1(a) \neq 0$. Thus, $h(z) = 1/g_1(z)$ is holomorphic in $B(a, r_2)$ for some $0 < r_2 < r$ and h has no zeros in $B(a, r_2)$. Arguing as before, we get that $g_1 \in H(B(a, r_2))$. Therefore,

$$f(z) - w = \frac{1}{g(z)} = (z - a)^{-m} \frac{1}{g_1(z)} = (z - a)^{-m} h(z)$$

for all $z \in B^o(a, r_2)$. As h is holomorphic at a, we can write

$$h(z) = \sum_{n=0}^{\infty} b_n (z - a)^n$$

where $b_0 \neq 0$, because $h(a) \neq 0$. Thus,

$$f(z) - w = (z - a)^{-m} \sum_{n=0}^{\infty} b_n (z - a)^n$$

and the case (2) holds.

Example 4.1 The function $f(z) = e^{1/z}$ is holomorphic in $\mathbb{C} \setminus \{0\}$ and has an **essential singularity** at $z = 0$.

We saw above how two holomorphic functions that agree on a set with limit points must be equal. A similar statement can be made about functions with poles. Suppose f and g are holomorphic in Ω except for a pole at a of order n. Then, $(z-a)^n f(z)$ and $(z-a)^n g(z)$ have removable singularities at a so there are functions $f_1, g_1 \in H(\Omega)$ such that $(z-a)^n f(z) = f_1(z)$ and $(z-a)^n g(z) = g_1(z)$ $\forall z \in \Omega \setminus \{a\}$. Both of these functions agree for all z in $\Omega \setminus \{a\}$ and so $f(z) = g(z)$ for all $z \in \Omega \setminus \{a\}$.

Exercises

1. Find the Laurent series expansion of

$$\frac{z+1}{z}$$

about $z = 0$ and $z = 1$.

2. Find the Laurent series expansion of

$$\frac{z}{z^2+1}$$

about $z = i$.

3. Suppose that $a \neq 0$. Write down the Laurent expansion of

$$\frac{1}{z-a}$$

about $z = 0$. If k is a natural number, what about the expansion of $1/(z-a)^k$ at $z = 0$?

4. Show that the function $(\sin z)/z$ has a removable singularity at $z = 0$.
5. Show that the function e^z/z does not have a removable singularity at $z = 0$.
6. Show that $z/(e^z - 1)$ has a removable singularity at $z = 0$.
7. Find the Laurent expansion of

$$f(z) = \frac{1}{z(z^2+1)}$$

that is valid for

(a) $0 < |z| < 1$.
(b) $|z| > 1$.

8. Suppose that $f(z)$ has an essential singularity at z_0. Let $w \in \mathbb{C}$. Show that there exists a sequence

$$z_1, z_2, \ldots$$

such that $z_n \to z_0$ and $f(z_n) \to w$. (This is called the **Casorati–Weierstrass theorem**. A more general theorem due to E. Picard was proved in 1879: If f has an essential singularity at z_0 and U is an arbitrarily small punctured ball around z_0, then given $w \in \mathbb{C}$, the equation $f(z) = w$ has infinitely many solutions in U, except perhaps for at most one value of w. This is called **Picard's theorem**. The reader can find a proof in the author's problem book on modular forms [1] which uses the theory of the j-function.)

4.5 The Maximum Modulus Principle

Sometimes called the **maximum principle**, the maximum modulus principle is one of the most important theorems in the theory of complex functions. We derive it below. The following theorem representing a nice fusion of Fourier series and complex analysis will serve as a catalyst for a collection of results leading to the maximum principle.

Theorem 4.13 *If*

$$f(z) = \sum_{n=0}^{\infty} c_n (z-a)^n$$

for all $z \in B(a, R)$ *and* $0 < r < R$, *then*

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} = \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{i\theta})|^2 \, d\theta.$$

Proof We have

$$f(a + re^{i\theta}) = \sum_{n=0}^{\infty} c_n r^n e^{in\theta}$$

, and the series is an absolutely convergent Fourier series. Thus, by Parseval's formula

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} = \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{i\theta})|^2 \, d\theta.$$

□

The following theorem of Liouville is one of the central theorems of complex analysis.

Corollary 4.5.1 (Liouville's theorem). *A bounded entire function is constant.*

Proof Suppose f is entire and $|f(z)| < M$ for all $z \in \mathbb{C}$. Writing

$$f(z) = \sum_{n=0}^{\infty} c_n z^n$$

for the power series expansion about $z = 0$, we see that the series converges for all $z \in \mathbb{C}$ since f is entire. By the theorem,

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} = \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{i\theta})|^2 \, d\theta < M^2.$$

If we let $r \to \infty$, we get a contradiction unless $c_1 = c_2 = \cdots = 0$. Thus, f is constant. □

Theorem 4.14 (The maximum modulus principle). *If Ω is a region and $f \in H(\Omega)$, then*

$$\max_{z \in \Omega} |f(z)| = \max_{z \in \partial\Omega} |f(z)|.$$

Moreover, if the maximum is attained at an interior point, then f is constant.

Proof Suppose that the maximum is attained at an interior point $a \in \Omega$. Then, there is an $r > 0$ such that $\overline{B(a, r)} \subseteq \Omega$ and

$$f(a) = \max_{z \in \Omega} |f(z)|.$$

Then, $|f(a + re^{i\theta})| \leqslant |f(a)|$ for all $\theta \in [0, 2\pi]$. Thus, writing

$$f(z) = \sum_{n=0}^{\infty} c_n (z - a)^n \quad \forall z \in B(a, r),$$

we see that

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} = \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{i\theta})|^2 \, d\theta \leqslant |f(a)|^2 = |c_0|^2,$$

which is a contradiction unless $c_1 = c_2 = \cdots = 0$. □

The minimum modulus theorem given below is an immediate consequence of the maximum principle.

Theorem 4.15 (The minimum modulus theorem). *If Ω is a region, $f \in H(\Omega)$ and $f(z) \neq 0$ for all $z \in \Omega$, then*

$$\min_{z\in\Omega} |f(z)| = \min_{z\in\partial\Omega} |f(z)|.$$

Moreover, if f is non-constant, then its minimum modulus cannot be attained at an interior point.

Proof Since $f(z) \neq 0$ for all $z \in \Omega$, we have $1/f \in H(\Omega)$. Applying the maximum modulus principle to $1/f$ yields the result. □

An important consequence of the maximum principle is the following proof of what has been called the fundamental theorem of algebra. Some mathematicians have expressed (fundamentalist) annoyance that all known proofs use some form of analysis!

Theorem 4.16 (The fundamental theorem of algebra). *If $f(z) \in \mathbb{C}[z]$ has degree $n \geqslant 1$, then there exist unique $\alpha_1, \ldots, \alpha_n \in \mathbb{C}$ such that*

$$f(z) = A(z - \alpha_1) \cdots (z - \alpha_n).$$

Proof Assume without loss of generality that f is monic:

$$f(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0.$$

Let $M = \max(|a_0|, \ldots, |a_{n-1}|)$. Then, for $|z| \geqslant \max(1, 2nM)$, we have

$$|f(z) - z^n| \leqslant |a_{n-1}||z|^{n-1} + \cdots + |a_1||z| + |a_0| \leqslant nM|z|^{n-1} \leqslant \frac{1}{2}|z|^n$$

since $|z| \geqslant 1$ and $|z| \geqslant 2nM$. Thus, for such z,

$$\frac{1}{2}|z|^n \geqslant |z|^n - |f(z)|$$

so that $|f(z)| \geqslant \frac{1}{2}|z|^n \geqslant \frac{1}{2}$, because $|z| \geqslant 1$. Now, if f has no zeros, then $1/f$ is entire and $\left|\frac{1}{f(z)}\right| \leqslant 2$ for $|z|$ sufficiently large, say $|z| \geqslant R$. Thus, $\frac{1}{f(z)}$ is bounded and entire. By Liouville's theorem, f is constant. Thus, f has a zero. Now by the division algorithm,

$$f(z) = (z - \alpha_1) f_1(z)$$

where $f_1(z)$ has degree $n - 1$. Applying induction completes the proof. □

The following theorem gives an estimate for coefficients of the power series representation of an analytic function.

Theorem 4.17 (Cauchy's estimates). *If $f \in H(\overline{B(a, R)})$ and $|f(z)| \leqslant M$ for $z \in B(a, R)$, then*

$$\frac{|f^{(n)}(a)|}{n!} \leqslant \frac{M}{R^n}.$$

Proof We have

$$f(z) = \sum_{n=0}^{\infty} c_n (z-a)^n$$

for $z \in B(a, R)$ where $c_n = \dfrac{f^{(n)}(a)}{n!}$. But

$$\sum_{n=0}^{\infty} |c_n|^2 R^{2n} = \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{i\theta})|^2 \, d\theta \leqslant M^2$$

so that

$$|c_n| \leqslant \frac{M}{R^n}.$$

□

The Schwarz lemma is an important tool for many results in geometric complex analysis.

Theorem 4.18 (Schwarz's lemma). *Let $f(z)$ be analytic in $|z| \leqslant 1$. Suppose that $f(0) = 0$, and $|f(z)| \leqslant M$ for all $|z| \leqslant 1$. Then,*

$$|f(z)| \leqslant M|z|$$

for all $|z| \leqslant 1$.

Proof Let

$$g(z) = \begin{cases} f(z)/z, & \text{for } z \neq 0; \\ f'(0), & \text{for } z = 0. \end{cases}$$

Then, $g(z)$ is analytic for $|z| \leqslant 1$.

By the maximum modulus theorem,

$$\max_{|z| \leqslant 1} |g(z)| = \max_{|z|=1} |g(z)| = \max_{|z|=1} \frac{|f(z)|}{|z|} = \max_{|z|=1} |f(z)| \leqslant M.$$

Thus, $|f(z)| \leqslant M|z|$. □

The following theorem illustrates how one can use complex analysis to study functions which are not analytic.

Theorem 4.19 *Let $f_1(z), \ldots, f_n(z)$ be analytic in a domain $\mathcal{D}$. Then,*

$$\phi(z) = |f_1(z)| + \cdots + |f_n(z)|$$

attains its maximum on $\partial\mathcal{D}$.

Proof We cannot apply the maximum modulus theorem directly to $\phi(z)$ because it is not an analytic function. So, we proceed as follows. Suppose the maximum is attained at an interior point z_0 of $\mathcal{D}$. Write

$$|f_j(z_0)| = c_j f_j(z_0) \text{ with } |c_j| = 1$$

for $1 \leqslant j \leqslant n$. Consider the function

$$F(z) = c_1 f_1(z) + \cdots + c_n f_n(z)$$

which is analytic in $\mathcal{D}$.

By the maximum modulus principle, there exists $z_1 \in \partial\mathcal{D}$ such that

$$|F(z_1)| \geqslant |F(z_0)| = \phi(z_0).$$

But as each c_j has absolute value 1, we see that

$$|F(z_1)| \leqslant |f_1(z_1)| + \cdots + |f_n(z_1)| = \phi(z_1)$$

so that $\phi(z_0) \leqslant \phi(z_1)$. □

We can sharpen the result of the previous theorem as follows. The maximum is attained only on the boundary unless all the f_j's are constant. This is easily seen from the formula:

$$c_j f_j(z_0) = |f_j(z_0)| \leqslant \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{i\theta})|^2 \, d\theta \leqslant \max_{z \in \partial\mathcal{D}} |f_j(z)|,$$

and the inequality is strict unless f_j is constant.

Next, we study the growth rates of the maximum modulus of analytic functions. If f is analytic in some domain containing $\overline{B(0, r)}$, we define

$$M_f(r) = \max_{|z|=r} |f(z)|.$$

Theorem 4.20 (Hadamard's three circle theorem). *Let* f *be analytic in the annulus* $0 < r_1 \leqslant |z| \leqslant r_3$. *Then,*

$$\log M_f(r)$$

is a convex function in $\log r$. *That is, if*

$$r_1 \leqslant r_2 \leqslant r_3,$$

then writing $r_2 = r_1^{\alpha} r_3^{\beta}$ *with* $\alpha + \beta = 1$, *we have*

$$M_f(r_2) \leqslant M_f(r_1)^{\alpha} M_f(r_3)^{\beta}.$$

Proof The function $g(z) = z^m f(z)^n$ is analytic in the annulus for any $m \in \mathbb{Z}$ and $n \in \mathbb{N}$. By the maximum modulus principle,

$$r_2^m M_f(r_2)^n \leqslant \max(r_1^m M_f(r_1)^n, r_3^m M_f(r_3)^n)$$

for such m, n. By homogeneity, both sides of the expansion can be raised to the power λ with $\lambda > 0$. Thus, the inequality holds for any $m \in \mathbb{Q}$ and $n \in \mathbb{Q}^+$. By continuity, this extends to $m \in \mathbb{R}$ and $n \in \mathbb{R}^+$. For

$$\lambda = \frac{\log M_f(r_1)/M_f(r_3)}{\log r_3/r_1},$$

we see that

$$\left(\frac{r_3}{r_1}\right)^{\lambda} = \frac{M_f(r_1)}{M_f(r_3)}$$

so that

$$r_1^{\lambda} M_f(r_1) = r_3^{\lambda} M_f(r_3).$$

Let us raise both sides to the power n, with $n \in \mathbb{R}^+$. Then setting $m = \lambda n$, we have

$$r_1^m M_f(r_1)^n = r_3^m M_f(r_3)^n.$$

Now for $\alpha + \beta = 1$,

$$(r_2^m M_f(r_2)^n)^{\alpha} \leqslant (r_1^m M_f(r_1)^n)^{\alpha}$$

$$(r_2^m M_f(r_2)^n)^{\beta} \leqslant (r_3^m M_f(r_3)^n)^{\beta}$$

Multiplying these two inequalities gives

$$r_2^m M_f(r_2)^n \leqslant r_1^{m\alpha} r_3^{m\beta} [M_f(r_1)^{\alpha} M_f(r_3)^{\beta}]^n.$$

Since $r_2 = r_1^{\alpha} r_3^{\beta}$, we get

$$r_2^m M_f(r_2)^n \leqslant r_2^m [M_f(r_1)^{\alpha} M_f(r_3)^{\beta}]^n$$

from which we deduce the desired result by taking nth roots. □

Hadamard noticed that the growth rate of the function yields new information about the function itself. Here is an example.

Theorem 4.21 *If f is entire and there is an $n \in \mathbb{N}$ such that*

$$M_f(r) \leqslant cr^n$$

for some constant $c > 0$, then $f(z)$ is a polynomial of degree at most n.

Proof By the local Cauchy theorem,

$$\frac{f^{(m)}(0)}{m!} = \frac{1}{2\pi i} \int_{C_r} \frac{f(z)}{z^{m+1}}\, dz$$

where C_r is the circle of radius r, centered at zero. By Cauchy's estimate,

$$\frac{|f^{(m)}(0)|}{m!} \leqslant \frac{M_f(r)}{r^{m+1}} \leqslant cr^{n-m-1}$$

which goes to zero if $m > n$. Thus, $f^{(m)}(0) = 0$ for all $m > n$ so f is a polynomial of degree $\leqslant n$. □

Exercises

1. Let f be analytic and nonzero in a region A. Show that $|f|$ has no strict local minima in A.
2. Show that the maximum of $|\sin(x + iy)|$ for $0 \leq x \leq 2\pi$ and $0 \leq y \leq 2\pi$ is $\cosh 2\pi$.
3. The functions $u(x, y)$ and $v(x, y)$ defined on $\mathbb{R} \times \mathbb{R}$ are said to be **harmonic conjugates** if $f(z) = u(x, y) + iv(x, y)$ with $z = x + iy$ is analytic. Find the harmonic conjugate of $u(x, y) = x^2 - y^2$.
4. Let g be analytic in the region $|z| < 1$ and assume that $|g(z)| = |z|$ for all $|z| < 1$. Show that $g(z) = e^{i\theta} z$ for some $\theta \in [0, 2\pi]$. [Hint: Use Schwarz's lemma.]
5. Consider the function $f(z) = e^{e^z}$ in the region $-\pi/2 \leq y \leq \pi/2$ where $y = \mathrm{Im}(z)$. Show that the function is bounded on the edges of this region, but is unbounded in the interior. (This exercise shows that the maximum modulus principle does not necessarily hold for unbounded regions. However, with suitable growth conditions, the Phragmén–Lindelöf theorem extends the principle to certain unbounded regions. See Sect. 4.12.)

4.6 The Global Cauchy Theorem

Suppose $\gamma_1, \ldots, \gamma_n$ are paths in $\mathbb{C}$. Each γ_i induces a linear function

$$\tilde{\gamma}_i(f) = \int_{\gamma_i} f(z)\, dz.$$

Define $\tilde{\Gamma} = \tilde{\gamma}_1 + \cdots + \tilde{\gamma}_n$. We have the formal sum $\Gamma = \gamma_1 + \cdots + \gamma_n$ called a **chain**. If each γ_i is a closed path, we call Γ a **cycle**. If each such γ_i is a path in some open set in Ω, then we say that Γ is a **chain in** Ω. We also define

$$\Gamma^* = \gamma_1^* \cup \cdots \cup \gamma_n^*.$$

If Γ is a cycle and $\alpha \notin \Gamma^*$, we define the **index** of α with respect to Γ by

$$\mathrm{Ind}_\Gamma(\alpha) := \frac{1}{2\pi i} \int_\Gamma \frac{dz}{z - \alpha}$$

and

$$\tilde{\Gamma}(f) = \int_\Gamma f(z)\, dz.$$

Theorem 4.22 (Cauchy's theorem). *Suppose $f \in H(\Omega)$. If Γ is a cycle such that $\mathrm{Ind}_\Gamma(\alpha) = 0$ for all $\alpha \notin \Omega$, then*

$$f(z)\mathrm{Ind}_\Gamma(z) = \frac{1}{2\pi i} \int_\Gamma \frac{f(w)\, dw}{w - z}$$

for all $z \in \Omega \setminus \Gamma^*$ and

$$\int_\Gamma f(z)\, dz = 0.$$

Thus, if Γ_1 and Γ_2 are cycles in Ω such that

$$\mathrm{Ind}_{\Gamma_1}(\alpha) = \mathrm{Ind}_{\Gamma_2}(\alpha) \quad \forall \alpha \notin \Omega,$$

then

$$\int_{\Gamma_1} f(z)\, dz = \int_{\Gamma_2} f(z)\, dz.$$

Remark 4.3 We had proved this theorem earlier for a *convex* set Ω. This is now more general.

Proof Define $g : \Omega \times \Omega \to \mathbb{C}$ by

$$g(z, w) = \begin{cases} \dfrac{f(w) - f(z)}{w - z}, & w \neq z \\ \\ f'(z), & w = z. \end{cases}$$

Then, g is a continuous function and we can define

$$h(z) = \frac{1}{2\pi i} \int_\Gamma g(z, w)\, dw.$$

To prove our theorem, we need to show that $h(z) = 0$ for all $z \in \Omega \setminus \Gamma^*$. To this end, we first show $h \in H(\Omega)$. Since g is uniformly continuous over any compact subset of $\Omega \times \Omega$, we see that if $z_n \to z \in \Omega$ then $g(z_n, w) \to g(z, w)$ uniformly in $w \notin \Gamma^*$. Thus, $h(z_n) \to h(z)$ so that h is continuous in Ω.

Now let Δ be a closed triangle in Ω. Then,

$$\int_{\partial\Delta} h(z)\,dz = \frac{1}{2\pi i}\int_\Gamma \int_{\partial\Delta} g(z, w)\,dz\,dw$$

by Fubini's theorem.

But for each $w \in \Omega$, the function

$$z \mapsto g(z, w)$$

can be thought of as a holomorphic function since the singularity $z = w$ is removable.

Thus, by the local Cauchy theorem,

$$\int_{\partial\Delta} g(z, w)\,dz = 0$$

for all $w \in \Gamma^*$. Hence,

$$\int_{\partial\Delta} h(z)\,dz = 0$$

for all triangles $\Delta \subseteq \Omega$. By Morera's theorem, $h \in H(\Omega)$.

Now let $\Omega_1 = \{z \in \mathbb{C} : \mathrm{Ind}_\Gamma(z) = 0\}$ and define

$$h_1(z) = \frac{1}{2\pi i}\int_\Gamma \frac{f(w)\,dw}{w - z}$$

for $z \in \Omega_1$.

Clearly, if $z \in \Omega \cap \Omega_1$, then $h_1(z) = h(z)$. Thus, there is a function $\phi \in H(\Omega \cup \Omega_1)$ whose restriction to Ω is h and to Ω_1 is h_1. By our hypothesis on Γ, $\mathrm{Ind}_\Gamma(\alpha) = 0$ for all $\alpha \notin \Omega$. That is, $\mathrm{Ind}_\Gamma(\alpha) = 0$ for all $\alpha \in \Omega^c$. Thus, $\Omega_1 \supseteq \Omega^c$. Therefore, ϕ is entire. Moreover, Ω_1 contains the unbounded component of $(\Gamma^*)^c$. Hence,

$$\lim_{|z|\to\infty} \phi(z) = \lim_{|z|\to\infty} h_1(z) = 0.$$

So by Liouville's theorem, $\phi \equiv 0$. But then $h \equiv 0$, as required.

Next, pick $a \in \Omega \setminus \Gamma^*$ and define $F(z) = (z - a)f(z)$. Then,

$$\frac{1}{2\pi i}\int_\Gamma f(z)\,dz = \frac{1}{2\pi i}\int_\Gamma \frac{F(z)\,dz}{z - a} = F(a)\mathrm{Ind}_\Gamma(a) = 0.$$

Applying this to the cycle $\Gamma = \Gamma_1 - \Gamma_2$ gives the last result. □

Exercises

1. Evaluate
$$\int_C \frac{2z^2 - 15z + 30}{z^3 - 10z^2 + 32z - 32} dz,$$
where C is the circle of radius 3 centered at zero and oriented counterclockwise.
2. Let $z_1, z_2, \ldots, z_k$ be complex numbers for which we have the estimate
$$\left| \sum_{j=1}^{k} z_j^n \right| \leq C B^n$$
for some absolute constant C and all n sufficiently large. Show that
$$|z_j| \leq B \qquad \text{for all } j = 1, 2, \ldots, k.$$
[Hint: Consider
$$\sum_{n=0}^{\infty} \left(\sum_{j=1}^{k} z_j^n \right) z^n = \sum_{k=1}^{n} \frac{1}{1 - z_j z}.$$
By the given estimate, the left-hand side is a power series that converges for $|z| < 1/B$. Hence, the right side must be analytic there.]
3. Let $f(z, w)$ be a continuous function of two complex variables z, w, with z in a region A and w on a curve γ. For each w on γ, assume that f is analytic in z. Let
$$F(z) = \int_\gamma f(z, w) dw.$$
Then, show that F is analytic and
$$F'(z) = \int_\gamma \frac{\partial f}{\partial z}(z, w) dw.$$
[Hint: Let $z_0 \in A$ and γ_0 a circle in A around z_0 whose interior also lies in A. Then,
$$F(z) = \frac{1}{2\pi i} \int_\gamma \left(\int_{\gamma_0} \frac{f(s, w)}{s - z} ds \right) dw.]$$

4.7 The Calculus of Residues

Definition 4.5 A function f is said to be **meromorphic** in an open set $\Omega \subseteq \mathbb{C}$ if there is a set $A \subseteq \Omega$ such that

(a) A has no limit point in Ω.
(b) $f \in H(\Omega \setminus A)$.
(c) f has a pole at each point of A.

Remark 4.4 The condition that A has no limit points implies that there is no compact subset of Ω containing infinitely many points of A. Thus, A is at most countable.

Let f be meromorphic with the set of poles A. For each $a \in A$, let

$$Q_a(z) = \sum_{k=1}^{m} c_k (z-a)^{-k}$$

be the principal part of f. We define c_1 as the **residue** of f at a and denote it by

$$\mathrm{Res}_{z=a} f(z) \text{ or } \mathrm{Res}(f; a).$$

If Γ is a cycle and $a \notin \Gamma^*$, then

$$\frac{1}{2\pi i} \int_\Gamma Q_a(z)\, dz = c_1 \mathrm{Ind}_\Gamma(a) = \big(\mathrm{Res}_{z=a} f(z)\big)\mathrm{Ind}_\Gamma(a),$$

since

$$\frac{1}{2\pi i} \int_C \frac{dz}{z^n} = \begin{cases} 0, & n \neq 1 \\ 1, & n = 1. \end{cases}$$

We generalize this as follows.

Theorem 4.23 (Residue theorem). *Let f be meromorphic on Ω, and let A be its set of poles. If Γ is a cycle in $\Omega \setminus A$ such that* $\mathrm{Ind}_\Gamma(\alpha) = 0$ *for all $\alpha \notin \Omega$, then*

$$\frac{1}{2\pi i} \int_\Gamma f(z)\, dz = \sum_{a \in A} \mathrm{Res}(f; a)\mathrm{Ind}_\Gamma(a).$$

Proof Let

$$B = \{a \in A : \mathrm{Ind}_\Gamma(a) \neq 0\}$$

, and let $W = (\Gamma^*)^c$ so that Ind_Γ is constant on each connected component V of W. Also, if V is unbounded or intersects Ω^c, then $\mathrm{Ind}_\Gamma(z) = 0$ for all $z \in V$.

Since A has no limit point and $B \subseteq A$, B has no limit point and as B is not in the unbounded component of W, B is actually a finite set. Thus, the sum in the statement

of the theorem is finite. Writing $B = \{a_1, \ldots, a_n\}$ with corresponding principal parts $Q_1, \ldots, Q_n$, put

$$g = f - (Q_1 + \cdots + Q_n)$$

(or $g = f$ if $B = \varnothing$). As g is analytic (i.e., has removable singularities at $a_1, \ldots, a_n$), we see

$$\frac{1}{2\pi i} \int_\Gamma g(z)\, dz = 0$$

from which the theorem follows immediately. □

In order to use the residue theorem, we need to know how to compute the residues of the functions at their poles.

Let f be meromorphic in Ω with a pole of order n at $a \in \Omega$, and let

$$Q_a(z) = \sum_{k=-1}^{n} c_{-k}(z-a)^{-k}$$

be its principal part. Then, $g = f - Q_a$ is holomorphic at a and has a power series expansion:

$$g(z) = \sum_{k=0}^{\infty} c_k (z-a)^k.$$

Thus,

$$f(z) = g(z) + Q_a(z) = \sum_{k=-n}^{\infty} c_k (z-a)^k$$

so that the function

$$h(z) := (z-a)^n f(z) = \sum_{k=0}^{\infty} c_{k-n}(z-a)^k.$$

In particular, $\operatorname{Res}(f; a) = c_{-1} = c_{(n-1)-n}$ so it is the coefficient of

$$(z-a)^{n-1}$$

in the above expression. Hence, it is given by

$$\frac{h^{(n-1)}(a)}{(n-1)!}.$$

Thus,

$$\operatorname{Res}_{z=a} f(z) = \frac{h^{(n-1)}(a)}{(n-1)!}.$$

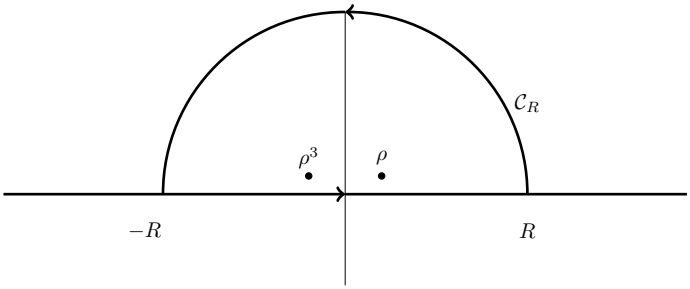


Fig. 4.6 Contour $\mathcal{C}_R$

In the case that the pole is **simple** (i.e., $n = 1$), we have

$$\operatorname{Res}(f; a) = \lim_{z \to a} (z - a) f(z).$$

Let us consider an example to see how this works.

Example 4.2 Let us compute

$$\int_{-\infty}^{\infty} \frac{dx}{x^4 + 1}$$

using residue calculus.

The integrand $f(z) = \dfrac{1}{z^4 + 1}$ has poles at $e^{\pi i k/4}$ for $k = 1, 3, 5, 7$. The poles $\rho = e^{\pi i/4}$, $\rho^3 = e^{3\pi i/4}$ lie in the upper half plane. For R large, they are contained inside the contour $\mathcal{C}_R$ depicted in Fig. 4.6. By our residue theorem,

$$\frac{1}{2\pi i} \int_{\mathcal{C}_R} \frac{dz}{z^4 + 1} = \operatorname{Res}_1 + \operatorname{Res}_2$$

where $\operatorname{Res}_k = \operatorname{Res}(f; e^{\pi i k/4})$. Let $z_k = e^{\pi i k/4}$. Then, as these are all *simple* poles,

$$\operatorname{Res}_k = \lim_{z \to z_k} \frac{(z - z_k)}{z^4 + 1} = \lim_{z \to z_k} \frac{z - z_k}{g(z)}$$

where $g(z) = z^4 + 1$. Hence, $\operatorname{Res}_k = \dfrac{1}{4z_k^3}$.

Therefore,

$$\frac{1}{2\pi i} \int_{\mathcal{C}_R} \frac{dz}{z^4 + 1} = \frac{1}{4}(e^{-3\pi i/4} + e^{-9\pi i/4}) = \frac{1}{4}(e^{-3\pi i/4} + e^{-\pi i/4}) = -\frac{i}{2\sqrt{2}}.$$

Letting $\mathcal{C}_R^*$ be the semicircular arc from R to $-R$, we have

$$\frac{1}{2\pi i}\int_{C_R} f(z)\,dz = \frac{1}{2\pi i}\int_{-R}^{R} f(z)\,dz + \frac{1}{2\pi i}\int_{C_R^*} f(z)\,dz.$$

We can estimate the last integral:

$$\left|\frac{1}{2\pi i}\int_{C_R^*} f(z)\,dz\right| \leqslant \frac{K}{R^3}$$

for some constant K. As $R \to \infty$, this goes to zero. Thus,

$$\frac{1}{2\pi i}\int_{-\infty}^{\infty} f(z)\,dz = -\frac{i}{2\sqrt{2}}.$$

Hence,

$$\int_{-\infty}^{\infty} \frac{dx}{x^4+1} = \frac{\pi}{\sqrt{2}}.$$

This example also illustrates how one can use complex analysis to calculate real definite integrals.

Exercises

1. Compute the residues of the following functions at their singularities:
   (a) $\dfrac{1}{(1-z)^3}$.
   (b) $\dfrac{e^z}{(1-z)^3}$.
   (c) $\dfrac{1}{z(1-z)^3}$.
   (d) $\dfrac{e^z}{z(1-z)^3}$.
2. Evaluate the integral
$$\int_C \frac{dz}{z^2+z+1}$$
where C is a circle oriented counterclockwise centered at zero of radius $1/2$.
3. Let K be a natural number. A collection of pairs of integers (a_j, n_j) with $0 \leq a_j < n_j$, $1 \leq j \leq K$ is called a **covering system** if every integer y satisfies the congruence $y \equiv a_j \pmod{n_j}$ for at least one value of j. It is called an **exact** system if for every integer y there is exactly one j with $1 \leq j \leq K$ such that $y \equiv a_j \pmod{n_j}$.

(a) Show that if the system is exact, then

$$\sum_{j=1}^{K} \frac{z^{a_j}}{1-z^{n_j}} = \frac{1}{1-z} \tag{4.1}$$

for $|z| < 1$.

(b) By computing the residues of both sides of (4.1) at $z = 1$, show that if the system is exact, then

$$\sum_{j=1}^{K} \frac{1}{n_j} = 1.$$

(c) Show that there is no exact covering system satisfying $n_1 \le n_2 \le \cdots \le n_{K-1} < n_K$ where the last inequality is strict. [Hint: Observe that the right-hand side of (4.1) has a simple pole at $z = 1$ only, whereas the left-hand side has simple poles at primitive n_Kth roots of unity. This result was conjectured by Erdös and was proved by L. Mirsky and D.J. Newman using this simple argument.]

4.8 Further Examples

We want to use residue calculus to evaluate

$$\lim_{A\to\infty} \int_{-A}^{A} \frac{\sin x}{x} e^{itx}\, dx,$$

an integral we met in our study of the Fourier transform. To do this, consider

$$f(z) = \frac{\sin z}{z} e^{itz}$$

which has a removable singularity at $z = 0$, since

$$\lim_{z\to 0} \frac{\sin z}{z} = 1.$$

Thus, $f(z)$ is entire.
Consider the path Γ_A from $-A$ to A indicated in Fig. 4.7.
By Cauchy's theorem,

$$\frac{1}{2\pi i} \int_{\Gamma_A} f(z)\, dz + \frac{1}{2\pi i} \int_{A}^{-A} f(z)\, dz = 0.$$

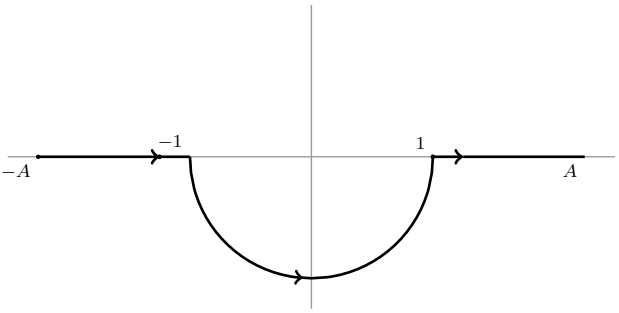


Fig. 4.7 Contour Γ_A

Thus,

$$\frac{1}{2\pi i}\int_{-A}^{A}\frac{\sin x}{x}e^{itx}\,dx=\frac{1}{2\pi i}\int_{\Gamma_A}f(z)\,dz. \tag{4.2}$$

Since $\sin z=\dfrac{e^{iz}-e^{-iz}}{2i}$, the integral on the right-hand side can be rewritten as

$$\frac{1}{2\pi i}\int_{\Gamma_A}\left(\frac{e^{iz}-e^{-iz}}{2iz}\right)e^{itz}\,dz=\frac{1}{2i}\left\{\frac{1}{2\pi i}\int_{\Gamma_A}\frac{e^{i(t+1)z}}{z}\,dz-\frac{1}{2\pi i}\int_{\Gamma_A}\frac{e^{i(t-1)z}}{z}\,dz\right\}.$$

Defining

$$\phi_A(s)=-\frac{1}{4\pi}\int_{\Gamma_A}\frac{e^{isz}}{z}\,dz\quad(s\in\mathbb{R}),$$

we see that our goal is to evaluate

$$\phi_A(t+1)-\phi_A(t-1).$$

Now, e^{isz}/z has a singularity at $z=0$. In fact, it is a simple pole. We complete Γ_A to a closed path in two ways and then apply Cauchy's theorem.

The closed path Γ_A+I (indicated in Fig. 4.8) avoids the singularity so that

$$\frac{1}{2\pi i}\int_{\Gamma_A+I}\frac{e^{isz}}{z}\,dz=0.$$

The closed path Γ_A+II (indicated in Fig. 4.9) encloses the singularity at $z=0$. As the residue is 1, we see by Cauchy's theorem

$$\frac{1}{2\pi i}\int_{\Gamma_A+II}\frac{e^{isz}}{z}\,dz=1.$$

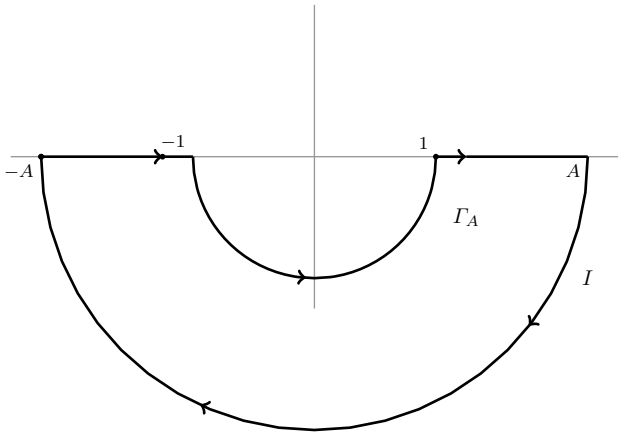


Fig. 4.8 Contour $\Gamma_A + I$

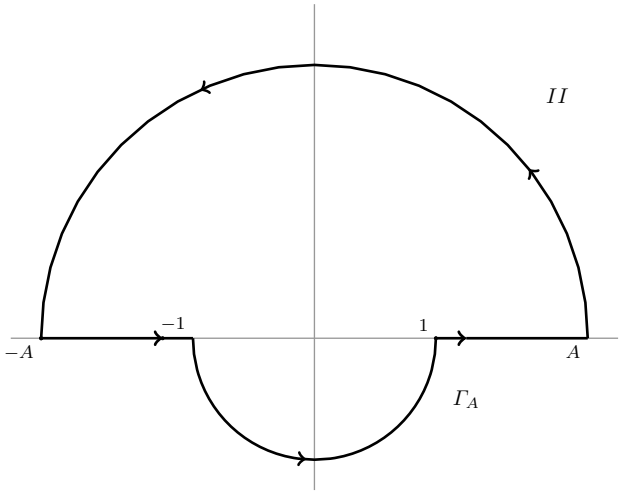


Fig. 4.9 Contour $\Gamma_A + II$

We parametrize the circular arc in each of these integrals. In our first integral,

$$0 = \frac{1}{2\pi i}\int_{\Gamma_A} \frac{e^{isz}}{z}\,dz + \frac{1}{2\pi i}\int_0^{-\pi} \frac{e^{isAe^{i\theta}}}{Ae^{i\theta}}(iAe^{i\theta})\,d\theta$$

by putting $z = Ae^{i\theta}$ so that $dz = iAe^{i\theta}d\theta$.

Now,

$$|e^{isAe^{i\theta}}| = |e^{isA(\cos\theta + i\sin\theta)}| = e^{-sA\sin\theta}$$

which is $\leqslant 1$ as $s \leqslant 0$ and $\sin\theta \leqslant 0$ for $\theta \in [-\pi, 0]$.

Letting $0 < \varepsilon < \pi$, we have $|\sin\theta| \leqslant |\sin\varepsilon|$ so that

$$\int_{-\varepsilon}^{-\pi+\varepsilon} |e^{isAe^{i\theta}}| d\theta \leqslant \pi e^{-|s|A|\sin\varepsilon|} \to 0$$

as $A \to \infty$. Thus,

$$\left| \int_0^{-\pi} e^{isAe^{i\theta}} d\theta \right| \leqslant \int_0^{-\varepsilon} |e^{isAe^{i\theta}}| d\theta + \int_{-\varepsilon}^{-\pi+\varepsilon} |e^{isAe^{i\theta}}| d\theta + \int_{-\pi+\varepsilon}^{-\pi} |e^{isAe^{i\theta}}| d\theta.$$

Since the integrand is bounded by 1, the first and last integrals are bounded by 2ε. Letting $\varepsilon \to 0$, we see that

$$\lim_{A\to\infty} \frac{1}{2\pi i} \int_{\Gamma_A} \frac{e^{isz}}{z} dz = 0 \tag{4.3}$$

if $s < 0$. Thus, $\lim_{A\to\infty} \phi_A(s) = 0$ if $s < 0$. This fact could have been deduced alternately by a simple application of the dominated convergence theorem (see Exercise 1 below).

We now consider the case if $s > 0$. Proceeding as before and parametrizing the arc, we have

$$1 = \frac{1}{2\pi i} \int_{\Gamma_A} \frac{e^{isz}}{z} dz + \frac{1}{2\pi i} \int_0^{\pi} i e^{isAe^{i\theta}} d\theta$$

and for $s > 0$,

$$|e^{isAe^{i\theta}}| = e^{-sA\sin\theta}.$$

As $\theta \in [0, \pi]$, $\sin\theta > 0$, so that as before

$$\lim_{A\to\infty} \frac{1}{2\pi i} \int_0^{\pi} i e^{isAe^{i\theta}} d\theta = 0.$$

Therefore,

$$\lim_{A\to\infty} \frac{1}{2\pi i} \int_{\Gamma_A} \frac{e^{isz}}{z} dz = 1$$

if $s > 0$.

Therefore,

$$\begin{aligned}\lim_{A\to\infty}\phi_A(s) &= \lim_{A\to\infty} -\frac{1}{4\pi}\int_{\Gamma_A}\frac{e^{isz}}{z}\,dz\\ &= \frac{1}{2i}\lim_{A\to\infty}\frac{1}{2\pi i}\int_{\Gamma_A}\frac{e^{isz}}{z}\,dz\\ &= \begin{cases}\frac{1}{2i}, & s>0\\ \\ 0, & s<0.\end{cases}\end{aligned}$$

That is,

$$\lim_{A\to\infty}\frac{1}{2\pi i}\int_{-A}^{A}\frac{\sin x}{x}e^{itx}\,dx = \phi_A(t+1)-\phi_A(t-1).$$

In other words,

$$\begin{aligned}\lim_{A\to\infty}\int_{-A}^{A}\frac{\sin x}{x}e^{itx}\,dx &= 2\pi i\{\phi_A(t+1)-\phi_A(t-1)\}\\ &= \begin{cases}\pi, & t+1>0 \text{ and } t-1<0;\\ -\pi, & t+1<0 \text{ and } t-1>0;\\ 0, & t+1<0 \text{ and } t-1<0.\end{cases}\end{aligned}$$

The first case means $|t|<1$. The second case is impossible. The third case means $|t|>1$. Therefore,

$$\lim_{A\to\infty}\int_{-A}^{A}\frac{\sin x}{x}e^{itx}\,dx = \begin{cases}\pi, \text{ if} |t|<1;\\ 0, \text{ if } |t|>1.\end{cases}$$

What happens at $t=\pm 1$? We need to study $\phi_A(0)$, which is

$$-\frac{1}{4\pi}\int_{\Gamma_A}\frac{dz}{z}.$$

But using $\Gamma_A + II$ (Fig. 4.9), we see

$$\frac{1}{2\pi i}\int_{\Gamma_A+II}\frac{dz}{z}=1$$

and

$$\frac{1}{2\pi i}\int_{II}\frac{dz}{z} = \frac{1}{2\pi i}\int_0^{\pi}\frac{iAe^{i\theta}}{Ae^{i\theta}}\,d\theta = \frac{1}{2}.$$

Thus,

$$\int_{\Gamma_A}\frac{dz}{z}=\pi i$$

so that $\phi_A(0) = -i/4$. Now by (4.2),

$$\frac{1}{2\pi i}\lim_{A\to\infty}\int_{-A}^{A}\frac{\sin x}{x}e^{ix}\,dx = \lim_{A\to\infty}\{\phi_A(2)-\phi_A(0)\} = \frac{1}{2i}+\frac{i}{4} = -\frac{i}{4}$$

if $t = 1$. We can compute for the case $t = -1$ in a similar way. Hence,

$$\lim_{A\to\infty}\int_{-A}^{A}\frac{\sin x}{x}e^{\pm ix}\,dx = -\frac{i}{4}\times 2\pi i = \frac{\pi}{2}.$$

Let us consider a related example. This was considered in the previous chapter and solved using the theorem related to differentiation under the integral sign. It is instructive to consider it now using complex analysis.

Let us compute $\displaystyle\int_0^\infty \frac{\sin x}{x}\,dx$.

$$\begin{aligned}\int_0^\infty \frac{\sin x}{x}\,dx &= \frac{1}{2}\int_{-\infty}^{\infty}\frac{\sin x}{x}\,dx\\ &= \frac{1}{2}\int_{-\infty}^{\infty}\frac{e^{ix}-e^{-ix}}{2i}\,\frac{dx}{x}\\ &= \frac{1}{2}\lim_{\substack{R\to\infty\\ \varepsilon\to 0}}\left\{\int_{-R}^{-\varepsilon}\frac{e^{ix}}{2ix}dx+\int_{\varepsilon}^{R}\frac{e^{ix}}{2ix}dx-\int_{-R}^{-\varepsilon}\frac{e^{-ix}}{2ix}dx-\int_{\varepsilon}^{R}\frac{e^{-ix}}{2ix}dx\right\}.\end{aligned}$$

Changing variables $x \to -x$ in the last two integrals, we see that this is

$$\lim_{\substack{R\to\infty\\ \varepsilon\to 0}}\left\{\int_{-R}^{-\varepsilon}\frac{e^{ix}}{2ix}dx+\int_{\varepsilon}^{R}\frac{e^{ix}}{2ix}dx\right\}.$$

This suggests we consider the following contour:

By Cauchy's theorem,

$$\frac{1}{2\pi i}\int_{\mathcal{C}_{\uparrow,\mathcal{R}}}\frac{e^{iz}}{z}dz = 0.$$

The integral along the circular arc of radius R is

$$\frac{1}{2\pi i}\int_0^\pi \frac{e^{iRe^{i\theta}}}{Re^{i\theta}}Rie^{i\theta}d\theta = \frac{1}{2\pi}\int_0^\pi e^{iR(\cos\theta+i\sin\theta)}d\theta \to 0 \text{ as } R\to 0$$

by the dominated convergence theorem via an argument similar to the one we used in our previous example.

It remains to consider the integral along the smaller circle $\mathcal{C}(\varepsilon)$ (Fig. 4.10):

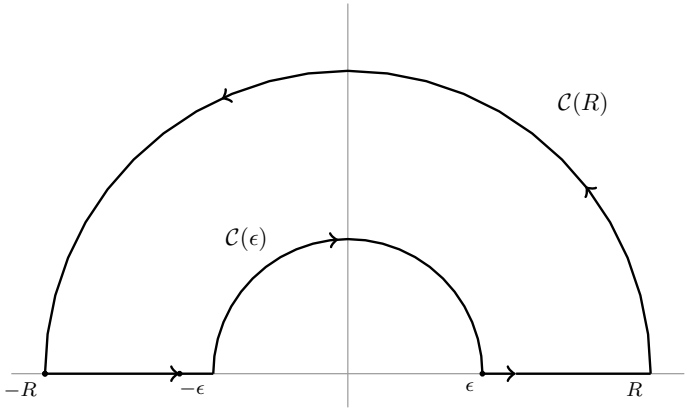


Fig. 4.10 Contour $\mathcal{C}_{\epsilon,R}$

$$\frac{1}{2\pi i}\int_{\mathcal{C}(\uparrow)}\frac{e^{iz}}{z}dz.$$

To evaluate this integral, we prove the following lemma.

Lemma 4.1 *Suppose $g(z)$ is holomorphic. Then,*

$$\lim_{\varepsilon\to 0}\int_{\mathcal{C}(\uparrow)}\frac{g(z)}{z}dz = \pi i g(0).$$

Proof We write $g(z) = g(0) + h(z)z$ where $h(z)$ is holomorphic. Then,

$$\lim_{\varepsilon\to 0}\int_{\mathcal{C}(\uparrow)}\left\{\frac{g(0)}{z} + h(z)\right\}dz = \lim_{\varepsilon\to 0}\int_{\mathcal{C}(\uparrow)}\frac{g(0)}{z}dz$$

because h being holomorphic, it is bounded on $\mathcal{C}(\uparrow)$. Now the integral on $\mathcal{C}(\varepsilon)$ is (taking into account the orientation)

$$-g(0)\int_0^{\pi}\frac{d(\epsilon e^{i\theta})}{\epsilon e^{i\theta}} = -g(0)\int_{0i}^{\pi}\frac{ie^{i\theta}d\theta}{e^{i\theta}} = -\pi i g(0).$$

□

Putting everything together, we get

$$\lim_{\varepsilon\to 0, R\to\infty}\frac{1}{2\pi i}\left(\int_{-R}^{-\epsilon}\frac{e^{iz}}{z}dz + \int_{\varepsilon}^{R}\frac{e^{iz}}{z}dz\right) - \frac{1}{2} = 0,$$

from which we deduce the formula:

$$\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}.$$

Exercises

1. Use the dominated convergence theorem to show that (4.3) holds.
2. Show that
$$\int_{-\infty}^{\infty} \frac{dx}{x^6+1} = \frac{2\pi}{3}.$$
3. For each natural number $n \geq 2$, show that
$$\int_0^\infty \frac{dx}{x^n+1} = \frac{\pi/n}{\sin \pi/n}.$$
[Hint: Consider the path from 0 to R, and then from R to $Re^{2\pi i/n}$ and then back to 0.]

4.9 Rouché's Theorem

The **logarithmic derivative** of a function f is defined as f'/f. If f is holomorphic, f'/f is meromorphic.

Theorem 4.24 (The argument principle). *Let γ be a closed curve in Ω such that $\mathrm{Ind}_\gamma(\alpha) = 0$ for all $\alpha \notin \Omega$ and $\mathrm{Ind}_\gamma(\alpha) = 0$ or 1 for $\alpha \in \Omega$. Let*

$$\Omega_1 = \{\alpha \in \Omega \,:\, \mathrm{Ind}_\gamma(\alpha) = 1\}.$$

If f is meromorphic in Ω, let N be the number of zeros and P the number of poles of f in Ω_1, counted with multiplicity. If none of these are in γ^*, then

$$\frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\,dz = N - P.$$

In other words, the difference between the number of zeros and poles of f can be counted by integrating the logarithmic derivative of f.

Proof If $\alpha \in \Omega_1$ is a zero of f of multiplicity m, then we can write

$$f(z) = (z-a)^m g(z)$$

with $g(a) \neq 0$. Also, g and $1/g$ are holomorphic in a neighborhood of a. In this neighborhood,

$$\frac{f'(z)}{f(z)} = \frac{m(z-a)^{m-1}g(z) + (z-a)^m g'(z)}{(z-a)^m g(z)} = \frac{m}{z-a} + \frac{g'(z)}{g(z)}.$$

Thus, $\text{Res}_{z=a}(f'/f)(z) = m$.
Similarly, if b is a pole in Ω_1 of order r, we can write

$$f(z) = (z-b)^{-r}h(z)$$

where $h(b) \neq 0$, and h and $1/h$ are holomorphic in a neighborhood of b. Thus,

$$\frac{f'(z)}{f(z)} = \frac{-r(z-a)^{-r-1}h(z) + (z-a)^{-r}h'(z)}{(z-a)^{-r}h(z)} = \frac{-r}{z-a} + \frac{h'(z)}{h(z)},$$

so $\text{Res}_{z=b}(f'/f)(z) = -r$.
The result now follows from the residue theorem. □

Theorem 4.25 (Rouché's theorem). *Suppose $f, g \in H(\Omega)$, and let γ be a closed path in Ω such that $\text{Ind}_\gamma(\alpha) = 0$ for all $\alpha \notin \Omega$. If $|f(z) - g(z)| < |f(z)|$ for all $z \in \gamma^*$, then f and g have the same number of zeros inside γ.*

Proof By the inequality in the hypothesis, neither f nor g vanishes on γ^*. Thus, the inequality can be rewritten as

$$\left|1 - \frac{g(z)}{f(z)}\right| < 1 \text{ for all } z \in \gamma^*.$$

That is, $g(z)/f(z)$ lies in the open unit disk centered at 1 for $z \in \gamma^*$.
So if γ is defined on $[a, b]$ and $F(z) = g(z)/f(z)$, then

$$F \circ \gamma : [a, b] \to B^o(1, 1).$$

Therefore,

$$\begin{aligned}
0 &= \text{Ind}_{F\circ\gamma}(0) \\
&= \frac{1}{2\pi i}\int_{F\circ\gamma} \frac{dz}{z} \qquad \text{Put } z = F \circ \gamma(t). \\
&= \frac{1}{2\pi i}\int_a^b \frac{F'(\gamma(t))\gamma'(t)\,dt}{F(\gamma(t))} \\
&= \frac{1}{2\pi i}\int_a^b \left\{\frac{f'(\gamma(t))}{f(\gamma(t))} - \frac{g'(\gamma(t))}{g(\gamma(t))}\right\}\gamma'(t)dt.
\end{aligned}$$

Since

$$\frac{F'(z)}{F(z)} = \frac{g'(z)f(z) - f'(z)g(z)}{f(z)^2} \cdot \frac{f(z)}{g(z)}.$$

Thus,

$$0 = \frac{1}{2\pi i}\int_\gamma \left(\frac{f'(z)}{f(z)} - \frac{g'(z)}{g(z)}\right) dz$$

which means f and g have the same number of zeros. □

Example 4.3 Let $p(z) = z^8 - 5z^3 + z - 2$, and let γ^* be the unit circle. If $f(z) = -5z^3$, then

$$|f(z) - p(z)| = |-z^8 - z + 2| \leqslant 4 < |f(z)|$$

for $z \in \gamma^*$. By Rouché's theorem, $p(z)$ has three zeros in the unit disk. □

We can also give an alternative proof of the fundamental theorem of algebra, using Rouché's theorem. Let

$$g(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0.$$

Then if $f(z) = z^n$, we see

$$|f(z) - g(z)| \leqslant (|a_{n-1}| + \cdots + |a_1| + |a_0|)R^{n-1}$$

for $|z| \leqslant R$. If we choose $R > |a_{n-1}| + \cdots + |a_1| + |a_0|$, we get

$$|f(z) - g(z)| < R^n = |f(z)|$$

for $|z| = R$. Thus, f and g have the same number of zeros in $|z| \leqslant R$. Thus, g has n zeros.

The proof also shows that all zeros lie in the disk

$$|z| < |a_{n-1}| + \cdots + |a_1| + |a_0|.$$

Another application of Rouché's theorem is Hurwitz's theorem.

Theorem 4.26 (Hurwitz). *Let f_n be a sequence of analytic functions on a region Ω converging uniformly on every closed disk in Ω to f. (By a theorem of Weierstrass, f is analytic.) Assume f is not identically zero. Let $z_0 \in \Omega$. Then, $f(z_0) = 0$ if and only if there is a sequence $z_n \to z_0$ such that $f_n(z_n) = 0$. That is, a zero of f is a limit of zeros of the function f_n.*

Proof Choose γ to be a small circle centered at z_0 and lying in Ω. By assumption, $f_n \to f$ uniformly on and inside γ. By choosing γ sufficiently small, we may suppose that z_0 is the only possible zero of f as the zeros of f are isolated. Let m be the minimum of $|f(z)|$ on γ. Then, $|f(z)| \geqslant m > 0$ on γ. Choose N large so that

$$|f_n(z) - f(z)| < m \text{ on } \gamma \text{ for } n \geqslant N.$$

This we can do by uniform convergence. But now, on γ^*,

$$|f_n(z) - f(z)| < |f(z)|$$

so that $f_n(z)$ and $f(z)$ have the same number of zeros in γ. The result now follows from this conclusion. □

Rouché's theorem allows us study one-to-one analytic functions and deduce the **open mapping theorem** which states that any non-constant analytic function f is an open map. That is, $f(U)$ is open for every open U. We relegate this to the exercises. A key tool needed is the following theorem which is of independent interest.

Theorem 4.27 *Let A be an open subset of $\mathbb{C}$, and suppose that $f : A \to \mathbb{C}$ is analytic and one to one. Then, $f'(z_0) \neq 0$ for all $z_0 \in A$.*

Proof Suppose that for some $z_0 \in A$, we have $f'(z_0) = 0$. Then, the function $f(z) - f(z_0)$ has a zero of order $k \geq 2$ at z_0. As f is one to one, and A is open, it is not constant, so the zeros of $f(z) - f(z_0)$ are isolated. So, there is a $\delta > 0$ such that $f(z) - f(z_0)$ has no zero and $f'(z) \neq 0$ for $0 < |z - z_0| \leq \delta$. We can also choose $\delta < m$. In particular, there is an $m > 0$ such that $|f(z) - f(z_0)| \geq m > 0$ and $f'(z) \neq 0$ for $0 < |z - z_0| \leq \delta$. For $0 < \eta < \delta$, we apply Rouché's theorem to the functions, $f(z) - f(z_0)$ and $f(z) - f(z_0) - \eta$ to conclude that $f(z) - f(z_0) - \eta$ has k zeros inside the disk $|z - z_0| = \delta$. Since $f'(z) \neq 0$ in the region $0 < |z - z_0| \leq \delta$, this cannot be a double zero. Therefore, $f(z)$ is not one to one, a contradiction. □

Exercises

1. Determine the number of zeros of the polynomial

$$z^{87} + 36z^{57} + 71z^4 + z^3 - z + 1$$

in the unit circle.

2. Determine the number of zeros of the polynomial

$$2z^5 - 6z^2 + z + 1$$

in the annulus $1 \leq |z| \leq 2$.

3. If $a > e$, show that the equation $e^z = az^n$ has n solutions inside the unit circle.
4. Let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

with $a_0 = 1, a_1 = 0$ and f not constant. Prove that f is one to one in the unit disk $|z| < 1$ if

$$\sum_{n=2}^{\infty} n|a_n| \leq 1.$$

[Hint: Fix z_0 in the unit disk. Let $g(z) = z - z_0$ and $h(z) = f(z) - f(z_0)$ and apply Rouché's theorem.]

5. By considering the example $f_n(z) = e^{z/n}$, show that the assumption that f be not identically zero is essential in Hurwitz's Theorem 4.26.
6. Suppose f is analytic and not constant on a region A. Show that f is an open map.

4.10 Infinite Products and Weierstrass Factorization

To what extent does the knowledge of the zeros and poles of a meromorphic function determine that function? Any polynomial is completely determined by the knowledge of its zeros and multiplicities. Any polynomial can be written as

$$Az^r \prod_{i=1}^{n} \left(1 - \frac{z}{a_i}\right)$$

where a_i are the nonzero roots of f. Does such a factorization exist for other functions?

To this end, we begin by studying convergence of infinite products.

Let $\{u_n\}_{n=1}^{\infty}$ be a sequence of complex numbers. Let

$$P_N = \prod_{n=1}^{N} (1 + u_n).$$

If $\lim_{N\to\infty} P_N$ exists and equals α, we say that the infinite product

$$\prod_{n=1}^{\infty} (1 + u_n) \tag{4.4}$$

exists and write

$$\alpha = \prod_{n=1}^{\infty} (1 + u_n).$$

Lemma 4.2 *Suppose $u_1, \ldots, u_N \in \mathbb{C}$. Let*

$$P_N = \prod_{n=1}^{N} (1 + u_n)$$

$$P_N^* = \prod_{n=1}^{N} (1 + |u_n|).$$

Then, $P_N^* \leqslant \exp(|u_1| + \cdots + |u_N|)$ *and* $|P_N - 1| \leqslant P_N^* - 1$.

Proof Recall that $1 + x \leqslant e^x$ for $x \geqslant 0$. This immediately implies the bound for P_N^*. To derive the bound for P_N, we proceed by induction.

For $N = 1$, this is trivially true. Assume the inequality holds for all subscripts $\leqslant N$. Then,

$$P_{N+1} - 1 = P_N(1 + u_{N+1}) - 1 = (P_N - 1)(1 + u_{N+1}) + u_{N+1}$$

so that

$$\begin{aligned} |P_{N+1} - 1| &\leqslant |P_N - 1|(1 + |u_{N+1}|) + |u_{N+1}| \\ &\leqslant |P_N^* - 1|(1 + |u_{N+1}|) + |u_{N+1}| \end{aligned}$$

by the induction hypothesis. Hence,

$$|P_{N+1} - 1| \leqslant P_N^*(1 + |u_{N+1}|) - (1 + |u_{N+1}|) + |u_{N+1}| = P_{N+1}^* - 1,$$

as claimed. □

We will say the product (4.4) **converges absolutely** if the product with u_n replaced by $|u_n|$ converges. It is tempting to reduce the study of infinite products to the study of infinite sums "by taking logarithms." This can, in fact, be done but one must proceed with some care to ensure the logarithm is well defined (see exercises below).

Theorem 4.28 *Suppose* $\{u_n(z)\}_{n=1}^{\infty}$ *is a sequence of bounded complex functions in* Ω *such that*

$$\sum_{n=1}^{\infty} |u_n(z)|$$

converges uniformly in Ω. *Then,*

$$\prod_{n=1}^{\infty} (1 + u_n(z))$$

converges uniformly to a function $f(z)$ *in* Ω. *Moreover, if* $f(z_0) = 0$ *for some* $z_0 \in \Omega$, *then* $u_n(z_0) = -1$ *for some* n.

Proof The hypothesis implies that $\sum_{n=1}^{\infty} |u_n(z)|$ is bounded in Ω. Indeed, let

$$f_N(z) = \sum_{n=1}^{N} |u_n(z)|,$$

the uniform convergence implies that for any $\varepsilon > 0$, $\exists N_0$ such that

$$|f_N(z) - f_M(z)| < \varepsilon$$

for $M, N \geqslant N_0$. In particular,

$$\begin{aligned} |f_N(z)| &= |f_{N_0}(z) + f_N(z) - f_{N_0}(z)| \\ &\leqslant |f_{N_0}(z)| + |f_N(z) - f_{N_0}(z)| \\ &\leqslant K N_0 + \varepsilon \end{aligned}$$

where $|u_n| \leqslant K$ for $n \leqslant N_0$.

By the previous lemma, the functions

$$P_N(z) = \prod_{n=1}^{N}(1 + u_n(z))$$

are uniformly bounded in Ω, say by K.

Since $\sum_{n=1}^{\infty} |u_n(z)|$ converges uniformly in Ω, given $\varepsilon > 0$, $\exists N_0$ such that

$$\sum_{n=N_0}^{\infty} |u_n(z)| < \varepsilon.$$

Thus, for $M \geqslant N \geqslant N_0$,

$$|P_M(z) - P_N(z)| = |P_N(z)| \left| \prod_{m=N+1}^{M} (1 + u_n(z)) - 1 \right| \leqslant |P_N(z)| \cdot (e^{\epsilon} - 1)$$

by the previous lemma. As the $P_N(z)$ are uniformly bounded by K, we see

$$|P_M(z) - P_N(z)| \leqslant O(\varepsilon).$$

Now if $z = z_0$ is a zero of f, then letting $M \to \infty$ above, we see from $|f(z) - P_N(z)| = O(\varepsilon)$ that $|f(z_0) - P_N(z_0)| = O(\varepsilon)$, which implies that $|P_N(z_0)| = O(\varepsilon)$ for any $\varepsilon > 0$. Thus, $P_N(z_0) = 0$. But $P_N(z)$ is a finite product, so one of the terms must be zero. This completes the proof. □

Example 4.4 The Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p} \left(1 - \frac{1}{p^s}\right)^{-1}$$

converges absolutely for $\operatorname{Re}(s) > 1$, and the infinite product is by virtue of unique factorization. We can apply our theorem to show $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$ since

$$\left(1-\frac{1}{p^s}\right)^{-1}=0 \Leftrightarrow \frac{p^s}{p^s-1}=0 \Leftrightarrow p^s=0$$

which cannot happen in the region $\mathrm{Re}\,(s)>1$. □

Given a sequence of complex numbers a_n, a series of the form

$$\sum_{n=1}^{\infty}\frac{a_n}{n^s}$$

is called a **Dirichlet series**. Such series play an important role in number theory. In particular, the study of the distribution of prime numbers hinges on a study of such series (see Sects. 4.19 and 4.20).

Let us now set

$$P_n(z)=\sum_{k=1}^{n}\frac{z^k}{k}.$$

Define $E_0(z)=1-z$ and

$$E_n(z)=(1-z)\exp(P_n(z)).$$

The only zero of $E_n(z)$ is at $z=1$. Note that

$$P_n'(z)=\sum_{k=1}^{n}z^{k-1}=\frac{z^n-1}{z-1}.$$

Lemma 4.3 *For* $|z|\leqslant 1$,

$$|1-E_n(z)|\leqslant |z|^{n+1}.$$

Proof This is trivially true for $n=0$. For $n\geqslant 0$,

$$E_n'(z)=-\exp(P_n(z))+(1-z)\left(\frac{z^n-1}{z-1}\right)\exp(P_n(z))=-z^n\exp(P_n(z)).$$

Therefore, $-E_n'(z)$ has a zero of order n at 0. Moreover, it has a power series expansion in z with non-negative coefficients because $P_n(z)$ has and $\exp(P_n(z))$ also does.

Now,

$$1-E_n(z)=-\int_{[0,z]}E_n'(w)\,dw.$$

Thus, $1-E_n(z)$ has a zero of order $n+1$ at $z=0$. Thus,

$$\phi(z)=\frac{1-E_n(z)}{z^{n+1}}$$

can be expressed as a power series

$$\sum_{m=0}^{\infty} a_m z^m$$

with $a_m \geqslant 0$. By the maximum modulus principle

$$\max_{|z|\leqslant 1} |\phi(z)| = \max_{|z|=1} |\phi(z)| \leqslant \sum_{m=0}^{\infty} |a_m| = \phi(1) = 1$$

and the result follows from this. □

Theorem 4.29 *Let z_n be a sequence of nonzero complex numbers such that $|z_n| \to \infty$ as $n \to \infty$. If p_n is a sequence of non-negative integers such that*

$$\sum_{n=1}^{\infty} \left(\frac{r}{|z_n|}\right)^{1+p_n} < \infty \tag{4.5}$$

for all $r > 0$, then

$$P(z) = \prod_{n=1}^{\infty} E_{p_n}(z/z_n)$$

is an entire function with a zero at each z_n and no other zeros in $\mathbb{C}$. Moreover, if w_0 occurs in the sequence z_n exactly m times, then $P(z)$ has a zero at $z = w_0$ of order exactly m.

Proof First, note that for any $r > 0$, $|z_n| > 2r$ for all n sufficiently large. So, the summation condition holds for $p_n = n - 1$ and is not necessary. Now for $|z| < r$, we have by the previous lemma,

$$|1 - E_{p_n}(z/z_n)| \leqslant \left|\frac{z}{z_n}\right|^{1+p_n} \leqslant \left(\frac{r}{|z_n|}\right)^{1+p_n}.$$

The convergence of the series (4.5) ensures that the product

$$\prod_{n=1}^{\infty} E_{p_n}(z/z_n)$$

converges absolutely to $P(z)$. Moreover, $P(z) = 0$ if and only if $E_{p_n}(z/z_n) = 0$ for some n, which occurs if and only if $z = z_n$ for some n. The last assertion is easily derived by noting that $|z_n| \to \infty$ so that any zero is of finite order and each zero contributed by $E_{p_n}(z/z_n)$ is a simple zero. □

We can use this to derive product expansion for entire functions.

Theorem 4.30 (Weierstrass factorization theorem). *Let f be entire and suppose $f(0) \neq 0$. Let $z_1, z_2, \ldots$ be zeros of f, listed with multiplicity. Then, there is an entire function g and a sequence $p_1, p_2, \ldots$ such that*

$$f(z) = e^{g(z)} \prod_{n=1}^{\infty} E_{p_n}(z/z_n).$$

Remark 4.5 If f has a zero of order k at $z = 0$, we can apply the above theorem to $f(z)/z^k$. The above product expansion is not necessarily unique.

Proof Let

$$P(z) = \prod_{n=1}^{\infty} E_{p_n}(z/z_n).$$

Note that necessarily $|z_n| \to \infty$. By the previous theorem, P and f have precisely the same zeros. Since f/P has no zeros, the next proposition tells us that $f(z)/P(z) = e^{g(z)}$. □

Proposition 4.10.1 *If f is entire and $f(0) \neq 0$ for all $z \in \mathbb{C}$, then there is an entire function g such that $f(z) = Ce^{g(z)}$, with C a constant.*

Proof Since f has no zeros, f'/f is entire. Then, there is an entire function h such that $h' = f'/f$.

Now let

$$F(z) = f(z)e^{-h(z)}.$$

Then,

$$\begin{aligned} F'(z) &= f'(z)e^{-h(z)} - f(z)h'(z)e^{-h(z)} \\ &= e^{-h(z)}\{f'(z) - f(z)h'(z)\} = 0. \end{aligned}$$

Therefore, $F(z) = C$ for some constant C. □

Example 4.5 Let us find a product expansion for the entire function $\sin z$. This has a zero of order 1 at $n\pi$, $n \in \mathbb{Z}$. By the theorem, the product

$$f(z) = \prod_{n \neq 0} \left(1 - \frac{z}{n\pi}\right) e^{z/n\pi}$$

is entire and has precisely the same zeros as $(\sin z)/z$.

Therefore,

$$\frac{\sin z}{z} = Ae^{g(z)} \cdot \prod_{n \neq 0} \left(1 - \frac{z}{n\pi}\right) e^{z/n\pi}$$

for some constant A and entire function $g(z)$.

Note that

$$\left(1-\frac{z}{n\pi}\right)\left(1+\frac{z}{n\pi}\right)=1-\frac{z^2}{n^2\pi^2}.$$

Thus,

$$\sin z = ze^{g(z)}\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2\pi^2}\right).$$

By logarithmic differentiation,

$$\frac{\cos z}{\sin z}=\frac{1}{z}+g'(z)+\sum_{n=1}^{\infty}\frac{2z}{z^2-n^2\pi^2}.$$

In particular,

$$\begin{aligned}\frac{\cos(ic/2)}{\sin(ic/2)} &= \frac{2}{ic}+g'(ic/2)-\sum_{n=1}^{\infty}\frac{4ic}{c^2+4n^2\pi^2}\\ &= -\frac{2i}{c}+g'(ic/2)-\sum_{n\neq 0}\frac{2ic}{c^2+4n^2\pi^2}\\ &= g'(ic/2)-\sum_{n=-\infty}^{\infty}\frac{2ic}{c^2+4n^2\pi^2}.\end{aligned}$$

Recall that we proved using the Poisson summation formula,

$$\frac{e^c+1}{e^c-1}=\sum_{n=-\infty}^{\infty}\frac{2c}{c^2+4\pi^2n^2}$$

so that

$$\frac{\cos(ic/2)}{\sin(ic/2)}=g'(ic/2)-i\cdot\frac{e^c+1}{e^c-1}.$$

On the other hand,

$$\frac{\cos(ic/2)}{\sin(ic/2)}=i\cdot\frac{e^{-c/2}+e^{c/2}}{e^{-c/2}-e^{c/2}}=-i\cdot\frac{e^c+1}{e^c-1}$$

so we deduce $g'(ic/2)=0$ for all $c\neq 2\pi in$. Thus, g is constant in an interval and hence constant everywhere by the analyticity. So, we can write

$$\sin z = Az\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2\pi^2}\right).$$

Now,

$$\lim_{z\to 0}\frac{\sin z}{z}=1$$

so that $A=1$. This proves:

Theorem 4.31

$$\sin \pi z=\pi z\prod_{n=1}^{\infty}\left(1-\frac{z^2}{n^2}\right).$$

An interesting corollary is that

$$\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}. \tag{4.6}$$

We leave the deduction of this from the theorem as an exercise to the reader.

Exercises

1. In some definitions of convergence of infinite products, we do not allow zero as a limit value for various reasons. (Sometimes, the authors write that the product diverges to zero to emphasize this.) Suppose that the product

$$\prod_{n=1}^{\infty}(1+u_n)$$

converges to a nonzero limit. Show that

$$\lim_{N\to\infty}\prod_{n=N}^{\infty}(1+u_n)=1.$$

2. For $z\in\mathbb{C}$, $z\neq 0$, write $z=|z|e^{i\theta}$ with $-\pi<\theta\leq\pi$. Define $\operatorname{Log} z$ as $\log|z|+i\theta$. If $b_n\neq 0$ for all $n\geq 1$, show that $\prod_{n=1}^{\infty}b_n$ converges to a nonzero limit if and only if $\sum_{n=1}^{\infty}\operatorname{Log} b_n$ converges.
3. Suppose that $c_n\neq -1$ for all $n\geq 1$. Show that $\prod_{n=1}^{\infty}(1+c_n)$ converges absolutely if and only if $\sum_{n=1}^{\infty}c_n$ converges absolutely.
4. Suppose a_n is a sequence of complex numbers with absolute value at most 1. Suppose further that a_n is **multiplicative**, that is, $a_{mn}=a_m a_n$. Show that the Dirichlet series

$$\sum_{n=1}^{\infty}\frac{a_n}{n^s}$$

is analytic in the region $\mathrm{Re}(s) > 1$ and is nonvanishing there.

5. Prove (4.6) using Theorem 4.31.
6. Using Theorem 4.31, deduce that $\zeta(2k)$ is a nonzero rational multiple of π^{2k}.

4.11 The Logarithm

Let X be a topological space. A **curve** in X is a continuous map

$$\gamma : [0, 1] \to X.$$

A curve is said to be **closed** if $\gamma(0) = \gamma(1)$.

Two closed curves γ_0, γ_1 in X are said to be **homotopic** in X if there is a continuous map

$$H : [0, 1] \times [0, 1] \to X$$

such that $H(s, 0) = \gamma_0(s)$ and $H(s, 1) = \gamma_1(s)$, for all $s \in [0, 1]$.

In this case, H is said to be a **homotopy** from γ_0 to γ_1.

One should think of this as giving mathematical expression to the idea that we can deform γ_0 to γ_1 continuously.

Recall that a space X is called **path-connected** if for any $a, b \in X$, there is a continuous map $\gamma : [0, 1] \to X$ such that $\gamma(0) = a$ and $\gamma(1) = b$.

Now suppose X is path-connected. We say X is **simply connected** if every closed path in X is homotopic to a point. That is, it is homotopic to the trivial closed curve $\gamma(s) = x_0$ for all $s \in [0, 1]$ and some $x_0 \in X$.

For example, the annulus is not simply connected.

One can show that if γ_0 and γ_1 are homotopic closed paths in a domain $\Omega \in \mathbb{C}$, then $\mathrm{Ind}_{\gamma_0}(\alpha) = \mathrm{Ind}_{\gamma_1}(\alpha)$ for all $\alpha \notin \Omega$.

The idea is that a homotopy H from γ_0 to γ_1 induces a family of paths $\gamma_t = H(\cdot, t)$ that depend continuously on t. Thus, the index $\mathrm{Ind}_{\gamma_t}(\alpha)$ depends continuously on t. But as the index is integer-valued, it is independent of t. In particular, if Ω is simply connected, then $\mathrm{Ind}_{\gamma}(\alpha) = 0$ for any closed path γ in Ω and any $\alpha \notin \Omega$.

Suppose $f \in H(\Omega)$ where Ω is a simply connected region. Fix $z_0 \in \Omega$ and define

$$g(z) = \int_{z_0}^{z} f(w)\, dw$$

for $z \in \Omega$, where $\int_{z_0}^{z}$ denotes the contour integral over a path in Ω from z_0 to z. By our earlier comments on simply connected regions, Cauchy's theorem (Theorem 4.5) tells us that the integral is path-independent. Thus, g is well defined and is holomorphic. Also, $g' = f$.

As a special case to this, let Ω be simply connected with $0 \notin \Omega$. Let $f(z) = 1/z$. Then, $f \in H(\Omega)$. Fixing $z_0 \in \Omega$, take w_0 such that $e^{w_0} = z_0$. This we can do in

several ways. For example, writing

$$z_0 = re^{i\theta}, \quad r > 0$$

we can take $w_0 = \log r + i\theta$.

Define

$$\log z = w_0 + \int_{z_0}^{z} \frac{dw}{w},$$

for $z \in \Omega$. Since there is freedom in choosing w_0, this definition is dependent on the choice of w_0. Each choice of w_0 gives a different **branch** of the logarithm.

In each case, $\log z$ is a holomorphic function for $0 \notin \Omega$.

Example 4.6 Define

$$\log z = \int_{1}^{z} \frac{dw}{w},$$

for $z \in \mathbb{C} \setminus \{x \in \mathbb{R} : x \leq 0\}$. This function is called the **principal branch** of the logarithm. Writing $z \in \Omega$, $z = re^{i\theta}$, $-\pi < \theta < \pi$, we see $\log z = \log r + i\theta$. In particular,

$$\log i = \frac{i\pi}{2} \text{ and } \log(-i) = -\frac{i\pi}{2}.$$

Example 4.7 If instead we take

$$\Omega = \mathbb{C} \setminus \{x \in \mathbb{R} : x \geqslant 0\}$$

, then the corresponding values of $\log i = \frac{i\pi}{2}$ and $\log(-i) = \frac{3i\pi}{2}$. □

We can use the logarithm to deal with more complicated integrals such as

$$\int_0^\infty f(x)x^{a-1}\,dx,$$

because we can define z^a as $e^{a\log z}$ provided we choose our domain properly.

Theorem 4.32 *Let $f(z)$ be analytic in $\mathbb{C}$ except for a finite number of poles α_i none of which lie on the positive real axis. Let $a \notin \mathbb{Z}$ and suppose*

(1) $\exists b > a$ such that $|f(z)| \leqslant \dfrac{K_1}{|z|^b}$ for some constant K_1 as $|z| \to \infty$.

(2) $\exists b'$ with $0 \leqslant b' < a$ such that $|f(z)| \leqslant \dfrac{K_2}{|z|^{b'}}$ as $|z| \to 0$.

Then,

$$\int_0^\infty f(x)x^{a-1}\,dx = -\frac{\pi e^{-\pi i a}}{\sin \pi a} \sum_i \text{Res}_{z=\alpha_i}(f(z)z^{a-1}),$$

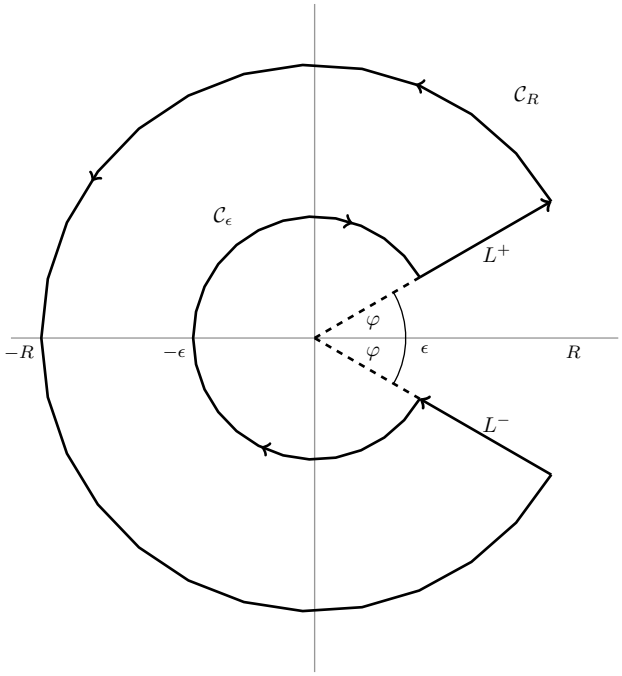


Fig. 4.11 Contour $\mathcal{C}$

where the sum is over the poles of f. (Since $a \notin \mathbb{Z}$, $\sin \pi a \neq 0$.)

Proof Write $z^{a-1} = e^{(a-1)\log z}$ where $\log z$ is the principal branch of the logarithm defined on the simply connected region

$$\mathbb{C} \setminus \{x \in \mathbb{R} : x \geqslant 0\}.$$

Let $\mathcal{C}$ be the closed contour indicated in Fig. 4.11.
By Cauchy,

$$\frac{1}{2\pi i} \int_{\mathcal{C}} f(z) z^{a-1}\, dz = \sum_{\alpha_i \text{ in } \mathcal{C}} \text{Res}_{z=\alpha_i}(f(z) z^{a-1}).$$

The contour has four segments : C_R, L^-, $C_\varepsilon and L^+$.
By hypothesis (1), $\int_{\mathcal{C}_\mathcal{R}} f(z) z^{a-1}\, dz \to 0$ as $R \to \infty$. Also, hypothesis (2) implies that $\int_{\mathcal{C}_\uparrow} f(z) z^{a-1}\, dz \to 0$ as $\varepsilon \to 0$ for similar reasons.
Now, we are left with $\int_{L^+}$ and $\int_{L^-}$. We parametrize L^+:

$$z = re^{i\varphi}, \quad \epsilon \leqslant r \leqslant R$$

so that

$$\log z = \log r + i\varphi.$$

Then,

$$\begin{aligned}\int_{L^+} f(z)z^{a-1}\,dz &= \int_\epsilon^R f(re^{i\varphi})e^{(a-1)(\log r+i\varphi)}\cdot e^{i\varphi}\,dr \\ &= e^{ia\varphi}\int_\epsilon^R f(re^{i\varphi})r^{a-1}\,dr.\end{aligned}$$

The function f is suitably bounded and by the dominated convergence theorem

$$\lim_{\epsilon\to 0}\lim_{\varphi\to 0}\int_\epsilon^R f(re^{i\varphi})r^{a-1}\,dr = \int_0^R f(r)r^{a-1}\,dr.$$

On L^-, $z = re^{i(2\pi-\varphi)}$ so that

$$\log z = \log r + i(2\pi-\varphi)$$

and

$$\int_{L^-} f(z)z^{a-1}\,dz = \int_R^\epsilon f(re^{i(2\pi-\varphi)})e^{(a-1)(\log r+i(2\pi-\varphi))}\cdot e^{i(2\pi-\varphi)}\,dr.$$

Now,

$$\begin{aligned}e^{(a-1)(\log r+i(2\pi-\varphi))}\cdot e^{i(2\pi-\varphi)} &= r^{a-1}\cdot e^{2\pi i(a-1)-\varphi i(a-1)}\cdot e^{-\varphi i} \\ &= r^{a-1}e^{2\pi ia}\cdot e^{-\varphi ia}\end{aligned}$$

so our integral is

$$\int_{L^-} f(z)z^{a-1}\,dz = -e^{2\pi ia}e^{-\varphi ia}\int_\epsilon^R f(re^{-i\varphi})r^{a-1}\,dr$$

so that letting $\varphi\to 0$ gives by the dominated convergence theorem that

$$\int_{L^-} f(z)z^{a-1}\,dz = -e^{2\pi ia}\int_\epsilon^R f(r)r^{a-1}\,dr.$$

Therefore,

$$\int_{L^+} f(z)z^{a-1}\,dz + \int_{L^-} f(z)z^{a-1}\,dz = (1-e^{2\pi ia})\int_\epsilon^R f(r)r^{a-1}\,dr.$$

As $\varepsilon\to 0$, this tends to

$$(1 - e^{2\pi i a}) \int_0^\infty f(r) r^{a-1}\, dr.$$

Putting everything together gives the result. □

Exercises

1. Let f be a meromorphic function and A its set of poles. Suppose it has a finite number of poles all of which lie in the upper half plane (i.e., have positive imaginary part). Suppose there is a constant $K > 0$ such that for some fixed $\delta > 1$, we have
$$|f(z)| \leq K/|z|^\delta,$$
for $|z|$ sufficiently large. Show that
$$\int_{-\infty}^{\infty} f(x)dx = 2\pi i \sum_{a \in A} \text{Res}(f; a).$$

2. Show that
$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx = \frac{\pi}{e}.$$

3. Let $f(z)$ have a simple pole at $z = 0$. Let $C(\epsilon)$ be the semicircular arc from $-\epsilon$ to ϵ of radius $\epsilon > 0$. Show that
$$\lim_{\epsilon \to 0} \int_{C(\epsilon)} f(z)dz = -\pi i \text{Res}(f; 0).$$
[Note: $C(\epsilon)$ is not a closed path.]

4. Suppose α is a complex number, $|\alpha| \neq 1$. Compute
$$\int_0^{2\pi} \frac{d\theta}{1 - 2\alpha \cos\theta + \alpha^2}.$$

5. Let C be the circle of radius 2 centered at the origin and oriented counterclockwise. Evaluate
$$\frac{1}{2\pi i} \int_C \frac{dz}{z^2(z-1)^3}.$$

4.12 The Phragmén–Lindelöf Theorem and Jensen's Theorem

An entire function $f(z)$ is said to be of **finite order** if for some $\alpha \geq 0$, we have

$$f(z) = O\left(e^{|z|^\alpha}\right) \tag{4.7}$$

as $|z| \to \infty$. If $\alpha = 0$, then $f(z)$ is constant by Liouville's theorem. The infimum of the numbers α such that the above estimate holds is called the **order** of $f(z)$. It is convenient to extend this notion as follows. Given a region A and f analytic on A, we say f has finite order on A if (4.7) is satisfied for some α and for all z in A.

In 1908, E. Phragmén (1863–1937) and E. Lindelöf (1870–1946) discovered an extension of the maximum modulus principle to vertical strips for functions of finite order.

Theorem 4.33 (Phragmén–Lindelöf theorem). *Let f be holomorphic in the vertical strip $a \leq \mathrm{Re}(z) \leq b$ and of finite order there. If f is bounded for z with $\mathrm{Re}(z) = a$ and $\mathrm{Re}(z) = b$, then it is bounded in the whole vertical strip.*

Proof Let B be a bound for f on the two vertical strips. Writing $z = x + iy$ with x, y real, we see that as f has finite order in the region, there are constants K and c such that for all y, we have

$$|f(x + iy)| \leq Ke^{|y|^c}, \qquad a \leq x \leq b.$$

Let m be an integer $\equiv 2 \pmod 4$ with $m > c$. For z in the vertical strip, $\arg z \to \pi/2$ as $|y| \to \infty$. Thus, there is a T_1 such that for $|y| > T_1$,

$$\left|\arg z - \frac{\pi}{2}\right| < \frac{\pi}{4m}.$$

Let $\epsilon > 0$ and consider the function

$$g_\epsilon(z) = e^{\epsilon z^m} f(z).$$

With our choice of m, it is now easily checked that (see Exercise 1) for $|y| \geq T_1$,

$$|g_\epsilon(z)| \leq Ke^{|y|^c} e^{-\epsilon|z|^m/\sqrt{2}} \tag{4.8}$$

so that $g_\epsilon(z) \to 0$ as $|z| \to \infty$ since $m > c$. Let T_2 be chosen so that $|g_\epsilon(z)| \leq B$ for $|\mathrm{Im}(z)| \geq T_2$. Thus,

$$|f(z)| \leq Be^{-\epsilon|z|^m \cos(m \arg z)} \leq Be^{\epsilon|z|^m},$$

for $|z| \geq T_2$. Applying the maximum modulus principle to the bounded region

$$a \le \mathrm{Re}(z) \le b, \quad 0 \le |\,\mathrm{Im}(z)| \le T_2,$$

we find $|f(z)| \le Be^{\epsilon|z|^m}$. Since this is valid for all $\epsilon > 0$, we can take $\epsilon \to 0$ to deduce the desired result. □

In the 1890s, Hadamard developed the theory of entire functions of finite order. He showed that, very much like polynomials, they can be factored into an infinite product over the zeros of $f(z)$.

In the next section, we will derive this factorization theorem of Hadamard for entire functions of order 1. In this section, we will derive Jensen's theorem which will be used in proving Hadamard's factorization theorem.

Let $f(z)$ be an entire function of finite order β. Jensen's theorem relates β to the distribution of the zeros of $f(z)$.

Theorem 4.34 *An entire function $f(z)$ of finite order β without any zeros must be of the form $f(z) = e^{g(z)}$, where $g(z)$ is a polynomial and $\beta = \deg g$.*

Proof Let $h(z) = \log f(z) - \log f(0)$. Then, $h(z)$ is entire, since $f(z)$ has no zeros. Also, for any $\epsilon > 0$,

$$\mathrm{Re}\, h(z) = \log |f(z)| \ll R^{\beta+\epsilon}.$$

Writing

$$h(z) = \sum_{n=0}^{\infty} (a_n + ib_n) z^n$$

with $a_n, b_n \in \mathbb{R}$, we see that for $z = Re^{i\theta}$,

$$\mathrm{Re}(h(z)) = \sum_{n=0}^{\infty} a_n R^n \cos n\theta - \sum_{n=0}^{\infty} b_n R^n \sin n\theta.$$

By Fourier analysis, we get

$$|a_n| R^n \ll \int_0^{2\pi} \left|\mathrm{Re}\left(h\left(Re^{i\theta}\right)\right)\right| d\theta.$$

Since $h(0) = 0$, we have $a_0 = 0$, and therefore,

$$\int_0^{2\pi} \mathrm{Re}\left(h\left(Re^{i\theta}\right)\right) d\theta = 0.$$

Observe that for $x \in \mathbb{R}$, we have

$$|x| + x = \begin{cases} 2x & \text{if } \ x \ge 0, \\ 0 & \text{if } \ x < 0. \end{cases}$$

Hence,

$$|a_n|R^n \ll \int_0^{2\pi} \Big\{ |\operatorname{Re}(h(Re^{i\theta}))| + \operatorname{Re} h(Re^{i\theta}) \Big\} d\theta$$
$$\ll R^{\beta+\epsilon}.$$

Letting $R \to \infty$ yields $a_n = 0$ if $n > \beta$. □

Notice that in this theorem the same result holds if the estimate

$$|f(z)| \ll e^{R_i^{\beta+\epsilon}}$$

holds for $|z| = R_i$ and R_i is a sequence tending to infinity. This observation will be used later in our proof of the Hadamard factorization theorem.

Theorem 4.35 (Jensen's theorem). *Let $f(z)$ be an entire function of order β such that $f(0) \neq 0$. If $z_1, z_2, \ldots, z_n$ are the zeros of $f(z)$ in $|z| < R$, counted with multiplicity, then*

$$\frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta = \log |f(0)| + \log \Big(\frac{R^n}{|z_1| \cdots |z_n|} \Big).$$

Proof We may assume, without loss of generality, that $f(0) = 1$. Also, it is clear that if the theorem is true for functions g and h, then it is also true for the product gh. Thus, it suffices to prove it for functions with either no zero or one zero in $|z| < R$. Indeed, if f has no zeros in $|z| < R$, the right-hand side is zero. The left-hand side is

$$\frac{1}{2\pi i} \int_{|z|=R} (\log f(z)) \frac{dz}{z},$$

which by Cauchy's theorem is zero. Taking real parts gives the desired result.

If f has one zero $z = z_1$ in $|z| < R$, we consider the contour $|z| = R$ taken in the counterclockwise direction and cut it from z_1 to the boundary. We deform the contour so that we go around z_1 in a clockwise direction along a circle of radius ϵ (say). Then, by Cauchy's theorem with $g(z) = \log f(z)$,

$$0 = \frac{1}{2\pi i} \int_{\mathcal{C}} g(z) \frac{dz}{z}$$

where $\mathcal{C}$ is the contour described above.

Since the argument changes by $-2\pi i$ when $g(z)$ goes around the zero $z = z_1$, we see that as $\epsilon \to 0$, we deduce

$$\frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta = \log \frac{R}{|z_1|},$$

as desired. This completes the proof. □

An alternative proof of Jensen's theorem can be given that avoids the use of cutting the plane. One considers

$$f(z) = \frac{R(z - z_1)}{R^2 - \overline{z}_1 z}.$$

Then, $f(z)$ is regular for $|z| \leq R$. Moreover, $|f(z)| = 1$ on $|z| = R$, and $|f(0)| = |z_1|/R$, as a simple calculation shows. Jensen's theorem is easily verified for this choice of f. But any holomorphic function on $|z| \leq R$ can be written as a function with no zeros in $|z| \leq R$ and a product of functions of the form

$$\frac{R(z - z_i)}{R^2 - \overline{z}_i z}.$$

Now Jensen's theorem easily follows.

Theorem 4.36 *Let f be as in Theorem 4.35. Then,*

$$\log\left(\frac{R^n}{|z_1| \cdots |z_n|}\right) \leq \max_{|z|=R} \log|f(z)| - \log|f(0)|.$$

Proof This is clear from Jensen's theorem. □

Exercises

1. Prove inequality (4.8).
2. Suppose $f(z)$ is analytic in $a \leq \mathrm{Re}(z) \leq b$ and of finite order there. Suppose that there is a constant $A \geq 0$ such that $|f(z)| = O(|y|^A)$ on the two vertical strips. Show that the same estimate holds for points in the interior. (This generalizes the Phragmén–Lindelöf theorem.)
3. Let f be a function of finite order and define $n_f(r) := n(r)$ to be the number of zeros of f in $|z| \leq r$. Show that

$$\int_0^R \frac{n(r)dr}{r} \leq \max_{|z|=R} \log|f(z)| - \log|f(0)|,$$

   with f as in Jensen's theorem.
4. If $f(z)$ is of order β, show that $n_f(r) = O(r^{\beta+\epsilon})$, for any $\epsilon > 0$.
5. Let $f(z)$ be an entire function of order β. Show that

$$\sum_{n=1}^{\infty} |z_n|^{-\beta-\epsilon}$$

   converges for any $\epsilon > 0$ (here, we have indexed the zeros z_i so that $|z_1| \leq |z_2| \leq \cdots$).

4.13 Entire Functions of Order 1

We will now derive a factorization theorem for entire functions of order 1. A similar result holds for entire functions of higher order.

Theorem 4.37 *Let $f(z)$ be an entire function of order 1 with zeros $z_1, z_2, \ldots$ arranged so that $|z_1| \leq |z_2| \leq \cdots$ and repeated with appropriate multiplicity. Then, f can be written as*

$$f(z) = e^{A+Bz} \prod_{n=1}^{\infty} \left(1 - \frac{z}{z_n}\right) e^{z/z_n},$$

where A and B are constants.

Proof The product

$$P(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{z_n}\right) e^{z/z_n}$$

converges absolutely for all z, since

$$(1 - z)e^z = 1 - z^2 + \cdots$$

and by Exercise 5 in the previous section. Thus, $P(z)$ represents an entire function. If we write

$$f(z) = P(z)F(z),$$

then $F(z)$ is an entire function without zeros. If F were of finite order, we could conclude by Theorem 4.34 that $F(z) = e^{g(z)}$, where $g(z)$ is a polynomial.

By the remark after Theorem 4.34, it suffices to show that

$$|F(z)| \ll e^{R_i^{1+\epsilon}}$$

to deduce that $F(z) = e^{g(z)}$ where $g(z)$ is of the form $A + Bz$ for certain constants A and B.

To this end, we will choose R_i satisfying

$$\Big|R_i - |z_n|\Big| > |z_n|^{-2}$$

for all n. This can be done, since the total measure of the intervals $(|z_n| - |z_n|^{-2}, |z_n| + |z_n|^{-2})$ is bounded by

$$2\sum_{n=1}^{\infty} |z_n|^{-2} < \infty,$$

since $f(z)$ has order 1.

We write

$$P(z) = P_1(z)P_2(z)P_3(z),$$

where in P_1, $|z_n| < \frac{1}{2}R_i$, in P_2, $\frac{1}{2}R_i \le |z_n| \le 2R_i$ and in P_3, $|z_n| > 2R_i$. For the factors of P_1, we have for $|z| = R_i$,

$$\left|\left(1 - \frac{z}{z_n}\right)e^{z/z_n}\right| \ge \left(\left|\frac{z}{z_n}\right| - 1\right)e^{-|z|/|z_n|} > e^{-R_i/|z_n|}.$$

Since

$$\sum_{|z_n|<\frac{1}{2}R} |z_n|^{-1} \le \left(\frac{1}{2}R\right)^{\epsilon} \sum_{n=1}^{\infty} |z_n|^{-1-\epsilon},$$

we get

$$|P_1(z)| > \exp(-R_i^{1+\epsilon}).$$

For $P_2(z)$,

$$\left|\left(1 - \frac{z}{z_n}\right)e^{z/z_n}\right| \ge e^{-2}|z - z_n|/2R_i \gg R_i^{-3}$$

by the way we have chosen R_i.

Since $n(R_i) = O(R_i^{1+\epsilon})$, we get

$$|P_2(z)| \gg (R_i^{-3})^{R_i^{1+\epsilon}} \ge \exp(-c_1 R_i^{1+2\epsilon}).$$

Finally, for $P_3(z)$, we have $|z/z_n| < 1/2$ so that

$$\left|\left(1 - \frac{z}{z_n}\right)e^{z/z_n}\right| \ge e^{-c_2 R_i^2/|z_n|^2}$$

and

$$\sum_{|z_n|>2R} |z_n|^{-2} < (2R)^{-1+\epsilon} \sum_{n=1}^{\infty} |z_n|^{-1-\epsilon}.$$

Thus, on $|z| = R_i$ we have

$$|P(z)| > \exp(-R^{1+3\epsilon}),$$

so that

$$|F(z)| < \exp(R^{1+4\epsilon}).$$

Hence, $F(z) = e^{g(z)}$, where $g(z)$ is a polynomial of degree at most 1. This completes the proof. □

There is an additive analogue of Hadamard's factorization theorem called the **Mittag-Leffler theorem**. A special case is given in the exercises below (see Exercise 4). The general version answers the question of whether a meromorphic function exists having a given principal part at a sequence of points a_n.

Theorem 4.38 (Mittag-Leffler theorem) *Let a_n be a sequence of distinct complex numbers such that $|a_n| \to \infty$. Let P_n be a sequence of polynomials without a constant term. Then, there exists a meromorphic function f whose only poles are at a_n with principal part $P_n(1/(z-a_n))$. The most general function of this kind can be written in the form*

$$f(z) = \sum_{n=1}^{\infty} \left[P_n\left(\frac{1}{z-a_n}\right) - Q_n(z) \right] + \phi(z),$$

where Q_n is a suitable polynomial and ϕ is entire. The series converges absolutely and uniformly on any compact set not containing the poles.

Proof Without loss of any generality, we may suppose that none of the a_n are equal to zero, since we can always add the principal part at zero to our function, if necessary. As the function

$$P_n\left(\frac{1}{z-a_n}\right)$$

is analytic at $z = 0$, we can write down its power series expansion. Let $Q_n(z)$ be the sum of the terms of degree less than d_n (with d_n being a natural number to be chosen) so that

$$\left| P_n\left(\frac{1}{z-a_n}\right) - Q_n(z) \right| \leq C_n \left| \frac{z}{a_n} \right|^{d_n},$$

valid for $|z| \leq |a_n|/2$ (say) and C_n is a suitable positive constant greater than 1. We choose d_n inductively so that $d_1 < d_2 < \cdots$ and

$$d_n > \max([\log C_n] + 1, d_1, d_2, \ldots, d_{n-1}).$$

Thus,

$$\frac{|C_n|^{1/d_n}}{|a_n|} \to 0,$$

because $|a_n| \to \infty$. We claim the series

$$\sum_{n=1}^{\infty} \left[P_n\left(\frac{1}{z-a_n}\right) - Q_n(z) \right]$$

converges absolutely and uniformly for z in any compact set not containing the a_n's. Indeed, for any given R, let N be such that $|a_N| \geq R$ and split the series as

$$\sum_{n=1}^{N}\left[P_n\left(\frac{1}{z-a_n}\right)-Q_n(z)\right]+\sum_{n=N+1}^{\infty}\left[P_n\left(\frac{1}{z-a_n}\right)-Q_n(z)\right].$$

The first part is a finite sum. If $|z|\le R/2$, the second part is bounded by

$$\sum_{n=N+1}^{\infty}(C_n/|a_n|^{d_n})z^{d_n},$$

which has a radius of convergence equal to infinity, by the ratio test. The finite sum has the desired polar part for $|z|\le R$. As this is true for every R, this completes the proof of the theorem. □

Exercises

1. Show that for $|z|<1$,
$$\prod_{n=0}^{\infty}(1+z^{2^n})=\frac{1}{1-z}.$$

2. Prove that
$$\cos\pi z=\prod_{n=1}^{\infty}\left(1-\frac{4z^2}{(2n-1)^2}\right).$$

3. Find the infinite product factorization of $\sinh z$ and $\cosh z$.
4. Let $f(z)$ be meromorphic with simple poles at $a_1,a_2,\ldots$, where $0<|a_1|<|a_2|<\cdots$. Let b_n be the residue of $f(z)$ at $z=a_n$. Suppose that f is uniformly bounded on a sequence of circles whose radii approach infinity. Prove that
$$f(z)=f(0)+\sum_{n=1}^{\infty}\left(\frac{b_n}{z-a_n}+\frac{b_n}{a_n}\right).$$

5. Using the previous exercise, show that
$$\cot z=\frac{1}{z}+\sum_{n\neq 0}\left(\frac{1}{z-n\pi}+\frac{1}{n\pi}\right),$$
where the sum is over all integers $n\neq 0$.

4.14 The Gamma Function

The Γ function is defined as follows.

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt, \qquad \mathrm{Re}(z) > 0.$$

It is easily verified that the integral represents an analytic function in the half plane $\mathrm{Re}(z) > 0$. Integration by parts shows

$$\Gamma(z+1) = z\Gamma(z). \tag{4.9}$$

For natural numbers n, we see that $\Gamma(n+1) = n!$ and so $\Gamma(z)$ is a complex analytic interpolation of the factorials. This functional equation allows us to derive an **analytic continuation** of $\Gamma(z)$ as a meromorphic function to the entire complex plane. Indeed, the above functional Eq. (4.9) can be rewritten as

$$\Gamma(z) = \frac{\Gamma(z+1)}{z},$$

and the right-hand side is analytic if $\mathrm{Re}(z) > -1$. Thus, by this inductive procedure, it is easy to see that $\Gamma(z)$ extends to a meromorphic function in the entire complex plane with simple poles at $z = 0, -1, -2, \ldots$.

We will prove that $1/\Gamma(z)$ is an entire function of order 1 and derive its Hadamard factorization. To this end, we first derive a second functional equation satisfied by the Γ function. The following result has the appearance of being a consequence of Theorem 4.32 but cannot be deduced from that theorem.

Theorem 4.39 *We have*

$$\int_0^\infty \frac{v^{x-1} dv}{1+v} = \frac{\pi}{\sin \pi x}$$

for $0 < x < 1$.

Proof Consider the integral

$$\int_{\mathcal{C}_\epsilon} \frac{z^{x-1} dz}{1+z},$$

where $\mathcal{C}_\epsilon$ is the contour taken along the real axis from ϵ to R, then in the positive direction along the circle c_1 of radius R centered at the origin and then back along the real axis to $z = \epsilon$ and finally around the circle c_2 of radius ϵ centered at the origin, taken in the negative direction.

The function

$$\frac{z^{x-1}}{1+z}$$

is regular except at $z = -1$, where it has a simple pole with residue

$$e^{\pi i(x-1)}.$$

We will take $\epsilon < 1 < R$ so that integrating the function along the contour indicated above shows by Cauchy's theorem

$$\int_\epsilon^R \frac{u^{x-1}du}{1+u} + \int_{c_1} \frac{z^{x-1}dz}{1+z} + \int_R^\epsilon \frac{(ue^{2\pi i})^{x-1}du}{1+u} + \int_{c_2} \frac{z^{x-1}dz}{1+z} = (2\pi i)e^{\pi i(x-1)}.$$

The two integrals along the real axis together give

$$(1-e^{2\pi i(x-1)})\int_\epsilon^R \frac{u^{x-1}du}{1+u} = -2ie^{\pi ix}(\sin \pi x)\int_\epsilon^R \frac{u^{x-1}du}{1+u}.$$

The other two integrals tend to 0 as $R \to \infty$ because on c_1,

$$\left|\frac{z^{x-1}}{1+z}\right| \leq \frac{R^{x-1}}{R-1},$$

so that

$$\left|\int_{c_1} \frac{z^{x-1}dz}{1+z}\right| \leq \frac{R^{x-1}}{R-1}2\pi R = \frac{2\pi R^x}{R-1},$$

which tends to 0, since $x < 1$.

Similarly,

$$\left|\int_{c_2} \frac{z^{x-1}dz}{1+z}\right| \leq \frac{2\pi\epsilon^x}{1-\epsilon},$$

which tends to 0, as $\epsilon \to 0$ since $x > 0$. Therefore,

$$-2ie^{\pi ix}(\sin \pi x)\int_0^\infty \frac{u^{x-1}du}{1+u} = -2\pi i e^{\pi ix},$$

which gives

$$\int_0^\infty \frac{u^{x-1}}{1+u} = \frac{\pi}{\sin \pi x}.$$

□

Theorem 4.40 *We have*

$$\Gamma(x)\Gamma(y) = 2\Gamma(x+y)\int_0^{\pi/2} (\cos\theta)^{2x-1}(\sin\theta)^{2y-1}d\theta$$

for $x, y > 0$.

Proof For $x, y > 0$, we have

$$\Gamma(x)\Gamma(y) = \left(\int_0^\infty e^{-t}t^{x-1}dt\right)\left(\int_0^\infty e^{-u}u^{y-1}du\right).$$

Putting $u = tv$ and inverting the order of integration, we obtain

$$\begin{aligned}\Gamma(x)\Gamma(y) &= \int_0^\infty e^{-t}t^{x-1}dt \int_0^\infty t^y v^y e^{-tv}\frac{dv}{v}\\ &= \int_0^\infty v^{y-1}dv \int_0^\infty e^{-t(1+v)}t^{x+y-1}dt\\ &= \Gamma(x+y)\int_0^\infty \frac{v^{y-1}dv}{(1+v)^{x+y}}.\end{aligned}$$

The interchanging of integrals is easily justified by Fubini's theorem. This last integral is

$$2\int_0^{\pi/2}(\cos\theta)^{2x-1}(\sin\theta)^{2y-1}d\theta,$$

where we have put $v = \tan^2\theta$. □

Making the substitution of $\lambda = \cos^2\theta$ transforms the integral of Theorem 4.40 to

$$\int_0^1 \lambda^{x-1}(1-\lambda)^{y-1}d\lambda,$$

which is called the **beta function** $B(x, y)$.

In particular, for $0 < x < 1$, we obtain

$$\Gamma(x)\Gamma(1-x) = \int_0^1 \lambda^{x-1}(1-\lambda)^{-x}d\lambda.$$

Putting

$$v = \frac{\lambda}{1-\lambda}$$

in the integral gives

$$\Gamma(x)\Gamma(1-x) = \int_0^\infty \frac{v^{x-1}dv}{1+v},$$

which by Theorem 4.39 is

$$\frac{\pi}{\sin\pi x}.$$

These observations prove the following theorem.

Theorem 4.41 *We have*

$$\Gamma(x)\Gamma(y) = \Gamma(x+y)\int_0^1 \lambda^{x-1}(1-\lambda)^{y-1}d\lambda.$$

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}$$

for $0 < x < 1$.

By the principle of analytic continuation, we deduce that for all complex z, we have

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}.$$

The theorem also provides an analytic continuation of the beta function.

Theorem 4.42 $1/\Gamma(z)$ *is an entire function with simple zeros at* $z = 0, -1, -2, \ldots$.

Proof From the functional equation

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z},$$

we see that $\Gamma(z)\Gamma(1-z)$ is regular except when z is an integer, in which case it has a simple pole.

We also see from this functional equation that since $\Gamma(z)$ is regular in $\mathrm{Re}(z) > 0$, $\Gamma(1-z)$ has simple poles at $z = 1, 2, 3, \ldots$. Therefore,

$$1/\Gamma(z) = \Gamma(1-z)(\sin \pi z)/\pi$$

is regular in $\mathrm{Re}(1-z) \geq 0$. If $\mathrm{Re}(z) \leq 0$, then $\mathrm{Re}(1-z) \geq 1$ and the right-hand side of the above equation is regular. This completes the proof. □

Exercises

1. Prove that
$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

2. **(Legendre's duplication formula)** Show that
$$\Gamma(2x)\Gamma\left(\frac{1}{2}\right) = 2^{2x-1}\Gamma(x)\Gamma\left(x+\frac{1}{2}\right)$$
for $x > 0$.

3. Let c be a positive constant. Show that as $x \to \infty$,

$$\Gamma(x+c) \sim x^c \Gamma(x).$$

4.15 Stirling's Formula

The student may be familiar with the classical Stirling's formula that

$$n! \sim \left(\frac{n}{e}\right)^n \sqrt{2\pi n},$$

as n tends to infinity. As the Γ function interpolates the factorials, it is prudent to see if the asymptotic extends to the complex domain. This is the goal of this section. The classical Stirling's formula will be a corollary of our discussion.

Theorem 4.43 (Stirling's formula) *We have*

$$\Gamma(x) \sim e^{-x} x^{x-1/2} \sqrt{2\pi}$$

as $x \to \infty$.

Proof By partial summation, we know that for a natural number n,

$$\log \Gamma(n) = \log(n-1)! = \left(n - \frac{1}{2}\right) \log n - n + c_1 + o(1)$$

as $n \to \infty$ (and with c_1 an absolute constant). If x is not an integer, let us write $x = n + c$ for some $0 < c < 1$. By Exercise 3 of the previous section, we have

$$\Gamma(n+c) \sim n^c \Gamma(n),$$

so that

$$\begin{aligned} \log \Gamma(x) &= \log \Gamma(n) + c \log n + o(1) \\ &= \left(x - \frac{1}{2}\right) \log n - n + c_1 + o(1). \end{aligned}$$

Also,

$$\log\left(\frac{n+c}{n}\right) = \log\left(1 + \frac{c}{n}\right) = \frac{c}{n} + O\left(\frac{1}{n^2}\right),$$

so that

$$\log x = \log n + \frac{c}{n} + O\left(\frac{1}{x^2}\right).$$

Inserting this observation above gives

$$\log \Gamma(x) = \left(x - \frac{1}{2}\right) \log x - x + c_1 + o(1).$$

We can use the duplication formula to evaluate c_1. Indeed, on the one hand we have from above

$$\log \Gamma(2x) = \left(2x - \frac{1}{2}\right) \log 2x - 2x + c_1 + o(1).$$

On the other hand, by the duplication formula (Exercise 2 of the last section) we have

$$\log \Gamma(2x) = (2x - 1) \log 2 + \log \Gamma(x) + \log \Gamma\left(x + \frac{1}{2}\right) - \frac{1}{2} \log \pi,$$

which is equal to

$$\left(2x - \frac{1}{2}\right) \log 2x - 2x - \frac{1}{2} \log 2 + 2c_1 - \frac{1}{2} \log \pi + o(1),$$

so that

$$c_1 = 2c_1 - \frac{1}{2} \log \pi - \frac{\log 2}{2}.$$

Thus, as required

$$c_1 = \log \sqrt{2\pi}.$$

□

Theorem 4.44 *For some constant K,*

$$\frac{\Gamma'(z)}{\Gamma(z)} = \int_0^1 \left\{1 - (1-t)^{z-1}\right\} \frac{dt}{t} - K.$$

Proof By Theorem 4.41, we have

$$\begin{aligned} \frac{\Gamma(z-h)\Gamma(h)}{\Gamma(z)} &= \int_0^1 t^{h-1}(1-t)^{z-h-1} dt \\ &= \frac{1}{h} + \int_0^1 \left\{(1-t)^{z-h-1} - 1\right\} t^{h-1} dt. \end{aligned}$$

The Taylor expansion of the left-hand side with respect to h is

$$\frac{1}{\Gamma(z)} \left\{\Gamma(z) - \Gamma'(z)h + \cdots\right\} \left\{\frac{1}{h} + K + \cdots\right\}$$

$$= \frac{1}{h} - \frac{\Gamma'(z)}{\Gamma(z)} + K + O(h).$$

The Taylor expansion of the right-hand side is

$$= \frac{1}{h} + \int_0^1 \left\{(1-t)^{z-1} - 1\right\} \frac{dt}{t} + O(h),$$

so that by equating the constant terms we get the desired result.

Theorem 4.45 *For z not a negative integer,*

$$\frac{\Gamma'(z)}{\Gamma(z)} = \sum_{n=0}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+z}\right) - K$$

for some constant K.

Proof First, for $z > 1$, we use Theorem 4.44 and expand

$$\frac{dt}{t} = \sum_{n=0}^{\infty} (1-t)^n dt$$

in the integrand and integrate term by term to obtain the result. The step is valid for $z > 1$ and by analytic continuation for all z unequal to a negative integer. This completes the proof. □

We are now ready to derive the Hadamard factorization of $1/\Gamma(z)$.

Theorem 4.46 *The Hadamard factorization of $1/\Gamma(z)$ is given by:*

$$\frac{1}{\Gamma(z)} = e^{\gamma z} z \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n},$$

where γ denotes Euler's constant.

Proof We integrate the formula

$$\frac{\Gamma'(z)}{\Gamma(z)} = \sum_{n=0}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+z}\right) - K$$

from $z = 1$ to $z = w$ and take exponentials, to obtain

$$\frac{1}{\Gamma(z)} = e^{Bz} z \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n}$$

for some constant B. Putting $z = 1$ and taking logarithms gives

$$\begin{aligned} 0 &= B + \sum_{n=1}^{\infty} \left\{ \log\left(1 + \frac{1}{n}\right) - \frac{1}{n} \right\} \\ &= B + \lim_{N\to\infty} \sum_{n=1}^{N} \left\{ \log\left(1 + \frac{1}{n}\right) - \frac{1}{n} \right\} \\ &= B - \gamma. \end{aligned}$$

□

Exercises

1. Show that

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log 2\pi + \int_0^{\infty} \frac{[u] - u + \frac{1}{2}}{u + z} du.$$

2. For any $\delta > 0$, show that

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log 2\pi + O\left(\frac{1}{|z|}\right)$$

   uniformly for $-\pi + \delta \le \arg z \le \pi - \delta$.
3. If σ is fixed, and $|t| \to \infty$, show that

$$|\Gamma(\sigma + it)| \sim e^{-\frac{1}{2}\pi|t|} |t|^{\sigma - \frac{1}{2}} \sqrt{2\pi}.$$

4. Show that $1/\Gamma(z)$ is of order 1.

5. Show that

$$\frac{\Gamma'(z)}{\Gamma(z)} = \log z + O\left(\frac{1}{|z|}\right)$$

   for $|z| \to \infty$ in the sector $-\pi + \delta < \arg z < \pi - \delta$ for any fixed $\delta > 0$.
6. Prove that $\Gamma(s)$ has poles only at $s = 0, -1, -2, \ldots$, and that these are simple, with

$$\mathrm{Res}_{s=-k} \Gamma(s) = (-1)^k / k!.$$

7. Show that

$$e^{-1/x} = \frac{1}{2\pi i} \int_{(\sigma)} x^s \Gamma(s) ds,$$

   for any $\sigma > 1$ and $x \ge 1$.

8. Let $f(s) = \sum_{n=1}^{\infty} a_n/n^s$ be an absolutely convergent Dirichlet series in the half plane $\mathrm{Re}(s) > 1$. Show that

$$\sum_{n=1}^{\infty} a_n e^{-n/x} = \frac{1}{2\pi i} \int_{(\sigma)} f(s) x^s \Gamma(s) ds$$

for any $\sigma > 1$.

9. With $\zeta(s)$ denoting the Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad \mathrm{Re}(s) > 1,$$

prove that for $\mathrm{Re}(s) > 1$,

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \int_0^{\infty} x^{s/2} \left(\sum_{n=1}^{\infty} e^{-n^2 \pi x} \right) \frac{dx}{x}.$$

10. Use formula 3.6 (with $a = 0$) to show that

$$\Xi(s) := \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \frac{1}{s(s-1)} + \int_1^{\infty} w(x) \left(x^{\frac{s}{2}} + x^{\frac{1-s}{2}} \right) \frac{dx}{x},$$

where

$$w(x) = \sum_{n=1}^{\infty} e^{-n^2 \pi x}.$$

Show that the integral converges absolutely for all $s \in \mathbb{C}$. This gives the analytic continuation and functional equation

$$\Xi(s) = \Xi(1-s),$$

of the Riemann zeta function. The proof is due to Riemann.

4.16 The Wiener–Ikehara Tauberian Theorem

In this section, we present an important application of complex analysis to the theory of Dirichlet series. This application, known as the Wiener–Ikehara Tauberian theorem, is motivated by the prime number theorem, which is the statement that the number of primes $p \le x$, denoted $\pi(x)$, satisfies the asymptotic formula

$$\pi(x) \sim \frac{x}{\log x}$$

as x tends to infinity. This theorem was proved independently in 1896 by J. Hadamard and C. de la Vallée Poussin. A critical ingredient in both proofs is the nonvanishing of the Riemann zeta function on the line $\mathrm{Re}(s) = 1$. Later through Tauberian theory, Wiener showed that in fact, the prime number theorem is *equivalent* to the assertion $\zeta(1 + it) \neq 0$ for all $t \in \mathbb{R}$. Thus, the general feeling in the 1940s was that the prime number theorem is essentially a theorem in complex analysis so that it was a surprise when an "elementary" proof of the prime number theorem, which avoided complex analysis, was discovered by A. Selberg (1917–2007) and P. Erdös (1913–1996) in 1948.

We motivate the notion of Tauberian theorems with a brief discussion of Abel's theorem which the reader may have already seen in an undergraduate course in analysis.

Let $\sum_{n=0}^{\infty} a_n x^n, x \in \mathbb{R}$ be a power series centered at 0 having radius of convergence 1. At the boundary of the region of convergence, i.e., at $|x| = 1$, the series may converge or diverge. In Chap. 1, we have already seen examples of this phenomenon. Abel's theorem states that if the series converges at a boundary point, then it is reasonably well behaved in the sense that it is continuous at that point. More precisely, if

$$\sum_{n=0}^{\infty} a_n = A, \tag{4.10}$$

then

$$\lim_{x \to 1^-} \sum_{n=0}^{\infty} a_n x^n = A. \tag{4.11}$$

Broadly speaking, Tauberian theorems are conditional converses of Abel's theorem. More precisely, any theorem in which conditions are imposed on the a_n's to ensure the converse of Abel's theorem holds is called a **Tauberian theorem**. They derive their name from a theorem of A. Tauber (1866–1942) [2] published in 1897, which states that if (4.11) is satisfied and we have the growth condition $a_n = o(1/n)$ on the coefficients of the power series, then (4.10) holds. These growth conditions were subsequently relaxed, most notably by Hardy and Littlewood and by K. Ananda Rau who showed in 1928 that if $a_n = O(1/n)$, then (4.10) holds.

Some of the most interesting applications of Tauberian theorems pertain to analytic number theory. In this context, Tauberian results can be thought of as estimates for the partial sums of coefficients of certain Dirichlet series. An important result of this type is the Wiener–Ikehara theorem. Introduced by Ikehara [3] in 1931, it generalizes a theorem of E. Landau (1877–1938) [4], by applying a Tauberian result obtained by Wiener. Proofs of this and other Tauberian theorems in the literature are usually found to be quite involved.

A well-known application of the Wiener–Ikehara theorem is to the derivation of the prime number theorem. In 1980, Newman [5] gave an ingenious short proof of the prime number theorem. We modify Newman's proof to derive the Wiener–Ikehara Tauberian theorem. That this can be done was also recognized by Korevaar

[6]. However, our presentation is simplified and our theorem is more general. We derive as a consequence an assortment of prime number theorems following the arrangement of Serre [7].

Exercises

1. Let $\Lambda(n) = \log p$ if $n = p^a$ for some prime p and zero otherwise. Show that the prime number theorem is equivalent to

$$\psi(x) := \sum_{n \le x} \Lambda(n) \sim x,$$

as $x \to \infty$.

2. Prove that

$$\sum_{p \le x} \log p = \psi(x) + O(x^{1/2} \log x),$$

where the summation is over primes p less than or equal to x.

4.17 The Analytic Theorem

The following analytic theorem of Newman [5] is the key result that will be used to prove the Tauberian theorem. The proof is an application of Cauchy's residue theorem. Newman's novel idea was the insertion of a new kernel into the relevant integral, playing a role similar to that of the Fejér kernel in standard proofs of the Tauberian theorem.

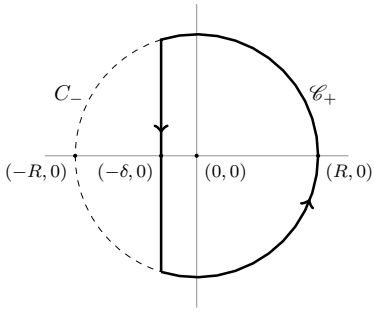
Theorem 4.47 *For $t \ge 0$, let $f(t)$ be a bounded and locally integrable function and let $g(s) := \int_0^\infty f(t)e^{-st}dt$ for $\mathrm{Re}(s) > 0$. If $g(s)$ has an analytic continuation to $\mathrm{Re}(s) \ge 0$, then $\int_0^\infty f(t)dt$ exists and equals $g(0)$.*

Proof For $T > 0$, let $g_T(s) = \int_0^T f(t)e^{-st}dt$. This integral converges for all values of s, and it is easy to see that $g_T(s)$ is an entire function. We need to show that

$$\lim_{T \to \infty} g_T(0) = g(0).$$

We will denote $\mathrm{Re}(s)$ by σ. Fix $R > 0$ and consider the positively oriented contour $\mathscr{C}$ shown in Fig. 4.12. Here, $\delta > 0$ (depending on R) is chosen small enough so that $g(s)$ is analytic on $\mathscr{C}$. Indeed, as $g(s)$ is analytic on the line $\sigma = 0$, one can cover the vertical strip from $(0, R)$ to $(0, -R)$ with open balls, on each of which $g(s)$ is analytic. Compactness of this strip allows one to obtain a finite subcover, which then gives the desired δ.

Fig. 4.12 Contour $\mathscr{C}$



We use the following notation :

$$\mathscr{C}_+ = \mathscr{C} \cap \{s : \sigma > 0\}, \quad \mathscr{C}_- = \mathscr{C} \cap \{s : \sigma < 0\}.$$

We also denote the semicircle of radius R to the left of the line $\sigma = 0$ by C_-. We will use the big O notation, treating everything other than the variables T, R and σ as constants.

Cauchy's theorem gives us

$$I_{\mathscr{C}} := \frac{1}{2\pi i} \int_{\mathscr{C}} (g(s) - g_T(s)) e^{sT} \left(1 + \frac{s^2}{R^2}\right) \frac{ds}{s} = g(0) - g_T(0), \tag{4.12}$$

as the integrand is analytic inside $\mathscr{C}$ except for a simple pole at $s = 0$. We denote the corresponding integrals over $\mathscr{C}_+$ and $\mathscr{C}_-$ as $I_{\mathscr{C}_+}$ and $I_{\mathscr{C}_-}$, respectively. Let $M = \sup_{t \geq 0} |f(t)|$. On $\mathscr{C}_+$, as $\sigma > 0$, we have

$$|g(s) - g_T(s)| = \left| \int_T^\infty f(t) e^{-st} dt \right| \leq M \int_T^\infty e^{-\sigma t} dt \ll \frac{e^{-\sigma T}}{\sigma}.$$

Using $s = Re^{i\theta}$ and $R \cos\theta = \sigma$ on $\mathscr{C}_+$, we obtain the following estimate for the kernel

$$\left| e^{sT} \frac{1}{s} \left(1 + \frac{s^2}{R^2}\right) \right| = e^{\sigma T} \left| \frac{1}{Re^{i\theta}} + \frac{e^{i\theta}}{R} \right| = e^{\sigma T} \left| \frac{2\cos\theta}{R} \right| \ll e^{\sigma T} \frac{|\sigma|}{R^2}. \tag{4.13}$$

Thus, the contribution to (4.12) from the path $\mathscr{C}_+$ of length πR is

$$|I_{\mathscr{C}_+}| \ll \frac{1}{R^2} \left| \int_{\mathscr{C}_+} ds \right| \ll \frac{1}{R}.$$

On $\mathscr{C}_-$, we examine $g_T(s)$ and $g(s)$ separately. Consider first the integral

$$I_1 := \frac{1}{2\pi i}\int_{\mathscr{C}_-} g_T(s)e^{sT}\left(1+\frac{s^2}{R^2}\right)\frac{ds}{s}$$

As $g_T(s)$ is entire and the rest of the integrand is analytic to the left of $\sigma = 0$, we have by Cauchy's theorem

$$I_1 = \frac{1}{2\pi i}\int_{C_-} g_T(s)e^{sT}\left(1+\frac{s^2}{R^2}\right)\frac{ds}{s}.$$

That is, we can integrate over the semicircle C_- instead of $\mathscr{C}_-$. Then, noting that $\sigma < 0$ in this case, we have

$$|g_T(s)| = \left|\int_0^T f(t)e^{-st}dt\right| \le M\int_0^T e^{-\sigma t}dt \ll \frac{e^{-\sigma T}}{|\sigma|},$$

and the estimate (4.13) holds on C_- exactly as it did on $\mathscr{C}_-$. We obtain $|I_1| \ll 1/R$ in the same way as done for $|I_{\mathscr{C}_+}|$ above. This leaves us with the integral

$$I_2 := \frac{1}{2\pi i}\int_{\mathscr{C}_-} g(s)e^{sT}\left(1+\frac{s^2}{R^2}\right)\frac{ds}{s}.$$

As $\mathscr{C}_-$ is contained in a compact set on which $g(s)$ is analytic, $|g(s)|$ can be bounded in terms of R on $\mathscr{C}_-$. As the estimate (4.13) holds on the arcs of $\mathscr{C}_-$, the integrand in this region is of the order of

$$|\sigma|e^{\sigma T} \text{ as } T \to \infty,$$

with the implicit constant depending on R. Recalling that $\sigma < 0$ in this region, the above quantity can be compared to the real-valued function xe^{-x}, which attains a global maximum of e^{-1} (as can be checked by standard derivative tests). Thus,

$$|\sigma|e^{\sigma T} \le e^{-1}/T,$$

giving a bound of $O_R(1/T)$ for the integrand over the arcs of $\mathscr{C}_-$. As the length of the arcs is again a function of R which gets absorbed into the implied constant, we see that the contribution to $|I_2|$ from the arcs of $\mathscr{C}_-$ is $O_R(1/T)$ as $T \to \infty$. On the vertical strip of $\mathscr{C}_-$, as $\sigma = -\delta$, we have

$$|e^{sT}| = e^{-\delta T}.$$

The rest of the integrand of I_2 is analytic in this region and hence absolutely bounded in terms of R. The contribution to $|I_2|$ from this strip is thus $O_R(e^{-\delta T})$. Putting everything together, we have obtained, as $T \to \infty$,

$$|g(0) - g_T(0)| = |I_{\mathscr{C}}| \le |I_{\mathscr{C}_+}| + |I_1| + |I_2|$$
$$\ll O\left(\frac{1}{R}\right) + O_R\left(\frac{1}{T}\right) + O_R(e^{-\delta T}).$$

As R is arbitrary, the right-hand side can be made as small as needed. This completes the proof. □

4.18 The Proof of the Tauberian Theorem

We establish the following version of the Tauberian theorem, applicable in many settings.

Theorem 4.48 *Let*

$$G(s) = \sum_{n=1}^{\infty} b_n/n^s$$

be a Dirichlet series with non-negative coefficients, satisfying

(a) $G(s)$ is absolutely convergent for $\mathrm{Re}(s) > 1$.
(b) The function $G(s)$ extends meromorphically to the region $\mathrm{Re}(s) \ge 1$, *having no poles except possibly a simple pole at $s = 1$ with residue R.*
(c) $B(x) := \sum_{n \le x} b_n = O(x)$.

Then, as $x \to \infty$,

$$B(x) = Rx + o(x).$$

We begin by making some elementary observations. For any $\epsilon > 0$,

$$\sum_{n \le x} b_n \le \sum_{n=1}^{\infty} b_n \left(\frac{x}{n}\right)^{1+\epsilon}.$$

The right-hand side is $x^{1+\epsilon}G(1+\epsilon)$ which is of the order of $x^{1+\epsilon}/\epsilon$ since $G(1+\epsilon) \ll 1/\epsilon$. Choosing $\epsilon = (\log x)^{-1}$ gives $B(x) \ll x \log x$. Note that this estimate does not use any information about the behavior of $G(s)$ on $\mathrm{Re}(s) = 1$, except at $s = 1$. Normally, (c) is not needed in the general Wiener–Ikehara Tauberian theorem. One can deduce it from the other assumptions, as indicated in the concluding remarks. However, in practically all applications, this condition is found to be readily available and we retain it for the sake of a shorter proof.

A natural starting point for this and indeed most proofs of the Tauberian theorem is what is known as Abel's trick: For $\mathrm{Re}(s) > 1$, we have

$$G(s) = s \int_1^{\infty} \frac{B(x)}{x^{s+1}} dx. \tag{4.14}$$

This can be derived using partial summation, as is done in Exercise 2.1.5 of [8]. We proceed to prove the above Theorem 4.48.

Proof Without loss of generality, we may suppose $R > 0$. Indeed, if $R \leq 0$, it is enough to prove the result for $G(s) + m\zeta(s)$, where ζ is the Riemann zeta function and m is an integer greater than $|R|$. For $R > 0$, replacing b_n by b_n/R if needed, we may assume $R = 1$. From our discussion above, we have for $\mathrm{Re}(s) > 1$,

$$\frac{G(s)}{s} - \frac{1}{s-1} = \int_1^\infty \frac{B(x) - x}{x^{s+1}} dx \tag{4.15}$$

After the change of variable x to e^u and then s to $s + 1$, we have for $\mathrm{Re}(s) > 0$,

$$\frac{G(s+1)}{s+1} - \frac{1}{s} = \int_0^\infty \frac{B(e^u) - e^u}{e^u} e^{-su} du,$$

which is suitable for application of Theorem 4.47 because the function

$$f(u) := (B(e^u) - e^u)/e^u$$

is bounded on account of (c) and the left-hand side has an analytic continuation to $\mathrm{Re}(s) \geq 0$ by (b). Hence, by Theorem 4.47, the integral

$$\int_0^\infty \frac{B(e^u) - e^u}{e^u} du = \int_1^\infty \frac{B(t) - t}{t^2} dt \tag{4.16}$$

converges. We will show that $B(x) \sim x$ as $x \to \infty$. Suppose not. Then, $\lim_{x\to\infty} B(x)/x$ either does not exist or does not equal 1 if it exists. In either case, we see that $\limsup_{x\to\infty} B(x)/x > 1$ or $\liminf_{x\to\infty} B(x)/x < 1$. Suppose the former inequality holds (the latter case can be treated similarly). Then, there exists some $\lambda > 1$ such that $B(x) \geq \lambda x$ for infinitely many x. As there exists x arbitrarily large with $B(x) \geq \lambda x$ and $B(x)$ is an increasing function, we have

$$\begin{aligned}\int_x^{\lambda x} \frac{B(t) - t}{t^2} dt &\geq \int_x^{\lambda x} \frac{\lambda x - t}{t^2} dt \\ &= \int_1^\lambda \frac{\lambda x - vx}{(vx)^2} x dv = \int_1^\lambda \frac{\lambda - v}{v^2} dv,\end{aligned}$$

which is a positive quantity $c(\lambda)$ (say) depending only on λ. This gives

$$\left| \int_x^\infty \frac{B(t) - t}{t^2} dt - \int_{\lambda x}^\infty \frac{B(t) - t}{t^2} dt \right| = c(\lambda)$$

For fixed λ, as $x \to \infty$, the above integrals are tails of the convergent integral (4.16) and can be made arbitrarily small, thereby giving a contradiction. This completes the proof. □

The result can be extended to Dirichlet series with complex coefficients as follows.

Corollary 4.1 *Let*

$$F(s) = \sum_{n=1}^{\infty} a_n/n^s$$

be a Dirichlet series with complex coefficients. Let $A(x)$ denote the partial sum of the coefficients:

$$A(x) = \sum_{n \le x} a_n.$$

Suppose there exists a Dirichlet series $G(s) = \sum_{n=1}^{\infty} b_n/n^s$ with non-negative coefficients, such that

(a) $|a_n| \le b_n$ *for all* n.
(b) $G(s)$ *is absolutely convergent for* $\mathrm{Re}(s) > 1$.
(c) The function $G(s)$ (resp. $F(s)$) extends meromorphically to the region $\mathrm{Re}(s) \ge 1$, *having no poles except for a simple pole at $s = 1$ with residue R (resp. r).*
(d) $B(x) := \sum_{n \le x} b_n = O(x)$.

Then, as $x \to \infty$,

$$A(x) = rx + o(x).$$

Proof If a_n's are real, we consider the series $G(s) - F(s)$, which has non-negative coefficients and satisfies the conditions of Theorem 4.48, giving

$$\sum_{n \le x} (b_n - a_n) = (R - r)x + o(x),$$

as $x \to \infty$. As $B(x) = Rx + o(x)$, this proves the result in the case of real coefficients. If the coefficients a_n are not real, we define

$$F^*(s) = \sum_{n=1}^{\infty} \bar{a}_n/n^s$$

so that

$$F = \frac{F + F^*}{2} + i\left(\frac{F - F^*}{2i}\right).$$

and apply the result for real coefficients separately to the real and imaginary part above after checking that the necessary conditions are satisfied. □

As remarked earlier, the added condition (c) in Theorem 4.48 is not restrictive for most practical purposes. However, it is possible to eliminate this condition altogether. We give a brief sketch of the argument. The key idea is to notice that the known bound $B(x) \ll x \log x$ implies that for any $\epsilon > 0$, the function

$$f_\epsilon(t) := \frac{f(t)}{e^{\epsilon t}} = \frac{B(e^t)}{e^{t(1+\epsilon)}} - \frac{1}{e^{\epsilon t}}$$

is bounded and satisfies the conditions of Theorem 4.47. Applying this theorem to $f_\epsilon(t)$ and following an elementary argument that exploits the increasing behavior of the function $B(e^t)/e^{t(1+\epsilon)}$, one obtains a uniform bound on $\sup_{t\geq 0} |f_\epsilon(t)|$. Letting $\epsilon \to 0$, we see that $f(t)$ must be bounded. A more detailed proof can be found in [6].

4.19 The Prime Number Theorem

The Riemann zeta function admits an analytic continuation to the entire complex plane, apart from a simple pole at $s = 1$. It satisfies a remarkable functional equation which we derived in an earlier chapter using the Poisson summation formula, essentially following the method of Riemann. However, for the purpose of deriving the prime number theorem, it is not necessary to have analytic continuation to the entire complex plane. It suffices to have continuation to $\mathrm{Re}(s) \geq 1$. We elaborate below.

Firstly, the series

$$\sum_{n=1}^{\infty} \frac{1}{n^s}$$

is analytic in the half plane $\mathrm{Re}(s) > 1$. Also, by an observation due to Euler, the series can be written as an infinite product over prime numbers:

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1},$$

again valid in the region $\mathrm{Re}(s) > 1$. In Example 4.4, we saw that this Euler product shows that $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$. This Euler product provides the fundamental link between $\zeta(s)$ and prime numbers. Indeed, if we define the **von Mangoldt function** by

$$\Lambda(n) := \begin{cases} \log p & \text{if } n = p^r,\ p \text{ prime} \\ 0 & \text{otherwise,} \end{cases}$$

then it is easy to see that

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}.$$

Since $\zeta(s)$ admits an analytic continuation to the entire complex plane, apart from a simple pole at $s = 1$, we see that the left-hand side is a meromorphic function. If $\zeta(s) \neq 0$, for $\mathrm{Re}(s) = 1$, then the left-hand side provides an analytic continuation of the series on the right-hand side for $\mathrm{Re}(s) \geq 1$, except for a simple pole at $s = 1$ with residue 1. The Wiener–Ikehara Tauberian theorem then implies that

$$\sum_{n \leq x} \Lambda(n) \sim x,$$

as x tends to infinity. It is a simple exercise (see Exercise 1 of Sect. 4.16) to show that this is equivalent to the prime number theorem.

To show that $\zeta(s) \neq 0$ for $\mathrm{Re}(s) = 1$, we begin by assuming to the contrary that $\zeta(1 + it) = 0$ for some $t \in \mathbb{R}$, $t \neq 0$. We apply the elementary inequality (see Exercise 3)

$$3 + 4\cos\theta + \cos 2\theta \geq 0, \tag{4.17}$$

and with $\sigma > 1$, we have from the Euler product (see Exercise 4)

$$|\zeta(\sigma)^3 \zeta(\sigma + it)^4 \zeta(\sigma + 2it)| = \exp\left(\sum_{p,r \geq 1} \frac{1}{r p^{r\sigma}} (3 + 4\cos(tr \log p) + \cos(2tr \log p) \right) \tag{4.18}$$

By (4.17), the term on the right-hand side of (4.18) is ≥ 1. This being valid for every $\sigma > 1$, we may take the limit of the left-hand side as $\sigma \to 1^+$. If $\zeta(1 + it) = 0$, the term on the left vanishes because $\zeta(\sigma)^3$ has a pole of order 3 and $\zeta(\sigma + it)^4$ has a zero of order 4 at $\sigma = 1$. This proves:

Theorem 4.49

$$\zeta(1 + it) \neq 0, \qquad \forall t \in \mathbb{R}, \quad t \neq 0.$$

The Tauberian theorem thus provides the simplest derivation of the prime number theorem once we understand the minimal requirements needed from the zeta function. That the zeta function admits an analytic continuation to $\mathrm{Re}(s) > 0$ is easily seen by noting that the alternating series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s} = (1 - 2^{1-s})\zeta(s), \tag{4.19}$$

provides immediately an analytic continuation of $\zeta(s)$ to $\mathrm{Re}(s) > 0$ (Exercise 1). Indeed, the left-hand side converges in that region and thus defines an analytic function.

Exercises

1. Suppose that a_n is a sequence of complex numbers such that

$$A(x) := \sum_{n \le x} a_n = O(1).$$

Show (by partial summation) that

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s} = s \int_1^{\infty} \frac{A(t)}{t^{s+1}} dt.$$

Deduce that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s}$$

defines an analytic function in $\mathrm{Re}(s) > 0$.

2. Show that for $\mathrm{Re}(s) > 1$,

$$(1 - 2^{1-s})\zeta(s) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s}.$$

3. Prove that for $\theta \in \mathbb{R}$,

$$3 + 4\cos\theta + \cos 2\theta \ge 0.$$

4. Show that for $t \in \mathbb{R}$,

$$|\zeta(\sigma)^3 \zeta(\sigma + it)^4 \zeta(\sigma + 2it)| \ge 1.$$

5. The **Möbius function** is defined as follows: $\mu(1) = 1$, and $\mu(n) = 0$ if n is not a product of distinct primes; otherwise, $\mu(n) = (-1)^k$ if n is a product of k distinct primes. Show that

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}.$$

6. Prove that

$$\sum_{n \le x} \mu(n) = o(x),$$

as n tends to infinity.

4.20 Further Applications

In this section, we demonstrate some applications of the Tauberian theorem, following the treatment of Serre [7] who gives a general setup for the same in the context of equidistribution.

We make this more precise in an abstract setting as follows. Let G be a compact group and X be the space of conjugacy classes of G. Let x_v be a family of elements of X, indexed by a countably infinite set $\mathcal{P}$. Let $N : \mathcal{P} \to \mathbb{Z}$ be a function taking integer values ≥ 2, ρ an irreducible complex representation of G with character χ. We define

$$\zeta_{\mathcal{P}}(s) = \prod_{v \in \mathcal{P}} \left(1 - \frac{1}{(Nv)^s}\right)^{-1}, \qquad L(s, \rho) = \prod_{v \in \mathcal{P}} \det\left(1 - \frac{\rho(x_v)}{(Nv)^s}\right)^{-1}.$$

Thus, for the trivial representation $\rho = 1$, $L(s, 1) = \zeta_{\mathcal{P}}(s)$.

Theorem 4.50 *Suppose $L(s, \rho)$ is absolutely convergent for $\mathrm{Re}(s) > 1$ and extends to a meromorphic function on $\mathrm{Re}(s) \geq 1$ with no zeros or poles except for a pole of order c_χ at $s = 1$. Then,*

$$\sum_{Nv \leq n} \chi(x_v) = (1 + o(1)) c_\chi \frac{n}{\log n}.$$

The proof of the above theorem follows by applying the Tauberian theorem to L'/L. We refer the reader to the appendix of Chap. 1 of [7] for the same. If Theorem 4.50 holds for all irreducible representations $\rho \neq 1$ with $c_\chi = 0$, then the Peter–Weyl theorem allows us to deduce that the x_v's are equidistributed with respect to the normalized Haar measure of G. Special cases of this theorem lead to important results, among them being the prime number theorem for arithmetic progressions (see exercises below), Chebotarev density theorem and the Sato–Tate theorem. An excellent reference for the interested reader wishing to delve deeper into these topics is [9].

Exercises

1. The Hurwitz zeta function is defined for $\mathrm{Re}(s) > 1$ by

$$\zeta(s, x) = \sum_{n=0}^{\infty} \frac{1}{(n + x)^s},$$

for $0 < x \leq 1$. Prove that

$$-\frac{1}{x^s} + \zeta(s, x) = \zeta(s) = \sum_{r=1}^{\infty} \binom{-s}{r} \zeta(s+r) x^r.$$

[Hint: Use the binomial theorem to write

$$\frac{1}{(n+x)^s} - \frac{1}{n^s} = \sum_{r=1}^{\infty} \binom{-s}{r} \frac{x^r}{n^{s+r}},$$

and then sum over n.]

2. Using the previous exercise, show that

$$(2^s - 2)\zeta(s) = 2^s + \sum_{r=1}^{\infty} \binom{-s}{r} \frac{\zeta(s+r)}{2^r}.$$

Deduce that $(2^s - 2)\zeta(s)$ extends analytically to $\mathrm{Re}(s) > 0$. [Hint: Put $x = 1/2$ in the identity of the previous exercise.]

3. For any natural number q, show that

$$(q^s - q)\zeta(s) = \sum_{a=1}^{q-1} \left(\zeta(s, \frac{a}{q}) - \zeta(s) \right).$$

4. Deduce that $(3^s - 3)\zeta(s)$ extends analytically to $\mathrm{Re}(s) > 0$. Combining this with Exercise 2, conclude that $\zeta(s)$ extends to $\mathrm{Re}(s) > 0$ apart from a possible pole at $s = 1$.
5. Show that $\lim_{s \to 1^+} (2^s - 2)\zeta(s) = 2 \log 2$. [Hint: Use question 2.] Deduce from the previous exercise that $\zeta(s)$ extends to an analytic function for $\mathrm{Re}(s) > 0$ except for a simple pole at $s = 1$ with residue 1.
6. For each character $\chi : (\mathbb{Z}/q\mathbb{Z})^* \to \mathbb{C}$, show that $L(s, \chi)$ can be written as

$$L(s, \chi) = q^{-s} \sum_{a=1}^{q} \chi(a) \zeta(s, a/q).$$

Deduce that $L(s, \chi)$ extends analytically to $\mathrm{Re}(s) > 0$ for every non-trivial character χ.

7. With notation as in the previous exercise, show that if $L(s, \chi)$ does not vanish on $\mathrm{Re}(s) = 1$, then for a and q coprime,

$$\sum_{n \le x, n \equiv a \pmod q} \Lambda(n) \sim \frac{x}{\phi(q)},$$

where $\phi(q)$ is Euler's function, equal to the order of the group $(\mathbb{Z}/q\mathbb{Z})^*$. Deduce that there are infinitely many primes in arithmetic progressions. (Dirichlet's theorem)

4.21 The Paley–Wiener Theorems

The theme of fusing Fourier transforms and analytic functions is the underlying goal of the Paley–Wiener theorems. It is sometimes the case that a Fourier transform of a function f on $\mathbb{R}$ can be extended to an analytic function in some region of the complex plane. For instance, we have already seen that if $f(t) = e^{-|t|}$, then (up to a constant factor) the Fourier transform $\widehat{f}(x)$ is

$$\frac{1}{1+x^2},$$

which is a rational function and thus defines a meromorphic function in the complex plane. Thus, it would be interesting to investigate under what conditions on f is it the case that $\widehat{f}$ is analytic in certain regions of the complex plane. This is the aim of the Paley–Wiener theorems.

For instance, let $F \in L^2(0, \infty)$ and define

$$f(z) = \int_0^\infty F(t)e^{itz}dt, \qquad z \in \mathfrak{H}, \tag{4.20}$$

where $\mathfrak{H}$ denotes the upper half plane (i.e., $z \in \mathbb{C}$ with $\mathrm{Im}(z) > 0$). If $\mathrm{Im}(z) > \delta > 0$, and z_n is a sequence of complex numbers with $\mathrm{Im}(z_n) > \delta$ tending to z , then a simple application of the dominated convergence theorem shows

$$\lim_{n\to\infty} \int_0^\infty |e^{itz_n} - e^{itz}|^2 dt = 0,$$

because the integrand is bounded by the L^1-function $4e^{-2\delta t}$ and tends to zero for every $t > 0$. The Cauchy–Schwarz inequality implies then that f is continuous in $\mathfrak{H}$. Furthermore, a direct application of Fubini's theorem and Cauchy's theorem shows that

$$\int_\gamma f(z)dz = 0,$$

for every closed path γ in $\mathfrak{H}$. By Morera's theorem, we can conclude that f is a holomorphic function on $\mathfrak{H}$. That is, $f \in H(\mathfrak{H})$. Thus, in this example, the Fourier transform extends to an analytic function on the upper half plane.

We can rewrite (4.20) with $x, y \in \mathbb{R}$ and y fixed as:

$$f(x+iy) = \int_0^\infty F(t)e^{-ty}e^{itx}dt.$$

Applying Plancherel's theorem gives

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} |f(x+iy)|^2 dx = \int_0^\infty |F(t)|^2 e^{-2ty}\, dt \le \int_0^\infty |F(t)|^2 dt$$

for every $y > 0$ (keeping in mind that our Lebesgue measure was normalized by $1/\sqrt{2\pi}$ in the earlier chapter). This proves:

Theorem 4.51 *If f is of the form (4.20), then f is holomorphic in $\mathfrak{H}$ and its restrictions to horizontal lines in $\mathfrak{H}$ form a bounded set in $L^2(\mathbb{R})$.*

Consider now a second class of functions f of the form

$$f(z) = \int_{-A}^{A} F(t)e^{itz}dt \tag{4.21}$$

where $0 < A < \infty$ and $F \in L^2(X)$ where $X = (-A, A)$. By the method of proof above, f is entire and it satisfies the growth condition

$$|f(z)| \le \int_{-A}^{A} |F(t)|e^{-ty}dt \le e^{A|y|}\int_{-A}^{A} |F(t)|dt.$$

If we denote by C this last integral, then $C < \infty$ and we have

$$|f(z)| \le Ce^{A|z|}. \tag{4.22}$$

Entire functions that satisfy (4.22) are said to be of **exponential type**. Our discussion shows that

Theorem 4.52 *Every f of the form (4.21) is an entire function that satisfies (4.22) and whose restriction to the real axis lies in L^2 (by the Plancherel theorem).*

It is remarkable that the converses of the two theorems above are true. This is the content of the Paley–Wiener theorems.

Theorem 4.53 *Suppose $f \in H(\mathfrak{H})$ and*

$$\sup_{0<y<\infty} \frac{1}{2\pi}\int_{-\infty}^{\infty} |f(x+iy)|^2 dx = C < \infty. \tag{4.23}$$

Then, there exists an $f \in L^2(0,\infty)$, such that

$$f(z) = \int_0^\infty F(t)e^{itz}dt, \qquad z \in \mathfrak{H},$$

and

$$\int_0^\infty |F(t)|^2 dt = C.$$

Proof To gain some intuition on what F should look like, we can apply the inversion theorem (without worrying whether the conditions are met or not). Thus, our desired F should be of the form

$$F(t) = e^{ty} \int_{-\infty}^{\infty} f(x+iy)e^{-itx} dx = \frac{1}{2\pi} \int_{\mathrm{Im}(z)=y} f(z)e^{-itz} dz,$$

and if this argument is correct, the last integral should not depend on which y is chosen suggesting that perhaps Cauchy's theorem should be applied. Motivated by this idea, let y be fixed with $0 < y < \infty$. For each $a > 0$, let γ_a be the rectangular path with vertices $\pm a + i$ and $\pm a + iy$. Since f is holomorphic in the upper half plane, we have by Cauchy's theorem

$$\int_{\gamma_a} f(z)e^{-itz} dz = 0.$$

We consider only real values of t. Let $\phi(b)$ be the integral of $f(z)e^{-itz}$ over the straight-line interval from $b + i$ to $b + iy$ ($b \in \mathbb{R}$). Set $I = [y, 1]$ if $y < 1$ and $I = [1, y]$ if $y > 1$. Then, by the Cauchy–Schwarz inequality

$$|\phi(b)|^2 = \left| \int_I f(b+iu)e^{-it(b+iu)} du \right|^2 \le \left(\int_I |f(b+iu)|^2 du \right) \left(\int_I e^{2tu} du \right). \tag{4.24}$$

Put

$$L(b) = \int_I |f(b+iu)|^2 du,$$

so that by (4.23) and an application of Fubini's theorem it gives

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} L(b)\, db \le C\mu(I).$$

Hence, there is a sequence of a_j with $a_j \to \infty$ such that

$$L(a_j) + L(-a_j) \to 0, \qquad j \to \infty.$$

By (4.24), this implies that

$$\phi(a_j) \to 0, \quad \phi(-a_j) \to 0, \qquad \text{as } j \to \infty.$$

This holds for every $t \in \mathbb{R}$, and the sequence a_j does not depend on t.

Let us now define

$$g_j(y,t) = \frac{1}{2\pi}\int_{-a_j}^{a_j} f(x+iy)e^{-itx}\,dx.$$

Then, we have from the above that

$$\lim_{j\to\infty}[e^{ty}g_j(y,t) - e^t g_j(1,t)] = 0, \qquad -\infty < t < \infty. \tag{4.25}$$

Writing $f_y(x)$ for $f(x+iy)$, we see that $f_y \in L^2(-\infty,\infty)$, by hypothesis. The Plancherel theorem shows that

$$\lim_{j\to\infty}\int_{-\infty}^{\infty} |\widehat{f_y}(t) - g_j(y,t)|^2\,dt = 0, \tag{4.26}$$

where $\widehat{f_y}$ is the Fourier transform of f_y. A subsequence of $\{g_j(y,t)\}$ converges therefore pointwise to $\widehat{f_y}(t)$ for almost all t. Thus, if we define

$$F(t) = e^t\,\widehat{f_1}(t), \tag{4.27}$$

then it follows from (4.25) that

$$F(t) = e^{ty}\,\widehat{f_y}(t). \tag{4.28}$$

Notice that (4.27) does not involve y and (4.28) holds for every $y > 0$. Plancherel's theorem can be applied to (4.28):

$$\int_{-\infty}^{\infty} e^{-2ty}|F(t)|^2dt = \int_{-\infty}^{\infty} |\widehat{f_y}(t)|^2dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |f_y(x)|^2dx \le C.$$

By taking y arbitrarily large in the above inequality, we deduce that $F(t) = 0$ almost all $t < 0$. On the other hand, letting $y \to 0$ in the penultimate inequality shows that

$$\int_0^{\infty} |F(t)|^2\,dt \le C.$$

From (4.28) and the Cauchy–Schwarz inequality, we deduce that

$$\int_{-\infty}^{\infty} |f_y(t)|\,dt \le \left(\int_{-\infty}^{\infty} e^{-2ty}\,dt\right)\left(\int_{-\infty}^{\infty} e^{2ty}|\widehat{f_y}(t)|^2dt\right) < \infty$$

for $y > 0$. Thus, $\widehat{f_y} \in L^1(\mathbb{R})$. The Fourier inversion theorem gives that

$$f_y(x) = \int_{-\infty}^{\infty} \widehat{f_y}(t)e^{itx}\,dt.$$

In other words,

$$f(z) = \int_0^\infty F(t)e^{-yt}e^{itx}\,dt = \int_0^\infty F(t)e^{itz}\,dt \qquad (z \in \mathfrak{H}).$$

Finally,

$$\int_0^\infty |F(t)|^2\,dt = C$$

follows from (4.23) and an application of Plancherel's theorem. This completes the proof. □

Theorem 4.54 *Suppose A and C are positive constants and f is an entire function such that*

$$|f(z)| \le Ce^{A|z|} \qquad \forall\, z \in \mathbb{C},$$

and

$$\int_{-\infty}^\infty |f(x)|^2\,dx < \infty.$$

Then, there exists an $F \in L^2(X)$ with $X = (-A, A)$ such that

$$f(z) = \int_{-A}^A F(t)e^{itz}\,dt,$$

for all $z \in \mathbb{C}$.

Proof Put $f_\epsilon(x) = f(x)e^{-\epsilon|x|}$ for $\epsilon > 0$ and $x \in \mathbb{R}$. We will show that

$$\lim_{\epsilon\to 0}\int_{-\infty}^\infty f_\epsilon(x)e^{-itx}\,dx = 0, \qquad t \in \mathbb{R}, \quad |t| > A. \tag{4.29}$$

Since

$$||f_\epsilon - f||_2 \to 0, \qquad \text{as} \quad \epsilon \to 0,$$

The Plancherel theorem would then imply that the Fourier transforms of f_ϵ converge in L^2 to the Fourier transform F of f. Then, (4.29) will imply that F vanishes outside of $[-A, A]$ so that the Fourier inversion theorem then implies the result. To prove (4.29), let for each real a, γ_a be the path defined by

$$\gamma_a(s) = se^{ia}, \qquad 0 \le s < \infty.$$

In other words, γ_a is the ray emanating from the origin to infinity with radial angle a. Put

$$P_a = \{w : \mathrm{Re}(we^{ia}) > A\},$$

and if $w \in P_a$, define

$$\Phi_a(w) = \int_{\gamma_a} f(z)e^{-wz}dz = e^{ia}\int_0^\infty f(se^{ia})\exp(-wse^{ia})ds.$$

By our hypothesis on f, the absolute value of the integrand above is at most

$$C\exp(-[\mathrm{Re}(we^{ia}) - A]s)$$

and it follows from our preliminary discussion that Φ_a is holomorphic in P_a. Now for $a = 0$ and $a = \pi$, we have

$$\Phi_0(w) = \int_0^\infty f(x)e^{-wx}dx, \qquad \mathrm{Re}(w) > 0,$$

$$\Phi_\pi(w) = -\int_{-\infty}^0 f(x)e^{-wx}dx, \qquad \mathrm{Re}(w) < 0.$$

Since $f \in L^2(\mathbb{R})$, we see that both Φ_0 and Φ_π are holomorphic in the indicated half planes. Now it is easily checked that for $t \in \mathbb{R}$,

$$\int_{-\infty}^\infty f_\epsilon(x)e^{-itx}dx = \int_{-\infty}^\infty f(x)e^{-\epsilon|x|}e^{-itx}dx = \Phi_0(\epsilon + it) - \Phi_\pi(-\epsilon + it),$$

so to prove our assertion, we need to show that the right-hand side of the above equation tends to zero as $\epsilon \to 0$ for $|t| > A$. We will do this by showing that any two of our functions Φ_a agree in the intersections of their domains of definition. That is, they are analytic continuations of each other. Once we have this, then we can replace Φ_0 and Φ_π by $\Phi_{\pi/2}$ when $T < -A$ and by $\Phi_{-\pi/2}$ for $t > A$ and it is then obvious that the difference tends to zero as $\epsilon \to 0$. To this end, suppose $0 < b - a < \pi$ and put

$$c = \frac{a+b}{2}, \quad d = \cos\frac{b-a}{2} > 0.$$

Putting $w = |w|e^{ic}$, we have

$$\mathrm{Re}(w^{ia}) = d|w| = \mathrm{Re}(we^{ib})$$

so that $w \in \mathbb{P}_a \cap \mathbb{P}_b$ as soon as $|w| > A/d$. Now consider the integral

$$\int_\gamma f(z)e^{-wz}dz, \tag{4.30}$$

where γ is the circular arc of radius r given by $\gamma(t)re^{it}$, for $a \le t \le b$. Since

$$\mathrm{Re}(-wz) = -|w|r\cos(t - c) \le -|w|rd,$$

the absolute value of the integrand in (4.30) is bounded by

$$C \exp\left((A - |w|d)r\right).$$

If $|w| > A/d$, it follows that the integral in (4.30) tends to zero as r tends to infinity. We now apply Cauchy's theorem:

$$\int_{[0,re^{ib}]} f(z)e^{-wz}dx + \int_{\gamma} f(z)e^{-wz}dz + \int_{[0,re^{ia}]} f(z)e^{-wz}dz = 0.$$

Since the middle integral tends to zero as r tends to infinity, we conclude that $\Phi_a(w) = \Phi_b(w)$ for $w = |w|e^{-ic}$ and $|w| > A/d$. As these two functions agree on an uncountable set, we conclude Φ_a and Φ_b agree on in the intersection of the half planes in which they were originally defined. This completes the proof. □

Exercises

1. Suppose f is an entire function of exponential type and

$$\varphi(y) = \int_{-\infty}^{\infty} |f(x + iy)|^2 dx.$$

   Prove that $f = 0$ if φ is a bounded function.
2. Suppose that f is an entire function of exponential type such that the restriction of f to two non-parallel lines belongs to L^2. Prove that $f = 0$.

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Chapter 5
Introduction to Algebraic Topology



5.1 A Very Brief Historical Introduction

The subject of algebraic topology was born in 1895, when Henri Poincaré introduced the notion of the fundamental group $\pi_1(X)$ attached to a topological space X. His idea was to study the topology of a space using group theory and more generally, algebra. The modern viewpoint is somewhat analogous (in some perspectives, identical) to the study of normal extensions of fields and their associated Galois groups.

Since the computations of fundamental groups of topological spaces often proved difficult, Poincaré also introduced homology groups to measure connectivity properties of topological spaces. These being abelian groups, one could invoke the machinery of $\mathbb{Z}$-modules (or more generally, module theory) to study the topological spaces. By this route, algebra entered the study of topology, and we call this branch of mathematics algebraic topology.

The subject developed rapidly in the early part of the twentieth century. For example, in 1910, L.E.J. Brouwer proved his famous fixed point theorem stating that any continuous map

$$f : D \to D,$$

where $D = \{z \in \mathbb{C} : |z| \leq 1\}$, has a fixed point. Later, in 1926, S. Lefschetz extended this to prove his celebrated fixed point theorem. In 1967, M. Atiyah and R. Bott gave a far-reaching generalization of Lefschetz's formulas. Their work led to the development of the Atiyah–Singer index theorem the following year.

A fundamental concept in algebraic topology is the notion of **homotopy**. If X and Y are two topological spaces, maps $f : X \to Y$ and $g : X \to Y$ are said to be **homotopically equivalent** if there is a continuous map

$$h : X \times [0, 1] \to Y$$

such that $h(x, 0) = f(x)$ and $h(x, 1) = g(x)$. Two spaces X and Y are said to be homotopically equivalent if there are maps $F : X \to Y$ and $G : Y \to X$ such that

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$F \circ G$ and $G \circ F$ are homotopically equivalent to the identity map. A space X is said to be a **contractible space** if it is homotopy equivalent to a point. Contractible spaces are the simplest to study, and the notion of homotopy creates a hierarchy of topological spaces that can now be explored systematically.

Further advances were made in the 1950s. All along, several notable conjectures motivated an accelerated development of the theory. The most famous of these conjectures is the Poincaré conjecture predicting that any simply connected closed manifold of dimension three is homeomorphic to the 3-sphere S^3. The generalized Poincaré conjecture is that any homotopy n-sphere (i.e., a topological space homotopically equivalent to the n-sphere) is homeomorphic to the n-sphere S^n. In other words, if $\pi_1(X) \simeq \pi_1(S^n)$, then $X \simeq S^n$. For $n \geq 5$, this was settled by S. Smale in 1961. In 1982, M. Freedman settled the case $n = 4$. But it was the case $n = 3$ that attained a celebrity status becoming a million dollar prize problem in the list of unsolved problems highlighted by the Clay Mathematical Institute in 2000. Finally, in 2003, G. Perelman solved the case $n = 3$. For this work, he was awarded the Fields Medal, but he refused to accept it, stating that "I am not interested in money or fame. I don't want to be on display like an animal in a zoo."

In 2010, the Clay Mathematics Institute decided to award its first million dollar prize to Perelman for his work. But this too was turned down by him saying that his work built on the work of many others and it was unfair to award it to a single person. Though he was awarded many accolades, including professorships at top universities, Perelman declined them all and was happy to work in his native Russia at the Steklov Institute in Saint Petersburg. Truly, this is a superb example of a man of renunciation, a mathematical ascetic.

Returning to our brief narrative, parallel developments occurred in the 1950s that hastened the development of algebraic geometry. Indeed, shortly after he formulated his famous conjectures regarding solutions of systems of equations over finite fields, André Weil fantasized that a theory of algebraic topology should exist over finite fields. If such a theory existed, then he indicated how many of his conjectures would follow from a (mod p) version of the Lefschetz fixed point theorem. This was realized in the magnificent edifice constructed by A. Grothendieck in the 1960s and 1970s. It culminated in the work of Pierre Deligne who proved all of the Weil conjectures, including the notorious Riemann hypothesis for zeta functions over finite fields.

Given the important role that algebraic topology plays in mathematics, as well as its central place of interconnectedness (no pun intended), it seemed relevant that any good course in analysis for graduate students should include this topic. This chapter is intended as a "crash course" and is meant as an elementary introduction to this fascinating subject. Hopefully, the student will be inspired to study more and with this goal, ample references for further reading are provided.

5.2 Homotopic Paths

Recall that a **topological space** is a pair $(X, \mathcal{O})$ where X is a set and $\mathcal{O}$ is a collection of subsets of X satisfying the following conditions:

(a) the empty set $\emptyset$ and X belong to $\mathcal{O}$;
(b) $A, B \in \mathcal{O}$ implies $A \cap B \in \mathcal{O}$;
(c) $A_i \in \mathcal{O}$ with i belonging to some index set I implies $\cup_{i \in I} A_i \in \mathcal{O}$.

Elements of $\mathcal{O}$ are called **open sets**. The complement of an open set is called a **closed set**.

If X and Y are topological spaces and we speak of maps from X to Y, we always mean continuous maps. Given two topological spaces X and Y, a map $f : X \to Y$ is called a **homeomorphism** if it is one to one, onto and $f(U)$ is open if and only if U is open. If such a map exists, then X and Y are said to be **homeomorphic**. The basic problem of topology is to classify topological spaces up to homeomorphism.

The real line $\mathbb{R}$ is an example of a topological space with open sets being disjoint union of open intervals. More generally, a metric space X with metric d is a topological space with open sets characterized by those subsets O of X with the property that for every $x \in O$, there is an $\epsilon > 0$ such that the ball $B(x, \epsilon) \subset O$. Thus, $\mathbb{R}^n$, $\mathbb{C}^n$ with the usual metric are examples of topological spaces. More generally, **n-manifolds** are topological spaces which are worthy of independent study. (Recall that X is said to be an **n-manifold** if X is a Hausdorff, second countable topological space and every element $x \in X$ belongs to an open set U which is homeomorphic to an open set of $\mathbb{R}^n$.)

Any subset S of a topological space X inherits a topology from X, called the **relative topology**. Thus, the open sets of S are simply of the form $S \cap O$ for some open set O of X. For instance, the **n-sphere** denoted S^n and defined by

$$S^n = \{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} : x_1^2 + \cdots + x_{n+1}^2 = 1\}$$

is an example of a topological space which plays an important role in algebraic topology. It is an example of an n-manifold. So also is the closed unit ball B^n defined by

$$B^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_1^2 + \cdots + x_n^2 \leq 1\}.$$

Notice that S^n is a subspace of $\mathbb{R}^{n+1}$ while B^n is a subspace of $\mathbb{R}^n$.

To determine if two given topological spaces are homeomorphic is not always easy. For instance, it is a non-trivial theorem that $\mathbb{R}^m$ and $\mathbb{R}^n$ are not homeomorphic whenever $m \neq n$. The same is true of the n-spheres. In some sense, algebraic topology is an approach to this classification problem through algebraic methods. It has been highly successful in this and thus has played a central role in the development of modern mathematics, not only in topology, but also in other branches of mathematics such as number theory, algebraic geometry, and differential geometry.

The first notions of algebraic topology were introduced by Poincaré in a series of papers from 1895 to 1905 where he attempts to understand the notion of "connectivity" of a space. This is where we will begin.

Recall that a space X is said to be **connected** if it cannot be expressed non-trivially as the disjoint union of two open sets. It is not hard to show that the continuous image of a connected space is connected (Exercise 4). Therefore, connectedness is preserved under homeomorphism. Thus, the 0-sphere S^0 consisting of $\{-1, 1\}$ which is disconnected is not homeomorphic to S^n for $n \geq 1$, for these latter spaces are connected.

A stronger property than connectedness is **path-connectedness**. Given a topological space X, with $a, b \in X$, a **path** from a to b is a continuous map $\alpha : I = [0, 1] \to X$ such that $\alpha(0) = a$ and $\alpha(1) = b$. A space X is said to be **path-connected** if any two points of X are connected by a path. If X is path-connected, it is also connected (Exercise 5).

The property of path-connectedness lies at the heart of algebraic topology. One of our goals in this chapter will be to prove:

Theorem 5.1 (Brouwer fixed point theorem, 1910) *Any continuous map $f : B^n \to B^n$, has a fixed point.*

For $n = 1$, this is straightforward. In this case, $B^1 = [-1, 1]$ and we may suppose $f(-1) \neq -1$, $f(1) \neq 1$, for otherwise, we are done. The path-connectedness of $[-1, 1] \times [-1, 1]$ now implies the result since the graph of f must intersect the diagonal at some point. Indeed, any continuous path joining $(-1, f(-1))$, which lies above the diagonal, to $(1, f(1))$, which lies below the diagonal, must cross the diagonal. The intersection of the path with the diagonal is our fixed point.

This simple proof does not seem to extend to higher dimensions. Already for $n = 2$, it is non-trivial and for $n \geq 3$, it requires more advanced notions.

The fixed point theorem has many applications in mathematics such as to the existence of solutions of systems of equations. Even in the simple case of $n = 1$, we have an interesting application to the problem of "how to slice a pancake evenly," or "how to slice two pancakes evenly."

In a series of remarkable papers, Poincaré introduced the notions of homotopy and fundamental group and thus initiated the study of modern algebraic topology. Given a topological space X, two paths α, β from a to b are said to be **homotopic** if there is a continuous map $F : I \times I \to X$ such that

$$F(s, 0) = \alpha(s), \quad F(s, 1) = \beta(s),$$

$$F(0, t) = a, \quad F(1, t) = b, \quad 0 \leq s \leq 1, \ 0 \leq t \leq 1.$$

The map F is called a **homotopy** of α into β. Intuitively, α is homotopic to β if we can continuously deform α into β. It is clear that this is an equivalence relation on the set of all paths from a to b. Thus we write $\alpha \sim \beta$ to indicate that α and β are homotopic. We already met this concept in our study of complex analysis.

A subset X of $\mathbb{R}^n$ is said to be **convex** if for any $x, y \in X$, the line joining x and y also lies in X.

Theorem 5.2 *Let X be a convex subset of $\mathbb{R}^n$, $a, b \in X$. Then any two paths from a to b are homotopic.*

Proof Let α and β be two paths from a to b. Define $F(s,t) = (1-t)\alpha(s) + t\beta(s)$. F is easily seen to be a homotopy between α and β. □

Thus, for any convex subset, there is only one equivalence class of homotopic paths from a to b.

Lemma 5.1 *Let α be a path in X from a to b. Let $\rho : I \to I$ satisfy $\rho(0) = 0$, $\rho(1) = 1$. Then, $\alpha \circ \rho$ is also a path in X from a to b which is homotopic to α.*

Proof Define $F(s,t) = \alpha((1-t)s + t\rho(s))$. Then, $F(s,0) = \alpha(s)$ and $F(s,1) = \alpha \circ \rho(s)$. As F is the composition of continuous functions, F is a homotopy between α and $\alpha \circ \rho$. □

If α is a path from a to b and β is a path from b to c, then we can define the "product" of α and β as a path from a to c by setting

$$(\alpha\beta)(s) = \begin{cases} \alpha(2s) & 0 \le s \le 1/2 \\ \beta(2s-1) & 1/2 \le s \le 1. \end{cases}$$

Lemma 5.2 *Let $a, b \in X$, with α_0, α_1 paths from a to b, and β_0, β_1 paths from b to c. If $\alpha_0 \sim \alpha_1$ and $\beta_0 \sim \beta_1$, then $\alpha_0\beta_0 \sim \alpha_1\beta_1$.*

Proof Clear. □

This lemma allows us to define products of homotopy classes. Let $[\alpha]$ be the homotopy class of α. If α is a path from a to b, β a path from b to c, we define

$$[\alpha][\beta] = [\alpha\beta],$$

which is well-defined by the previous lemma.

Now consider three paths: α, β, γ from a to b, b to c c to d, respectively. Then $(\alpha\beta)\gamma$ and $\alpha(\beta\gamma)$ are both defined. More precisely

$$((\alpha\beta)\gamma)(s) = \begin{cases} \alpha(4s) & 0 \le s \le 1/4 \\ \beta(4s-1) & 1/4 \le s \le 1/2 \\ \gamma(2s-1) & 1/2 \le s \le 1 \end{cases}$$

and

$$(\alpha(\beta\gamma))(s) = \begin{cases} \alpha(2s) & 0 \le s \le 1/2 \\ \beta(4s-2) & 1/2 \le s \le 3/4 \\ \gamma(4s-3) & 3/4 \le s \le 1 \end{cases}$$

so we see that it is easy to give examples of α, β, γ such that $(\alpha\beta)\gamma \neq \alpha(\beta\gamma)$ (see Exercise 6). Thus the product of paths (when defined) is not necessarily associative. However, we can prove that they are homotopically equivalent.

Lemma 5.3

$$(\alpha\beta)\gamma \sim \alpha(\beta\gamma).$$

Proof We see from the above formulas

$$(\alpha(\beta\gamma))(s) = ((\alpha\beta)\gamma)(\rho(s))$$

where

$$\rho(s) = \begin{cases} s/2 & 0 \le s \le 1/2 \\ s - 1/4 & 1/2 \le s \le 3/4 \\ 2s - 1 & 3/4 \le s \le 1. \end{cases}$$

By Lemma 5.2, we are done. □

The constant path at b will be simply denoted b.

Lemma 5.4 *If α is a path from a to b,*

$$a\alpha \sim \alpha \sim \alpha b.$$

That is, $[a\alpha] = [\alpha] = [\alpha b]$ and $[a][\alpha] = [\alpha][b]$.

Proof We have

$$(a\alpha)(s) = \begin{cases} a & 0 \le s \le 1/2 \\ \alpha(2s-1) & 1/2 \le s \le 1. \end{cases}$$

If we set

$$\rho(s) = \begin{cases} 0 & 0 \le s \le 1/2 \\ 2s - 1 & 1/2 \le s \le 1 \end{cases}$$

then $a\alpha = \alpha \circ \rho$ so that by Lemma 5.2, $a\alpha \sim \alpha$ and similarly $\alpha \sim \alpha b$.

If α is a path from a to b, we define the inverse path α^{-1} from b to a by

$$\alpha^{-1}(s) = \alpha(1-s).$$

We insert a word of caution to the reader that α^{-1} is not to be confused with the inverse function of α, which may not be defined.

Lemma 5.5 *If α_0 and α_1 are paths from a to b with $\alpha_0 \sim \alpha_1$, then $\alpha_0^{-1} \sim \alpha_1^{-1}$.*

Proof If F is a homotopy between α_0 and α_1, define $G(s,t) = F(1-s,t)$ which is a homotopy between α_0^{-1} and α_1^{-1}. □

Lemma 5.5 allows us to define the inverse homotopy class by setting

$$[\alpha]^{-1} := [\alpha^{-1}].$$

Lemma 5.6 *If α is a path from a to b, then $\alpha\alpha^{-1}$ is homotopic to the constant path at a. That is, $[\alpha\alpha^{-1}] = [a]$.*

Proof Define

$$F(s,t) = \begin{cases} \alpha(2s) & 0 \le s \le t/2 \\ \alpha(t) & t/2 \le s \le 1 - t/2 \\ \alpha(2-2s) & 1 - t/2 \le s \le 1. \end{cases}$$

Then $F(s,0) = \alpha(0) = a$ and $F(s,1) = (\alpha\alpha^{-1})(s)$ so that F is the required homotopy. □

A path α is called a **loop** based at a if $\alpha(0) = \alpha(1) = a$. The above discussion enables us to conclude that the set of all homotopy classes of loops based at a can be given a group structure. This is what we shall take up in the next section.

Exercises

1. Recall that a **basis** $\mathcal{B}$ for a topological space X consists of open sets such that every open set of X is a union of elements of $\mathcal{B}$. Prove that the collection of open intervals in $\mathbb{R}$ is a basis for $\mathbb{R}$ with the usual Euclidean topology.
2. Let $\mathcal{B}$ be a non-empty collection of subsets of a set X. If $\mathcal{B}$ is closed under finite intersections and every element of X belongs to some element of $\mathcal{B}$, show that $\mathcal{B}$ is a basis for some topology on X.
3. Consider $\mathbb{Z}$ with the **arithmetic progression topology** where a basis of open sets is given by $\{an + b : n \in \mathbb{Z}\}$ where a, b range over all integers satisfying $(a,b) = 1$. Prove that for each prime p, the set $p\mathbb{Z}$ is closed. Deduce that there are infinitely many primes. (This topology was introduced by H. Furstenberg.) [Hint: if there are only finitely many primes p, the union $\cup_p (p\mathbb{Z})$ is closed. The complement consists of $\{-1, 1\}$ and is not open.]
4. Prove that the continuous image of a connected space is connected.
5. If X is path-connected, show that it is connected.
6. Give an example to show that the product of paths, when defined is not necessarily associative.
7. Show that $\mathbb{R}$ and $\mathbb{R}^n$ for $n \ge 2$ are not homeomorphic.

8. Let X be a topological space and U a connected subset. If U is dense in X, show that X is connected.
9. Consider the **topologist's sine curve** given by the following set of points in $\mathbb{R}^2$: $\{(x, \sin 1/x) : \quad x > 0\} \cup \{(0, y) : \quad -1 \leq y \leq 1\}$. Show that X is connected but not path-connected.
10. Let $f : S^1 \to \mathbb{R}$ be a continuous map. Show that there is a point x_0 such that $f(x_0) = f(-x_0)$. [Hint: consider $h(x) = f(x) - f(-x)$ and the map $H : B^1 \to \mathbb{R}$ given by $H(x) = h(e^{\pi i x})$.]

5.3 The Fundamental Group

Let X be a topological space, $b \in X$ a **base point**. A path $\alpha : I \to X$ is called a **loop** at b if $\alpha(0) = \alpha(1) = b$. The set of all loops based at b (modulo homotopy) forms a group, by the results of the previous section. We call this group **the fundamental group** based at b and denote it by the notation $\pi(X, b)$. By Theorem 5.2, we deduce:

Theorem 5.3 *If X is a convex subset of $\mathbb{R}^n$, $b \in X$, then $\pi(X, b)$ is trivial.*

How does the fundamental group depend upon the base point? The next result answers this question.

Theorem 5.4 *Let $b, c \in X$ and let α be a path from b to c. Given a homotopy class γ, based at c, the function*

$$\alpha_*(\gamma) := \alpha\gamma\alpha^{-1}$$

is well-defined and gives an isomorphism from $\pi(X, c)$ to $\pi(X, b)$.

Proof Since the initial and terminal points match, α_* is well-defined. (Recall that α^{-1} is a path from c to b.) Suppose β, γ are loops at c. Then

$$[\beta][\gamma] = [\beta][c][\gamma] = [\beta][\alpha]^{-1}[\alpha][\gamma].$$

Consequently,

$$\alpha_*([\beta][\gamma]) = [\alpha][\beta][\gamma][\alpha]^{-1} = [\alpha][\beta][\alpha]^{-1}[\alpha][\gamma][\alpha]^{-1} = \alpha_*(\beta)\alpha_*(\gamma).$$

Thus, α_* is a homomorphism. If $\alpha_*(\beta) = \alpha_*(\gamma)$, then

$$\beta \sim \alpha^{-1}\alpha\beta\alpha^{-1}\alpha = \alpha^{-1}\alpha\gamma\alpha^{-1}\alpha \sim \gamma,$$

so that α_* is one to one. α_* is also onto because if λ is a loop based at b, then $\beta = (\alpha^{-1}\lambda)\alpha$ is a loop based at c that satisfies $\alpha_*(\beta) = \alpha\alpha^{-1}\lambda\alpha\alpha^{-1} = \lambda$. □

Therefore, if X is path-connected, the various groups $\pi(X, b)$ are all isomorphic. Thus, we can speak of the **fundamental group** of X for path-connected spaces, without specifying the base point. For such spaces, we denote by $\pi(X)$ this fundamental

group determined up to isomorphism. The map $X \mapsto \pi(X)$ is an example of a **functor** from the set of path-connected topological spaces to the category of groups. We do not go into the detailed definition here but refer the reader to the book by Fulton given in the section on suggestions for further reading. Suffice it to say now that this functor moves the topological study of X into the realm of group theoretic study of $\pi(X)$. The role of functors in general is similar.

A path-connected space X is called **simply connected** if $\pi(X)$ is trivial. Thus, any convex subset of $\mathbb{R}^n$ is simply connected.

Exercises

1. Show that the punctured complex plane $\mathbb{C} \setminus \{0\}$ is path-connected but not simply connected.

5.4 Examples of Some Fundamental Groups

1. A subset W of $\mathbb{R}^n$ is called **star-shaped** with respect to a point $w \in W$ if, whenever $y \in W$, the straight line segment from w to y is contained in W. Thus, W is simply connected.
2. By stereographic projection, we can show that $S^2 \setminus \{0\}$ is homeomorphic to $\mathbb{R}^2$. Thus, $\pi(S^2 \setminus \{0\})$ is trivial. The same logic can be applied to show that $\pi(S^n \setminus \{0\})$ is trivial for $n \geq 2$. (See Exercise 2)
3. It is not hard to see that $\pi(X \times Y) \simeq \pi(X) \times \pi(Y)$. Thus, $X \times Y$ is simply connected whenever X and Y are. (See Exercise 3)

If f is a map between two pointed topological spaces, then f induces a map between their fundamental groups. More precisely, let $f : (X, b) \to (Y, c)$ be a map of topological spaces with $f(b) = c$. We will define a homomorphism between $\pi(X, b)$ and $\pi(Y, c)$.

Lemma 5.7 *Let X and Y be topological spaces and let $f : X \to Y$ be a continuous map. Let f_0 and f_1 be paths in X which are homotopic with endpoints fixed. Then $f \circ f_0$ and $f \circ f_1$ are paths in Y that are homotopic with endpoints fixed.*

Proof If $\{f_t\}_{0 \leq t \leq 1}$ is a homotopy of f_0 and f_1, then the paths $\{f \circ f_t\}_{0 \leq t \leq 1}$ form a homotopy between $f \circ f_0$ and $f \circ f_1$. □

Suppose now that $b_0, b_1 \in X$ and that f is a map from X to Y. If α is a path in X from b_0 to b_1, then $f_*([\alpha])$ is defined to be the homotopy class of paths in Y from $f(b_0)$ to $f(b_1)$ that includes $f \circ \alpha$:

$$f_*([\alpha]) = [f \circ \alpha].$$

By Lemma 5.7, this definition does not depend on the choice of the representative in the homotopy class $[\alpha]$. If the product path $\alpha\beta$ is defined, then $f \circ (\alpha\beta) = (f \circ \alpha)(f \circ \beta)$, so that

$$f_*([\alpha][\beta]) = f_*([\alpha])f_*([\beta]).$$

Since the inverse path of $f \circ \alpha$ is $f \circ \alpha^{-1}$ we have

$$f_*([\alpha]^{-1}) = f_*([\alpha])^{-1}.$$

Finally, if b is a constant path in X, then

$$f_*([b]) = [f(b)].$$

We can now prove:

Theorem 5.5 *Let (X, b), (Y, c) be pointed topological spaces and let $f : (X, b) \to (Y, c)$ be a map. Then, f_* is a homomorphism of $\pi(X, b)$ and $\pi(Y, c)$. If furthermore $g : (W, d) \to (X, b)$ is a map, then $(f \circ g)_* = f_* \circ g_*$. If $X = Y$, $b = c$ and f is the identity map, then f_* is the identity morphism.*

Proof That f_* is a homomorphism follows from Lemma 5.7. The others follow directly from the definitions, and we leave them to the reader as an exercise.

Corollary 5.1 *If $f : X \to Y$ is a homeomorphism, $b \in X$ and $c = f(b)$, then $\pi(X, b)$ is isomorphic to $\pi(Y, c)$.*

Proof Since $f \circ f^{-1}$ and $f^{-1} \circ f$ are the identity maps, the induced maps are identity isomorphisms. □

There is one important application of the notion of induced homomorphisms to the computation of the fundamental groups of topological groups. Recall that a **topological group** G is a group G endowed with a topology such that the map

$$m : G \times G \to G$$

given by

$$m(g, h) = gh^{-1}$$

is continuous. For instance, $\mathbb{R}$ with the usual topology, (or more generally, $\mathbb{R}^n$) is an example of a topological group. An important example is given by S^1, where we identify it with the group of complex numbers of absolute value 1.

As indicated in example 3 above, the fundamental group of the product space $X \times Y$ consists of pairs of loops (α, β) with $\alpha \in \pi(X)$ and $\beta \in \pi(Y)$. In the case of a topological group, we will consider loops based at the identity. Since $m : G \times G \to G$, the induced map

$$m_* : \pi(G) \times \pi(G) \to \pi(G)$$

is given by

$$[(\alpha, \beta)] \mapsto [m(\alpha, \beta)] = [\alpha \cdot \beta]$$

where the product on the right is multiplication in the group. That is $\alpha \cdot \beta$ indicates the path $\alpha(s)\beta(s)$ for $0 \leq s \leq 1$. On the one hand, m_* is a homomorphism, so that

$$[(\alpha, \beta)] = [(\alpha, 1)(1, \beta)] \mapsto [\alpha][\beta].$$

On the other hand,

$$[(\alpha, \beta)] = [(1, \beta)(\alpha, 1)] \mapsto [\beta][\alpha].$$

We deduce:

Theorem 5.6 *If G is a topological group, and $\pi(G)$ consists of loops based at the identity, then $\pi(G)$ is abelian.*

The computation of fundamental groups is often not easy. The usual technique is to look for subspaces about which we know the structure of their fundamental group and then try "to paste" together this information. Here is the simplest case of this occurrence.

Theorem 5.7 *Suppose X is the union of two open sets U and V which are simply connected. Suppose further that $U \cap V$ is non-empty and path-connected. Then X is simply connected.*

The idea of the proof is to write any loop of X as a product of loops contained in U or V. The reader will be easily convinced by drawing a picture to see what is going on. The formal proof will involve the following lemma.

Lemma 5.8 (Lebesgue's lemma) *Let X be a compact metric space and $\mathcal{F}$ an open cover. Then, there is a $\delta > 0$ such that any subset of X of diameter $< \delta$ is contained in some member of $\mathcal{F}$.*

Proof If the lemma is false, we can find a sequence of subsets $A_1, A_2, \ldots$ none of which is contained in a member of $\mathcal{F}$ and whose diameters tend to zero. For each n, choose $x_n \in A_n$. As X is compact, the sequence of x_n's must have a limit point p (say). Let U be an element of $\mathcal{F}$ containing p. Choose $\epsilon > 0$ such that $B(p, \epsilon) \subseteq U$. Choose N so that $\text{diam}(A_N) < \epsilon/2$ and $x_N \in B(p, \epsilon/2)$. Then $d(x_N, p) < \epsilon/2$ and $d(x, x_N) < \epsilon/2$ for all $x \in A_N$. Hence, $d(x, p) < \epsilon$ for all $x \in A_N$. Thus, $A_N \subseteq U$, a contradiction. □

Proof of Theorem 5.7 We show that any loop in X is homotopic to a product of loops each of which is contained in U or V. This is enough to prove the theorem since U and V are simply connected. Choose $p \in U \cap V$. Let $f : I \to X$ be a loop based at p. By Lebesgue's lemma, we can find N such that $f([(j-1)/N, j/N])$ is always contained in U or V for $1 \leq j \leq N$. If we write f_j for the path

$$s \mapsto f((s+j-1)/N), \qquad 0 \le s \le 1,$$

we can join p to $f(j/N)$, $1 \le j \le N-1$ by a path γ_j which lies in U if $f(j/N)$ is in U, or which lies in V if $f(j/N)$ lies in V. If $f(j/N)$ lies in $U \cap V$, we can ensure γ_j to lie in $U \cap V$ since $U \cap V$ is path-connected. The loop f can thus be factored as

$$(f_1\gamma_1^{-1})(\gamma_1 f_2 \gamma_2^{-1})(\gamma_2 f_3 \gamma_3^{-1}) \cdots (\gamma_N f_N)$$

each member of which is a loop contained in U or V. This completes the proof. □

We can apply this theorem to compute the fundamental groups of S^n for $n \ge 2$. Since S^n is the union of the two hemispheres, each of which is simply connected, and whose intersection is S^{n-1} which is path-connected, we deduce immediately that $\pi(S^n) = 1$ for $n \ge 2$. Notice that this proof does not work for S^1 for in that case, when we write S^1 as the union of the upper and lower semi-circles, their intersection is $\{-1, 1\}$ which is not path-connected. By Theorem 5.6, we know that $\pi(S^1)$ is abelian. We will see in the next two sections that $\pi(S^1) \simeq \mathbb{Z}$.

A subset A of X is called a **retract** of X if there exists a continuous map $r : X \to A$ (called a **retraction**) such that $r(a) = a$ for all $a \in A$. For example, S^1 is a retraction of $\mathbb{R}^2 \backslash \{0\}$. The retraction is simply the point of intersection with S^1 obtained by joining the origin to $x \in \mathbb{R}^2 \backslash \{0\}$. Clearly r is continuous and $r(x) = x$ for all $x \in S^1$.

If $r : X \to A$ is a retraction and $i : A \to X$ is the inclusion map, we get the induced maps

$$i_* : \pi(A, a) \to \pi(X, a)$$

$$r_* : \pi(X, a) \to \pi(A, a)$$

for any $a \in A$. Because $r \circ i = 1$, $r_* \circ i_* = 1$. We conclude i_* is a monomorphism, r_* is an epimorphism. One can show that if $i_*\pi(A)$ is a normal subgroup of $\pi(X)$, then $\pi(X)$ is isomorphic to the direct product of $i_*\pi(A)$ and the kernel of r_*.

However, if we impose a further restriction on r, we can show that $\pi(A, a)$ and $\pi(X, a)$ are isomorphic for $a \in A$. A subset A of X is a **deformation retract** of X if there is a retraction $r : X \to A$ and a homotopy $F : X \times I \to X$ such that

$$\begin{aligned} F(x, 0) &= x, \qquad \forall\, x \in X \\ F(x, 1) &= r(x) \\ F(a, t) &= a \quad \forall\, a \in A,\ t \in I. \end{aligned}$$

Thus, r can be deformed to the constant map keeping A fixed throughout.

Theorem 5.8 *If A is a deformation retract of X, the inclusion map $i : A \to X$ induces an isomorphism of $\pi(A, a)$ onto $\pi(X, a)$ fo all $a \in A$.*

Proof $r_* \circ i_*$ is the identity map of $\pi(A, a)$. Since $i \circ r$ is homotopic to the identity map, $i_* \circ r_*$ is the identity map. □

We shall use this theorem in two ways. First, we use it to determine when two spaces have isomorphic fundamental groups. Second, we use it to show that a subspace is not a deformation retract by proving that certain retracts are not deformation retracts.

A space X is called **contractible to a point** if there is an $x_0 \in X$ such that x_0 is a deformation retract of X.

Any convex subset X of $\mathbb{R}^n$ is contractible to a point. To see this, let x_0 be X and define $f : X \times I \to X$ by

$$f(x, t) = (1 - t)x + tx_0.$$

Clearly, f is continuous and $f(x, 0) = x$. As $f(x, 1) = x_0$, x_0 is a deformation retract of X.

Theorem 5.9 *If X is contractible to a point, then X is simply connected.*

Example. S^{n-1} is a deformation retract of $B^n \backslash \{0\}$. Indeed, consider the map $f : X \times I \to X$ where $X = B^n \backslash \{0\}$ given by

$$f(x, t) = (1 - t)x + t\frac{x}{|x|}.$$

Thus, they have the same fundamental groups.

Exercises

1. Show that a star-shaped subset of $\mathbb{R}^n$ is contractible to a point.
2. Find an explicit homeomorphism between $S^2 \backslash (0, 0, 1)$ and $\mathbb{R}^2$ given by stereographic projection, namely the intersection with the plane of the line joining $x \in S^2$ with $(0, 0, 1)$.
3. Let (X, x_0) and (Y, y_0) be pointed topological spaces. Show that

   $$\pi(X \times Y, (x_0, y_0))$$

   is isomorphic to the direct product $\pi(X, x_0) \times \pi(Y, y_0)$.
4. Show that the fundamental group of $\mathbb{R}^n \backslash \{0\}$ is trivial for $n \geq 3$.
5. A pointed space (X, x_0) is called an **H-space** (after H. Hopf) if there exists a pointed map

   $$m : (X \times X, (x_0, x_0)) \to (X, x_0)$$

   such that each of the (necessarily pointed) maps $m(x_0, \cdot)$ and $m(\cdot, x_0)$ on (X, x_0) are homotopic to the identity loop based at x_0. Show that $\pi(X, x_0)$ is abelian.
6. Let Δ_n denote the **n-simplex** defined by

   $$\{(x_0, x_1, \ldots, x_n) : \ x_0 + x_1 + \cdots + x_n = 1, \ x_i \geq 0\}.$$

This is a subset of $\mathbb{R}^{n+1}$ with the induced topology. For instance, a 0-simplex is a point, a 1-simplex is homeomorphic to the interval $[0, 1]$ and so on. Show that $\pi(\Delta_n)$ is trivial.

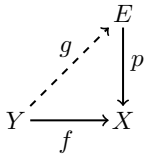
5.5 Covering Spaces

Let E and X be topological spaces and $p : E \to X$ a map. An open subset U of X is said to be **evenly covered** by p if the inverse image $p^{-1}(U)$ is a union of disjoint open sets of E, each of which is homeomorphically mapped onto U by p. We call p a **covering map** if $p : E \to X$ is continuous, surjective and every point of X has an open neighborhood which is evenly covered by p. In such a case, we say that E is a **covering space** of X.

A prototypical example to keep in mind is the map $p : \mathbb{R} \to S^1$ given by the exponential function: $x \mapsto e^{2\pi i x}$, where we are viewing S^1 as a subset of $\mathbb{C}$ consisting of numbers of absolute value 1. If $e^{2\pi i x_0} \in S^1$ and $0 < \epsilon < 1/2$, let $U = \{e^{2\pi i x} : |x - x_0| < \epsilon\}$. Then, $p^{-1}(U)$ is the disjoint union of intervals $(m + x_0 - \epsilon, m + x_0 + \epsilon)$, $m \in \mathbb{Z}$, and each of these intervals is mapped homeomorphically onto U by p.

If $p : E \to X$ is a covering map of topological spaces, we call the set $p^{-1}(x)$ the **fiber** over x. Thus, each $x \in X$ has an open neighborhood U such that $p^{-1}(U)$ is homeomorphic to $p^{-1}(x) \times U$. The subsets of $p^{-1}(U)$ mapped homeomorphically onto U are called **sheets** of $p^{-1}(U)$.

Now let $p : E \to X$ be a covering map. Suppose $f : Y \to X$ is a map of topological spaces. We want to consider the **lifting problem**: when is there a map $g : Y \to E$ such that $f = p \circ g$. In other words, when is there a map g making the following diagram commute:



Such a map g is called a **lift** of f. We begin by showing that if Y is a connected space, such a lift, if it exists, is essentially unique.

Lemma 5.9 *Let $p : E \to X$ be a covering map, and let Y be a connected Hausdorff topological space. Let $f : Y \to X$ be a map, and let $g, h : Y \to E$ be two lifts of f. If $g(y) = h(y)$ for some $y \in Y$, then $g = h$.*

Proof Let

$$S = \{y \in Y : \quad g(y) = h(y)\},$$

$$T = \{y \in Y : \quad g(y) \neq h(y)\}.$$

Clearly $Y = S \cup T$. If we can show that S is both open and closed, then by the connectedness of Y, we get $S = Y$ because, by hypothesis, S is non-empty.

Let $y \in Y$ and U an open neighborhood of $f(y)$ that is evenly covered by p. Let V and W be sheets of $p^{-1}(U)$ such that $g(y) \in V$, $h(y) \in W$. If $g(y) = h(y)$, then $V = W$ since p maps homeomorphically onto U. If $g(y) \neq h(y)$, then V and W are disjoint for the same reason.

Since g and h are continuous at y, there is an open neighborhood N of y such that $g(N) \subset V$, $h(N) \subset W$. For $y \in T$, $V \cap W = \phi$, so that $g(z) \neq h(z)$ for $z \in N$ and hence $N \subset T$. Thus, T is open. On the other hand, if $y \in S$, then $V = W$. Moreover, for each $z \in N$, $g(z)$ must be the unique point $v \in V$ such that $p(v) = f(z)$ and $h(z)$ must be the unique point $w \in W$ such that $p(w) = f(z)$. Thus, $g = h$ on N and $N \subset S$. Hence S is open. This completes the proof. □

We now address the problem of lifting of paths. For this purpose, we will use Lebesgue's lemma (Lemma 5.8).

Theorem 5.10 (Path lifting theorem) *Let $p : E \to X$ be a covering map. Let $\gamma : I \to X$ be a path and let $e_0 \in E$ satisfy $p(e_0) = \gamma(0)$. Then, there exists a unique path $\alpha : I \to E$ such that $\alpha(0) = e_0$ and $p \circ \alpha = \gamma$.*

Proof For each $x \in X$, choose an open neighborhood U_x of x that is evenly covered by p. The sets $\gamma^{-1}(U_x)$, $x \in X$, form an open cover of the compact space $[0, 1]$. Thus, we can find a finite subcover. By Lebesgue's lemma, we can find $0 = s_0 < s_1 < \cdots < s_m = 1$ so that we have evenly covered open sets $U_1, \ldots, U_m$ satisfying

$$[s_{j-1}, s_j] \subset \gamma^{-1}(U_j), \quad \text{for } 1 \leq j \leq m.$$

Thus, $\gamma([s_{j-1}, s_j]) \subset U_j$ for $1 \leq j \leq m$. The lifting of α is performed in m steps. Since $p(e_0) = \gamma(0) \in U_1$, there is an open neighborhood V_1 of e_0 mapped homeomorphically onto U_1 by p. Define α on $[0, s_1]$ such that $\alpha(s)$ is the unique point of V_1 covering $\gamma(s)$. That is,

$$\alpha = (p|_{V_1})^{-1} \circ \gamma \qquad \text{on } [0, s_1].$$

Then, $\alpha(0) = e_0$ and $p \circ \alpha = \gamma$ on $[0, s_1]$. Now perform the same procedure with $e_0 = \alpha(0)$ replaced by $\alpha(s_1)$ and U_1 replaced by U_2, to extend α to the interval $[s_1, s_2]$. After m steps, we have lifted the entire path α. The uniqueness follows from Lemma 5.9. □

We can actually lift homotopies. That is, if we have a family of paths depending continuously on a parameter, the lifted paths also depend continuously on the parameter.

Theorem 5.11 (Homotopy lifting theorem) *Let $p : E \to X$ be a covering map, and let $F : I \times I \to X$ be a map. Let $e_0 \in E$ satisfy $p(e_0) = F(0, 0)$. Then, there exists a unique lift G of F such that $G : I \times I \to E$ and $G(0, 0) = e_0$ and $p \circ G = F$.*

Proof The uniqueness follows from Lemma 5.9 since $I \times I$ is connected. According to the path lifting theorem, there is a unique path $t \mapsto e_t$, $0 \le t \le 1$ in E such that $p(e_t) = F(0,t)$, $0 \le t \le 1$. By the same theorem, there exists for each t, a unique path $s \mapsto G(s,t)$, $0 \le s \le 1$, such that $G(0,t) = e_t$ and $(p \circ G)(s,t) = F(s,t)$, $0 \le s \le 1$. This defines the lift G of F. We must show it is continuous.

Let $\gamma : [0,1] \to X$ be the path defined by

$$\gamma(s) = F(s,0), \qquad 0 \le s \le 1.$$

As in the proof of Theorem 5.10, we can find open sets $U_1, \ldots, U_m$ and $0 = s_0 < s_1 < \cdots < s_m = 1$ and $\epsilon > 0$ such that

$$F([s_{j-1}, s_j] \times [0, \epsilon]) \subset U_j, \qquad 1 \le j \le m.$$

Since $e_0 \in V_1$, we can assume also that the initial points e_t belong to V_1, for $0 \le t \le \epsilon$, since they depend continuously on t. As before, we can define G in m steps. The first step is to define

$$G = (p|_{V_1})^{-1} \circ F \qquad \text{on } [0, s_1] \times [0, \epsilon].$$

In particular, G is continuous on $[0, s_1] \times [0, \epsilon]$. Proceeding inductively, we find G is continuous on $[s_{j-1}, s_j] \times [0, \epsilon]$, for $1 \le j \le m$ so that G is continuous on $[0,1] \times [0, \epsilon]$. The same proof shows that for each t_0 satisfying $0 < t_0 \le 1$, there exists an $\epsilon > 0$ such that G is continuous on $[0,1] \times [t_0 - \epsilon, t_0 + \epsilon]$ (replacing $t_0 + \epsilon$ by t_0 if $t_0 = 1$). Consequently, G is continuous on $[0,1] \times [0,1]$. □

Now let (E, e) and (X, b) be pointed spaces and let $p : (E, e) \to (X, b)$ be a covering map. That is, $p : E \to X$ is a covering map with $p(e) = b$. Let $\gamma : I \to X$ be a loop based at b. By the path lifting theorem, there is a unique lift $\alpha : I \to E$ such that $\alpha(0) = e$. The lift need not be a loop. However, the terminal point $\alpha(1)$ satisfies $p(\alpha(1)) = \gamma(1) = b$, so that $\alpha(1)$ lies in the fiber $p^{-1}(b)$ over b.

Suppose now that γ_1 is another loop in X based at b such that $\gamma_1 \sim \gamma$. Let $\{\gamma_t : 0 \le t \le 1\}$ be the homotopy so that $\gamma_0 = \gamma$. By the homotopy lifting theorem applied to $F(s,t) = \gamma_t(s)$, we obtain a mapping $G : I \times I \to E$ such that $G(0,0) = e$ and

$$p(G(s,t)) = \gamma_t(s), \quad 0 \le s, t \le 1.$$

Then the path α_t in E defined by $\alpha_t(s) = G(s,t)$, $0 \le t \le 1$, is a lift of γ_t and $\alpha_0 = \alpha$. We claim that the α_t's all start at e. To see this, observe that the map $t \mapsto G(0,t)$ is the unique lift to E starting at e of the constant path at b. Hence the lift coincides with the constant path at e and $e = G(0,t) = \alpha_t(0)$, $0 \le t \le 1$. Similarly, the map $t \mapsto G(1,t)$ is the unique lift to E starting at $G(1,0) = \alpha(1)$ of the constant path at b. Hence $\alpha(1) = G(1,t) = \alpha_t(1)$ for $0 \le t \le 1$ so that all the paths α_t terminate at $\alpha(1)$. In particular, the lift α_t of γ_1 of E starts at e and terminates at $\alpha(1)$. Therefore, the terminal point $\alpha(1)$ is the same for all loops in the same homotopy class of γ.

Corollary 5.2 (Monodromy theorem) *Let $p : (E, e) \to (X, b)$ be a covering map of pointed spaces. If γ_0 and γ_1 are two homotopic loops based at b and α_0, α_1 are the liftings of γ_0 and γ_1, respectively, with $\alpha_0(0) = \alpha_1(0) = e$, then $\alpha_0(1) = \alpha_1(1)$. Moreover, α_0 and α_1 are homotopic paths in E.*

This allows us to define a function

$$\Phi : \pi(X, b) \to p^{-1}(b)$$

so that $\Phi([\gamma])$ is the terminal point of the lift of γ that starts at e.

Theorem 5.12 *Let $p : (E, e) \to (X, b)$ be a covering map and suppose that E is simply connected. Then Φ is a one-to-one correspondence of $\pi(X, b)$ and the fiber of $p^{-1}(b)$.*

Proof Suppose that $y \in p^{-1}(b)$. Let α be a path in E from e to y and set $p \circ \alpha = \gamma$. Then, α is the lift of γ to E that starts at e so that $\Phi([\gamma]) = \alpha(1) = y$. Hence the function is onto. Suppose that γ_0 and γ_1 are loops in X based at b such that $\Phi([\gamma_0]) = \Phi([\gamma_1])$. Let α_0, α_1 be lifts of γ_0 and γ_1 respectively that starts at e. Then α_0, α_1 have the same terminal point, so that $\alpha_0\alpha_1^{-1}$ in E is a loop based at e. Since E is simply connected, there is a homotopy

$$F : I \times I \to E$$

of $\alpha_0\alpha_1^{-1}$ to the point e. Then, $p \circ F : I \times I \to X$ is a homotopy of the loop $\gamma_0\gamma_1^{-1}$ to the point b. Hence $[\gamma_0][\gamma_1]^{-1} = [b]$ so that $[\gamma_0] = [\gamma_1]$ and Φ is one to one. □

Let us now consider the covering map $p : (\mathbb{R}, 0) \to (S^1, 1)$ given by

$$p(t) = e^{2\pi i t}, \quad t \in \mathbb{R}.$$

Since $p^{-1}(1)$ coincides with $\mathbb{Z}$, we deduce from Theorem 5.12 that the elements of $\pi(S^1, 1)$ are in one-to-one correspondence with the integers. (In the next section, we shall see that $\pi(S^1, 1) \simeq \mathbb{Z}$.)

Exercises

1. Show that $p : \mathbb{C} \to \mathbb{C}^*$ given by $p(z) = e^z$, $z \in \mathbb{C}$ is a covering map.
2. Prove that the elements of $\pi(\mathbb{C}^*)$ are in one-to-one correspondence with the integers.
3. Show that the map $p : \mathbb{R} \to S^1$ given by

$$p(t) = (\cos t, \sin t)$$

for $t \in \mathbb{R}$ is a covering map.

4. Is the map $f : (0, 3) \to S^1$ given by $f(x) = e^{2\pi ix}$ a covering map?
5. If $\tilde{X}$ is a covering space of X and $\tilde{Y}$ is a covering space of Y, show that $\tilde{X} \times \tilde{Y}$ is a covering space of $X \times Y$.
6. Show that the map from $\mathbb{C}^*$ to $S^1 \times \mathbb{R}$ given by $z \mapsto (z/|z|, |z| - 1)$ if $|z| \geq 1$ and $(z/|z|, (|z| - 1)/|z|)$ for $0 < |z| \leq 1$. Deduce that $\mathbb{C}^*$ and S^1 have the same fundamental group.

5.6 Applications

We will apply the theory of covering spaces to compute $\pi(S^1, 1)$. We view S^1 as the subset of $\mathbb{C}$ whose absolute value is 1. Recall that the map $p : \mathbb{R} \to S^1$ given by $p(t) = e^{2\pi it}$ is a covering map. Given any loop γ of S^1 based at 1, we can lift it uniquely to a path $\alpha : I \to \mathbb{R}$ with $\alpha(0) = 0$, by the path lifting theorem. This allows us to define the degree of the loop γ based at 1 by setting

$$\deg \gamma = \alpha(1),$$

which is necessarily an integer. In particular, each of the loops

$$\gamma_n(x) = e^{2\pi inx}$$

has degree n. Moreover, none of these loops are homotopically equivalent, for otherwise (by the monodromy theorem) their lifts would have the same endpoint, which is not the case. In addition, if γ is any loop of S^1, let $n = \deg \gamma$ and consider $\gamma\gamma_n^{-1}$ whose lift is homotopic to the constant loop at zero. Thus, $\gamma \sim \gamma_n$. This proves:

Theorem 5.13

$$\pi(S^1, 1) \simeq \mathbb{Z}.$$

This gives us the first example of a topological space with a non-trivial fundamental group. For any map $f : S^1 \to S^1$, we have the induced map

$$f_* : \pi(S^1, 1) \ \to \ \pi(S^1, f(1)).$$

If α is a map sending $f(1)$ to 1, we have, by Theorem 2.2, another map

$$\pi(S^1, f(1)) \ \stackrel{\alpha_*}{\to} \ \pi(S^1, 1).$$

Thus, γ_1 is sent to an integer multiple of γ_1 under the composition of these maps. We designate this multiple by $\deg f$. This observation allows us to give a topological proof of the fundamental theorem of algebra:

Theorem 5.14 *Any polynomial $f(z) \in \mathbb{C}[z]$ has a root in $\mathbb{C}$.*

Proof Without loss of generality, we can suppose f is monic: $f(z) = z^n + c_{n-1} z^{n-1} + \cdots + c_1 z + c_0$. Assume $f(z) \neq 0$ for all $z \in \mathbb{C}$. Define a map $\tilde{f} : S^1 \to S^1$ by

$$\tilde{f}(z) = \frac{f(z)}{|f(z)|},$$

which is well-defined as the denominator does not vanish. We proceed to calculate deg $\tilde{f}$. Consider the map

$$F : S^1 \times I \to S^1$$

given by

$$F(x, t) = f(tx)/|f(tx)|.$$

Thus, $\tilde{f}$ is homotopic to the constant map. Since degree is preserved under homotopy, $\tilde{f}$ has degree zero. On the other hand, the homogenized polynomial

$$G(x, t) = t^n f(x/t) = x^n + c_{n-1} t x^{n-1} + \cdots + c_1 t^{n-1} x + c_0 t^n,$$

can be used to give a homotopy between $\tilde{f}$ and z^n by

$$H(x, t) = G(x, t)/|G(x, t)|.$$

Since z^n has degree n we deduce that $\tilde{f}$ has degree n, which is a contradiction. Thus, for some $z \in \mathbb{C}$, $f(z) = 0$. □

Another application of the determination of $\pi(S^1)$ is to prove the case $n = 2$ of the Brouwer fixed point theorem. We need:

Lemma 5.10 *There is no continuous map $r : B^2 \to S^1$ such that $r \circ i = 1$, where i is the inclusion map.*

Proof If there is such a map, we have the composite homomorphism

$$\pi(S^1, 1) \xrightarrow{i_*} \pi(B^2, 1) \xrightarrow{r_*} \pi(S^1, 1)$$

equal to the identity. But $\pi(B^2, 1)$ is trivial, so that r_* is trivial and the composition cannot be the identity. □

Theorem 5.15 (Brouwer fixed point theorem for $n = 2$) *Any continuous map $f : B^2 \to B^2$ has a fixed point.*

Proof Suppose that $f(x) \neq x$ for all $x \in B^2$. Define $r(x) \in S^1$ to be the intersection with S^1 of the ray that starts at $f(x)$ and passes through x. Certainly $r(x) = x$ for $x \in S^1$. By writing an equation for r in terms of f (see Exercise 1), we see that r is continuous. But this contradicts Lemma 5.10. □

One can prove the higher dimensional version of Brouwer's theorem using higher homotopy groups. Given a pointed topological space (X, b) we may consider the loop space $(\Omega X, \tilde{b})$ of all loops based at b. By abuse of notation, we will denote this space by $(\Omega X, b)$. One can endow this space with the compact-open topology: namely the sets

$$(K; \emptyset) = \{f \in \Omega X : f(K) \subset \mathcal{O}\}, \quad K \subset I$$

with K a compact subset of I, and $\mathcal{O}$ an open set in X, are used as a subbase for a topology on ΩX. If X were a metric space with metric d, this topology agrees with the topology induced by the metric d^* on ΩX given by

$$d^*(f, g) = \sup_{t \in I} d(f(t), g(t)).$$

We can now define the higher homotopy groups recursively as follows: $\pi_1(X, b) := \pi(X, b)$, $\pi_2(X, b) := \pi_1(\Omega X, b)$ and generally

$$\pi_k(X, b) = \pi_{k-1}(\Omega X, b).$$

We say two spaces are homotopically equivalent if there exist maps $f : X \to Y$ and $g : Y \to X$ such that $f \circ g$ is homotopic to the identity on Y, and $g \circ f$ is homotopic to the identity on X. We leave as an exercise (see Exercise 3) to show that if X and Y are homotopically equivalent, then $\pi_k(X) \simeq \pi_k(Y)$ for every $k \geq 1$. In particular, if X is contractible to a point x_0, then X is homotopically equivalent to $\{x_0\}$. In such a case, we conclude that $\pi_k(X)$ is trivial for $k \geq 1$. Thus, if X is a convex subset of $\mathbb{R}^n$, we conclude that $\pi_k(X)$ is trivial. Hence, $\pi_k(B^n)$ is trivial for every $k \geq 1$.

On the other hand, one can show that $\pi_k(S^n)$ is trivial for $k < n$ and $\pi_n(S^n) \simeq \mathbb{Z}$. This fact allows us to give a proof of the Brouwer fixed point theorem. Indeed, if $f : B^n \to B^n$ did not have a fixed point, we can join x to $f(x)$ by a straight line which when extended intersects the boundary S^{n-1} at a point $r(x)$ (say). Note that $r(x) = x$ for all $x \in S^{n-1}$. We leave as an exercise (see Exercise 4) to show that r is a continuous map. This gives us a retraction:

$$r : B^n \to S^{n-1}.$$

Together with the inclusion map, we obtain

$$S^{n-1} \xrightarrow{i} B^n \xrightarrow{r} S^{n-1}.$$

Applying the π_{n-1} functor, we get

$$\mathbb{Z} \xrightarrow{i_*} 0 \xrightarrow{r_*} \mathbb{Z}.$$

Since $r \circ i = 1$, we must have $r_* \circ i_* = 1$ which is a contradiction. This completes our sketch of proof of the Brouwer fixed point theorem in the general case.

Not much is known about $\pi_k(S^n)$ for $k > n$. Serre showed (in his doctoral thesis) that $\pi_k(S^n)$ is a finite group unless n is even and $k = 2n - 1$, in which case we have

$$\pi_{2n-1}(S^n) \simeq \mathbb{Z} \oplus (\text{finite group}).$$

We give several applications of the Brouwer fixed point theorem. The first is to prove:

Theorem 5.16 (Brouwer, 1911)

$$\mathbb{R}^m \simeq \mathbb{R}^n$$

if and only if $m = n$.

Proof If $\mathbb{R}^n \simeq \mathbb{R}^m$, then

$$\mathbb{R}^n \backslash \{0\} \simeq \mathbb{R}^m \backslash \{0\}.$$

By stereographic projection, we know that $S^{n-1} \simeq \mathbb{R}^n \backslash \{0\}$ so that we deduce that $S^n \simeq S^m$. By applying the π_n functor, we deduce that $n = m$. □

This theorem (called the theorem of the invariance of domain), though not a direct application of the fixed point theorem, has the same essential idea in its proof as the fixed point theorem. Its significance is the corollary that the notion of dimension is well-defined. That is, an n-manifold cannot be at the same time be an m-manifold unless $m = n$.

Our next application is to linear algebra which plays a fundamental role in the Google PageRank algorithm.

Theorem 5.17 (Perron) *If $A = (a_{ij})$ is an $n \times n$ matrix with $a_{ij} > 0$, then A has a positive eigenvalue λ. Moreover, there is a corresponding eigenvector v with all coordinates positive such that $Av = \lambda v$.*

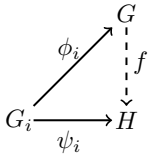
Proof For any vector $x \in \mathbb{R}^n$ define $\sigma(x)$ to be the sum of its coordinates. Let Δ_{n-1} denote the $(n-1)$-simplex. Now consider the map $f : \Delta_{n-1} \to \Delta_{n-1}$ given by

$$f(x) = \frac{Ax}{\sigma(Ax)}.$$

It is not difficult (see Exercise 6) to see that this is indeed a map into Δ_{n-1}. As Δ_{n-1} is homeomorphic to B^n, we can apply the fixed point theorem to deduce the result. □

The computation of the fundamental group of S^1 allows us to deduce the fundamental groups of other topological spaces. For example, it is immediate that the fundamental group of the torus $T = S^1 \times S^1$ is $\mathbb{Z} \times \mathbb{Z}$.

A theorem of Seifert and van Kampen allows us to write down more fundamental groups from the fact that $\pi(S^1) \simeq \mathbb{Z}$. This theorem is a vast generalization of Theorem 5.7, though essentially based on the same ideas. To state it precisely, we recall the definition of free products of groups. Given any collection of groups G_i for $i \in I$, an index set, and a fixed group G with homomorphisms $\phi_i : G_i \to G$, we say that a group G is the **free product** of the groups G_i if and only if the following condition holds: if H is any other group and we have homomorphisms $\psi_i : G_i \to H$, there exists a unique homomorphism $f : G \to H$ making the following diagram commute.



One can show that given any collection of groups, their free product exists. Intuitively, the free product is to be thought of as a group consisting of "words" formed using the "alphabet" of elements of the G_i for $i \in I$. More precisely, elements of the free product consist of finite sequences $(x_1, x_2, \ldots, x_n)$ where each x_k belongs to some G_i, any two successive terms belong to different groups and no term is the identity element of any G_i. One can define multiplication of "words" in the obvious way. The essential point is to ensure that all of this is well-defined and that G exists.

If G_1 and G_2 are two groups, we denote their free product as $G_1 * G_2$. Now suppose that X is a topological space with subspaces U and V. We have the induced homomorphisms

$$\pi(U) \to \pi(X), \qquad \pi(V) \to \pi(X)$$

as well as

$$\pi(U \cap V) \xrightarrow{i} \pi(U), \qquad \pi(U \cap V) \xrightarrow{j} \pi(V),$$

arising from the inclusion maps. Then:

Theorem 5.18 (Seifert and van Kampen, 1933) *Suppose that X is a topological space with path-connected subspaces U and V. If $U \cap V$ is path-connected and $a \in U \cap V$, then*

$$\pi(X, a) \simeq \pi(U, a) * \pi(V, a)/H$$

where H is the normal subgroup generated by the words $i_*(g) j_*(g)^{-1}$ where $g \in \pi(U \cap V, a)$.

This theorem allows us to compute, for instance, the fundamental group of the "figure eight" as the free product $\mathbb{Z} * \mathbb{Z}$ since $\pi(S^1)$ is $\mathbb{Z}$ and the intersection consists of a point in this case. More generally, the same logic leads us to conclude that the fundamental group of a flower with n petals is the n-fold free product of the integers.

In particular, if X is the union of two path-connected subspaces U and V with $U \cap V$ simply connected, we deduce from Theorem 5.18 that $\pi(X)$ is isomorphic to the free product of $\pi(U)$ and $\pi(V)$.

Exercises

1. Prove that the map constructed in Theorem 5.15 is continuous.
2. Show that any map $f : S^1 \to S^1$ such that $\deg f \neq 1$ has a fixed point.
3. Prove that if X and Y are homotopically equivalent, then $\pi_k(X) \simeq \pi_k(Y)$ for every $k \geq 1$.
4. Suppose that $f : B^n \to B^n$ has no fixed point. Define $r(x)$ as the point of intersection of the ray joining x to $f(x)$ and S^{n-1}. Prove that r is a continuous function.
5. Suppose X is homeomorphic to B^n. Show that any continuous map $f : X \to X$ has a fixed point.
6. Let A be an $n \times n$ matrix with positive entries. For any $v \in \mathbb{R}^n$ define $\sigma(v)$ to be the sum the coordinates of v. Show that

$$f(x) = \frac{Ax}{\sigma(Ax)}$$

defines a map $f : \Delta_{n-1} \to \Delta_{n-1}$ where Δ_{n-1} denotes the $(n-1)$-simplex.

5.7 Group Actions and Orbit Spaces

Recall that an **action** of a group G acting on a set X is a map $G \times X \to X$, which we denote by $(g, x) \mapsto g \cdot x$, satisfying

(i) $1 \cdot x = x$ for all $x \in X$ and
(ii) $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$.

Two elements x, y of X are said to be in the same **orbit** if there is an element g of G so that $g \cdot x = y$. Clearly, this is an equivalence relation and thus, an action of G on X partitions X into disjoint orbits.

Now let G be a group acting on a topological space X. We also insist that in addition to (i) and (ii) above, this action satisfies:

(iii) the map $x \mapsto g \cdot x$ is a homeomorphism of X for all $g \in G$.

That is, G defines a group of homeomorphisms of X. We can endow the space of orbits, which we will denote by the notation, X/G, with a topology as follows. Let $p : X \to X/G$ be the map that sends each point to its orbit. A subset U of X/G is defined to be open if $p^{-1}(U)$ is open in X. (This is a special case of a more general construction of the quotient topology induced on a set Y by a map $f : X \to Y$ from a topological space X.)

For example, $\mathbb{Z}$ acts on $\mathbb{R}$ by translation: $n \cdot x = x + n$ for $n \in \mathbb{Z}$ and $x \in \mathbb{R}$. The orbit space is homeomorphic to S^1. Another example is given by the action of $\mathbb{Z}/2\mathbb{Z}$ on S^n given by $(\pm 1) \cdot x = \pm x$. Thus the non-trivial element of $\mathbb{Z}/2\mathbb{Z}$ takes x to its **antipodal** point. The quotient space is called the real projective space and denoted $\mathbb{P}^n(\mathbb{R})$.

Another important example concerns the following action of $\mathbb{Z} \times \mathbb{Z}$ on $\mathbb{C}$. Take $\tau \in \mathbb{C} \backslash \mathbb{R}$ and define $(m, n) \cdot z = z + m + n\tau$. The quotient space $\mathbb{C}/\mathbb{Z}^2$ is then homeomorphic to a torus.

We say that G acts **evenly** or **properly discontinuously** if any point in X has a neighborhood N such that $g \cdot N$ and $h \cdot N$ are disjoint for $g \neq h$. (The latter terminology is more prevalent in the literature, but here we opt for the terser adjective as introduced by Fulton, Algebraic Topology, p. 159.) The importance of even action is given by the following theorem.

Theorem 5.19 *If G acts evenly on X, then the projection map*

$$p : X \to X/G$$

is a covering map.

Proof The map p is continuous by construction of X/G. If N is as in the definition of even action, then $p^{-1}(p(N))$ is a union of open sets $g \cdot N$ for $g \in G$. We claim that $p(N)$ is evenly covered. This is obvious from the definition of even action since all the $g \cdot N$ are disjoint and homeomorphic to N which is in turn homeomorphic to $p(N)$. Indeed, p provides a one-to-one correspondence between N and $p(N)$ because if $p(x) = p(y)$ for $x, y \in N$, then there is some $g \in G$ so that $g \cdot x = y$. This means $y \in N \cap g \cdot N$, contrary to the definition of even action. □

Observe that in both of the examples above, we have an even action. Thus, in both cases,

$$p : X \to X/G$$

is a covering map. If X is simply connected, by what we have shown in the previous chapter, there is a one-to-one correspondence between $\pi(X/G)$ and elements of the fiber $p^{-1}(b)$ for some $b \in X/G$. In the second example above, there are two preimages for every element of $\mathbb{P}^n(\mathbb{R})$. We deduce that $\pi(\mathbb{P}^n(\mathbb{R})) \simeq \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$ as S^n is simply connected in this case.

An interesting example can be given with the group

$$SL_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : \ a, b, c, d \in \mathbb{Z}, \ ad - bc = 1 \right\}$$

acting on the upper half-plane $H = \{z \in \mathbb{C} : \ Im(z) > 0\}$ given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}.$$

This defines an action (see Exercise 1). This, however, is not an even action because even actions are fixed point free (see Exercise 2). That is, for an even action, $g \cdot x = x$ for any element x implies $g = 1$ (see Exercise 2). In this example, $z = i = \sqrt{-1}$ is fixed by

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Therefore, $H \to H/SL_2(\mathbb{Z})$ is **not** a covering map.

However, if we consider the group $PSL_2(\mathbb{R})$ acting on the upper half-plane, there are situations where a discrete subgroup Γ of $PSL_2(\mathbb{R})$ acts evenly on H.

We now consider a path-connected space X with an even action by a group G. Thus, $p : X \to X/G$ is a covering map. Fix $e \in X$ and let $b = p(e)$. Our goal now is to find a relationship between G and $\pi(X/G, b)$. Notice that

$$p^{-1}(b) = \{g \cdot e : \ g \in G\}.$$

We will define an action of $\pi(X/G, b)$ on the fiber $p^{-1}(b)$. If $\gamma \in \pi(X/G, b)$ then there is a unique lift $\tilde{\gamma}$ of γ that begins at e and we have $\tilde{\gamma}(1) \in p^{-1}(b)$ and so there is a unique element $g_\gamma \in G$ such that $\tilde{\gamma}(1) = g_\gamma \cdot e$. Clearly, g_γ depends only on the path class of γ since the endpoint of the lift depends only on the path class. Thus, we have a map

$$\phi : \pi(X/G, b) \to G$$

given by $\gamma \mapsto g_\gamma$.

Theorem 5.20 *The map ϕ is a homomorphism of groups.*

Proof Consider two loops γ, η in X/G, based at b. Let $\widetilde{\gamma\eta}$ be the unique lift of $\gamma\eta$ that begins at e. Then, $\widetilde{\gamma\eta} = \tilde{\gamma}(\eta_a)$ where $a = \tilde{\gamma}(1)$ and η_a is the unique lift of η that begins at a. This is because $\tilde{\gamma}(\eta_a)$ is also a lift of $\gamma\eta$ that begins at e. Let $\tilde{\eta}$ be the unique lift of η that begins at e. Since $g_\gamma \cdot \tilde{\eta}$ is also a lift that begins at $g_\gamma \cdot e$ and $a = \tilde{\gamma}(1) = g_\gamma \cdot e$, it follows that $\eta_a = g_\gamma \cdot \tilde{\eta}$. Hence,

$$\widetilde{\gamma\eta}(1) = (\tilde{\gamma}\eta_a)(1) = \eta_a(1) = g_\gamma \cdot \tilde{\eta}(1) = g_\gamma \cdot (g_\eta \cdot e) = (g_\gamma g_\eta) \cdot e.$$

It follows that ϕ is a homomorphism. □

Lemma 5.11 *The kernel of ϕ is the subgroup $p_*\pi(X, e)$.*

Proof The kernel consists of elements $\gamma \in \pi(X/G, b)$ with $\phi(\gamma) = 1$. These are the elements for which $\tilde{\gamma}(1) = e$ which are loops based at e. □

Corollary 5.3 *The groups*

$$\pi(X/G, b)/p_*\pi(X, e)$$

and G are isomorphic.

In particular, if X is simply connected, we have the isomorphism

$$\pi(X/G, b) \simeq G.$$

This result allows us to deduce (yet again) the fact that $\pi(S^1) \simeq \mathbb{Z}$ from the even action of $\mathbb{Z}$ on $\mathbb{R}$ given by translation. Similarly, we can deduce the fundamental group of the torus.

We give one more example involving the Möbius strip. Let

$$X = \{z \in \mathbb{C} : z = x + iy,\ x \in \mathbb{R},\ 0 \leq y \leq 1\}.$$

Define a transformation T on X by setting

$$T(z) = \overline{z} + 1 + i$$

and let G be the cyclic group generated by T. Then, X/G is homeomorphic to the Möbius strip. Moreover, the action is even and

$$X \to X/G$$

is a covering map. We deduce immediately that the fundamental group of the Möbius strip is $\mathbb{Z}$.

We will compute the fundamental group of another important topological space, namely the Klein bottle. This will turn out to be a non-abelian group.

Define the following transformations of the plane:

$$A \cdot (x, y) = (x + 1, y); \quad B \cdot (x, y) = (-x, y + 1).$$

Notice that

$$(AB) \cdot (x, y) = A \cdot (-x, y + 1) = (-x + 1, y + 1)$$

and

$$(BA) \cdot (x, y) = B \cdot (x + 1, y) = (-x - 1, y + 1).$$

Thus, the group of tranformations of the plane generated by A and B is non-abelian. We will leave as an exercise to show that this action is even. As $\mathbb{R}^2$ is simply connected, we deduce that the fundamental group of the quotient space is the group generated by A and B. It is infinite (since A, for example, has infinite order) and non-abelian since $AB \neq BA$. The quotient space is called the Klein bottle.

The importance of $\mathbb{P}^2(\mathbb{R})$ arises in the classification theorem of surfaces. It was defined above as a quotient space of S^2 where we identify antipodal points. In other words, the southern hemisphere is identified with the northern hemisphere, which is

$$\{(x, y, z) :\ x^2 + y^2 \leq 1,\ z \geq 0\}.$$

It is easy to see that this is homeomorphic to the unit disk with antipodal points on the boundary identified. Thus,

$$\mathbb{P}^2(\mathbb{R}) \simeq B^2/R$$

where R is the relation $x \sim y$ if and only if $x = y$ or $x, y \in S^1 = \partial B^2$ and $x = -y$. The reader should be able to see that this is really the "Möbius strip sewn onto a disk." It turns out that $\mathbb{P}^2(\mathbb{R})$ is essentially the only closed "non-orientable surface" in the sense that all such surfaces are built out of it. We clarify this below.

Compact connected 2-manifolds are called **surfaces**. The 2-sphere S^2, the torus $T = S^1 \times S^1$ and the real projective plane $\mathbb{P}^2(\mathbb{R})$ are examples of surfaces. It turns out that every compact connected 2-manifold can be obtained from these three surfaces by forming what is called the "connected sum."

More precisely, let S_1 and S_2 be two disjoint surfaces. Their connected sum is constructed as follows. Choose $D_1 \subseteq S_1$ and $D_2 \subseteq S_2$ so that D_1, D_2 are each homeomorphic to B^2. Let $h_i : D_i \to B^2$ be homeomorphisms. Define

$$S_1 \# S_2 = (S_1 \backslash D_1^o) \cup (S_2 \backslash D_2^o)/R$$

where R is the equivalence relation given by $(x, y) \in R$ if and only if $x, y \in \partial D_1 \cup \partial D_2$ and $h_1(x) = h_2(y)$. We can now state:

Theorem 5.21 (Classification of surfaces) *Any compact surface S is homeomorphic to precisely one of the following:*

(a)

$$S^2 \# \underbrace{T \# T \# \cdots \# T}_{g \text{ times}} \qquad g \geq 0;$$

(b)

$$S^2 \# \underbrace{\mathbb{P}^2(\mathbb{R}) \# \cdots \# \mathbb{P}^2(\mathbb{R})}_{n \text{ times}} \qquad n \geq 1;$$

In case (a), we say the surface is **orientable** and in the second, **non-orientable**. g (and sometimes n) is called the **genus** of the surface.

Theorem 5.21 was initiated and carried through in the orientable case by A.F. Möbius (1790–1868) in a paper he submitted for the Grand Prix de Mathématiques of the Paris Academy of Sciences. He was 71 at the time. The jury did not consider any of the submitted papers worthy of the prize and so the work of Möbius appeared as just another mathematical paper in their proceedings. It is not clear who finally proved the theorem in its full generality. Some ascribe it to H.R. Brahana, whose paper appeared in the Annals of Mathematics in 1922.

Exercises

1. Show that the action of $SL_2(\mathbb{Z})$ on the upper half plane H given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}$$

is indeed an action.
2. If G acts evenly on a topological space, and $g \cdot x = x$ for some $x \in X$ and $g \in G$, show that $g = 1$.
3. Notice that S^3 is homeomorphic to

$$X = \{(z, w) : \ z, w \in \mathbb{C}, \ |z|^2 + |w|^2 = 1\}.$$

We define an action of $G = \mathbb{Z}/q\mathbb{Z}$ on X by $a \cdot (z, w) = (\zeta^a z, \zeta^a w)$ where $\zeta = e^{2\pi i/q}$ is a primitive qth root of unity. Show that this gives an even action of $\mathbb{Z}/q\mathbb{Z}$ on X. Deduce that $\pi(X/G) \simeq \mathbb{Z}/q\mathbb{Z}$. (This space is called the Lens space.)
4. Show that every finitely generated abelian group can be realized as the fundamental group of some path-connected space.
5. Show that the fundamental group of the cylinder is $\mathbb{Z}$.
6. Prove that the action defined on $\mathbb{R}^2$ describing the Klein bottle as a quotient space is an even action.
7. With the action on $\mathbb{R}^2$ described by

$$A \cdot (x, y) = (x + 1, y); \quad B \cdot (x, y) = (-x, y + 1)$$

show that $ABA = B$. Deduce that the fundamental group of the Klein bottle is the group generated by the two elements A and B with only the relation $ABA = B$.

5.8 Automorphisms of Covering Spaces

Let $p : (E, e) \to (X, b)$ be a covering map of topological spaces. We will now consider the group designated $Aut(E/X, p)$ of all **covering transformations** or sometimes called **deck transformations** defined as homeomorphisms ϕ of E to E satisfying $p \circ \phi = p$. We begin with:

Lemma 5.12 *Let E be path-connected and $p : (E, e) \to (X, b)$ a covering map. Suppose $\phi : E \to E$ is a continuous map with $p \circ \phi = p$. If for some $x_1 \in E$, we have $\phi(x_1) = x_1$, then $\phi(x) = x$ for all $x \in E$.*

Proof Let x be any point of E and let $\alpha : I \to E$ be a path from x_1 to x. Since $\phi(x_1) = x_1$, the paths α and $\phi \circ \alpha$ both begin at x_1. Moreover, $p \circ \alpha = p \circ \phi \circ \alpha$ so that α and $\phi \circ \alpha$ are lifts of the path $(p \circ \alpha) : I \to X$. By the uniqueness lemma (Lemma 5.9), $\alpha = \phi \circ \alpha$. In particular, the endpoints coincide so that $\phi(x) = x$. □

It is clear that $Aut(E/X, p)$ acts on E simply by $\phi \cdot x = \phi(x)$. We would like to make this into an even action (or equivalently, a properly discontinuous action) as defined in the previous section. To ensure this, we need to assume that E is **locally path-connected**. That is, each point of E has a neighborhood which is path connected. It is not hard to see that the continuous image of a locally path-connected space is also locally path-connected (Exercise 1).

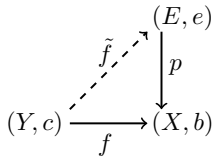
Theorem 5.22 *If E is connected and locally path-connected, then the action of $Aut(E/X, p)$ on E is an even action.*

Proof Let $x \in E$. Let U be an evenly covered neighborhood of $p(x)$. Thus, $p^{-1}(U) = \cup_j V_j$ and $x \in V_k$ for some k. Let $\phi \in Aut(E/X, p)$. If $\phi(x) = x$, then ϕ is the identity map by Lemma 5.12. Therefore, if $\phi \neq 1$, we have $\phi(x) \neq x$. Since $p \circ \phi(x) = p(x)$, it follows that $\phi(x) \in V_j$ for some j. If $V_j = V_k$, then $\phi(x) = x$, which is not the case. Thus, $V_j \cap V_k = \emptyset$. We may insist that U is path-connected since E (and hence X, by Exercise 1) is locally path-connected. Therefore, each of the sets V_i are path-connected and $p \circ \phi(V_k) = U$ so that $\phi(V_k) \subseteq \cup_i V_i$. As the V_j are path-connected and $\phi(x) \in V_j$ for $x \in V_k$ we deduce that $\phi(V_k) \subseteq V_j$ and $V_k \cap \phi(V_k) = \emptyset$. Hence the action is even. □

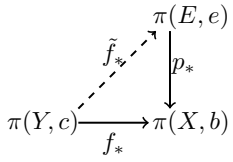
Let $p : (E, e) \to (X, b)$ be a covering of topological spaces. In Sect. 3, we proved that if Y is a connected space and

$$f : Y \to X$$

is a continuous map, then any lift $\tilde{f} : Y \to (E, e)$ if it exists is essentially unique. We will now consider the question of when such a lift exists. Thus, given a map $f : (Y, c) \to (X, b)$ of pointed spaces, does there exist an $\tilde{f} : (Y, c) \to (E, e)$ making the following diagram commute,



The diagram above induces the following diagram of fundamental groups:



The homomorphism p_* is injective (Exercise 4) and $p_* \circ \tilde{f}_* = f_*$ so that

$$f_*\pi(Y, c) = p_*\tilde{f}_*\pi(Y, c) \subseteq p_*\pi(E, e).$$

Therefore, this is a necessary condition for the lift $\tilde{f}$ to exist. It turns out that this is also a sufficient condition provided Y is locally path-connected. Though the proof of this is not difficult, it is rather long and we omit it. For future reference, we state it as:

Theorem 5.23 (Lifting criterion) *Let $p : (E, e) \to (X, b)$ be a covering map. Suppose Y is connected, locally path-connected and*

$$f : (Y, c) \to (X, b)$$

is a continuous map. Then, there is a lift

$$\tilde{f} : (Y, c) \to (E, b)$$

such that $f(c) = b$ *if and only if*

$$f_*\pi(Y, c) \subseteq p_*\pi(E, e).$$

We use this lifting criterion to determine when we can have "isomorphic" coverings of a given topological space X.

Lemma 5.13 *Let* $p_1 : (E_1, e_1) \to (X, b)$ *and* $p_2 : (E_2, e_2) \to (X, b)$ *be two coverings of* X*, with* E_1, E_2 *both connected and locally path-connected. If*

$$p_{1*}\pi(E_1, e_1) = p_{2*}\pi(E_2, e_2)$$

then there is a homeomorphism $\phi : (E_1, e_1) \to (E_2, e_2)$ *so that* $p_2 \circ \phi = p_1$.

Proof We have the diagram:

$$\begin{array}{ccc} & & (E_1, e_1) \\ & & \downarrow{\scriptstyle p_1} \\ (E_2, e_2) & \xrightarrow[p_2]{} & (X, b) \end{array}$$

Since E_2 is connected and locally path-connected, p_2 has a lift $\tilde{p}_2$ by Theorem 5.23. By the uniqueness lemma (Lemma 5.9), this is unique and $p_1 \circ \tilde{p}_2 = p_2$. Similarly, we have a unique lift $\tilde{p}_1 : (E_1, e_1) \to (E_2, e_2)$ so that $p_2 \circ \tilde{p}_1 = p_1$. We thus have

$$(E_1, e_1) \overset{\tilde{p}_1}{\to} (E_2, e_2) \overset{\tilde{p}_2}{\to} (E_1, e_1)$$

so we can consider the composition

$$\psi := (\tilde{p}_2 \circ \tilde{p}_1) : (E_1, e_1) \to (E_1, e_1).$$

Note that

$$p_1 \circ \psi = p_1 \circ (\tilde{p}_2 \circ \tilde{p}_1) = (p_1 \circ \tilde{p}_2) \circ \tilde{p}_1 = p_2 \circ \tilde{p}_1 = p_1.$$

As $\psi(e_1) = e_1$, we have by Lemma 6.1 that ψ is the identity map. Similarly, by reversing the roles of E_1 and E_2, we deduce $\tilde{p}_1 \circ \tilde{p}_2 = 1$. Thus, $\tilde{p}_1$ and $\tilde{p}_2$ are homeomorphisms and we may take $\phi = \tilde{p}_1$ to deduce the result. □

This lemma has a converse, the proof of which we leave as an exercise (Exercise 5).

Lemma 5.14 *Let* X *be a Hausdorff space and let*

$$p_1 : (E_1, e_1) \to (X, b), \quad p_2 : (E_2, e_2) \to (X, b)$$

be covering maps with E_1, E_2 connected and locally path-connected. If there is a homeomorphism

$$\phi : (E_1, e_1) \to (E_2, e_2)$$

satisfying $p_2 \circ \phi = p_1$, then

$$p_{1*}\pi(E_1, e_1) = p_{2*}\pi(E_2, e_2).$$

These lemmas allow us to make the definition that two coverings

$$p_1 : (E_1, e_1) \to (X, b) \quad \text{and} \quad p_2 : (E_2, e_2) \to (X, b)$$

are **equivalent** if there is a homeomorphism

$$\phi : E_1 \to E_2$$

satisfying $p_2 \circ \phi = p_1$. (Note that base points of E_1 and E_2 are not necessarily preserved by ϕ.)

By Theorem 5.4, we immediately deduce:

Theorem 5.24 *Let $p_1 : (E_1, e_1) \to (X, b)$ and $p_2 : (E_2, e_2) \to (X, b)$ be two coverings with E_1, E_2 connected and locally path-connected. The two coverings are equivalent if and only if*

$$p_{1*}\pi(E_1, e_1) \quad \text{and} \quad p_{2*}\pi(E_2, e_2)$$

are conjugate subgroups in $\pi(X, b)$.

By combining Theorem 5.22 with Theorem 5.24, we will deduce the important result:

Theorem 5.25 *Let $p : (E, e) \to (X, b)$ be a covering with E connected and locally path-connected. If*

$$p_*\pi(E, e) \lhd \pi(X, b)$$

then X is homeomorphic to E/G where $G = Aut(E/X, p)$.

Remark. Such coverings are called **Galois** coverings (or **normal** or **regular** coverings in the literature).

Proof Since $p_*\pi(E, e) \lhd \pi(X, b)$, we see that

$$p_*\pi(E, e_1) = p_*\pi(E, e_2)$$

for any $e_1 \in p^{-1}(b)$. By Lemma 5.13, there is an element $\phi \in Aut(E/X, p)$ with $\phi(e_2) = e_1$. Thus, if $p(e_1) = p(e_2)$, there is an element $\phi \in Aut(E/X, p)$ such that

$\phi(e_1) = e_2$. Conversely, it is clear that if $\phi(e_1) = e_2$ for some $\phi \in Aut(E/X, p)$, then $p(e_1) = p(e_2)$. Therefore, $Aut(E/X, p)$ identifies points in E the same way p does. This gives a one to one correspondence say ψ between X and E/G. These two spaces are also homeomorphic as topological spaces because U in X is open if and only if $p^{-1}(U)$ is open in E if and only if $\psi(U)$ is open in E/G. This completes the proof. □

Applying Corollary 5.3, we obtain

Theorem 5.26 *Let $p : (E, e) \to (X, b)$ be a covering map with E connected and locally path-connected. If $p_*\pi(E, e) \lhd \pi(X, b)$ then*

$$\pi(X, b)/p_*\pi(E, e) \simeq Aut(E/X, p).$$

In particular, if E is simply connected, we deduce:

Theorem 5.27 *If $p : (E, e) \to (X, b)$ is a covering with E simply connected and locally path-connected, then*

$$\pi(X, b) \simeq Aut(E/X, p).$$

This theorem allows us to compute the fundamental group of any topological space X by first finding a covering E which is simply connected and locally path-connected and then computing its automorphism group. The question of when a given space possesses such a covering will be discussed in the next chapter.

For now, let us observe the similarity of Theorem 5.27 with the main theorem in Galois theory. The fundamental group has been identified as the group of automorphisms of the simply connected covering E. This analogy has other features that parallel Galois theory, and these we take up in the next chapter.

Exercises

1. Show that the continuous image of a locally path connected space is locally path-connected.
2. Show that a connected space which is locally path connected is path-connected.
3. Given an example of a space which is connected but not locally path-connected.
4. If $p : (E, e) \to (X, b)$ is a covering map, show that $p_* : \pi(E, e) \to \pi(X, b)$ is injective.
5. Prove Lemma 5.14.

5.9 The Universal Covering Space

In the last section, we showed that a covering $p : (E, e) \to (X, b)$ is determined up to equivalence by the conjugacy class of the subgroup $p_*\pi(E, e)$. This raises the question of whether for a given conjugacy class of subgroups of $\pi(X, b)$ there exists a covering $p : (E, e) \to (X, b)$ with $p_*\pi(E, e)$ belonging to the conjugacy class. At one extreme, there is always the trivial covering $(X, b) \to (X, b)$ given by the identity map and corresponding to the entire fundamental group. At the other extreme, the covering $p : (E, e) \to (X, b)$ corresponding to the trivial subgroup is very interesting since then, we may realize $\pi(X, b)$ as $Aut(E/X, p)$ by the results of the last section. Such a covering E is called the universal covering space of X, when it exists.

Suppose $p : (E, e) \to (X, b)$ is a covering with E simply connected. If $x \in X$ and $y \in p^{-1}(x)$, then there is an evenly covered neighborhood U of x with $p^{-1}(U) = \cup_j V_j$, $y \in V_k$ for some k and each V_j is homeomorphic to U. Let $V = V_k$. We have the diagram:

$$\begin{array}{ccc} V & \longrightarrow & E \\ {\scriptstyle p|_V}\downarrow & & \downarrow{\scriptstyle p} \\ U & \xrightarrow{i} & X \end{array}$$

which induces the following diagram of fundamental groups:

$$\begin{array}{ccc} \pi(V, y) & \longrightarrow & \pi(E, y) \\ {\scriptstyle (p|_V)_*}\downarrow & & \downarrow{\scriptstyle p_*} \\ \pi(U, x) & \xrightarrow{i_*} & \pi(X, x) \end{array}$$

Since $p|_V$ is a homeomorphism, $(p|_V)_*$ is an isomorphism. Since $\pi(E, y)$ is trivial, we deduce that i_* must be the trivial homomorphism. In other words, if X has a universal covering, every point $x \in X$ has a neighborhood U such that the homomorphism $\pi(U, x) \to \pi(X, x)$ is trivial. In such a situation, we say that X is **semilocally simply connected**.

Thus, combining the results of the last section, we see that for a space X that is connected and locally path-connected to have a universal covering space, it is necessary that it be semilocally simply connected. This condition turns out to be sufficient.

Most spaces we encounter will satisfy these conditions. However, there are spaces which do not have universal coverings. For example,

$$X = \cup_{n \geq 1} C_n$$

where C_n is the circle in $\mathbb{R}^2$ of radius $1/n$ and center $(1/n, 0)$ is not semilocally simply connected at $(0, 0)$ (see Exercise 1). Thus, this space fails to have a universal covering.

We will now sketch the proof of:

Theorem 5.28 *Let X be connected and locally path connected. Then X has a universal covering if and only if X is semilocally simply connected.*

Proof (Sketch) We saw above that the condition is necessary. Now we show that it is sufficient. We construct the universal covering E as follows. Fix a base point $b \in X$. Let E be the equivalence classes of homotopic paths that begin at b. Thus,

$$E = \{[\alpha] : \ \alpha : I \to X, \ \alpha(0) = b, \ [\alpha] = [\beta] \iff a \sim \beta\}.$$

Define a map $p : E \to X$ by setting $p([\alpha]) = \alpha(1)$. By the theorem on lifting of homotopies (Theorem 5.11), this is well-defined and independent of the representative we choose from the class. We have to put a topology on E to make it simply connected. Let U be an open set in X and α a path from b to c (say) in U. Define

$$[U, \alpha] = \{[\alpha\beta] : \ \beta : I \to X, \ \beta(0) = \alpha(1), \ \beta(I) \subseteq U\}.$$

In other words, $[U, \alpha]$ consists of equivalence classes of paths $\alpha\beta$ with β lying in U and beginning at $\alpha(1)$. We use the sets $[U, \alpha]$ as a basis for a topology on E and leave to the reader (Exercises 2 and 3) the verifications that they indeed generate a topology. With this topology, the map $p : E \to X$ is continuous. Indeed, if U is open in X and $p^{-1}(U)$ is empty, we are done. If $[\alpha] \in p^{-1}(U)$, then

$$\begin{aligned} p([U, \alpha]) &= \{(\alpha\beta)(1) : \ [\alpha\beta] \in [U, \alpha] \\ &= \{\beta(1) : \ [\alpha\beta] \in [U, \alpha]\} \\ &\subseteq U \end{aligned}$$

so that

$$p^{-1}(U) = \cup_{[\alpha] \in p^{-1}(U)} [U, \alpha]$$

which is open in E. Thus, p is continuous.

It remains to show that $p : E \to X$ is a covering. Thus, for each $x \in X$, we have to find a neighborhood which is evenly covered. This is where the semilocally simply connected property of X is used. Let V be an open neighborhood of x which is path-connected and for which every closed path based at x is equivalent to the constant path at x. Now

$$p^{-1}(V) = \cup_{[\alpha] \in p^{-1}(V)} [V, \alpha].$$

We claim that the $[V, \alpha]$ are either equal or disjoint. Indeed, if $[\gamma] \in [V, \alpha] \cap [V, \beta]$, by Exercise 2,

$$[V, \gamma] = [V, \beta], \quad [V, \gamma] = [V, \alpha]$$

so that $[V, \alpha] = [V, \beta]$. Thus, the $[V, \alpha]$'s are disjoint. The map $p_\alpha = p|[V, \alpha] : [V, \alpha] \to V$ is continuous and surjective. To prove it is injective, suppose $p_\alpha(\alpha\beta) = p_\alpha(\alpha\gamma)$. Then β and γ have the same endpoints. The path $\beta\gamma^{-1}$ is a closed path in

V and thus equivalent to the constant path by our choice of V. Thus, $\beta \sim \gamma$ and $[\alpha\beta] = [\alpha\gamma]$. To complete the proof, we need to check that p_α^{-1} is continuous. We leave this to the reader (Exercise 4).

Finally, one needs to check that E is simply connected. First we show that E is path-connected. Let $\tilde{e}$ denote the class of the constant path at b. Let $[\alpha] \in E$. Define $\tilde{\alpha} : I \to E$ by $\tilde{\alpha}(s) = [\alpha_s]$ for $s \in I$ and $\alpha_s(t) = \alpha(st)$, for $t \in I$. Then, $\tilde{\alpha}$ is a path in E from $\tilde{e}$ to $[\alpha]$. This establishes path-connectedness.

If β is a closed path in E based at $\tilde{e}$, by uniqueness of liftings

$$\beta = \widetilde{p \circ \beta}$$

so that

$$[p \circ \beta] = [p \circ (\widetilde{p\beta})] = [\widetilde{p\beta}(1)] = \tilde{e}$$

which is the constant path. Hence E is simply connected. This completes the proof. □

We can now state:

Theorem 5.29 (Fundamental Theorem of Covering Spaces) *Let X be connected, locally path-connected and semilocally simply connected. If H is a subgroup of $\pi(X, b)$, then there is a covering*

$$p_H : (X_H, b_H) \to (X, b)$$

unique up to equivalence of coverings, such that

$$H = p_{H*}\pi(X_H, b_H).$$

This correspondence is one to one. Moreover, if H is normal in $\pi(X, b)$, then X_H is a Galois covering and conversely.

Proof Let $p : E \to X$ be a universal covering of X, guaranteed by Theorem 5.28. Let $G = Aut(E/X, p)$ be the group of covering transformations. Since

$$G \simeq \pi(X, b)$$

by Theorem 5.27, we may take $X_H = E/H$ and p_H the map induced by p. This space has the required property.

We can apply this to study complex manifolds of dimension one which are called **Riemann surfaces**. Roughly speaking, it is a topological space X for which every point has a neighborhood homeomorphic to the unit disk in $\mathbb{C}$.

The simplest example is of course $\mathbb{C}$ itself. Another example is the projective line over $\mathbb{C}$, $\mathbb{P}^1(\mathbb{C})$ called the **Riemann sphere**. This is the one point compactification

of $\mathbb{C}$. A third example is given by the upper half-plane H introduced in an earlier section.

A famous theorem of complex analysis states that any simply connected Riemann surface is homeomorphic to one of these three spaces. This theorem is called **the uniformization theorem**.

This allows us to classify all compact connected Riemann surfaces. As 2-manifolds, we can show they are orientable and we can speak of their genus via the classification of surfaces. A Riemann surface of genus zero is homeomorphic to the Riemann sphere. If it is of genus one, it is homeomorphic to $\mathbb{C}/L$ for some lattice L in $\mathbb{C}$ and we call it an **elliptic curve**. If it has genus ≥ 2, one can show it must be of the form H/Γ for some discrete subgroup $\Gamma \subseteq PSL_2(\mathbb{R})$. Moreover, every such discrete group gives rise to a Riemann surface. Every curve defined by a polynomial equation

$$f(x, y) = 0 \quad f \in \mathbb{C}[x, y]$$

determines a compact connected Riemann surface M where we identify the $\mathbb{C}$-points of the curve with points of M. Conversely, every Riemann surface determines a plane algebraic curve defined over the complex numbers. (We are assuming here that our curves have no singularities.)

Exercises

1. Prove that
$$X = \cup_{n\geq 1}\{(x, y) : \ (x - 1/n)^2 + y^2 = 1/n^2\}$$
is not semilocally simply connected at $(0, 0)$.
2. In the proof of Theorem 5.28, show that if $[\gamma] \in [U, \alpha]$, then $[U, \gamma] = [U, \alpha]$.
3. Prove that the intersection of two elements of the form $[U, \alpha]$ is again a union of sets of this form.
4. In the proof of Theorem 5.28, check that p_α^{-1} is continuous.

5.10 Suggestions for Further Reading

Here is a brief guide to some books in algebraic topology. It is by no means exhaustive, but a cursory glance ranging from elementary texts to the more advanced ones. I have ranked the books in what I perceive as their level of difficulty.

T.W. Gamelin and R.E. Greene, *Introduction to Topology*, Saunders Series, 1983.

This is a beautiful and brief introduction to the subject. One of the nice features is that it treats the Borsuk–Ulam theorem without introducing homology or cohomology. It is highly recommended.

C. Kosniowski, *A First Course in Algebraic Topology*, Cambridge University Press, 1980.

This has the features of the previous book and more. A full semester course can be based on the book since the chapters are short. The emphasis is on fundamental groups and covering spaces though there is a brief chapter at the end on singular homology. This is highly recommended for the serious student of algebraic topology.

K. Jänich, *Topology*, Undergraduate Texts in Mathematics, Springer-Verlag, 1984.

This is a very informal introduction to the subject with more emphasis on the intuitive aspects. Its drawback is that in some cases there are no proofs and the treatment is sloppy. But its colloquial style makes up for it. It is recommended reading on the bus or for bedtime.

W. Massey, *Algebraic Topology, An Introduction*, 1967.

This is a standard text for the subject. But it is lengthy and it is probably impossible to cover all of it in a semester. Homology and cohomology are not discussed at all, nor is higher homotopy theory.

W. Fulton, *Algebraic Topology, A First Course*, Graduate Texts in Mathematics, Springer-Verlag, 1995.

This is definitely an excellent text. I would have preferred an alternate arrangement of contents. Covering spaces and fundamental groups are treated in the middle of the book after a discussion of homology and cohomology. The beautiful aspect of the book is its discussion of Riemann surfaces seldom found in introductory texts in algebraic topology.

M. Greenberg, *Lectures on Algebraic Topology*, Benjamin, 1967.

This book was later reprinted as *Algebraic Topology, A first course* by M. Greenberg and J. Harper. The first part, consisting of about 30 pages on elementary homotopy theory, is very accessible and can be covered within a month.

M. Armstrong, *Basic Topology*, Undergraduate Texts in Mathematics, Springer-Verlag, 1983.

This book is highly recommended with a gentle introduction to simplicial homology. The book ends with a proof of the Lefschetz fixed point formula and a discussion of knot theory.

J. Dieudonné, *A History of Algebraic and Differential Topology*, 1900-1960, Birkhäuser, 1989.

This is a voluminous text full of interesting material. Reading it linearly is probably not a good idea. Part 3 of Chap. 1 of this book is relevant to our chapter. One gains a historical perspective on the evolution of these concepts by reading it. But definitely, this book is not bedtime reading. It is a serious text.

G. Bredon, *Topology and Geometry*, Graduate Texts in Mathematics, Springer-Verlag, 1993.

This is an excellent compendium with emphasis on manifold theory. Chapter 3 covers the theory of the fundamental group and covering spaces. It is rich in examples, especially to the study of topology of Lie groups.

J. Rotman, *An Introduction to Algebraic Topology*, Graduate Texts in Mathematics, Springer-Verlag, 1988.

Rotman has a strong algebraic flavor, and this appeals to many. Surprisingly, he does the theory of covering spaces near the end of the book. Chapters 1–3, Chaps. 10 and 11 are relevant to our chapter. The intuitive style in some places of the book makes for pleasant reading.

H. Sato, *Algebraic Topology, An Intuitive Approach*, Translations of Mathematical Monographs, Vol. 183, American Math. Society, 1996.

Despite its title, this is not an easy book to read. Its merit is a painless introduction to fiber bundles and spectral sequences.

M. Waldschmidt, et al., *From Number Theory to Physics*, Springer-Verlag, 1992.

This is a collection of articles. The article by E. Reyssat entitled "Galois theory for coverings and Riemann surfaces" is excellent and should be read by every student of number theory. I would also recommend the article by J-B. Bost, entitled "Introduction to Compact Riemann Surfaces, Jacobians and Abelian Varieties."

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